



NEPHROLOGY · DIALYSIS · INTERNAL MEDICINE

SK Nefro Báza **1** Compendium

*Abstracts and summaries of specialist
articles from nefro.polasci.net · English
edition*

April 25, 2026 to October 2, 2026 · 387 articles

Ľubomír Polaščin, MD

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Bratislava 2026

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Scope: abstracts and summaries of 387 specialist articles in 16 chapters and an appendix, with chapter syntheses. Full text: SK Nefro Báza 1, English edition (2026).

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Information on drugs, indications, dosing, and reimbursement reflects the situation as of the publication date of each article and refers primarily to Slovakia; it may differ in other countries and at the time of reading. It should be verified in primary sources before use, in particular in the current SmPC or prescribing information, in the database of the national medicines agency (in Slovakia, the State Institute for Drug Control, ŠÚKL), and in the guidelines of professional societies.

Online version of the articles (Slovak): nefro.polasci.net

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Introduction to the compendium

The full edition of *SK Nefro Báza 1* contains 387 specialist articles from the nefro.polascin.net portal on more than 1,700 pages. It is a collection for study and reference. In the outpatient clinic, on ward rounds, or before a seminar, however, a physician more often needs to find out quickly what an article showed, how strong the evidence is, and what it means for a particular patient.

The compendium was created for this purpose. Each article is condensed into an abstract and a short summary with the key figures, the limits of the evidence, and a practical conclusion. Each chapter opens with a synthesis that links the individual articles into a coherent picture and refers to them by number. The volume has thus shrunk from more than 1,700 pages to roughly one seventh, while the essential data have been preserved.

Condensation has its price. The compendium does not reproduce reference lists, tables, figures, or detailed reasoning. It is a map, not a substitute for the full text. For clinical decisions, citation, or teaching, the full version of the article in *SK Nefro Báza 1* or on the portal, together with the primary sources listed there, should be consulted.

The texts reflect the state of knowledge as of the publication date of each article, that is, between April 25, 2026 and October 2, 2026. Summaries of product characteristics, reimbursement conditions, and the recommendations of professional societies change continuously, so they should be verified in current primary sources before being applied in practice.

This English edition is a translation of the Slovak compendium. The articles were written for physicians practicing in Slovakia, so references to Slovak legislation, institutions, and reimbursement rules have been kept. The online versions linked from each article are in Slovak; the full English text is available in the English edition of *SK Nefro Báza 1*.

Bratislava, October 4, 2026

Lubomír Polaščin, MD

How to use the compendium

Organization. The compendium follows the structure of the full edition: 16 thematic chapters and an appendix with cheat sheets and algorithms. Article numbers (for example, 3.12) are identical to those in the English edition of *SK Nefro Báza 1*, so the full version can be found immediately. Appendix articles are numbered A.1 to A.11.

Key messages of the chapter. Each chapter begins with a synthesis of about two pages. Numbers in parentheses, for example (3.12), refer to the articles on which the individual statements are based.

Abstract. A single paragraph that states the question, the type of evidence, the main finding, and its relevance for practice.

Summary. Four to seven points with the most important figures, limitations, and a practical recommendation. Short articles and cheat sheets have fewer points.

Strength of evidence. The texts distinguish randomized trials and meta-analyses from observational data, conference abstracts, preprints, and hypotheses. Where a point reports an association, it is not evidence of causation.

Terminology and units. The abbreviations CKD, AKI, KRT, eGFR, and others are explained in the list of abbreviations at the end. Values are given in SI units with a decimal point; spelling follows US English. Numerical ranges are written with “to”.

Links. In the electronic editions, a number in parentheses in a synthesis links directly to the corresponding article, and the “online version” link leads to the full Slovak text on the portal.

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CHAPTER 1

Chronic kidney disease: diagnosis, risk, and kidney protection

Estimating and measuring glomerular filtration rate, albuminuria, risk stratification, and layered kidney protection. The chapter follows how the treatment of chronic kidney disease is moving from a single drug to the concurrent use of several pillars. Number of articles: 29.

Key messages of the chapter

The assessment of chronic kidney disease is shifting from staging to individualized risk estimation. KDIGO 2024 builds on the four-variable KFRE equation (age, sex, eGFR, UACR): a 5-year risk of kidney failure of 3% to 5% is the criterion for referral to a nephrologist, and a 2-year risk above 40% signals the need to plan kidney replacement therapy (1.3, 1.28). Without albuminuria, stratification is inadequate, so UACR belongs to the standard workup (1.3, 1.27). A preserved glomerular filtration rate does not rule out kidney damage; CKD can be diagnosed even at an eGFR ≥ 60 mL/min/1.73 m² if UACR ≥ 30 mg/g persists (1.22). The eGFR estimate itself depends on the equation and on body composition: compared with CKD-EPI 2021, EKFC moves some patients with diabetes from category G1 to G2 (1.15), and in the population-based SHIP cohort, higher fat mass was associated with lower eGFR (1.25). For borderline decisions, the combined equation with cystatin C helps (1.3, 1.15).

Observational data reveal a large pool of unaddressed risk. In REVEAL-CKD and in a conference abstract from the TriNetX database, a diagnostic code for CKD was missing in 68.4% to 68.7% of patients with hypertension or cardiovascular disease without diabetes (1.27). In a primary care poster, almost 8% of never-referred patients with stage 3 to 5 disease had a significant risk according to KFRE (1.6). The dynamics of proteinuria add prognostic information: a 30% decrease in UPCR or UACR was associated with a lower risk of kidney replacement therapy, although at an eGFR below 15 mL/min/1.73 m², UACR is preferable (1.14). Time in the target UACR range (1.7) and low fasting urine osmolality, which was only partially replicated (1.20), remain research markers.

The strongest evidence in this chapter comes from an individual participant data meta-analysis of 46 randomized trials: a 5 mm Hg reduction in systolic BP lowered major cardiovascular events by 9% to 10%, equally in CKD including stages 4 to 5, regardless of drug class (1.1). Kidney protection rests on layering RAS blockade, SGLT2 inhibitors, finerenone, and GLP-1 receptor agonists; each class has outcome RCTs, but no randomized trial has demonstrated the benefit of a universal four-drug combination, and the gain of 5.5 years free of kidney events is a modeled estimate (1.19, 1.24). For finerenone, a pooled analysis of 14,574 participants showed a 24% reduction in kidney failure or progression regardless of diabetes, and the FIND-CKD trial in nondiabetic CKD showed a 23% reduction in the composite endpoint (1.11). The hypertensive nephropathy subgroup had an HR of 0.61 with intervals close to no effect, outside the European indication (1.29). The phase 2 CONFIDENCE trial showed a 52% decrease in UACR with simultaneous initiation of finerenone and empagliflozin, of which the early change in BP explained less than 10%, but hard outcomes are lacking (1.18). The observational rise in eGFR after finerenone is only a signal (1.10).

Other directions have a lower level of evidence. A retrospective cohort linked protein intake below 1.0 g/kg/day with a lower risk of the composite outcome (HR 0.77) and of dialysis initiation (HR 0.65), but it does not demonstrate causality (1.9). The gut-kidney axis links TMAO and indoxyl sulfate to progression, cardiovascular events, and cognitive impairment only by association (1.12, 1.13); neither regression of fibrosis after TMAO inhibition in a mouse model (1.21) nor a slower eGFR de-

cline with lubiprostone in a small phase II study (1.4) justifies clinical use. Direct anti-inflammatory strategies have failed in RCTs or caused adverse effects, so an elevated hsCRP is a prompt to search for the cause, not an indication for treatment (1.26). Frequent cognitive impairment (1.23) and frailty (1.17) also require assessment of functional reserve.

This yields a framework for the outpatient clinic. In at-risk individuals with diabetes, hypertension, cardiovascular disease, or obesity, or aged over 60 years, measure both eGFR and UACR and record the diagnosis (1.2, 1.27). Refer according to KFRE, trend, and albuminuria, not only according to an eGFR below 60 mL/min/1.73 m² (1.28, 1.6). Lower BP without delay even in advanced stages (1.1), layer kidney protection according to indication, for example an SGLT2 inhibitor at an eGFR ≥ 20 mL/min/1.73 m², and follow patients after AKI in a structured way (1.24). In type 2 diabetes with persistent albuminuria on maximal RAS blockade, add a nonsteroidal MRA with a predefined potassium monitoring plan because of the risk of hyperkalemia (1.8). In the JOINT-KD study, collaboration with the general practitioner did not delay dialysis but was associated with a lower risk of hospitalization for infection (HR 0.36) (1.16); data from multidisciplinary programs come from the organizations themselves, and the term remission should be used with caution (1.5).

1.1 Blood pressure lowering in patients with chronic kidney disease: a meta-analysis

Published May 8, 2026 · [online version \(Slovak\)](#)

Abstract. An individual participant data meta-analysis of 46 randomized trials (285,124 participants, 20.7% with CKD, median follow-up 4.4 years), published in The Lancet in April 2026, examined whether blood pressure lowering confers the same cardiovascular benefit in patients with CKD as in those without it. Each 5 mm Hg reduction in systolic blood pressure lowered the risk of major cardiovascular events by approximately 9% to 10% (HR 0.91 with CKD, 0.90 without CKD), including in stages 4 to 5 and at blood pressure below 120/70 mm Hg. Treatment should not be deferred even in advanced CKD.

Summary

- The effect was the same in patients with CKD (HR 0.91; 95% CI, 0.87 to 0.94) and without CKD (HR 0.90; 95% CI, 0.88 to 0.93), with no heterogeneity by CKD stage, including stages 4 to 5, or by proteinuria; the benefit persisted even at a baseline blood pressure below 120/70 mm Hg.
- A network meta-analysis showed a similar relative effect across the main antihypertensive drug classes (ACE inhibitors, angiotensin receptor blockers, calcium channel blockers, diuretics, and beta-blockers), indicating that the benefit derives from blood pressure reduction itself rather than from the drug class.
- The exception was the combination of CKD and diabetes, with an HR of 0.96 (95% CI, 0.90 to 1.02) versus 0.88 (95% CI, 0.84 to 0.93) in CKD without diabetes (p for interaction = 0.044); according to the authors, this signal requires further confirmation.
- In practice, blood pressure should be lowered in CKD regardless of stage, baseline blood pressure, or drug class, with an individualized approach in CKD with diabetes; the authors emphasize that treatment should not be deferred even in stages 4 to 5.

1.2 Chronic kidney disease: a silent problem worth detecting early

Published May 13, 2026 · [online version \(Slovak\)](#)

Abstract. This educational overview for patients points out that chronic kidney disease is painless in its early stages and that many patients see a physician only once kidney damage is severe. People with diabetes, hypertension, a family history of kidney disease, obesity, or cardiovascular disease, and those over 60 years of

age, are at higher risk. A urine test, serum creatinine, and eGFR are sufficient for early detection. Early diagnosis makes it possible to slow or halt progression and also protects the cardiovascular system.

Summary

- Risk groups include patients with diabetes mellitus, hypertension, a family history of kidney disease, obesity, or cardiovascular disease, and people over 60 years of age; edema, fatigue, changes in urination, foamy urine, pruritus, cramps, and nausea appear only in later stages.
- Basic screening consists of three measures: a urine test for protein or blood, the blood creatinine level, and the estimated glomerular filtration rate (eGFR).
- Repeated proteinuria or a reduced eGFR warrants evaluation by a nephrologist; the article emphasizes the importance of preventive checkups with the general practitioner (GP).

1.3 KDIGO 2024: from CKD stages to personalized risk estimation

Published May 15, 2026 · [online version \(Slovak\)](#)

Abstract. The KDIGO 2024 guideline shifts the assessment of chronic kidney disease from staging by eGFR and albuminuria alone toward individual risk estimation using externally validated prediction models. The key tool is the four-variable Kidney Failure Risk Equation (KFRE) (age, sex, eGFR, and UACR), which estimates the 2-year and 5-year risk of kidney failure. A 5-year risk of 3% to 5% may prompt referral to a nephrologist, and a 2-year risk above 40% may prompt planning for kidney replacement therapy (KRT). The calculator, however, does not replace clinical judgment.

Summary

- KDIGO 2024 recommends the KFRE particularly in CKD G3 to G5 (eGFR below 60 mL/min/1.73 m²); the four-variable version uses age, sex, eGFR, and UACR, data available from a routine evaluation.
- Thresholds: a 5-year kidney failure risk of approximately 3% to 5% as a criterion for nephrology referral, a 2-year risk above 10% as grounds for more intensive multidisciplinary care, and a 2-year risk above 40% as a signal to plan KRT.
- Planning includes education on hemodialysis, peritoneal dialysis, conservative management, and transplantation; preparation of vascular access or a peritoneal catheter; transplant eligibility evaluation; and, where appropriate, placement on the waiting list.
- For earlier stages (G1 to G3), the CKD-PC models are cited for estimating a significant eGFR decline, for example by 40%; for IgA nephropathy, the International IgAN Prediction Tool; and for ADPKD, models based on total kidney volume (TKV) and the Mayo classification.
- In adults, equations without a race coefficient (CKD-EPI 2021, EKFC) are preferred, and for decisions requiring greater accuracy, such as drug dosing, kidney donation, or low or high muscle mass, the combined eGFR_{cr-cys}; in individuals up to 25 years of age, the CKiD U25 equations are also used.
- The author emphasizes that risk stratification is inadequate without albuminuria, so UACR should be a standard test; the outpatient clinic should record creatinine, eGFR and its trajectory, blood pressure, comorbidities, and the underlying diagnosis, and interpret the calculator result in the clinical context.

1.4 A constipation drug may slow the decline in kidney function in chronic kidney disease

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. LUBI-CKD, a placebo-controlled, multicenter phase II trial from Japan (150 patients with moderate CKD, 24 weeks), investigated lubiprostone, a drug for chronic constipation. At both 8 µg and 16 µg, eGFR declined more slowly than with placebo, with a dose-dependent effect and the strongest signal at 16 µg. Ac-

cording to the analysis, the mechanism appears related to changes in the microbiome and spermidine production rather than to a reduction in uremic toxins. This is a signal, not definitive evidence.

Summary

- The LUBI-CKD TRIAL was conducted at nine institutions in Japan and enrolled 150 patients with moderate CKD who received lubiprostone or placebo; kidney function was assessed by eGFR over 24 weeks.
- Lubiprostone at both 8 µg and 16 µg slowed eGFR decline compared with placebo; the effect appeared dose-dependent, and the 16 µg dose showed the strongest signal of function preservation.
- The original hypothesis of uremic toxin reduction was largely not confirmed; the analysis showed changes in the gut microbiome and in bacterial polyamine synthesis pathways, with increased production of spermidine, which is associated with healthier mitochondrial function, although this remains only a possible explanation of the mechanism.
- An advantage is that lubiprostone is already an approved drug, so if the effect is confirmed, the path to clinical use could be shorter than for new molecules.
- Limitations include the small number of patients and the short follow-up; phase III trials are needed to confirm efficacy and safety, and the team is searching for biomarkers to identify the patients most likely to benefit.
- The author emphasizes that the finding does not justify off-label use of lubiprostone to slow CKD, but it supports further study of the gut-kidney axis as a potential therapeutic direction.

1.5 From slowing progression to remission: a quiet revolution in nephrology

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. Based on a Becker's Hospital Review report, this commentary describes the model of the US organization Monogram Health, which views chronic kidney disease as part of the cardiovascular-kidney-metabolic syndrome and combines home-based care, evidence-based algorithms, and a multidisciplinary team. According to the reported data, the program achieved a 35% reduction in hospitalizations, a 44% reduction in readmissions, and more planned dialysis starts. In diabetic kidney disease, combining four drug pillars opens the prospect of functional remission in some patients, although this term should be used with caution.

Summary

- Monogram Health delivers care directly in the patient's home, where the team sees actual medication use, diet, social circumstances, and barriers to adherence.
- The algorithms are meant to support, not replace, the physician; in resistant hypertension, they prompt verification of blood pressure measurement, adherence, drug availability, sodium intake, drug interactions, and secondary causes, which in the case described revealed hyperaldosteronism.
- According to the reported data, the program reduced hospitalizations by 35% and readmissions by 44% and improved the transition to dialysis, with fewer urgent starts via a central venous catheter (CVC) and more planned starts, permanent vascular accesses, and home dialysis.
- In diabetic kidney disease, the approach combines SGLT2 inhibitors, GLP-1 receptor agonists, ACE inhibitors or angiotensin receptor blockers, and nonsteroidal mineralocorticoid receptor antagonists such as finerenone, which act on glomerular hemodynamics, inflammation, fibrosis, albuminuria, and cardiovascular risk.
- The author cautions that not every patient will achieve an improvement in kidney function and that not every stabilization represents reversal of the disease; more ambitious goals are realistic mainly in appropriately selected patients at earlier stages.
- The program's results are presented as data reported by the organization; the practical message is early identification of at-risk patients, use of data, adherence monitoring, and multidisciplinary collaboration.

1.6 Proactive nephrology consultations may reduce the risk of adverse outcomes in patients with CKD

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. An observational analysis from the University of California, San Francisco, presented as a poster at the NKF Spring Clinical Meetings 2026, evaluated patients with CKD in primary care who had never been referred to a nephrologist. Of 5,270 patients, 58.6% had intermediate to high cardiovascular risk according to the PREVENT equations, and almost 8% of patients with CKD stages 3 to 5 had a significant 2-year risk of kidney failure according to the KFRE. The results support a proactive, targeted nephrology consultation service guided by risk scores.

Summary

- Eligible patients were aged 30 to 79 years, had evidence of CKD, were actively followed in safety-net primary care from July 1, 2021, to June 20, 2025, and had no prior nephrology referral; cardiovascular risk was estimated with the PREVENT equations and kidney risk with the KFRE.
- The cohort of 5,270 patients (53% men, ethnically diverse) had diabetes in 50.1%, lipid-lowering therapy in 58.6%, and antihypertensive therapy in 76.0%; 3,086 patients (58.6%) had a PREVENT score in the intermediate to high risk category.
- Of 1,383 patients with stages 3 to 5, 4.4% had a 2-year KFRE risk of 3.0% to 9.9%, and a further 3.8% had a risk above 10%.
- This is an observational risk assessment that does not prove that consultation itself reduces mortality or prevents kidney failure; it does, however, show the size of the group with unaddressed risk.
- The author recommends actively identifying at-risk patients from laboratory data and PREVENT and KFRE scores rather than passively waiting for referral; early consultation can optimize blood pressure, diabetes management, albuminuria, SGLT2 inhibitor use, RAS blockade, lipids, drug dosing, and AKI prevention.

1.7 UACR “time in target range” may help in the management of chronic kidney disease

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. An analysis of the prospective All of Us cohort (306,645 adults with at least 3 UACR measurements) evaluated time in target range (TTR), the proportion of time during which UACR remains within the target range. In the CKD group, TTR below 30 mg/g was associated with a reduced risk of MACE ($P = 0.002$ with the traditional calculation and 0.007 with the Rosendaal method), whereas mean UACR did not reliably capture risk. Greater adherence to target predicted a 15% to 18% reduction in lifetime MACE risk. Guidelines remain unchanged for now.

Summary

- From the All of Us cohort (849,000 adults in the United States), 306,645 participants with at least 3 UACR measurements were selected; mean age was 59 years, 62% were women, 8% had diabetes, 22% hypertension, and 10% CKD, and mean UACR was 117 mg/g.
- In multivariable analysis, TTR below 30 mg/g in CKD was associated with a reduced risk of MACE (myocardial infarction, stroke, or death), with $P = 0.002$ and 0.007 depending on the calculation method, and a similar trend was seen without CKD ($P = 0.001$); mean UACR had a hazard ratio close to 1.00.
- NKF President Kirk Campbell sees value in the approach for defining endpoints in future trials; guidelines are unlikely to change in the short term, and TTR could eventually be implemented in electronic health records (EHR) for laboratory values with a predefined target range.

1.8 When and how to add a nonsteroidal MRA (nsMRA) in a patient with T2D and CKD with persistent albuminuria

Published May 30, 2026 · [online version \(Slovak\)](#)

Abstract. This practical algorithm, based on educational material, addresses a patient with type 2 diabetes and CKD who has persistent albuminuria despite maximally tolerated RAS inhibition. In this situation, the logical next step is to add a nonsteroidal mineralocorticoid receptor antagonist (nsMRA), such as finerenone, to protect both the kidneys and the heart. The article does not specify thresholds; the key to safety is a predefined plan for monitoring potassium and kidney parameters according to the local protocol.

Summary

- The indication framework consists of T2D and CKD, persistent albuminuria, and already maximally tolerated therapy with an ACE inhibitor or ARB; other pillars, such as SGLT2 inhibition in suitable patients, should be addressed according to the standard of care.
- Before initiation, creatinine, eGFR, and K⁺ should be measured, along with acid-base status if the local protocol requires it, and the risk of hyperkalemia should be assessed: higher baseline potassium, more advanced renal insufficiency, potassium-raising drugs, and conditions carrying a risk of dehydration.
- Steps: confirm persistent albuminuria by the trend of repeated measurements, ensure foundational therapy, initiate the nsMRA with a clear monitoring plan, and explain to the patient the need for K⁺ monitoring.
- After initiation, K⁺ and kidney parameters in particular are checked shortly after starting and then according to risk; the article does not specify exact timing and refers to the local protocol.
- If K⁺ rises, local rules apply (dose reduction, temporary interruption, adjustment of concomitant medication, management of precipitating factors), and the clinic should have a plan B ready.
- A model patient (52 years old) on empagliflozin and ramipril with persistent proteinuria receives semaglutide for a metabolic target and finerenone, so the nsMRA is an add-on to the overall strategy rather than an isolated intervention.

1.9 Moderate protein restriction may improve prognosis in chronic kidney disease

Published June 5, 2026 · [online version \(Slovak\)](#)

Abstract. A retrospective cohort study in JAMA Network Open evaluated 1,441 patients with nondialysis CKD stages III and IV whose protein intake was estimated from 24-hour urinary nitrogen excretion (nDPI). In a matched cohort of 530 patients, an nDPI below 1.0 g/kg/day was associated with a 23% lower risk of the composite outcome (HR 0.77) and a 35% lower risk of dialysis initiation (HR 0.65), without deterioration in nutritional parameters. The study does not prove causality but supports individualized moderate restriction.

Summary

- Adults with an eGFR of 15 to 60 mL/min/1.73 m² who were not on dialysis and had not received a transplant were included (mean age 67.2 years, 35.2% women); median nDPI was 1.18 g/kg/day, and after propensity score matching, 265 patients had an intake below 1.0 and 265 patients an intake of 1.0 g/kg/day or more.
- The primary outcome comprised an eGFR decline of at least 50% confirmed after at least 28 days, dialysis initiation, or death; median follow-up was 67 months, with a maximum of 15 years.
- Lower protein intake had an HR of 0.77 (95% CI, 0.62 to 0.97) for the composite outcome and an HR of 0.65 (95% CI, 0.42 to 0.99) for dialysis initiation; the association persisted after adjustment for age, sex, diabetes, BMI, eGFR, serum albumin, and UACR.
- eGFR declined in both groups, numerically more slowly with lower intake, and nutritional parameters did not differ significantly between the groups.
- Limitations include the retrospective design without proof of causality, possible selection bias, and missing data on dietary counseling, adherence, and the proportion of animal and plant protein.

- The author recommends controlled, individualized moderate restriction below 1.0 g/kg/day with nutritional assessment, particularly in older patients and those at risk of protein-energy malnutrition, rather than arbitrary dietary restriction.

1.10 Finerenone: a kidney-protective effect in CKD stages 3 to 4 regardless of diabetes

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. A Physicians Weekly report describes a prospective evaluation of 180 outpatients in Iraq with CKD stages 3 to 4 (eGFR 20 to 60 mL/min, UACR above 30 mg/g), 78% of whom had diabetes. After 1 year of finerenone therapy, eGFR rose from approximately 32 to 47 mL/min and UACR fell from 542 to 180 mg/g, with similar findings in patients without diabetes and no rise in potassium. An improvement in eGFR is atypical for this drug class, so the results should be viewed as a signal.

Summary

- In the overall group, eGFR rose from 32 ± 10 to 47 ± 15 mL/min and UACR fell from 542 ± 400 to 180 ± 170 mg/g; in patients without diabetes, eGFR rose from 33 ± 10 to 50 ± 16 mL/min and UACR fell from 592 ± 431 to 203 ± 197 mg/g.
- In the subgroup with an eGFR of 20 to 25 mL/min (36% of patients), eGFR rose from 21 ± 2 to 36 ± 8 mL/min, and potassium was 4.8 ± 0.7 mmol/L at baseline and 4.8 ± 0.6 mmol/L after 1 year.
- The report does not provide the full design or statistics, and treatment was evidently outside the approved indication; the author recommends viewing the results as a signal and, when initiating therapy, following local guidelines and monitoring potassium.

1.11 Finerenone: potential as foundational therapy in CKD without diabetes and in glomerular diseases

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. A ScienceDaily popular science summary presents the results of several large finerenone trials. In a pooled analysis of 14,574 participants with diabetic and nondiabetic CKD (The Lancet), finerenone reduced the risk of kidney failure or CKD progression by 24%, of hospitalization for heart failure or cardiovascular death by 20%, and of death from any cause by 12%. According to the authors, the benefit was consistent regardless of diabetes status. Hyperkalemia was more frequent than with placebo, so monitoring is essential.

Summary

- In the FIND-CKD trial (1,584 participants with nondiabetic CKD, 24 countries), finerenone added to standard therapy slowed the decline in kidney function and reduced the composite endpoint by 23%.
- In glomerular diseases (analysis in JAMA), finerenone reduced the risk of kidney failure or CKD progression by 26% compared with placebo and reduced albuminuria at 12 months by 42%.
- Treatment discontinuation and hospitalization for hyperkalemia were infrequent; because this is a summary without full protocols, the author recommends following local guidelines and rigorously monitoring potassium and kidney parameters.

1.12 Indoxyl sulfate and cognitive impairment in chronic kidney disease: what we know about tryptophan derivatives

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. A cross-sectional analysis of a French cohort of 2,389 adults with CKD stages 2 to 5 who were not on dialysis and had not received a transplant, summarized on Medscape, examined the relationship of tryptophan-derived uremic solutes to cognitive impairment assessed by the MMSE. Cognitive impairment was present in 35.9% of patients. Higher indoxyl sulfate levels were independently associated with cognitive

impairment (adjusted OR 1.11), whereas kynurenine and indole-3-acetate were not. The finding is a signal for research on biomarkers and risk stratification, not a therapeutic recommendation.

Summary

- Free serum toxin levels were measured by a validated LC-MS/MS method in a cohort enrolled from 2013 to 2016; cognitive impairment, defined as an MMSE score $\leq 26/30$, was present in 858 patients (35.9%).
- Higher indoxyl sulfate was associated with a greater likelihood of impairment (adjusted OR 1.11; $P = 0.03$), and the highest versus the lowest group had an adjusted OR of 1.43 ($P = 0.02$); kynurenine and indole-3-acetate showed no significant association.
- Limitations: the cross-sectional design precludes conclusions about causality or about whether lowering indoxyl sulfate improves cognition, the MMSE may not capture executive function, and diet and nutrition were not taken into account.
- If the association is confirmed in prospective studies, indoxyl sulfate could be used for risk stratification and for planning cognitive monitoring; for now, this remains a hypothesis.

1.13 Trimethylamine N-oxide (TMAO): a gut metabolite as a uremic toxin in chronic kidney disease

Published July 3, 2026 · [online version \(Slovak\)](#)

Abstract. This review summarizes current knowledge on trimethylamine N-oxide (TMAO), which is formed from choline, L-carnitine, and betaine through the action of the gut microbiome and hepatic oxidation by the enzyme FMO3 and is excreted predominantly by the kidneys. In CKD, it accumulates because of reduced clearance and gut dysbiosis and is associated with disease progression, cardiovascular events, and mortality. The evidence is predominantly associative and mechanistic, and no intervention targeting TMAO has yet shown benefit on hard outcomes.

Summary

- TMAO precursors come mainly from red meat, egg yolk, dairy products, and fish; gut bacteria convert them to trimethylamine, which the liver oxidizes to TMAO, and because the kidneys are the main route of elimination, a decline in filtration translates directly into higher levels.
- In CKD, the gut-kidney axis comes into play: reduced excretion and dysbiosis with impaired gut barrier function reinforce each other, as with indoxyl sulfate and p-cresyl sulfate.
- Experimental and clinical studies link TMAO to renal fibrosis, tubular injury, atherosclerosis, vascular calcification, and myocardial fibrosis, with inflammation and oxidative stress as a common denominator; a direct causal contribution is still being investigated.
- In heart failure, splanchnic congestion disrupts the gut barrier, which is why TMAO has been proposed as a potential therapeutic target in cardiorenal syndrome.
- Six months of L-carnitine administration markedly increased TMAO in older women; fish are a source of both omega-3 fatty acids and TMAO, and their overall effect in CKD remains unresolved, while microbiome modulation and inhibitors such as DMB are being studied experimentally.
- Targeted interventions against TMAO are not part of standard care; in practice, management remains appropriate adjustment of protein intake in patients not on dialysis, a balanced diet, and comprehensive cardiovascular risk control.

1.14 Urinary protein vs. albumin for assessing kidney failure risk in CKD: does UPCR have the same predictive value as UACR?

Published July 3, 2026 · [online version \(Slovak\)](#)

Abstract. A prospective observational analysis of Japanese patients with predialysis CKD, summarized on ReachMD, compared the prognostic value of total proteinuria (UPCR) and albuminuria (UACR) for kidney failure with replacement therapy (KFRT). A 30% reduction was associated with a lower risk of KFRT for both

UPCR (HR 0.52) and UACR (HR 0.58), and the models showed similar discrimination and calibration. At an eGFR below 15 mL/min/1.73 m², the association for UPCR was weaker, so UACR is preferable in that setting.

Summary

- UPCR also captures nonalbumin components of proteinuria, and the proportion of albumin in total proteinuria may decrease as kidney function declines; the outcome was initiation of hemodialysis or peritoneal dialysis, or kidney transplantation.
- A 30% reduction in UPCR had an HR of 0.52 (95% CI, 0.38 to 0.71), and a 30% reduction in UACR an HR of 0.58 (95% CI, 0.41 to 0.82); a 30% reduction in UPCR over 1 year in 831 patients had an HR of 0.56 (95% CI, 0.43 to 0.73).
- Models with UPCR and with UACR had approximately the same Harrell C-index and good overall calibration.
- At an eGFR below 15 mL/min/1.73 m², the association of UPCR with KFRT was weaker, which the authors attribute to the lower proportion of albumin, and estimates were uncertain at very low baseline UACR, for example below 30 mg/g creatinine.
- This is an observational analysis without proof of causality, and the comparability applies within this cohort, not as a general substitution for every phenotype and population.
- In practice, a 30% reduction in UPCR or UACR over 1 to 2 years signals a better prognosis; in the common stages, UPCR can serve as an approximately equivalent marker, whereas in advanced CKD it is more reasonable to prefer UACR.

1.15 More accurate eGFR estimation in patients with diabetes: when the equation changes the CKD stage

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. A short commentary in the Journal of Nephrology analyzed a longitudinal cohort of 4,591 Chinese patients with type 2 diabetes (median follow-up approximately 1.01 years) and compared the creatinine-based EKFC and CKD-EPI 2021 equations for assignment to G categories. CKD-EPI assigned 37.44% of patients to G1, whereas EKFC shifted a substantial proportion to G2 (33.09% vs. 25.90%). EKFC may therefore be more conservative in patients with diabetes. The finding supports interpreting eGFR together with albuminuria and trend rather than mechanically.

Summary

- The composite kidney outcome comprised an eGFR decline of more than 40%, doubling of serum creatinine, or kidney failure; the authors examined shifts between G categories (stage migration) and the prognostic value of the equations.
- According to the authors, after the EKFC implementation was refined with the correct exponential factors for Scr/Q ≥ 1 , the distribution across more advanced categories better reflected the clinical reality of a high-risk diabetic population.
- An overestimated eGFR can lead to underestimation of risk, less frequent monitoring, and delayed kidney protection; however, a lower estimate is not automatically more accurate, and a G1 or G2 category without a marker of kidney damage does not by itself establish a diagnosis of CKD.
- Creatinine is affected by muscle mass, diet, age, sex, and medications, so confirmation with cystatin C or a combined equation is valuable in cases of uncertainty, borderline decisions, or dosing of high-risk drugs.
- The source is a short commentary without full methodology, so the statistics, the representation of albuminuria, and confounders cannot be assessed, and it does not follow that any single equation is universally best.
- For the outpatient clinic: know the laboratory's equation, follow the eGFR trend, measure UACR, and avoid decisions based solely on the 90 mL/min/1.73 m² threshold; care includes SGLT2 inhibitors, RAAS blockade, consideration of finerenone, and timely nephrology referral.

1.16 Collaboration between the general practitioner and the nephrologist in CKD G5: lessons from the JOINT-KD study

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. JOINT-KD, a retrospective cohort study from nine Japanese nephrology centers, evaluated whether collaboration between a primary care physician and a nephrologist confers clinical benefit in 570 patients with CKD G5. Collaboration (16.0% of patients) did not delay dialysis initiation (adjusted subdistribution HR 0.89; 95% CI, 0.64 to 1.23) but was associated with a lower risk of hospitalization for infection (HR 0.36; 95% CI, 0.15 to 0.87). This is an observational signal that supports a coordinated, shared care plan.

Summary

- Median follow-up was 1.4 years; 399 patients (70.0%) started dialysis, 11 (1.9%) underwent preemptive transplantation, and 53 (9.3%) died, and the analysis accounted for competing risks.
- Collaboration did not significantly affect CKD-related or cardiovascular hospitalizations but reduced the risk of infection-related hospitalization (adjusted subdistribution HR 0.36; 95% CI, 0.15 to 0.87).
- The author attributes this to early detection of common outpatient infections by the primary care physician, who is more accessible to the patient between nephrology visits; infection-related hospitalization in CKD G5 can accelerate functional decline and lead to urgent dialysis initiation.
- Limitations include the retrospective observational design without proof of causality, possible differences between groups, short follow-up, and a Japanese health care system that differs from the Slovak one; the cited source is a correction of the ethics statement that does not change the main results from 2025.
- Primary care does not replace the nephrologist, who manages preparation for dialysis or transplantation, anemia, mineral and bone disorder, acidosis, hyperkalemia, and volume status; the author recommends a clear division of responsibilities.
- The shared plan should include the current nephrology status, medications, and limits for potassium, blood pressure, and volume; warning signs (fever, dyspnea, confusion, oliguria, edema, hypotension); escalation rules; and prevention, including vaccination.

1.17 Frailty in chronic kidney disease: why monitoring eGFR alone is not enough

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. A longitudinal cohort study from southern Taiwan published in the Journal of Nephrology followed 12-month trajectories of multidimensional frailty in 120 patients with CKD not on dialysis and their relationship to nutritional status, physical activity, and handgrip strength. Detailed numerical results are not available from the accessible extract. Frailty has physical, psychological, and social components, is dynamic, and reaches a prevalence of approximately 70% in some groups with advanced CKD. Patients should be assessed not only by eGFR but also by functional reserve.

Summary

- The study enrolled 120 patients with CKD aged over 20 years from a nephrology outpatient clinic between November 2022 and May 2024; because only an extract is available, the exact changes in individual frailty domains cannot be reliably summarized.
- Frailty is a syndrome of reduced physiological reserve that impairs tolerance of infections, hospitalizations, procedures, and treatment changes; it can worsen or improve, so tracking its course over time is worthwhile.
- Malnutrition in CKD has multiple causes and leads to loss of muscle mass, falls, and infections; dietary recommendations should be safe yet nutritionally adequate so that the patient does not develop protein-energy malnutrition.
- Handgrip strength is a quick, inexpensive, and repeatable marker of sarcopenia and physical reserve that should be interpreted in light of age, sex, hand dominance, and neurologic or orthopedic status, and it is most useful as a trend indicator.

- The author proposes an outpatient checklist: unintentional weight loss, appetite and protein intake, falls and fear of falling, walking and stairs, weekly activity, fatigue, depression, social isolation, and handgrip strength.
- Frailty is not a reason to withhold dialysis but calls for individualized decision-making; in addition to access, predialysis preparation should include nutrition, exercise, rehabilitation, treatment of anemia and acidosis, fall prevention, and psychosocial support.

1.18 Finerenone with empagliflozin in CKD and type 2 diabetes: does blood pressure explain the reduction in albuminuria?

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. CONFIDENCE, a randomized, double-blind phase 2 trial (800 patients with CKD and type 2 diabetes analyzed), showed that simultaneous initiation of finerenone and empagliflozin reduced UACR at 180 days by approximately 52%, more than finerenone (32%) or empagliflozin (29%) alone. An editorial in Nephrology Dialysis Transplantation suggested that this was mainly an effect of blood pressure lowering; however, a more recent exploratory mediation analysis found that the early change in systolic blood pressure explained less than 10% of the UACR reduction. Long-term clinical benefit remains unproven.

Summary

- Participants had an eGFR of 30 to 90 mL/min/1.73 m², a UACR of 100 to 5,000 mg/g, and RAS blockade (98.4%); they received finerenone 10 or 20 mg, empagliflozin 10 mg, or the combination, with no placebo arm, at a median UACR of 579 mg/g.
- The combination reduced UACR by 29% more than finerenone (ratio 0.71; 95% CI, 0.61 to 0.82) and by 32% more than empagliflozin (ratio 0.68; 95% CI, 0.59 to 0.79); this is a surrogate endpoint at six months, not a reduction in the risk of dialysis or death.
- With the combination, systolic blood pressure fell by approximately 7.4 mm Hg and eGFR by 5.6 mL/min/1.73 m² by day 30, and after discontinuation, blood pressure, UACR, and eGFR partially returned toward baseline; the early eGFR decline may be an expected pharmacodynamic response.
- The mediation analysis in Hypertension was exploratory, used office blood pressure, and relied on a single time point, so a hemodynamic contribution cannot be excluded; synergy in the strict statistical sense was not demonstrated.
- Hyperkalemia was reported in 9.3% with the combination, 11.4% with finerenone, and 3.8% with empagliflozin, and AKI in 5, 3, and 0 patients, respectively; the results do not apply to a baseline potassium above 4.8 mmol/L or to symptomatic heart failure with reduced ejection fraction.
- ADA 2026 states that simultaneous initiation may be considered at a UACR of at least 100 mg/g, an eGFR of 30 to 90 mL/min/1.73 m², and treatment with a RAS inhibitor; the author recommends an individualized approach with monitoring of blood pressure, potassium, and eGFR, at approximately four weeks for finerenone.

1.19 Combination therapy for chronic kidney disease: four pillars, evidence, and open questions

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. An editorial in Nephrology Dialysis Transplantation, with Kaiyu He as first author, describes the shift from supportive care to combination therapy for CKD based on RAS inhibitors (RASi), SGLT2 inhibitors, finerenone, and GLP-1 receptor agonists; the article supplements it with the primary trials. Each class has outcome evidence, for example DAPA-CKD with a 39% relative risk reduction, but an outcome RCT of

the full four-pillar regimen is lacking, and the gain of 5.5 years free of kidney events is only a model-based estimate. Early, individualized layering of therapy is recommended.

Summary

- Outcome trials: CREDENCE (canagliflozin) reduced the relative risk of the primary endpoint by 30%, DAPA-CKD by 39%, EMPA-KIDNEY by 28%, FIDELIO-DKD (finerenone) by 18%, and FLOW (semaglutide) by 24%; populations and endpoints differ, so these percentages cannot be mechanically compared or added together.
- A model for a 50-year-old patient with type 2 diabetes and albuminuria estimated a 5.5-year extension of the kidney event-free period (95% CI, 4.0 to 6.7), assuming additivity and lifelong treatment; this is not an observed result, and the term synergy is too strong in the absence of evidence.
- SGLT2 inhibitors were associated with a lower risk of hyperkalemia and, in a pooled analysis of CREDENCE and DAPA-CKD, with less frequent RASi interruption (HR 0.85; 95% CI, 0.74 to 0.99), but the risk of hyperkalemia is not eliminated.
- FIND-CKD (1,584 patients without diabetes) showed a difference in annual eGFR slope of 0.7 mL/min/1.73 m² and an HR of 0.77 for the kidney-cardiovascular endpoint, with hyperkalemia in 17.0% versus 13.3%; vicadrost and zibotentan have so far reduced only albuminuria and remain investigational agents.
- An early hemodynamic eGFR decline is not automatically kidney injury, but a decline greater than approximately 30%, hypotension, or volume depletion requires evaluation; monitoring of blood pressure, eGFR, potassium, and volume is suggested at approximately 1 to 4 weeks.
- Practical framework: determine the CKD phenotype and cause, optimize foundational therapy, do not combine an ACE inhibitor with an ARB, do not delay an indicated SGLT2 inhibitor, add finerenone according to albuminuria, eGFR, and potassium, and adjust the pace to risk and frailty.

1.20 Fasting urine osmolality and the risk of kidney disease progression in patients with type 2 diabetes

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. A prospective observational study in Nephrology Dialysis Transplantation examined, in two cohorts of patients with type 2 diabetes (SMART2D, 1,711 participants; SURDIAGENE, 1,097), whether lower fasting urine osmolality is associated with kidney disease progression beyond eGFR and UACR. The lowest tertile had a higher risk of the primary composite in SMART2D (aHR 2.94) and higher odds of rapid kidney function decline in SURDIAGENE (aOR 1.84). Replication was only partial; this is a prognostic signal without a validated threshold.

Summary

- The primary outcome was a composite of ESKD or doubling of serum creatinine, and the secondary outcome was rapid kidney function decline (RKFD), defined as an annual eGFR decline of at least 5 mL/min/1.73 m²; mean follow-up was 6.6 and 7.4 years, with 239 and 82 primary events, respectively.
- For the primary outcome, the aHR was 2.94 (95% CI, 1.12 to 7.69) in SMART2D and 1.74 (95% CI, 0.85 to 3.58) in SURDIAGENE, and for RKFD the aOR was 1.47 (95% CI, 0.95 to 2.28) and 1.84 (95% CI, 1.06 to 3.19), respectively; the direction was consistent, but statistical significance was not replicated.
- The association persisted after accounting for copeptin and KIM-1, but this does not prove specific injury to the distal nephron; osmolality is nonspecific and is influenced by nephron loss, vasopressin, fluid intake, glycosuria, and treatment.
- SGLT2 inhibitors alter interpretation through glycosuria, so the prognostic value must be verified in contemporary treated populations, and osmolality must not be a reason to discontinue or initiate this therapy.

- Limitations include the observational design, a single baseline measurement, tertiles as relative categories without a universal threshold, and no evidence of improved discrimination, calibration, or risk reclassification.
- eGFR and UACR remain the foundation; values such as 350 or 400 mOsm/kg H₂O are not a validated threshold, refractometric specific gravity is not an equivalent substitute, and the study does not support changing fluid intake or treatment.

1.21 Targeted inhibition of microbial TMAO production in a CKD model: regression of fibrosis, not clinical evidence

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. An experimental study by DiDonato and coauthors in the Journal of the American Society of Nephrology tested targeted inhibition of microbial TMAO production with fluoromethylcholine (FMC) in CD1 mice only after CKD had developed. Twelve weeks after nephrectomy on a choline diet, measured GFR was 48% lower, and over the following 8 weeks FMC almost completely suppressed TMAO, increased GFR by 39% relative to pretreatment values, and reduced fibrosis by 41%. This represents regression of the phenotype in an animal model, not clinical evidence.

Summary

- FMC is a mechanism-based inhibitor of microbial choline TMA-lyase that limits TMA production in the gut and thereby indirectly hepatic TMAO production, without broad antibiotic suppression of the microbiome.
- Mice after unilateral nephrectomy on a high-fat diet with 1% choline had approximately 15-fold higher TMAO and 3.5-fold greater tubulointerstitial fibrosis after 12 weeks and were then randomized to 8 weeks with or without FMC.
- With FMC, measured GFR was more than 50% higher than in the untreated group, in which TMAO, functional impairment, and fibrosis continued to progress; creatinine, cystatin C, pseudouridine, UACR, and expression of profibrotic genes also decreased.
- Indoxyl sulfate and phenylacetylglycine also decreased, so it cannot be claimed that the entire effect was mediated exclusively by the reduction in TMAO.
- Limitations include a small animal model with an induced phenotype and a high choline load, 8-week treatment, and no data on safety and efficacy in humans or on kidney failure, hospitalizations, or mortality.
- Standard kidney-protective therapy remains unchanged, and TMAO is not a routine therapeutic biomarker; the author advises against arbitrarily eliminating sources of choline, an essential nutrient, because doing so could increase the risk of malnutrition.

1.22 Renal functional reserve: why a normal eGFR does not exclude significant kidney damage

Published July 16, 2026 · [online version \(Slovak\)](#)

Abstract. This review is based on a sponsored educational interview on ReachMD and compares it with KDIGO 2024 and the peer-reviewed literature. It explains that a preserved eGFR does not exclude CKD or significant kidney damage, because the remaining nephrons can maintain total GFR through compensatory hyperfiltration. CKD can be diagnosed even at an eGFR ≥ 60 mL/min/1.73 m² if a marker of damage persists, for example a UACR ≥ 30 mg/g. Testing of renal functional reserve does not yet have a standardized routine role.

Summary

- Renal functional reserve is the difference between basal GFR and GFR after a stimulus, such as a protein load or intravenous amino acids; it lacks a uniform protocol, a reference range, and a validated threshold, and a reduced response cannot be translated into a specific loss of nephrons.

- Compensatory single-nephron hyperfiltration is an effective short-term adaptation, but in the long term, increased intraglomerular pressure and podocyte stress can promote albuminuria and glomerulosclerosis; the claim of maladaptation after loss of more than half of the nephrons does not represent a universal threshold.
- According to KDIGO, CKD is an abnormality of kidney structure or function lasting at least three months; markers include UACR ≥ 30 mg/g (≥ 3 mg/mmol), abnormal urinary sediment or hematuria of renal origin, tubular disorders, histologic or structural abnormalities, and a history of transplantation, whereas a G1 or G2 category alone does not meet the criteria.
- eGFR does not reflect nephron number or histology, and creatinine is affected by muscle mass, diet, and medications; for critical decisions, KDIGO recommends a combined estimate with cystatin C or, where appropriate, measured GFR.
- In glomerular diseases, such as IgA nephropathy or lupus nephritis, eGFR may be preserved despite active findings; biopsy is not a screening test and is indicated when it may change the diagnosis, prognosis, or treatment.
- Practical approach: confirm the finding and its chronicity, assess UACR and sediment, evaluate the trend, refine the filtration estimate if results are discordant, determine the cause, and treat the proven disease rather than a hypothetical loss of reserve; the source does not provide population data.

1.23 Chronic kidney disease also affects the brain: cognitive impairment, vascular damage, and clinical consequences

Published August 2, 2026 · [online version \(Slovak\)](#)

Abstract. Chronic kidney disease increases the risk of stroke, cerebral small vessel disease, and cognitive impairment, which affects adherence as well as decisions about dialysis or transplantation. The review bases its conclusions on meta-analyses and observational studies. The pooled prevalence of cognitive impairment is 40% in CKD and 53% in patients on hemodialysis, but this is not the prevalence of dementia. The mechanism is multifactorial, and neither any single toxin nor blood-brain barrier disruption is a confirmed therapeutic target. In practice, targeted screening, medication review, recognition of delirium, and individualization of dialysis are key.

Summary

- A meta-analysis of 50 studies with 25,289 patients estimated the prevalence of cognitive impairment at 40% (95% CI, 33% to 46%): 32% without dialysis, 53% on hemodialysis, 39% on peritoneal dialysis, and 26% after transplantation, with high heterogeneity in instruments and thresholds.
- A meta-analysis of 63 observational studies (57,030 individuals) linked an eGFR below 60 mL/min/1.73 m² and proteinuria with microbleeds, white matter hyperintensities, and lacunar infarcts; these data do not prove causality.
- For each 10 mL/min/1.73 m² decrease in GFR, the relative risk of stroke rises by 7%, and for each 25 mg/mmol increase in the albumin-to-creatinine ratio, by 10%; however, these are population-level associations, not individual predictions.
- The BREIN study (15 patients, 14 controls) found increased blood-brain barrier permeability and lower MoCA scores, but given the small sample this is only a preliminary signal; in a cohort of 121 older patients on hemodialysis, a greater intradialytic decrease in cerebral blood flow was associated with cognitive decline over 12 months.
- Drug neurotoxicity (cefepime, acyclovir, gabapentinoids, baclofen, morphine, sedatives) should be actively sought; in a review of 135 cases of cefepime neurotoxicity, 80% of patients had renal dysfunction, and about a quarter of cases occurred despite appropriate dosing.
- The MoCA performed in a stable state is recommended for screening, but a positive result is not a diagnosis of dementia, and acute confusion should be considered delirium until proven otherwise.

- In practice, treatment should be simplified, understanding verified, and a family member or caregiver involved; in patients on dialysis, the ultrafiltration rate and hemodynamic intolerance should be limited, and decision-making capacity should be assessed for each specific decision.

1.24 Treatment of chronic kidney disease in 2026: layered kidney protection, precise risk stratification, and follow-up after AKI

Published August 4, 2026 · [online version \(Slovak\)](#)

Abstract. Modern CKD treatment offers RAS blockade, SGLT2 inhibitors, finerenone, and GLP-1 receptor agonists; the open questions concern patient selection, sequencing, and monitoring. The article compares an HCPLive expert synthesis from July 2026 with KDIGO guidelines and pivotal RCTs. Each class has its own target population, for example an SGLT2 inhibitor at an eGFR ≥ 20 mL/min/1.73 m² and UACR ≥ 200 mg/g. No trial has demonstrated the benefit of a universal four-drug combination across all phenotypes. Practice should rely on indication-based layering and structured follow-up after AKI.

Summary

- An ACEi or ARB is titrated to the maximum tolerated dose, with a follow-up check within 2 to 4 weeks; according to KDIGO, a reversible cause should be sought particularly if creatinine rises by more than 30% within 4 weeks, and combining an ACEi with an ARB is not recommended.
- KDIGO 2024 recommends an SGLT2 inhibitor at an eGFR ≥ 20 mL/min/1.73 m² and UACR ≥ 200 mg/g (≥ 20 mg/mmol), in heart failure, or in type 2 diabetes; at an eGFR of 20 to 45 mL/min/1.73 m² with lower albuminuria, the recommendation is weaker.
- Finerenone is suggested in type 2 diabetes with an eGFR > 25 mL/min/1.73 m², normal potassium, and albuminuria > 30 mg/g despite maximally tolerated RAS blockade; the European indication is restricted to type 2 diabetes.
- The CONFIDENCE trial showed a greater UACR reduction at 180 days with the combination of finerenone and empagliflozin, but the primary outcome was a surrogate endpoint, not a clinical event.
- A GLP-1 receptor agonist (semaglutide in the FLOW trial) is indicated in type 2 diabetes and CKD when the glycemic target is not achieved, not automatically in every patient with CKD; aldosterone synthase inhibitors are not yet part of standard care.
- After AKI, the discharge summary should state the baseline, peak, and most recent creatinine, the cause, and the plan; a high-risk patient needs earlier follow-up, and KDIGO recommends evaluation no later than three months after the episode.
- A previous AKI episode is not in itself a reason to permanently discontinue RAS blockade or other prognostically effective therapy.

1.25 Body composition and estimated glomerular filtration rate: what a population-based analysis shows and where the informative value of eGFR ends

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. The relationship between obesity and kidney function is not straightforward. A cross-sectional analysis of 4,211 individuals from the German population-based SHIP Trend-0 cohort assessed body composition by anthropometry, bioimpedance, and magnetic resonance imaging and compared it with eGFR based on both creatinine and cystatin C. Higher body and fat mass were associated with lower eGFR, more markedly in women and older individuals, whereas in younger individuals the direction was reversed. Part of the association may reflect the effect of body composition on the biomarkers rather than on filtration. A normal eGFR in a young patient with obesity is not reassuring.

Summary

- Higher measures of body and fat mass were associated with lower eGFR in all equations regardless of fat distribution, and a higher proportion of relative lean mass with higher eGFR; the effects were more pronounced in women.

- In younger age groups, higher fat mass was associated with higher eGFR, which the author attributes to obesity-related hyperfiltration that over time progresses to loss of function; however, the cross-sectional design allows neither description of changes over time nor causal conclusions.
- Creatinine depends on muscle mass, and cystatin C on fat mass, inflammation, thyroid function, and glucocorticoids, so distinguishing these effects would require measured GFR (for example, with iohexol); further limitations include selection of healthier participants and a single regional population.
- The publicly available abstract does not report the numbers of participants with each imaging measurement, the versions of the eGFR equations, or the age at which the association reverses, so the article does not report them either.
- In a young patient with obesity, adding albuminuria testing is more important, and at extremes of body composition, a combined creatinine and cystatin C equation is more reliable; the findings do not change management.

1.26 Inflammation in chronic kidney disease: from risk biomarker to therapeutic target

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. Chronic inflammation links CKD with cardiovascular disease, anemia, and wasting; the question is whether it is also a therapeutic target. A review in *Nature Reviews Nephrology* (September 2026) is supplemented here with the results of RCTs of direct anti-inflammatory strategies. In CANTOS, canakinumab reduced MACE (HR 0.85) at the cost of fatal infections; in ZEUS, ziltivekimab did not reduce MACE (HR 0.99); and bardoxolone increased the risk of heart failure. The strongest evidence supports RAAS blockade, SGLT2 inhibitors, GLP-1 receptor agonists, and finerenone. An elevated hsCRP should prompt a search for the cause, not serve as an indication for treatment.

Summary

- Higher hsCRP in CKD is associated with faster functional decline, kidney failure, cardiovascular events, and mortality, but this is predominantly a prognostic association, and lowering hsCRP does not automatically improve prognosis.
- CANTOS (10,061 patients after myocardial infarction): canakinumab 150 mg reduced MACE with an HR of 0.85 (95% CI, 0.74 to 0.98) without changing all-cause mortality (HR 0.94) and with a higher incidence of fatal infections; kidney analyses were only secondary.
- ZEUS (6,376 patients with ASCVD, CKD, and hsCRP ≥ 2 mg/L): ziltivekimab reduced free IL-6 and hsCRP but, according to the topline results announced by the sponsor, did not reduce MACE (HR 0.99; 0.88 to 1.11); serious infections were more frequent, and a peer-reviewed publication is not yet available.
- BEACON (2,185 patients with CKD G4) was terminated early: there was no effect on the primary endpoint (HR 0.98), but the HR for heart failure was 1.83, and eGFR, albuminuria, and blood pressure all increased.
- Selonsertib improved the eGFR slope by 1.20 mL/min/1.73 m² per year ($p = 0.14$), with kidney events in 17% versus 12%; baricitinib reduced UACR by 41%, with anemia in 32.0% versus 3.7%; and pentoxifylline in the open-label PREDIAN trial slowed eGFR decline by 4.3 mL/min/1.73 m² over 2 years, without hard endpoints.
- In patients on dialysis, inflammation increases hepcidin and reduces the response to ESAs; a modifiable cause (vascular access, catheter, biofilm, water, periodontitis) should be sought first rather than empirically administering an immunosuppressant.
- Routine direct anti-inflammatory therapy based solely on elevated hsCRP cannot currently be recommended.

1.27 Chronic kidney disease often remains unrecognized in patients with hypertension and cardiovascular disease

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. CKD often remains unrecorded, particularly in patients without diabetes. A conference abstract from Kidney Week using TriNetX data and the peer-reviewed REVEAL-CKD study in BMJ Open evaluated patients with two eGFR values of 30 to less than 60 mL/min/1.73 m². A diagnostic code was missing in 68.4% to 68.7% of patients with hypertension or cardiovascular disease without diabetes, and even more often in Europe, reaching 95.5% in France. This reflects administrative nonrecording, not necessarily clinical nonrecognition. In at-risk patients, both eGFR and UACR should be measured and the diagnosis recorded.

Summary

- In the US TriNetX analysis (2015 to 2020), a CKD code was missing in 68.4% of patients with hypertension without diabetes, 68.7% of those with hypertension or atherosclerotic disease, 68.6% of those with hypertension or heart failure, and 51.7% of those with type 2 diabetes.
- REVEAL-CKD found unrecognized CKD G3 in 95.5% of patients in France, 92.1% in Japan, 84.3% in Germany, 77.0% in Italy, and 64.3% and 61.6% in the United States; the differences may also reflect coding and reimbursement practices.
- A missing diagnosis was associated with female sex (OR 1.29 to 1.77), category G3a versus G3b (OR 1.81 to 3.66), and absence of diabetes (OR 1.26 to 2.77) and of hypertension (OR 1.35 to 1.78), and its frequency increased with age.
- Limitations include dependence on record completeness, equating a missing code with nonrecognition, missing data on albuminuria and etiology, possible transient eGFR decline, and funding by AstraZeneca; the US analysis is only a conference abstract.
- The label “early CKD” is inaccurate because the analyses concerned categories G3a and G3b; categories G1 and G2 were not captured, so the true extent may be greater.
- In patients with hypertension, heart failure, or atherosclerotic disease, the author recommends measuring both eGFR and UACR, confirming chronicity, recording the diagnosis, and translating it into treatment, including RAS blockade and an SGLT2 inhibitor.

1.28 The nephrologist is not just the doctor for dialysis: when and why early nephrology evaluation matters

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. The nephrologist does not deal only with dialysis but primarily with diagnosing the causes of CKD, slowing its progression, and preparing treatment. This review bases its referral criteria on the KDIGO 2024 guideline, which replaces a single eGFR threshold with the calculated risk of kidney failure. Referral is appropriate at a 5-year risk of 3% to 5% according to the KFRE, which requires age, sex, eGFR, and UACR. An eGFR below 60 mL/min/1.73 m² alone is not an automatic indication. The practical benefit is targeted referral based on risk, trend, and albuminuria.

Summary

- Nephrology consultation is appropriate for an eGFR below 30 mL/min/1.73 m², a rapid or unexplained eGFR decline, A3 albuminuria, hematuria with suspected glomerular disease, resistant hypertension, severe electrolyte disorders, and unclear or hereditary etiology.
- KDIGO 2024 recommends referral at a calculated 5-year kidney failure risk of 3% to 5%, multidisciplinary care at a 2-year risk above 10%, and preparation for KRT at a 2-year risk above 40%.
- The KFRE is validated for CKD G3 to G5; at an eGFR of 40 mL/min/1.73 m², patients with A1 and A3 albuminuria may have fundamentally different risks, so urgency cannot be determined without UACR.
- A diagnosis of CKD requires an abnormality lasting at least three months, and neither a single reduced eGFR nor a single elevated UACR confirms it; the trend, for example a decline from 75 to 55 mL/min/1.73 m² over a year, is often more important than an isolated value.

- Potassium should not be routinely restricted in everyone, excessive water intake is not a universal preventive measure, and NSAIDs are risky particularly in combination with a RAS blocker and a diuretic.
- Earlier referral is associated with better preparation for KRT, fewer urgent dialysis starts, and lower mortality, but the evidence is predominantly nonrandomized.
- Most patients with CKD never reach dialysis, and the goal of nephrology care is to protect kidney function and reduce cardiovascular risk.

1.29 Finerenone in hypertensive nephropathy without diabetes: what does the FIND-CKD analysis show?

Published September 13, 2026 · [online version \(Slovak\)](#)

Abstract. Randomized data on finerenone by etiology in nondiabetic CKD have been lacking. A prespecified subgroup analysis of the FIND-CKD RCT evaluated 459 patients with hypertensive nephropathy on maximally tolerated RAS blockade. Finerenone slowed the annual eGFR slope by 0.65 mL/min/1.73 m² and reduced the composite kidney-cardiovascular endpoint (12.0% vs. 18.7%; HR 0.61). The confidence intervals approach the null effect, and hypertensive nephropathy is not a European indication. The result is a supportive signal, not a reason for widespread prescribing.

Summary

- In the subgroup, the annual eGFR slope was -3.17 versus -3.82 mL/min/1.73 m², a difference of 0.65 mL/min/1.73 m² per year (95% CI, 0.02 to 1.29; $p = 0.044$), and the HR for the composite endpoint was 0.61 (95% CI, 0.38 to 0.99; $p = 0.045$).
- In the overall trial (1,584 patients), the slope difference was 0.7 mL/min/1.73 m² per year and the HR for the composite was 0.77; the subgroup had a mean eGFR of 44 mL/min/1.73 m² and a median UACR of 797 mg/g, with etiology determined by the investigator without histologic confirmation.
- At six months, systolic blood pressure had fallen by 3.5 mm Hg and UACR by 33%; however, the absence of an interaction with baseline blood pressure does not prove that the effect is independent of blood pressure reduction.
- Hyperkalemia occurred in 17.1% versus 9.8% of patients and led to treatment discontinuation in 1.3% versus 0%; the abstract does not report control for multiplicity or a test of interaction between etiologies.
- Only 16.9% of participants were taking an SGLT2 inhibitor at baseline, so the benefit with fully layered therapy has not been precisely established; the trial was funded by Bayer.
- In CKD, finerenone is not initiated at a potassium above 5.0 mmol/L or an eGFR below 25 mL/min/1.73 m², and follow-up occurs after four weeks; in the EU, it is approved only for CKD with type 2 diabetes and for heart failure with an ejection fraction $\geq 40\%$.

CHAPTER 2

Cardiovascular-kidney-metabolic syndrome and heart failure

The heart, kidneys, and metabolism as one clinical framework. Guidelines, stages of cardiovascular-kidney-metabolic syndrome, and heart failure from a nephrologist's perspective. Number of articles: 20.

Key messages of the chapter

Obesity, diabetes, CKD, and cardiovascular disease are beginning to be treated within the unified framework of CKM syndrome. The first multidisciplinary AHA, ACC, ADA, and ASN guideline from June 2026 introduces CKM stages 0 to 4, risk quantification with the PREVENT equations, and routine screening for CKD using eGFR and UACR (2.2, 2.3, 2.8). In type 2 diabetes at stage 2 or 3, it sets a PREVENT-CVD risk of at least 7.5% as the threshold for starting a GLP-1 receptor agonist or an SGLT2 inhibitor (2.2), and it defines stage 3, among other criteria, by a coronary calcium score above 100 or a PREVENT risk above 20% (2.8). KDIGO reviews show that existing recommendations for CKD, BP, diabetes, and lipids already form a common framework with a systolic BP target below 120 mm Hg under standardized measurement, although its components do not carry the same methodological weight (2.7, 2.16). The ESC and ERA European guidelines from August 2026 call for active screening for CKD in cardiovascular disease, recommend a systolic BP of 120 to 129 mm Hg at an eGFR of at least 30 mL/min/1.73 m² (class I, level A), and prefer DOACs over vitamin K antagonists (2.18).

The framework also has limitations. Staging does not replace a diagnosis, and adiposity is not the only cause of risk (2.8). In the FF-CHAYA cohort, 79.2% of young adults met the criteria for at least stage 1, mostly because of adiposity, yet the difference in intima-media thickness at stage 1 was only 0.014 mm, which supports prevention rather than a diagnosis of disease (2.15). The interpretation of CKM as a mitochondrial systemic disorder is a working hypothesis without a validated biomarker (2.20). An expert commentary, not new data, names typical errors in practice: omitting UACR, ignoring stage 0, a siloed approach, and giving up after reimbursement is denied (2.4).

Treatment decisions should rest on drugs with hard outcomes. SGLT2 inhibitors had kidney hazard ratios of 0.61 to 0.72 in three randomized trials, the FLOW trial demonstrated a kidney benefit of semaglutide (HR 0.76), and finerenone complements therapy in persistent albuminuria (2.12, 2.1). A post hoc analysis of FIDELITY with 12,990 patients showed a consistent reduction in both cardiovascular and kidney events with finerenone across CKM stages II to IV, with stage IV carrying approximately twice the risk of stage II (2.5). Combinations, however, are supported only by albuminuria data: CONFIDENCE reduced UACR by 52%, a surrogate endpoint, while neither triple therapy nor the sequence of initiation has been tested (2.10, 2.12), and FINE-ONE in type 1 diabetes did not demonstrate hard clinical outcomes (2.10).

Heart failure is a distinct axis of this chapter. Heart failure affects 10% to 30% of patients with CKD, and CKD affects 30% to 60% of patients with heart failure; in the CRIC cohort, hospitalization for heart failure was associated with both CKD progression (HR 1.93) and death (HR 2.20), although these are observational associations (2.13). In the Swedish registry, CKD was most strongly associated with HFpEF (OR 2.46), and the association strengthened as the glomerular filtration rate declined (2.9). Natriuretic peptides have rule-out thresholds of BNP below 35 pg/mL and NT-proBNP below 125 pg/mL, but no universal threshold applies in CKD, and an elevated value is a reason for

echocardiography, not proof of the diagnosis (2.14). The ESC 2026 guidelines abolish the HFmrEF category, recommend an SGLT2 inhibitor and an MRA regardless of LVEF (class I, level A), and direct that changes in kidney function be managed by monitoring rather than by withholding prognostic therapy (2.19), consistent with KDIGO, which calls for maintaining GDMT despite a hemodynamic decline in eGFR (2.13). The observational association of time spent at a systolic BP of 110 to 130 mm Hg with a smaller decline in kidney function (OR 0.74) is not a treatment target (2.17). Subcutaneous furosemide in an autoinjector may be part of a planned decongestion protocol, but its supporting data are pharmacokinetic, and it does not replace urgent intravenous therapy (2.11).

This yields a practical framework for the outpatient clinic. In every patient with obesity, diabetes, hypertension, or cardiovascular disease, measure eGFR and UACR together and automate screening (2.3, 2.4, 2.18). Determine the CKM stage and absolute risk with the PREVENT and KFRE equations (2.3, 2.7), and layer therapy according to the dominant risk phenotype with drugs of proven effect on kidney and cardiovascular outcomes (2.12, 2.20). In advanced CKD with dyspnea, actively consider HFrEF even with a normal ejection fraction (2.9), and do not discontinue prognostic therapy because of an expected decline in eGFR (2.13, 2.19). Guidelines are best converted into an internal protocol that maps all three risks at once and coordinates specialties (2.6, 2.7).

2.1 The cardio-nephro-metabolic revolution is running at full speed

Published April 25, 2026 · [online version \(Slovak\)](#)

Abstract. Treatment of chronic kidney disease is evolving rapidly in the cardio-nephro-metabolic field. This short commentary, without original data, summarizes the state of therapy: SGLT2 inhibitors are a standard pillar, and following the FLOW trial, the indication for semaglutide was expanded in 2025 and 2026 to include slowing CKD progression. Together with finerenone and new aldosterone synthase inhibitors, a comprehensive approach is emerging that, according to the author, may favorably affect both diabetic and non-diabetic nephropathy.

Summary

- SGLT2 inhibitors are a standard pillar of CKD therapy.
- Following the FLOW trial, the indication for semaglutide was expanded to include slowing CKD progression.
- The addition of finerenone and aldosterone synthase inhibitors forms a comprehensive approach.

2.2 ACC/AHA issue the first guideline on cardiovascular-kidney-metabolic syndrome

Published June 11, 2026 · [online version \(Slovak\)](#)

Abstract. Obesity, metabolic disorders, CKD, and cardiovascular disease have until now been managed separately. The ACC and AHA, in collaboration with the ADA, the Obesity Association, and the ASN, have issued the first joint guideline on CKM syndrome, which replaces the 2013 AHA/ACC obesity guideline. It introduces a four-stage model and, in type 2 diabetes at stage 2 or 3, specifies a PREVENT-CVD risk of at least 7.5% as the threshold for initiating treatment with a GLP-1 receptor agonist or an SGLT2 inhibitor. For practice, this means earlier risk identification and coordinated care.

Summary

- The guideline views obesity as the central pathophysiological driver of multiorgan progression, in which dysfunctional adipose tissue leads to metabolic risk factors, CKD, and cardiovascular disease.
- Stage 1 comprises overweight, obesity, or prediabetes; stage 2, a metabolic risk factor or moderate- to high-risk CKD; stage 3, subclinical cardiovascular disease, a 10-year PREVENT-CVD risk of at least 20%, or very high-risk CKD according to KDIGO; and stage 4, clinical cardiovascular disease.

- Management is graded by stage, from intensive lifestyle modification through obesity pharmacotherapy to metabolic and bariatric surgery in selected patients; for the first time, GLP-1 receptor agonists are included to reduce the risk of cardiac events.
- In type 2 diabetes and CKM stage 2 or 3, the threshold for initiating a GLP-1 receptor agonist, an SGLT2 inhibitor, or their combination is a 10-year PREVENT-CVD risk of at least 7.5%.
- The guideline rejects a specialty-fragmented approach and calls for interdisciplinary models with a designated care coordinator.
- An accompanying editorial notes that implementation may be hindered by lack of infrastructure, unequal access to antiobesity medications, and reimbursement mechanisms that favor episodic care.

2.3 The new multidisciplinary CKM strategy (AHA/ACC/ADA/ASN, 2026): the first guideline for cardiovascular-kidney-metabolic syndrome

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. Obesity, type 2 diabetes, CKD, and cardiovascular disease are interconnected as a single continuous syndrome. On June 9, 2026, the AHA, ACC, ADA, and ASN published the first multidisciplinary guideline for CKM syndrome, which replaces and expands the 2013 framework. It introduces CKM stages 0 to 4, risk quantification with the PREVENT equations, and routine CKD screening with eGFR and UACR. For nephrology practice, this means viewing the kidney as a central component of the syndrome and acting as early as the preventive phase.

Summary

- The guideline introduces a standardized classification of CKM stages from 0 to 4, intended to enable earlier identification of patients and selection of treatment according to absolute cardiovascular risk.
- In stages 0 to 3, the PREVENT equations are recommended to estimate the 10-year and 30-year risks of ASCVD, heart failure, and total cardiovascular disease, which determine the intensity of prevention.
- CKD should be screened for actively and regularly using eGFR and UACR; for CKD, the guideline lists RAAS blockade and SGLT2 inhibitors and, for persistent albuminuria, depending on context, an nsMRA or GLP-1-based therapy.
- Key therapies are SGLT2 inhibitors and GLP-1-based therapy because of their concurrent cardiovascular and kidney benefits; in heart failure, RAAS inhibitors and MRAs should also be integrated according to phenotype.
- Obesity is assessed by both BMI and waist circumference, and antiobesity medications and bariatric surgery are considered in suitable patients; screening for conditions such as pre-heart failure, MASLD, and OSA is included.
- The guideline requires interdisciplinary care with a coordinator and systematic attention to social determinants of health, such as access to food, housing, and transportation.

2.4 5 critical mistakes in managing CKM syndrome and practical steps for nephrology

Published June 27, 2026 · [online version \(Slovak\)](#)

Abstract. Although the concept of CKM syndrome is rapidly gaining ground, routine practice is hampered by fragmented thinking. This commentary on a Medscape article describes five typical mistakes that delay diagnosis and intervention in patients at risk for CKM: omitting UACR, ignoring stage 0, a siloed approach, assuming that GLP-1 receptor agonists are unavailable, and giving up after a reimbursement denial. This is expert opinion, not new data. For practice, it recommends automating UACR screening and actively addressing access to treatment.

Summary

- According to the new recommendations, albuminuria screening is indicated in hypertriglyceridemia, hypertension, metabolic syndrome, and type 2 diabetes, yet a large proportion of at-risk patients do not

receive a UACR test, even though the risk of CKD progression and cardiovascular mortality rises continuously even at low values.

- The author recommends configuring electronic health records to automatically prompt UACR or a CKM laboratory panel in at-risk situations.
- Stage 0 without risk factors does not mean absence of future risk, so prevention according to AHA Life's Essential 8 and education about CKM are recommended for younger patients.
- The CKM framework is intended to support treatment with multiple targets; in suitable patients, the article recommends starting with consideration of GLP-1 receptor agonists or SGLT2 inhibitors.
- GLP-1 receptor agonists should not be ruled out in advance because of high cost; discount programs should be sought, and a reimbursement denial need not be final if the justification or the assessment of the disease changes.
- For nephrology practice, the decisive steps are connecting workflows, particularly automated UACR screening, timing interventions correctly, including at stage 0, and using combined, risk-guided therapy.

2.5 Finerenone across stages of cardiovascular-kidney-metabolic (CKM) syndrome in patients with T2D and CKD: what the FIDELITY post hoc analysis shows

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. Patients with type 2 diabetes and CKD have high cardiovascular and kidney risk. A post hoc pooled analysis of the FIDELITY RCTs with 12,990 patients evaluated the effect of finerenone by baseline CKM stage II to IV. Compared with stage II, stage IV had a higher risk of the cardiovascular composite (aHR 1.87) and of the kidney composite (aHR 1.96), and finerenone reduced both outcomes consistently across stages. The post hoc design precludes definitive conclusions, but the result supports the use of finerenone regardless of CKM stage.

Summary

- The analysis included patients with a mean age of 64.8 years, 70% men, potassium ≤ 4.8 mmol/L, and maximally tolerated RAAS inhibition; symptomatic HFrEF was excluded, and median follow-up was 3 years.
- Distribution by stage: II (n = 3,864), III (n = 3,275), and IV (n = 5,851); stage IV had an aHR of 1.87 (95% CI, 1.56 to 2.24) for the cardiovascular composite and an aHR of 1.96 (95% CI, 1.43 to 2.69) for the kidney composite.
- Finerenone reduced the cardiovascular composite (p for interaction = 0.86) and the kidney composite (p for interaction = 0.65) independently of stage.
- With finerenone, regression from stage III to II was more likely (OR 1.66; p < 0.001), and progression was less frequent, although only at the margin of significance (aOR 0.89; p = 0.05); safety, including hyperkalemia, was similar across stages.
- The FIDELITY trials were not designed to evaluate CKM, some subgroups were small, and not all variables needed for staging were collected, so the analysis confirms consistency rather than providing definitive evidence.
- From a nephrology perspective, the result supports viewing finerenone as a drug with combined kidney and cardiovascular benefit.

2.6 CKM syndrome finally as “one framework”: what the new ACC/AHA/ADA/ASN guidelines mean for nephrology practice

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. CKD rarely occurs in isolation, and the patient's risk increases across organs. This commentary on a Medscape report presents the new AHA/ACC/ADA/ASN guidelines on CKM syndrome, published as a review in JACC and as a comprehensive guideline on prevention, detection, evaluation, and management. In these guidelines, CKD is not the renal end of a chain but part of the cardiometabolic picture. The author

proposes turning the guidelines into an internal nephrology protocol that maps kidney, cardiovascular, and metabolic risk simultaneously.

Summary

- The guidelines create an explicit framework for CKM syndrome instead of isolated management of individual diagnoses in separate specialties.
- For the nephrologist, preventing CKD progression often also prevents cardiovascular events, and focusing solely on a kidney endpoint may overlook the dominant component of risk.
- Proposed approach: in detection, look for the cardiometabolic axis in every patient with CKD; in evaluation, unify kidney and cardiovascular goals in a single plan; and in follow-up, assess albuminuria, blood pressure, and metabolic parameters, not just the eGFR trend.

2.7 The kidney at the center of cardiovascular-kidney-metabolic syndrome: how KDIGO connects nephrology, diabetology, and cardiology

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. CKD overlaps with obesity, diabetes, hypertension, and cardiovascular disease within CKM syndrome. A review by Levin and coauthors in *Kidney International* (May 2026) shows that the KDIGO guidelines on CKD, blood pressure, diabetes, and lipids form a common framework for this syndrome. The key elements are detection with eGFR and UACR, absolute risk estimation with the KFRE, and layering of therapy, for example an SGLT2 inhibitor at an eGFR ≥ 20 mL/min/1.73 m². This is not a new study; what matters for practice is early recognition and coordination across specialties.

Summary

- Very high kidney risk is equivalent to high cardiovascular risk, so a patient with CKD may belong to an advanced CKM stage even without a history of myocardial infarction or stroke.
- The KFRE is recommended in CKD G3 to G5: a 5-year risk of 3% to 5% may support nephrology referral, a 2-year risk above 10% involvement of a multidisciplinary team, and a risk above 40% preparation for KRT; these thresholds are not automatic instructions.
- KDIGO suggests a target systolic blood pressure below 120 mm Hg with standardized measurement; an ACEi or ARB is titrated to the maximum tolerated dose with a check within 2 to 4 weeks and continued unless creatinine rises by more than 30% within four weeks.
- An SGLT2 inhibitor is recommended in type 2 diabetes with an eGFR ≥ 20 mL/min/1.73 m², at a UACR ≥ 20 mg/mmol, or in heart failure, and if tolerated it may be continued below 20 mL/min/1.73 m² until KRT is initiated.
- Finerenone is indicated in type 2 diabetes with an eGFR > 25 mL/min/1.73 m², normal potassium, and albuminuria > 3 mg/mmol despite maximal RAS blockade; a GLP-1 receptor agonist is indicated when the glycemic target is not reached despite metformin and an SGLT2 inhibitor.
- At age ≥ 50 years with an eGFR below 60 mL/min/1.73 m², without dialysis or transplantation, a statin or a statin with ezetimibe is recommended; initiation after the start of dialysis has weaker evidence support, which, however, does not mean automatic discontinuation.
- A mild rise in creatinine during effective decongestion in heart failure does not necessarily indicate kidney injury.

2.8 CKM syndrome: stages 0 to 4, screening, and treatment according to the 2026 AHA/ACC/ADA/ASN guideline

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. The 2026 AHA, ACC, ADA, and ASN guideline, published in *Circulation*, integrates obesity, diabetes, CKD, and cardiovascular disease into a single risk continuum. This reference overview systematically describes the five CKM stages from 0 to 4, the specific stage 3 thresholds, such as a calcium score above 100 or a PREVENT risk above 20%, and a screening panel with concurrent assessment of eGFR and UACR. The

author cautions that adiposity is not the only cause and that staging does not replace a diagnosis. The benefit is early risk identification and targeted treatment.

Summary

- The full classification has five stages: 0, no risk factors; 1, excess or dysfunctional adiposity and prediabetes; 2, metabolic risk factors or CKD; 3, subclinical cardiovascular disease or a risk equivalent; and 4, clinical cardiovascular disease.
- Stage 3 is defined by a coronary artery calcium score above 100, NT-proBNP ≥ 125 pg/mL with echocardiographic evidence of dysfunction, CKD at very high risk according to KDIGO or in G4 to G5, and a 10-year PREVENT risk above 20%.
- In stage 2, annual UACR testing is emphasized because albuminuria may precede eGFR decline; chronicity requires the abnormality to persist for at least three months.
- According to the author, adiposity is a central and often modifiable mechanism, not the only cause, and in a lean patient with a CKM phenotype, risk may be underestimated.
- Treatment includes structured obesity intervention, RAAS blockade for albuminuria without combining an ACEi with an ARB, SGLT2 inhibitors, GLP-1 receptor agonists, nonsteroidal MRAs in type 2 diabetes with albuminuria, and statins; aspirin should not be given automatically.
- Staging is not biologically absolute, progression need not be linear, the stage does not replace an etiologic diagnosis, and the calcium score, NT-proBNP, and echocardiography are not tools for population-wide screening.
- For nephrology, it is important to prevent therapeutic inertia and to reduce residual kidney and cardiovascular risk through combined therapy.

2.9 Chronic kidney disease and incident heart failure: the strongest association points to HFpEF

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. The relationship between CKD and heart failure is taken for granted, but it is not the same for all phenotypes. A retrospective case-control study from the Swedish Heart Failure Registry compared 51,848 patients with 51,848 matched controls. CKD was associated with approximately twofold odds of heart failure, most strongly with HFpEF (OR 2.46), and the association strengthened as eGFR declined. These are observational data without proof of causality. In advanced CKD with dyspnea, a normal ejection fraction is not sufficient reassurance, and HFpEF should be actively considered.

Summary

- When CKD was defined by ICD-10 codes, the odds ratio was 2.46 for HFpEF, 1.52 for HFrEF, and 1.30 for HFmrEF (p for interaction = 0.001); categories G3b to G5 versus G2 were more often associated with HFpEF than with HFrEF.
- The finding of 8% higher odds of HFpEF versus HFrEF at an eGFR below 60 mL/min/1.73 m² compares phenotypes only among patients with heart failure and therefore must not be confused with the odds ratio of 2.46 versus controls.
- An odds ratio is neither a relative risk nor an incidence, and the design does not allow calculation of absolute individual risk; mechanisms such as inflammation, microvascular dysfunction, volume overload, or anemia are only a contextual hypothesis.
- Limitations include differential capture of CKD by codes among controls, which inflates the odds ratios, missing albuminuria data, possible effects of congestion on eGFR (reverse causality), and a study period from 2005 to 2021 that predates widespread use of SGLT2 inhibitors.
- Patients with CKD, particularly in stages G3b to G5, should be actively evaluated from a cardiac standpoint, with targeted questions about exertional dyspnea, orthopnea, and congestion and with echocardiography as indicated.

- NT-proBNP is elevated in CKD even without decompensation and should be interpreted mainly by its trajectory in the same patient.
- In prevention, an SGLT2 inhibitor has a special place, as it reduced both CKD progression and hospitalizations for heart failure in the DAPA-CKD trial.

2.10 Chronic kidney disease in diabetes: early screening and layered cardiorenal therapy

Published August 14, 2026 · [online version \(Slovak\)](#)

Abstract. According to ADA 2026, CKD affects approximately 20% to 40% of people with diabetes and long remains asymptomatic. This updated expert synthesis compares an educational program supported by the manufacturer of finerenone with the ADA 2026 and KDIGO guidelines, randomized trials, and the European marketing authorization. Screening requires eGFR and UACR together, and treatment in type 2 diabetes is layered with RAS blockade, an SGLT2 inhibitor, finerenone, and a GLP-1 receptor agonist. CONFIDENCE reduced UACR by 52%, but this is a surrogate endpoint. FINE-ONE in type 1 diabetes did not demonstrate hard clinical outcomes.

Summary

- ADA 2026 recommends measuring eGFR and UACR at least once a year in type 2 diabetes and, in type 1 diabetes, after five years' duration; albuminuria should be confirmed by at least two abnormal results from three samples collected over three to six months.
- With an active sediment, nephrotic syndrome, or rapid eGFR decline, another cause of CKD should be sought; nephrology consultation is also needed at an eGFR below 30 mL/min/1.73 m².
- The blood pressure target for most patients is below 130/80 mm Hg; an ACE inhibitor or ARB is titrated to the maximum tolerated dose, the two are not combined, and a creatinine rise of up to 30% without hypovolemia is not an automatic reason to discontinue.
- In type 2 diabetes and CKD, an SGLT2 inhibitor is initiated at an eGFR of at least 20 mL/min/1.73 m², and finerenone in the presence of albuminuria at an eGFR of at least 25 mL/min/1.73 m², with a potassium check after approximately four weeks; in FLOW (3,533 patients), semaglutide reduced the primary composite by 24%.
- CONFIDENCE (818 patients) showed a 52% UACR reduction at 180 days with simultaneous initiation of finerenone and empagliflozin; ADA 2026 lists this as a level B recommendation, not mandatory quadruple therapy, because hard events were not evaluated.
- FINE-ONE (242 patients with type 1 diabetes) reduced UACR by 34% versus 12% with placebo, with hyperkalemia in 10.1% versus 3.3%; the European CKD indication for finerenone remains restricted to type 2 diabetes.
- The author recommends a written plan for acute illness and prescheduled checks of potassium, eGFR, blood pressure, and volume status.

2.11 Subcutaneous furosemide in an autoinjector: possibilities and limits of home treatment of edema in heart failure and CKD

Published August 14, 2026 · [online version \(Slovak\)](#)

Abstract. In July 2026, the FDA approved the Furoscix ReadyFlow autoinjector, which delivers 80 mg of furosemide subcutaneously in approximately 10 seconds to adults with edema due to chronic heart failure or CKD. A review of the registration data shows a relative bioavailability of 107.3% compared with two 40 mg intravenous boluses in 19 healthy volunteers. The pilot AT HOME-HF study tested a different, five-hour sys-

tem and did not meet its primary endpoint. The autoinjector may be part of a planned outpatient decongestion protocol, not a substitute for urgent intravenous therapy.

Summary

- ReadyFlow is a single-use mechanical autoinjector containing 80 mg of furosemide in 1 mL for administration into the abdomen at least 5 cm from the navel, for adults only; it is not intended for chronic use and should be replaced by an oral diuretic as soon as possible.
- In a pharmacokinetic comparison in 19 healthy adults, relative bioavailability was 107.3% (90% CI, 103.9% to 110.8%), maximum concentration was approximately 4,530 ng/mL versus 10,100 ng/mL, and the diuretic effect lasted about 6 hours; similar exposure does not mean an identical concentration profile.
- AT HOME-HF (51 patients, On-body Infusor system) showed a 2.02 kg weight difference on day 3 in favor of subcutaneous therapy, but the primary composite endpoint was not significant (win ratio 1.11; 95% CI, 0.48 to 2.50), and the study was small, open-label, and manufacturer-funded.
- The fixed 80 mg dose may be insufficient in markedly reduced kidney function, and the effect of subcutaneous edema on absorption is unknown; the drug is contraindicated in anuria.
- The author considers only a stable adult suitable, with a predefined plan, the ability to monitor weight, blood pressure, and urine output, and assured follow-up; in hypoxemia, hypotension, oliguria, or suspected AKI, home administration must not delay urgent evaluation.
- After administration, weight, blood pressure, urine output, creatinine, and electrolytes should be monitored, in high-risk patients within 24 to 72 hours according to the author; FDA approval does not imply marketing authorization or reimbursement in Slovakia.

2.12 Selecting and combining SGLT2 inhibitors, GLP-1 receptor agonists, and dual agonists in type 2 diabetes with cardiorenal risk

Published August 20, 2026 · [online version \(Slovak\)](#)

Abstract. In type 2 diabetes with CKD, heart failure, or atherosclerotic disease, the choice of glucose-lowering drug is guided by hard cardiorenal outcomes, not by HbA1c. A review of randomized trials shows the most consistent kidney effect for SGLT2 inhibitors, with hazard ratios of 0.61 to 0.72 in three trials. FLOW demonstrated a kidney benefit of semaglutide (HR 0.76), and finerenone complements therapy in persistent albuminuria. Combinations are supported only by albuminuria data; triple therapy and the sequence of initiation have not been tested. The decision framework is based on the dominant risk phenotype.

Summary

- DAPA-CKD achieved an HR of 0.61 (NNT 19), EMPA-KIDNEY 0.72, and CREDENCE 0.70, with 0.66 for the kidney composite; the effect of SGLT2 inhibitors was consistent even in patients without diabetes.
- In patients with type 2 diabetes and CKD, FLOW showed that semaglutide reduced the composite (HR 0.76), the kidney composite (0.79), cardiovascular death (0.71), and all-cause mortality by 20%, and slowed the annual eGFR decline by 1.16 mL/min/1.73 m².
- In FIDELIO-DKD, finerenone achieved an HR of 0.82, with discontinuation for hyperkalemia in 2.3% versus 0.9% with placebo; it complements, rather than replaces, RAS blockade and SGLT2 inhibitors.
- In CKD with albuminuria, the foundation is an SGLT2 inhibitor on maximally tolerated RAS blockade; in heart failure, an SGLT2 inhibitor; in ASCVD, a GLP-1 receptor agonist with proven benefit; and in obesity, a GLP-1 receptor agonist or a dual GIP/GLP-1 receptor agonist.
- CONFIDENCE showed a 29% and 32% greater UACR reduction with the combination of finerenone and empagliflozin than with either component alone, but this is only a surrogate endpoint; triple therapy, the sequence of initiation, and kidney protection by tirzepatide have not been demonstrated, and retatrutide is not approved.
- The initial eGFR dip after starting an SGLT2 inhibitor is hemodynamic and not a reason to discontinue; diuretics should be reassessed, potassium monitored after finerenone, and gastrointestinal tolerance assessed with GLP-1 receptor agonists, and patients should be instructed on rules for acute illness.

2.13 Heart failure and CKD: conclusions of the KDIGO Controversies Conference on bidirectional burden, GDMT, and coordinated care

Published September 1, 2026 · [online version \(Slovak\)](#)

Abstract. Heart failure affects approximately 10% to 30% of patients with CKD, and CKD affects 30% to 60% of patients with heart failure. The consensus report of the 2024 KDIGO Controversies Conference, published in 2026, is not a new guideline but a review of evidence and practical principles. In the CRIC cohort, hospitalization for HF was associated with both CKD progression (HR 1.93) and death (HR 2.20), although these are observational associations. The key message is to measure both eGFR and UACR, to maintain GDMT despite a hemodynamic decline in eGFR, and to provide coordinated interdisciplinary care.

Summary

- In the CRIC cohort (3,791 adults with CKD), the rate of HF hospitalization was 5.8 per 100 person-years, with an adjusted rate ratio of 2.2 at eGFR below 30 mL/min/1.73 m² and 2.6 at UACR of at least 300 mg/g; after a single hospitalization, the HR was 1.93 for CKD progression and 2.20 for death.
- General natriuretic peptide thresholds (outpatient BNP of at least 35 pg/mL, NT-proBNP of at least 125 pg/mL) are not specific in CKD; a normal value helps exclude HF, whereas in stages G4 to G5 and on dialysis echocardiography should be added, in patients on hemodialysis preferably on a non-dialysis day.
- In the FIDELITY analysis (13,026 patients), finerenone reduced both the cardiovascular composite (HR 0.86) and the kidney composite (HR 0.77), with permanent discontinuation for hyperkalemia in 1.7% vs. 0.6%; in FLOW, semaglutide reduced the kidney composite by 24%.
- A hemodynamic eGFR decline of up to 30% after GDMT initiation should not by itself lead to treatment discontinuation, whereas a decline exceeding 30% warrants a search for another cause of AKI; in CRE-DENCE, only 0.5% of patients had a decline exceeding 30%.
- At eGFR below 60 mL/min/1.73 m², the conference supports initiating an ACEi/ARB or MRA when potassium is below 5 mmol/L; hyperkalemia should be managed with an SGLT2 inhibitor, potassium binders, correction of acidosis, and diuretics rather than reflexive discontinuation.
- Limitations: the report is a synthesis of discussions, the CRIC data come from the United States, evidence in G5 and G5D is weak, and the review of coordinated CKM care is heterogeneous, permitting only associative conclusions.

2.14 Five steps toward earlier detection of heart failure in primary care: a practical diagnostic framework

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. Heart failure, particularly with preserved ejection fraction, often remains undiagnosed for a long time in primary care. This Slovak adaptation of a practical five-step framework from Medscape describes diagnostic triage: a targeted history, BNP or NT-proBNP, surveillance of high-risk groups, early echocardiography, and assessment of comorbidities. The ESC 2021 rule-out thresholds are BNP below 35 pg/mL and NT-proBNP below 125 pg/mL. In CKD, atrial fibrillation, older age, and obesity, no universal threshold suffices; an elevated peptide is a reason for echocardiography, not proof of the diagnosis.

Summary

- The history should actively seek exertional and nocturnal dyspnea, orthopnea, fatigue, ankle edema, and unexplained weight gain, which older patients often attribute to age.
- According to ESC 2021, outpatient BNP below 35 pg/mL and NT-proBNP below 125 pg/mL make HF unlikely, with corresponding acute thresholds of 100 pg/mL and 300 pg/mL; in HFpEF, supplementary thresholds are higher in atrial fibrillation (NT-proBNP above 365 pg/mL), and up to approximately 20% of patients with HFpEF have values below the thresholds.
- The HFA 2023 consensus proposes age-specific NT-proBNP rule-in thresholds (at least 125, 250, and 500 pg/mL) and, in obesity, lowering the threshold by 25% to 40% according to BMI; the NICE threshold of at least 400 ng/L is not an ESC threshold.

- CKD raises peptide levels, so a low value rules out better than a high value rules in; in G4 to G5 and on dialysis, specificity is poor, and evidence of elevated filling pressures, especially on echocardiography, is decisive.
- Targeted attention is warranted in patients with hypertension, ischemic heart disease, diabetes, CKD, obesity, atrial fibrillation, and prior cardiotoxic cancer therapy; with typical clinical features and an elevated peptide, echocardiography should not be delayed.
- Limitations: the primary source is a practical article, the new ESC guideline of August 2026 was not available in full text, and the TOPCAT analysis describes prognosis, not diagnostic thresholds.

2.15 Cardiovascular-kidney-metabolic risk begins in youth. What does the finding in 79% of young adults really mean?

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. The AHA cardiovascular-kidney-metabolic (CKM) syndrome is a framework of five risk stages, not a standalone diagnosis. A cross-sectional analysis of 1,283 young adults from the US FF-CHAYA cohort with a mean age of 22.9 years showed that 79.2% met the criteria for at least stage 1, mostly because of excess adiposity. A higher stage was associated with a worse Life's Essential 8 score and greater intima-media thickness, although in stage 1 the difference was only 0.014 mm. The finding does not confirm a diseased heart or kidneys, but it supports early prevention.

Summary

- Stage 0 included 20.7%, stage 1 40.2%, stage 2 36.2%, and stage 3 only 2.8% of participants; none had clinically manifest cardiovascular disease.
- Compared with stage 0, the Life's Essential 8 score was lower by 7.6 points in stage 1, by 13.9 points in stage 2, and by 13.2 points in stage 3, with overlapping intervals for stages 2 and 3.
- Compared with stage 0, intima-media thickness was greater by 0.014 mm in stage 1, by 0.020 mm in stage 2, and by 0.100 mm in stage 3; according to the author, the difference in stages 1 and 2 is a population-level signal below the threshold of individual measurability.
- The abstract does not report the prevalence of albuminuria or reduced eGFR, so it cannot be claimed that 79% had impaired kidney function; in young people, eGFR is influenced by muscle mass and creatine use.
- Limitations: a cross-sectional design that cannot assess causality or progression, a US urban population, and only 36 participants in stage 3.
- The author recommends measuring blood pressure, weight, and waist circumference and, in the presence of hypertension, diabetes, obesity, or a family history, testing eGFR and UACR; the study does not justify population-wide carotid ultrasound in asymptomatic young people.

2.16 The kidney in the middle: how KDIGO guidelines connect cardiovascular-kidney-metabolic syndrome

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. Cardiovascular-kidney-metabolic (CKM) syndrome, formally named by the AHA in 2023, links CKD with obesity, diabetes, hypertension, and cardiovascular disease. A review by Levin and colleagues in *Kidney International* (2026) shows that KDIGO provides a comprehensive framework composed of graded guidelines and conference reports that do not carry the same methodological weight. The key elements are active screening of both eGFR and albuminuria, a systolic blood pressure target below 120 mm Hg with

standardized measurement, and drugs with both kidney and cardiovascular benefit. It is not a systematic synthesis presenting new evidence.

Summary

- The KDIGO framework consists of graded guidelines (CKD 2024, blood pressure 2021, diabetes in CKD 2022, lipids 2013) and conference reports on obesity, heart failure, and CKD prevention, which are not graded guidelines and should not be cited as such.
- CKD should be actively sought in diabetes, hypertension, obesity, heart failure, and a family history using eGFR and UACR, adding eGFRcr-cys where useful and using validated kidney failure prediction models in G3 to G5.
- Recommendations include physical activity of at least 150 minutes per week, sodium intake below approximately 2 g/day, protein intake of about 0.8 g/kg/day in suitable patients, and a systolic blood pressure below 120 mm Hg with standardized measurement when tolerated, with caution in frail patients and in orthostatic hypotension.
- The foundation remains an ACE inhibitor or ARB titrated to the maximum tolerated dose, followed by SGLT2 inhibitors, GLP-1 receptor agonists, an nSMA for persistent albuminuria with potassium monitoring, a statin particularly from age 50 years, and, in atrial fibrillation in G1 to G4, a DOAC in preference to warfarin.
- The obesity conference emphasizes cumulative exposure, the benefit of pharmacotherapy including GLP-1 receptor agonists and of bariatric surgery, and nonstigmatizing communication; the heart failure conference states that a mild eGFR decline after treatment initiation generally does not require discontinuation.
- Limitations: it is a review without meta-analysis, the lipid guideline dates from 2013, and no guideline for heart failure in CKD existed at the time of publication.

2.17 Time in target range for blood pressure in heart failure: association with eGFR decline and mortality

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. A prospective multicenter Chinese cohort of patients after hospitalization for heart failure assessed the estimated time with systolic blood pressure in the range of 110 to 130 mm Hg. A one standard deviation increase in this measure was associated with lower odds of kidney function decline (OR 0.74) and lower mortality (HR 0.90). These are observational data with sparse blood pressure measurements that do not demonstrate causality. The research range therefore cannot automatically be adopted as a treatment target, but the result underscores the importance of long-term blood pressure monitoring.

Summary

- The study enrolled patients from 52 hospitals in China (2016 to 2018), with 1,529 patients in the kidney analysis and 2,195 in the mortality analysis; blood pressure was measured at months 1, 6, and 12, and time in range was estimated by linear interpolation, not by continuous monitoring.
- The kidney outcome required an eGFR decline of at least 20% and an eGFR below 60 mL/min/1.73 m² at month 12; it represents neither kidney failure nor confirmed CKD.
- A one standard deviation increase, approximately 35 to 36 percentage points, corresponded to an OR of 0.74 (0.60 to 0.93) for kidney function decline and an HR of 0.90 (0.83 to 0.98) for death over a median follow-up of 1.0 and 4.2 years, respectively; absolute risks are not reported.
- Limitations: observational design with residual confounding, three measurements per year, overlap between the blood pressure measurement period and the eGFR change, different analytic populations, unavailable full text, and no validated decision threshold.
- The author notes that KDIGO 2024 suggests a systolic blood pressure below 120 mm Hg in CKD with standardized measurement when tolerated, which is not identical to the range used in the study; adjust-

ments of diuretics and other therapy should be guided by the clinical status, not by an effort to increase time in range.

2.18 The 2026 ESC/ERA guidelines: a shared plan of care for the heart and kidneys

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. The first ESC guidelines developed with the ERA for cardiovascular disease and CKD were published on August 28, 2026. In patients with cardiovascular disease, they call for active detection of CKD using eGFR and UACR and for organizing care according to the STAMP framework. A systolic blood pressure of 120 to 129 mm Hg is recommended at eGFR of at least 30 mL/min/1.73 m² (class I, level A), and DOACs are preferred over vitamin K antagonists. The guidelines do not prescribe a uniform drug combination but rather individual indications and coordination across specialties.

Summary

- The STAMP framework comprises screening with eGFR and UACR, risk triage, modification of CKD risk, adjustment of cardiovascular therapy to kidney function, and planning of coordinated care; chronicity should be confirmed by repeated measurement over at least three months without delaying evaluation of acute deterioration.
- In type 2 diabetes and CKD, an SGLT2 inhibitor is initiated at eGFR of at least 20 mL/min/1.73 m² regardless of glycemia, whereas in CKD without diabetes the strength of the recommendation depends on UACR; finerenone is tied to type 2 diabetes, albuminuria, and adequate eGFR with potassium monitoring.
- The systolic blood pressure target of 120 to 129 mm Hg applies at eGFR of at least 30 mL/min/1.73 m² when tolerated (I A); below this threshold, the target is individualized.
- In atrial fibrillation, DOACs are preferred at eGFR of at least 30 mL/min/1.73 m² (I A), an oral factor Xa inhibitor should be considered at eGFR of 15 to 29 (IIa B1), and below 15, including dialysis, decisions are individualized; this does not mean intensified anticoagulation.
- With an ACE inhibitor or ARB, KDIGO recommends checking blood pressure, creatinine, and potassium within 2 to 4 weeks, and a creatinine rise exceeding 30% requires reassessment; after an SGLT2 inhibitor or GLP-1 receptor agonist, ESC/ERA do not automatically require more frequent monitoring.
- The 2026 US AHA/ACC/ADA/ASN guidelines call for eGFR and UACR at least once yearly in CKM stage 2 and higher, but their thresholds should not be confused with those of ESC/ERA.

2.19 ESC 2026 and heart failure: what changes from the nephrology perspective

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. The ESC 2026 heart failure guidelines change the classification, diagnosis, and treatment nomenclature. The HFmrEF category has been eliminated, HFrEF now includes LVEF below 50%, and treatment is divided into FMT, AMT, and GDIT. SGLT2 inhibitors and MRAs are recommended regardless of LVEF (class I, level A), and eGFR and UACR are part of the initial laboratory evaluation. For the nephrologist, the key point is that changes in kidney function should be managed with planned monitoring, not by long-term withholding of prognostic therapy.

Summary

- HFrEF is defined as LVEF below 50% and HFpEF as LVEF of at least 50%, so a patient with LVEF of 41% to 49% is classified as having HFrEF; the diagnosis nevertheless requires a clinical syndrome and objective evidence of cardiac dysfunction.
- FMT in HFrEF consists of renin-angiotensin system inhibition, a beta-blocker, an MRA, and dapagliflozin or empagliflozin, and in HFpEF an SGLT2 inhibitor and an MRA; titration is generally recommended every 1 to 2 weeks.
- The initial evaluation includes a complete blood count, eGFR and UACR, electrolytes, HbA1c, lipids, and iron status (I C); NT-proBNP thresholds for probable HF are at least 125, 250, and 500 pg/mL according to age, and HF is less likely below 125 pg/mL.

- CKD by itself is not a reason to withhold FMT; a hemodynamic change in eGFR is assessed together with volume status, blood pressure, and potassium, and in AKI, severe hyperkalemia, or hypotension, temporary adjustment with active reassessment is possible.
- In decompensation, loop diuretics remain the foundation; with an inadequate response, intravenous acetazolamide or hydrochlorothiazide can be added, urinary sodium-guided management may be considered, and an SGLT2 inhibitor should be initiated during hospitalization.
- The author notes that the class of recommendation is not equivalent to drug approval, reimbursement, or prescribing restrictions, and that the new LVEF cutoff does not turn the disease into two homogeneous groups.

2.20 CKM syndrome as a mitochondrial systems disorder: inflammation, energetics, and interorgan connections

Published September 27, 2026 · [online version \(Slovak\)](#)

Abstract. A mechanistic review in Nature Reviews Nephrology interprets cardiovascular-kidney-metabolic syndrome as a disorder of an interconnected system in which failure of mitochondrial homeostasis, oxidative stress, inflammation, and interorgan communication create amplifying loops. The model is biologically plausible but rests on experimental and tissue evidence of unequal strength. It is a working hypothesis, not a randomized trial or a screening recommendation. No validated mitochondrial biomarker exists, and clinical decisions should rely on drugs with proven effects on kidney and cardiovascular outcomes.

Summary

- The model describes a loop in which metabolic or hemodynamic overload disrupts mitochondria, production of reactive oxygen species increases, inflammation and fibrosis are activated, and the damaged tissue further reinforces the loop.
- The proximal tubule requires large amounts of ATP and relies mainly on fatty acid oxidation, which makes it vulnerable; the heart loses metabolic flexibility, nitric oxide availability declines in the vasculature, and adipose tissue and MASLD amplify the systemic burden.
- The RAAS is an important amplifier but not the only axis, and the benefit of ACE inhibitors, ARBs, or MRAs derives from clinical trials, not from evidence of mitochondrial repair; signals such as GDF15, FGF21, or ceramides are not standard biomarkers.
- As research priorities, the authors list biomarker validation, better demonstration of causality in humans, and interventional studies targeting mitochondrial homeostasis.
- A mitochondrial panel, hs-CRP or IL-6 as standalone decision tests, urate lowering without another indication, and mitochondrial transplantation do not yet belong in routine practice.
- In practice, the priority remains assessment of eGFR, UACR, and the cause of CKD, risk factors, heart failure, and MASLD, together with organ-protective therapy according to established indications; RAAS blockers, SGLT2 inhibitors, GLP-1 receptor agonists, and nonsteroidal MRAs should be chosen according to their proven effect on kidney and cardiovascular outcomes, not according to mechanistic plausibility.

CHAPTER 3

Diabetes, obesity, and incretin-based therapy

Type 1 and type 2 diabetes, obesity, and the new generation of incretin-based drugs. The emphasis is on kidney outcomes, safety, and the limits of evidence for rapidly expanding indications. Number of articles: 56.

Key messages of the chapter

Type 2 diabetes is ceasing to be a disease of glycemia and is becoming a model of cardiorenal-metabolic protection. The ADA and EASD 2026 consensus recommends SGLT2 inhibitors and GLP-1-based therapy potentially from diagnosis (3.56), NICE starts with extended-release metformin and adds an SGLT2 inhibitor (3.11), and HbA1c alone is insufficient for risk estimation (3.8). SGLT2 inhibitors are not interchangeable: in the EU and in Slovakia, only dapagliflozin and empagliflozin are indicated for heart failure and CKD (3.35). A typical error in practice is discontinuing an SGLT2 inhibitor because of an expected eGFR decline of 10% to 15% (3.1).

GLP-1 receptor agonists are becoming part of kidney protection. In the randomized FLOW trial, semaglutide reduced the primary kidney composite by 24% (HR 0.76), and the kidney analysis of SELECT showed an HR of 0.78 even without diabetes (3.37, 3.55). For tirzepatide, the kidney signal comes from an exploratory analysis of SURPASS-CVOT (HR 0.77) (3.24); the primary cardiovascular outcome achieved noninferiority to dulaglutide (HR 0.92), and the CHMP did not recommend an indication for MACE reduction in the EU (3.42); the risk of retinopathy did not increase (3.15). Observational within-class comparisons do not demonstrate superiority: semaglutide did not differ from other agonists in the primary kidney composite (3.27), and lower mortality with tirzepatide in the TriNetX database is an association (3.3). A meta-analysis of head-to-head comparisons showed greater weight loss with tirzepatide by 4.28 percentage points, but also more serious adverse events (RR 1.83) (3.48). In type 1 diabetes, the data are weaker: a target trial emulation suggested a lower risk of MACE and ESKD with GLP-1 receptor agonists (3.9), and the FDA approval of finerenone rests only on a 25% decrease in UACR in the FINE-ONE trial, with hyperkalemia in 10.1% versus 3.3% (3.52). For combinations, the only direct evidence is the phase 2 CONFIDENCE trial on albuminuria (3.36).

Obesity is a chronic disease with a direct kidney impact through glomerular hyperfiltration (3.49), and even a weight loss of 5% to 10% improves BP and glycemia (3.7). Body mass index is insufficient; the distribution and function of fat also matter (3.30, 3.45). Weight returns after treatment is stopped: in STEP 1, 11.6 of the 17.3 percentage points of weight loss were regained, and in SURMOUNT-MAINTAIN, continued tirzepatide maintained most of the loss (3.50), which conference data from ECO 2026 also confirm (3.4, 3.5). A conference abstract suggested a decrease of 560 steps per day after treatment initiation (3.40). New molecules, such as retatrutide (3.13, 3.31), efcoglipron (3.17), HTD1801 (3.43), and liver-targeted insulin (3.16), so far have only metabolic or glycemic results without organ protection. Gray-market retatrutide is associated with unknown product purity and, in a preprint, with a higher burden of cardiovascular symptoms (3.26, 3.41).

Safety signals call for informed vigilance, not panic. GLP-1 receptor agonists are not routinely discontinued before surgery (3.2), an association with NAION has not been causally proven (3.14), and Wernicke encephalopathy is a pharmacovigilance signal that calls for early parenteral thiamine (3.29). The proportion of off-label prescriptions rose from 4.5% to 17%, so screening for eating disorders is advisable (3.19). The main kidney risk during titration and switching between agonists is

volume depletion with gastrointestinal symptoms (3.55, 3.48). Biomarkers and predictors remain hypothesis-generating, for example acylcarnitines (3.10), podocalyxin (3.33), ketone bodies (3.34), genotypes (3.6, 3.28), differences among DPP-4 inhibitors (3.51), pentoxifylline (3.18), and concerns about TMAO-raising supplements (3.23). A post hoc analysis of ACCORD linked stable maintenance of HbA1c and BP at target with a lower kidney risk (HR 0.71) (3.44), and the manufacturer-funded CON-NECT RCT showed a benefit of CGM even without insulin (3.12).

This yields a framework for the outpatient clinic: in type 2 diabetes, regularly measure eGFR and albuminuria (3.8); start an SGLT2 inhibitor and a GLP-1 receptor agonist early according to the cardiorenal phenotype and the registered indication (3.56, 3.35, 3.37); in persistent albuminuria, do not unnecessarily delay adding another pillar (3.36); and with finerenone, monitor for hyperkalemia (3.20, 3.52). Plan obesity treatment for the long term with attention to nutrition, muscle mass, physical activity, and volume status (3.50, 3.21, 3.40), and offer patients approved therapy instead of unregistered peptides (3.38, 3.41).

3.1 The 10 most common mistakes in prescribing SGLT2 inhibitors and GLP-1 receptor agonists

Published May 11, 2026 · [online version \(Slovak\)](#)

Abstract. SGLT2 inhibitors and GLP-1 receptor agonists are now prescribed for organ protection in type 2 diabetes, obesity, CKD, and heart failure, not only to lower HbA1c. This practical review, based on Medscape, summarizes ten recurring mistakes in outpatient practice. They include discontinuing an SGLT2 inhibitor because of the expected eGFR decline of 10% to 15%, underestimating euglycemic ketoacidosis, routinely stopping GLP-1 receptor agonists before procedures, and combining them with a DPP-4 inhibitor. Correct treatment management preserves the cardiorenal benefit and reduces unnecessary risk.

Summary

- In a patient with CKD, albuminuria, obesity, or heart failure, the decision should be driven by the need for organ protection, not by HbA1c alone; SGLT2 inhibitors are often indicated regardless of glycemia and of the presence of diabetes.
- The initial eGFR decline of approximately 10% to 15% after starting an SGLT2 inhibitor is hemodynamic, a decline of up to 30% during the first three months is acceptable, and a larger or progressive decline should prompt a search for hypovolemia, hypotension, excessive diuretic therapy, or NSAID use.
- Genital mycotic infections affect up to approximately 10% of patients and are usually treated without interrupting the drug; euglycemic ketoacidosis is a risk with fasting, a ketogenic diet, alcohol, infection, surgery, and insufficient insulin therapy, so the patient must know the sick day rules.
- An SGLT2 inhibitor is withheld 3 days before planned surgery and resumed once normal food intake returns; GLP-1 receptor agonists are not routinely stopped before procedures, and interruption is considered in gastroparesis, vomiting, or recent dose escalation.
- With a rapid HbA1c decline on semaglutide, worsening retinopathy should be considered; a GLP-1 receptor agonist should not be combined with a DPP-4 inhibitor, and with concomitant sulfonylurea or basal insulin, a dose reduction of approximately 50% should be considered.
- Semaglutide should be discontinued at least 2 months before a planned pregnancy, and a personal or family history of medullary thyroid carcinoma or MEN 2 syndrome is an absolute contraindication.

3.2 GLP-1 receptor agonists before surgery: stop or continue

Published May 11, 2026 · [online version \(Slovak\)](#)

Abstract. GLP-1 receptor agonists slow gastric emptying and may increase residual gastric volume before anesthesia. This review, based on Medscape, shows that meta-analyses have not demonstrated a clear increase in the risk of aspiration, whose incidence is approximately 0.05% to 0.20%, and short-term discon-

tinuation has no documented benefit. The 2025 French guidelines do not recommend routinely stopping long-acting GLP-1 receptor agonists within 7 days before surgery in low-risk patients. The decision should be individualized, in collaboration between the anesthesiologist and the prescribing physician.

Summary

- Higher residual gastric volume has been described with both short-acting and long-acting agents, more markedly with weekly formulations, and delayed gastric emptying may persist for more than 7 days after interruption.
- The risk is higher with higher doses, during titration, with gastrointestinal symptoms, diabetic autonomic neuropathy, and concomitant opioids.
- A clear relationship between the duration of interruption and the reduction in gastric volume is lacking, and discontinuation may worsen glycemia and deprive a patient with CKD of part of the cardiorenal benefit; some studies suggest a higher risk of aspiration, but the evidence is limited.
- Routine prolongation of preoperative fasting is not recommended because it may lead to dehydration and nausea; in case of uncertainty, preoperative gastric ultrasound and adjustment of the anesthetic strategy, such as rapid sequence induction, are helpful.
- The decision should take into account the type of drug and regimen, the titration phase, symptoms of gastroparesis, neuropathy, obesity, concomitant therapy, and the type of procedure and anesthesia.
- The author concludes that continuation of therapy may be appropriate in low-risk patients, whereas with symptoms of gastroparesis, during dose titration, or with other risk factors, specific anesthetic measures or ultrasound assessment of the stomach should be considered.

3.3 Tirzepatide versus semaglutide: lower mortality and fewer gastrointestinal complications

Published May 11, 2026 · [online version \(Slovak\)](#)

Abstract. An observational analysis of the TriNetX database presented at Digestive Disease Week 2026 compared patients with type 2 diabetes and overweight or obesity who initiated tirzepatide or semaglutide. After propensity score matching (2 groups of 126,971), all-cause mortality was 0.9% vs. 1.6% (relative risk 0.59). Tirzepatide was also associated with less frequent ileus and aspiration pneumonitis, whereas the incidence of gastroparesis was similar. This is an association, not proof of causality, and the results support individualized selection rather than a universal preference for tirzepatide.

Summary

- There were 1,182 deaths in the tirzepatide group and 1,998 in the semaglutide group, corresponding to a 41% lower relative risk of all-cause mortality over follow-up from 1 day to 5 years.
- Paralytic ileus or intestinal obstruction occurred in 0.9% vs. 1.0% (relative risk 0.84), aspiration pneumonitis in 0.3% vs. 0.4% (0.69), and gastroparesis in 1.0% vs. 1.1% (0.95).
- Limitations: observational design, possible residual confounding even after matching, incomplete or erroneous coding, and no accounting for doses, adherence, or the reason for drug selection; the authors call for confirmation in prospective studies.
- The principal investigator described it as surprising that the dual GIP/GLP-1 receptor agonist had fewer adverse events, and the poster authors suggest a more favorable benefit-risk ratio of tirzepatide in routine practice; this is their interpretation, not evidence from a randomized trial.
- The choice between tirzepatide and semaglutide should consider the indication, reimbursement, kidney function, cardiovascular risk, metabolic goals, gastrointestinal tolerability, perioperative risk, and patient preferences.

3.4 Five key advances in obesity treatment from ECO 2026

Published May 12, 2026 · [online version \(Slovak\)](#)

Abstract. This summary of conference data from ECO 2026 in Istanbul presents five advances in the pharmacologic treatment of obesity. The most striking are the SURMOUNT-MAINTAIN results: continuing tirzepatide 10 to 15 mg maintained a weight loss of 22.4%, whereas switching to placebo led to weight regain. ATTAIN-MAINTAIN showed maintenance of most of the weight loss after switching to oral orforglipron, and further data concern semaglutide 7.2 mg, oral semaglutide 25 mg, and body composition. These are conference presentations that support long-term and personalized treatment.

Summary

- In SURMOUNT-MAINTAIN, after 60 weeks of tirzepatide treatment (weight loss of 22.1%), continuing the 10 to 15 mg dose resulted in a weight loss of 22.4%, reducing the dose to 5 mg in a weight loss of 16.6% with a regain of about 5.4 kg, and switching to placebo in a weight loss of 10.1% with a regain of about 13.6 kg; 67% of the placebo group required rescue therapy.
- In ATTAIN-MAINTAIN, after switching from Wegovy to orforglipron, the weight loss was 15.1% with a regain of about 0.9 kg, and after switching from Zepbound it was 16.8% with a regain of about 5.0 kg over 52 weeks, with significantly greater regain in the placebo groups.
- In STEP UP, early responders on semaglutide 7.2 mg achieved a weight loss of 27.7% at week 72 (24.8% with 2.4 mg), and in OASIS 4, early responders on oral semaglutide 25 mg achieved 21.6% vs. 2.7% with placebo.
- An MRI substudy of STEP UP showed that approximately 84% of the weight loss consisted of fat mass, visceral fat decreased by more than 30%, and lean mass by approximately 10%, with preserved muscle function.
- The main messages are long-term treatment, flexible dosing, and new oral options; the data come from conference presentations, and early responders are a selected subgroup.

3.5 ECO 2026 brought new data on long-term obesity treatment

Published May 14, 2026 · [online version \(Slovak\)](#)

Abstract. This conference report from ECO 2026 in Istanbul summarizes findings on long-term pharmacologic treatment of obesity. In SURMOUNT-MAINTAIN, patients who continued tirzepatide maintained nearly all of their weight loss, whereas those switched to placebo regained weight. ATTAIN-MAINTAIN suggests that oral orforglipron maintains a large part of the weight loss achieved with injectable therapy, and visceral fat decreased by more than 30% with semaglutide. The results support viewing obesity as a chronic disease with relevance to kidney risk as well.

Summary

- Continuing tirzepatide maintained nearly all of the weight loss, whereas switching to placebo led to substantial regain, supporting continuous treatment of obesity.
- Most of the weight loss with semaglutide 7.2 mg and oral semaglutide 25 mg came from fat mass with preserved muscle function, and oral orforglipron may be an alternative for patients who prefer tablets.
- For nephrology, the key point is that weight reduction affects cardiovascular, diabetes-related, and kidney risk; the author recommends individualizing treatment and monitoring long-term sustainability and body composition.

3.6 Vitamin D receptor genotyping: a new path to type 2 diabetes prevention?

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. In the randomized D2d trial, vitamin D3 at 4,000 IU daily nonsignificantly reduced progression from prediabetes to type 2 diabetes (HR 0.88; 95% CI, 0.75 to 1.04). A secondary genetic analysis published in JAMA Network Open assessed vitamin D receptor polymorphisms. In carriers of the Apal CC or AC geno-

type, supplementation reduced the risk of diabetes by 19% (HR 0.81), whereas it had no effect in the AA genotype. ApaI genotyping is a promising tool for personalized prevention but requires validation before routine use.

Summary

- The D2d trial enrolled adults with prediabetes who received 4,000 IU of vitamin D3 daily or placebo, with a median follow-up of 2.5 years; the ApaI, BsmI, and FokI polymorphisms were assessed, and ApaI yielded the most consistent results.
- The HR was 0.81 (0.66 to 0.99) for the ApaI CC or AC genotype and 1.02 (0.72 to 1.44) for the AA genotype.
- In the discovery phase, patients with the CC genotype had an HR of 0.29 at 25(OH)D of 40 to 50 ng/mL and 0.17 at 50 ng/mL or higher, and those with the AC genotype 0.51 and 0.26, respectively; these levels and doses cannot be mechanically translated into a routine recommendation.
- In D2d, the effect was more pronounced at BMI below 30, and similarly in VITAL among people without obesity; the link between higher BMI and a genetic profile of weaker response remains an unconfirmed hypothesis.
- Limitations: the study was not large enough for analysis by ethnic group, did not clarify the mechanism, and type 2 diabetes is associated with more than 500 genetic loci.
- Supplementation should be considered individually, taking into account 25(OH)D, serum calcium, kidney function, and the risk of nephrolithiasis; weight reduction, physical activity, and diet remain the foundation of prevention.

3.7 Obesity in general practice: why it must be treated as a chronic disease

Published May 18, 2026 · [online version \(Slovak\)](#)

Abstract. Obesity is a chronic metabolic disease that increases the risk of type 2 diabetes, hypertension, dyslipidemia, chronic kidney disease, cardiovascular disease, and other complications. This educational review, based on a STREAMED UP educational video, describes obesity management in the general practitioner's office: diagnosis beyond BMI, sensitive communication, a comprehensive plan, pharmacotherapy, and indications for specialist referral. A weight loss of only 5% to 10% can improve blood pressure, glycemia, triglycerides, and hepatic steatosis. For nephrology, weight reduction is part of preventing hyperfiltration, albuminuria, and CKD progression.

Summary

- In addition to BMI, patient assessment should include waist circumference, blood pressure, glycemic and lipid profiles, liver tests, kidney function, and albuminuria, as well as screening for sleep apnea, joint pain, depression, anxiety, and weight-promoting medications.
- It is crucial to determine whether organ or metabolic complications are already present, because at the same BMI the risk varies substantially according to the presence of diabetes, hypertension, albuminuria, hepatic steatosis, or cardiovascular disease.
- A conversation about weight should begin by asking whether the patient agrees to discuss the topic and should use neutral language about health risk and goals; a weight loss of 5% to 10% can improve blood pressure, glycemia, triglycerides, hepatic steatosis, and musculoskeletal load.
- Treatment rests on dietary modification, physical activity, sleep, stress management, treatment of comorbidities, and psychological support; goals should be modest, measurable, and achievable, and the author considers weight regain not a failure of the patient but a reason to continue treatment.
- Pharmacotherapy, including incretin-based drugs, does not replace lifestyle measures; drug selection should take into account BMI, comorbidities, cardiometabolic risk, contraindications, tolerability, and availability and should follow current guidelines and approved indications.
- Specialist referral is appropriate in severe obesity, rapid weight gain, suspected endocrine cause, complicated diabetes, advanced CKD, heart failure, severe hepatic steatosis, eating disorders, or when bariatric treatment is being considered.

- Treatment success should be measured not only by weight but by improvement in risk factors, kidney parameters, functional capacity, and quality of life; in some patients, weight stabilization is a realistic goal.

3.8 Type 2 diabetes at the center of cardiometabolic prevention: from lipids to organ health

Published May 18, 2026 · [online version \(Slovak\)](#)

Abstract. Type 2 diabetes mellitus is a complex cardiometabolic disease affecting the blood vessels, heart, kidneys, liver, and lipid metabolism. This review, based on an educational livestream on streamed-up.com, emphasizes that HbA1c alone is insufficient to estimate risk and that treatment should be guided by the overall cardiometabolic profile. It presents no new trial data but summarizes the roles of statins, ezetimibe, PCSK9 inhibitors, SGLT2 inhibitors, GLP-1 receptor agonists, and finerenone. In practice, regular testing of albuminuria and eGFR is essential, because albuminuria is often the first warning sign and early intervention can slow the progression of kidney damage.

Summary

- A patient with acceptable HbA1c may simultaneously have a high LDL cholesterol burden, hypertension, albuminuria, obesity, hepatic steatosis, or subclinical atherosclerosis; cardiovascular disease remains the leading cause of morbidity and mortality.
- Atherogenic dyslipidemia with elevated triglycerides, low HDL cholesterol, and small dense LDL particles is typical; moderate- to high-intensity statins are the foundation of treatment, and ezetimibe or PCSK9 inhibitors may be considered when targets are not achieved, particularly at very high risk.
- Albuminuria is a marker of both vascular damage and cardiovascular risk; a patient with albuminuria and declining eGFR requires rigorous blood pressure control, optimization of glucose-lowering therapy, and treatment with a proven kidney-protective effect.
- SGLT2 inhibitors favorably affect CKD progression, hospitalizations for heart failure, and cardiovascular outcomes in selected patients; in albuminuric CKD, finerenone is used when indicated, and GLP-1 receptor agonists reduce atherosclerotic events in selected high-risk patients.
- Heart failure may long remain unrecognized in patients with diabetes because dyspnea, fatigue, or edema are attributed to age or obesity; care should also include assessment of metabolic dysfunction-associated steatotic liver disease and the risk of fibrosis.
- Metformin remains a suitable foundation if tolerated, but further therapy should be chosen according to the presence of atherosclerotic disease, heart failure, CKD, obesity, and the risk of hypoglycemia, not only according to HbA1c; the author recommends not waiting for a marked eGFR decline.

3.9 Cardiac and kidney benefits of GLP-1 analogues may extend to type 1 diabetes

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. People with type 1 diabetes have increased cardiovascular and kidney risk but were generally excluded from the pivotal RCTs of GLP-1 receptor agonists. An observational target trial emulation study using the Optum Labs Data Warehouse included 174,678 people with T1D, of whom 14,488 initiated a GLP-1 receptor agonist. The 5-year risk of MACE was 4.3% vs. 5.0% and the risk of ESKD 1.6% vs. 1.9%, with no increase in diabetic ketoacidosis or severe hypoglycemia. The results need to be verified in large randomized trials.

Summary

- Median follow-up was 38 months; in the intention-to-treat analysis, the relative risk was approximately 15% lower for MACE and approximately 19% lower for ESKD, which the authors consider comparable to the effect in T2D.
- Hospitalizations for heart failure decreased by approximately 18% and severe liver events by approximately 28%, and more patients achieved weight loss of 5%, 10%, and 15%.

- Hospitalizations for DKA or severe hypoglycemia did not increase, gastrointestinal symptoms were more frequent without clear statistical significance, and results were consistent across age and baseline HbA1c.
- These are observational data, so caution is warranted; the authors link safety in practice to glucose monitoring technologies, appropriate patient selection, and close follow-up.

3.10 Serum acylcarnitines may predict kidney function decline in type 2 diabetes

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. The rate of eGFR decline in patients with type 2 diabetes varies widely, and clinical models predict it only partially. A prospective observational study used targeted metabolomics to measure 40 serum acylcarnitines in a discovery cohort of 575 patients (Gargano Mortality Study) and a validation cohort of 252 patients (Joslin Kidney Study). Eleven acylcarnitines were independently associated with faster eGFR decline, and a score based on 6 metabolites improved the performance of established clinical models. These are associations with incomplete external validation, not a test ready for routine practice.

Summary

- Baseline serum levels of 40 short-, medium-, and long-chain acylcarnitines were measured by targeted metabolomics, and their association with the rate of eGFR decline over time and their added predictive value beyond existing clinical models were assessed.
- In the discovery cohort, with a median follow-up of approximately 9 years, 11 of 40 acylcarnitines were independently associated with accelerated eGFR decline, most strongly certain short-chain species, whereas other metabolites showed an inverse association.
- In the validation cohort, with a median follow-up of approximately 10 years, tiglylcarnitine and methylglutaryl carnitine in particular were confirmed as predictors of eGFR decline.
- A multimarker score of 6 acylcarnitines (methylglutaryl carnitine, hydroxyvaleryl carnitine, hexenoyl carnitine, decadienyl carnitine, dodecanedioyl carnitine, tetradecadienyl carnitine) improved the identification of individuals at risk of rapid eGFR decline beyond clinical models.
- Limitations: external validation was only partial, the performance of the 6-metabolite score could not be externally verified, and data on diet, which may influence acylcarnitine levels, were lacking.
- The results are associations, not proof of causation, and do not clarify the origin of the metabolites or their causal effect on the disease.
- In practice, this is not yet a clinical test; acylcarnitines are a promising biomarker for early risk stratification that needs to be confirmed in broader and fully external cohorts.

3.11 Extended-release metformin and an SGLT2 inhibitor as the new first-line treatment of type 2 diabetes

Published June 5, 2026 · [online version \(Slovak\)](#)

Abstract. The updated guideline from the UK's NICE, reported by Medscape, shifts the treatment of type 2 diabetes from glycemic control toward cardiorenal-metabolic protection. In most patients, treatment should begin with extended-release metformin, followed by the addition of an SGLT2 inhibitor and, in atherosclerotic cardiovascular disease, also subcutaneous semaglutide at doses up to 1 mg weekly. Drugs should be added sequentially after titration to the highest tolerated dose. For nephrology practice, the guideline confirms that type 2 diabetes calls for cardiorenal thinking from the outset.

Summary

- Extended-release metformin is preferred because of fewer gastrointestinal adverse effects and better adherence; a patient who tolerates standard metformin well and has good control need not be switched automatically.

- An SGLT2 inhibitor is added for its cardiovascular and kidney benefits, particularly fewer hospitalizations for heart failure and slower CKD progression; in CKD, the NICE guideline specifies an eGFR threshold above 30 mL/min/1.73 m².
- In atherosclerotic cardiovascular disease, NICE recommends triple therapy with extended-release metformin, an SGLT2 inhibitor, and subcutaneous semaglutide up to 1 mg weekly, naming a specific drug for the first time rather than only a drug class.
- Drugs should not be started simultaneously: one is initiated and titrated to the highest tolerated dose before the next is added, with active involvement of the patient in decision making.
- Before initiating an SGLT2 inhibitor, contraindications, kidney function, and the risk of dehydration, urogenital infections, and ketoacidosis should be considered, as should the benefit-risk ratio in frail patients; temporary interruption may be necessary during acute illness, vomiting, diarrhea, or before surgery.
- GLP-1 receptor agonists or tirzepatide have a role in early-onset diabetes and in obesity if glycemic targets are not achieved after initial therapy; intensification, including insulin, should be explained to the patient as a consequence of disease progression, not as a personal failure.

3.12 Continuous glucose monitoring may also help patients with type 2 diabetes not on insulin

Published June 9, 2026 · [online version \(Slovak\)](#)

Abstract. Continuous glucose monitoring (CGM) is the standard in insulin-treated diabetes, but large randomized evidence in type 2 diabetes without insulin has been lacking. The multicenter CONNECT RCT, presented at the ADA 2026 Scientific Sessions, randomized 283 adults from 22 primary care practices to Dexcom G7 or a standard glucose meter. After 26 weeks, HbA1c decreased by 1.6% vs. 0.7%, and time in range increased from 29% to 62%. Patients with marked hyperglycemia benefited most; the trial was funded by the device manufacturer.

Summary

- The population comprised 145 patients with CGM and 138 in usual care, with a mean age of 60 years, median diabetes duration of 10 years, mean HbA1c of 8.8%, and BMI of 33 kg/m²; 37% were taking an SGLT2 inhibitor and 40% incretin-based therapy.
- The between-group difference in HbA1c reduction was 0.9% and statistically significant; with baseline HbA1c of at least 10%, HbA1c decreased by 3.1% with CGM vs. 1.2%, whereas the difference was smaller with HbA1c below 8%.
- Time in the 70 to 180 mg/dL range (3.9 to 10.0 mmol/L) increased from 29% to 62% with CGM and from 31% to 42% in the control group, corresponding to approximately 5 more hours per day in range.
- The benefit was observed in patients with and without GLP-1 receptor agonists and SGLT2 inhibitors; severe hypoglycemia did not occur in either group, and patients with CGM reported greater satisfaction and less diabetes distress.
- The trial was funded by Dexcom, the manufacturer of the system used, which should be considered in interpretation, and the results need to be confirmed in further studies and in routine practice.
- CGM may help patients with elevated HbA1c despite treatment, low time in range, or a need for feedback on the effects of diet, physical activity, and medications, provided the team can interpret the data and translate them into decisions.

3.13 Retatrutide delivers substantial weight loss and metabolic benefits, but the new data must be read with caution

Published June 9, 2026 · [online version \(Slovak\)](#)

Abstract. Retatrutide is an investigational triple agonist of the GLP-1, GIP, and glucagon receptors. Randomized trials presented at the ADA 2026 congress showed, in TRANSCEND-T2D-1 (537 patients with type 2 dia-

betes, published in *The Lancet*), HbA1c reductions of up to 1.94% and weight loss of 15.3% after 40 weeks, and in TRIUMPH-1 (2,339 adults with obesity), weight loss of 28.3% after 80 weeks with the 12 mg dose. Urinary tract infections are a new safety signal. The drug remains unapproved, and its role in practice will be determined by further data.

Summary

- In TRANSCEND-T2D-1 (HbA1c 7.0% to 9.5%, BMI of at least 23 kg/m²), HbA1c decreased by 1.69%, 1.86%, and 1.94% with the 4, 9, and 12 mg doses vs. 0.81% with placebo; with 12 mg, weight decreased by 15.3% (approximately 16.6 kg) vs. 2.6%.
- In TRIUMPH-1, the 9 mg dose led to weight loss of approximately 26%; in the extension to 104 weeks, weight loss ranged from 19.2% to 30.3%, and more than one quarter of participants lost at least 35% of their body weight.
- In nested analyses of TRIUMPH-1, knee osteoarthritis pain according to WOMAC decreased by up to 73.1%, and the AHI in moderate to severe sleep apnea by up to 60.6%.
- In TRIUMPH-1, triglycerides decreased by up to 41.0%, non-HDL cholesterol by up to 24.2%, systolic blood pressure by up to 12.3 mm Hg, and waist circumference by up to 24.1 cm; in TRANSCEND-T2D-1, the corresponding values were 39.6%, 19.8%, 6.4 mm Hg, and 12.4 cm.
- In TRIUMPH-1, urinary tract infections occurred in 7.5% to 8.8% vs. 5.3% with placebo, with an unclear mechanism, and discontinuation for adverse events, mainly gastrointestinal, was dose-dependent at 4.1%, 6.9%, and 11.3% vs. 4.9%.
- Experts caution that maximal weight loss is not an appropriate goal for everyone; data on long-term safety, durability of effect, and patient selection are lacking.

3.14 GLP-1 drugs and ischemic optic neuropathy: a signal to monitor, not a reason to panic

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. With the expanding use of GLP-1 receptor agonists, a possible association with nonarteritic anterior or ischemic optic neuropathy (NAION), whose incidence is estimated at 2 to 10 cases per 100,000 persons per year, is under discussion. This review evaluates available observational studies, pharmacovigilance data, and case reports and concludes that a causal relationship has not been demonstrated and that the risk factors for NAION overlap substantially with the indications for treatment. The absolute risk appears small. The author recommends informed vigilance: do not routinely discontinue the drugs, but manage sudden visual disturbance urgently.

Summary

- NAION is the most common acute optic neuropathy in adults older than 50 years, typically presenting with sudden painless, mostly unilateral vision loss and visual field defects, often noticed on awakening; the damage may be permanent.
- Risk factors for NAION (diabetes, hypertension, obstructive sleep apnea, older age, vascular disease, a crowded optic disc, and probably nocturnal hypotension) are common precisely among treated patients, so the signal may reflect overlapping risk populations.
- Proposed mechanisms, such as rapid correction of glycemia and changes in blood pressure and volume status, are biologically possible but unproven; the stronger pharmacovigilance signal with Wegovy may theoretically relate to the higher semaglutide dose but does not prove causality.
- Observational and pharmacovigilance data cannot exclude confounding by indication, the influence of the underlying disease, or selective reporting, and with millions of treated patients, even rare events appear more often in absolute numbers.
- The author recommends asking about visual disturbances, educating the patient about warning symptoms, taking diabetes, hypertension, and sleep apnea into account, and avoiding inappropriately rapid metabolic correction in high-risk patients.

- With sudden visual disturbance, treatment should be individually reassessed and the patient referred for urgent ophthalmologic examination; routine discontinuation of GLP-1 receptor agonists solely out of concern about NAION is not justified.

3.15 Tirzepatide did not increase the risk of diabetic retinopathy in high-risk patients

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Rapid improvement in glycemia may transiently worsen diabetic retinopathy in some patients, so investigators examined whether the highly effective tirzepatide affects it. The retina substudy of the randomized SURPASS-CVOT trial, presented at the ADA 2026 congress, compared tirzepatide with dulaglutide in 920 high-risk patients with type 2 diabetes. Despite a greater HbA1c reduction, progression by two or more ETDRS steps at 36 months was similar, 21.3% vs. 23.3% (OR 0.89, nonsignificant difference). Regular eye screening remains essential.

Summary

- The substudy included 920 of 13,165 participants with type 2 diabetes and cardiovascular disease who had retinopathy or macular edema in at least one eye, or diabetes for 15 years or longer and HbA1c of at least 8.0%; 449 received tirzepatide up to 15 mg and 471 dulaglutide 1.5 mg weekly.
- With the same baseline HbA1c of 8.69%, the reduction at six months was 2.18 percentage points with tirzepatide vs. 1.28 with dulaglutide, and the difference was statistically significant.
- Time to first progression by two or more ETDRS steps did not differ significantly between groups, and the retina was assessed with standardized fundus photographs at 12, 18, 24, and 36 months.
- The composite endpoint of serious eye events (photocoagulation, vitreous hemorrhage, anti-VEGF therapy, vitrectomy, significant loss of visual acuity) had an HR of 1.1 with no significant difference, regardless of the presence of retinopathy at baseline.
- The comparator was active dulaglutide, not placebo; a commenting ophthalmologist stated that transient worsening of retinopathy with a rapid HbA1c decline is more common in type 1 diabetes and only rarely clinically significant.
- With long diabetes duration, higher HbA1c, existing retinopathy, or rapid glycemic improvement, regular retinal screening remains essential, and with sudden visual disturbance the patient should seek medical attention without delay.

3.16 Liver-directed insulin: a possible path to lower hypoglycemia risk in type 1 diabetes

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Hypoglycemia remains the main barrier to treating type 1 diabetes, even with continuous glucose monitoring. The randomized, double-blind phase 2b OPTI-2 trial, presented at the ADA 2026 congress, compared liver-directed HDV insulin combined with lispro with standard lispro in 226 adults. HDV-lispro was noninferior for HbA1c and, in the maintenance phase, reduced 24-hour level 2 hypoglycemia below 54 mg/dL (3.0 mmol/L) by 28%. This is an investigational therapy that must be confirmed in a phase 3 trial, and it does not yet change practice.

Summary

- HDV (hepatic-directed vesicle) insulin uses a phospholipid matrix bound to short-acting insulin to direct its effect to the liver and promote glycogen storage; protection against hypoglycemia is so far a mechanistic hypothesis.
- All participants used once-daily degludec and Dexcom G7, median HbA1c was approximately 7.3% to 7.4%, and the trial had a 12-week optimization phase and a 13-week maintenance phase.
- HbA1c decreased by 0.34 vs. 0.42 percentage points at 12 weeks and by 0.31 vs. 0.38 percentage points at 25 weeks, without a significant difference, meeting the FDA noninferiority requirement; superiority for the primary endpoint during titration was narrowly missed.

- In the maintenance phase, HDV-lispro reduced daytime level 2 hypoglycemia episodes by 33%, time below 54 mg/dL by 28% over 24 hours and by 32% during the day, and prolonged episodes by 36%; over the entire trial, the incidence of level 2 hypoglycemia was 25% lower.
- No level 3 hypoglycemia occurred with HDV-lispro, compared with five cases with standard lispro, although the difference narrowly missed statistical significance; adverse events occurred in 53.6% vs. 50% of patients, with no deaths, DKA, or hepatic signals.
- The key question for phase 3 will be long-term hepatic safety, which a smaller phase 2 trial may not capture.

3.17 Elecglipton, a new oral GLP-1 receptor agonist: phase 2 results with improved HbA1c and weight loss

Published June 13, 2026 · [online version \(Slovak\)](#)

Abstract. Elecglipton is a new oral small-molecule GLP-1 receptor agonist. In two phase 2 trials over 26 weeks, it achieved dose-dependent reductions in HbA1c of approximately 0.91% to 1.88% and in weight of 2.54% to 7.61% in SOLSTICE among 406 patients with type 2 diabetes, and weight loss of 2.6% to 10.5% vs. 0.6% with placebo in VISTA among 310 people with obesity or overweight without diabetes. The adverse effect profile was consistent with the class. Long-term kidney outcomes are unknown and will be determined by phase 3 trials.

Summary

- SOLSTICE (phase 2b) randomized patients with type 2 diabetes to elecglipton 5, 15, or 25 mg once daily with different titration schedules, to placebo, or, on an exploratory basis, to oral semaglutide titrated to 14 mg.
- In SOLSTICE, at the highest dose a greater proportion of patients achieved weight loss of at least 5% compared with placebo, and some blood pressure markers and liver biomarkers also improved.
- In VISTA (obesity or overweight with at least one comorbidity), 88.8% achieved weight loss of at least 5% vs. 15.6% with placebo, and by week 36 the effect on some measures continued to increase without a clear plateau.
- Adverse events were typical of the GLP-1 class, mainly nausea, constipation, diarrhea, and vomiting, and some patients discontinued treatment, particularly at higher doses and with some titration schedules.
- The weight loss is comparable to that with other oral small-molecule GLP-1 receptor agonists but lower than with some injectable incretin therapies; an advantage is oral administration without strict dosing restrictions.
- Improvements in HbA1c and weight have been demonstrated only in phase 2; whether they translate into long-term kidney outcomes remains to be shown by phase 3 trials with longer follow-up and a broader population.

3.18 Pentoxifylline in diabetic kidney disease: does it still have a place? (AJKD mini review)

Published June 17, 2026 · [online version \(Slovak\)](#)

Abstract. Diabetic kidney disease progresses despite RAAS blockade, SGLT2 inhibitors, nonsteroidal mineralocorticoid receptor antagonists, and GLP-1 receptor agonists, which is why complementary anti-inflammatory strategies are being sought. A mini review in AJKD evaluates pentoxifylline as a repurposed drug. The strongest data come from the PREDIAN trial, in which pentoxifylline added to RAAS blockade reduced proteinuria and significantly slowed eGFR decline, whereas other studies and meta-analyses rely on surrog-

ate endpoints. Pentoxifylline remains a candidate for add-on therapy, not a standard, until the large VA PTXRx trial reports.

Summary

- The biological rationale is the role of inflammation in DKD progression: the proinflammatory cytokines TNF- α , IL-1, and IL-6, oxidative stress, and interstitial macrophages are associated with worsening proteinuria, tubulointerstitial injury, fibrosis, and declining kidney function.
- Pentoxifylline is a nonspecific inhibitor of the phosphodiesterases PDE3 and PDE4; increased cAMP and PKA activation are thought to suppress production of IL-1, IL-6, and TNF- α and, via the NF- κ B axis, indirectly affect oxidative stress and fibrosis.
- In the PREDIAN trial, pentoxifylline also reduced urinary TNF- α , with no increase in serious adverse events but with a twofold incidence of gastrointestinal symptoms compared with placebo.
- Most supporting studies were small and short and assessed proteinuria, albuminuria, or changes in eGFR; neither PREDIAN nor the other trials were powered for ESKD or mortality.
- Meta-analyses suggest reductions in albuminuria and proteinuria without a signal of serious adverse effects, but because the studies are small, heterogeneous, and methodologically weaker, they are insufficient to change practice.
- The answer should come from the pragmatic, randomized, placebo-controlled, multicenter VA PTXRx trial (NCT03625648), with time to ESKD or death as the endpoint; until then, there is no legitimate reason for routine add-on use in patients on modern foundational therapies.

3.19 GLP-1 and eating disorders: growing concerns about inappropriate use and how to prevent it

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. Easier access to GLP-1 receptor agonists through oral formulations and telemedicine increases the risk of prescribing to people outside approved indications. According to an analysis of community pharmacies, the proportion of such prescriptions rose from 4.5% in 2018 to 17% in 2023. This review of clinical experience highlights underrecognized eating disorders, aesthetic goals, and the risk of excessive weight loss or loss of muscle mass. It recommends a targeted history before prescribing, screening when a disorder is suspected, for example with BEDS-7, and framing the treatment goal as metabolic health.

Summary

- Warning signs include a desire to lose weight even after reaching a healthy weight, a purely aesthetic goal, a concealed history of an eating disorder, binge eating, purging behavior, restriction, an intense fear of weight gain, mental preoccupation with food (food noise), and a request for rapid weight loss without medical need.
- When initiating treatment, clinicians should ask about loss of control over eating and any past eating disorder and, with a positive response, add a validated screening tool, such as the Binge Eating Disorder Screener-7, together with a detailed history of body image.
- Appetite suppression may make adequate protein intake difficult and contribute to loss of muscle mass, so nutritional education is needed, possibly with a food diary and the involvement of a psychologist or dietitian.
- Clinicians describe more frequent inappropriate use among middle-aged women who want to lose the last few kilograms, among patients with a history of an eating disorder, and in those with existing loss of muscle mass or declining bone density.
- The evidence is not unequivocal: some studies suggest a benefit of semaglutide in binge eating disorder, so the problem may lie not in the drug's mechanism but in the indication and the treatment goal.
- When a disorder is suspected, a mental health professional should be consulted, and caution is warranted with abbreviated telemedicine prescribing; some experts call for an in-person assessment at least at initiation and at follow-up visits.

3.20 Beyond hemodynamics: new evidence and perspectives in nondiabetic chronic kidney disease (CKD)

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. In nondiabetic CKD, a residual risk of progression persists despite RAAS inhibition and SGLT2 inhibitors. A Medscape educational activity summarizes data from the ERA congress, particularly the randomized FIND-CKD trial of finerenone in 1,584 patients with nondiabetic CKD and albuminuria of at least 200 mg/g. Finerenone attenuated the eGFR decline from approximately 4 to 3.3 mL/min per year and reduced the key secondary cardiorenal composite by approximately 23%. Hyperkalemia was more frequent, but discontinuation or hospitalization because of it was rare. The results support targeting nonhemodynamic mechanisms.

Summary

- According to the source, albuminuria is the main driver of residual risk and a trigger of glomerular and tubular injury leading to fibrosis; even subclinical aldosterone excess may predict eGFR decline and adverse outcomes.
- FIND-CKD enrolled patients with eGFR above 25 and below 75 mL/min/1.73 m² on stable RAAS inhibition, randomized to finerenone or placebo; the primary endpoint was the total GFR slope over 3 years.
- The absolute difference in eGFR slope was approximately 0.7 mL/min per year, which may appear modest, but the source translates it into hard outcomes.
- The key secondary composite of kidney failure or a 57% decline in GFR, hospitalization for heart failure, and cardiovascular death was reduced by approximately 23% in relative terms, a statistically significant result.
- Finerenone was well tolerated, with no imbalance in serious adverse events; hyperkalemia occurred in approximately 17% vs. 13.3%, but permanent discontinuation or hospitalization because of it occurred in less than 2%.
- The source recommends supplementing hemodynamic therapy with treatments targeting the aldosterone axis and inflammatory signaling; it describes glomerular diseases as a group with fewer treatment options and faster progression.

3.21 The GLP-1 era demands a new model of obesity care (and what it means for nephrology)

Published June 27, 2026 · [online version \(Slovak\)](#)

Abstract. Growing pressure to prescribe GLP-1 receptor agonists reveals a mismatch between patients' long-term needs, healthcare infrastructure, and the societal perception of obesity. A Medscape article from the HLTH Europe 2026 conference summarizes the views of panelists, not trial results. According to them, after treatment interruption patients often regain most of their weight within approximately 18 months, and structured treatment reduced sick leave by approximately 50% within 6 months. For nephrology, this implies the need for a long-term multidisciplinary program with emphasis on nutrition, muscle mass, and kidney function.

Summary

- The treatment goal is shifting from weight loss alone to metabolic health, including adequate protein and fiber intake and retention of muscle mass; weight reduction without regard to muscle may impair functional reserve.
- Barriers to wider use include stigma portraying obesity as a lifestyle choice, unequal access in the public and private systems, and economics, for example in the United Kingdom, with limited access through the NHS and a high number of private prescriptions.
- Because relapse is likely after treatment ends, GLP-1 receptor agonists should be viewed as chronic therapy with a plan for continuation, a maintenance phase, and ongoing outcome measurement.

- The panelists consider economic models, such as the NICE model with a high number of primary care visits, outdated and regard digital care as a path to lower costs without loss of safety.
- Within 5 to 10 years, prices are expected to fall owing to competition and generics, and according to Kai Eberhardt, obesity drugs could become a basic preventive tool, similar to statins.
- For nephrology, the author proposes program-based treatment: monitoring nutrition, body composition, kidney function, and hydration, coordinating treatment with comorbidities, and addressing access barriers; however, the article is not a nephrology guideline.

3.22 Obesity as a “cost” diagnosis in healthcare: what it means for access to GLP-1 therapy and, indirectly, for nephrology

Published July 2, 2026 · [online version \(Slovak\)](#)

Abstract. Reimbursement for obesity treatment is both a clinical and an economic question. An opinion piece by Amy Faith Ho on Medscape discusses US health policy, the BALANCE model linking GLP-1 drugs with behavioral support, and the temporary Medicare GLP-1 Bridge program, whose extension leaves a permanent solution unclear. It presents no clinical data, and the savings estimates cited are only indicative. For nephrology, unstable access to treatment threatens adherence and the long-term cardiorenal benefit, so continuity of treatment should be planned early and its indication documented.

Summary

- According to the text, obesity leads to hypertension, type 2 diabetes, dyslipidemia, sleep apnea, and systemic inflammation, which in nephrology translates into faster CKD progression, higher cardiovascular risk, and a larger number of patients on dialysis.
- The BALANCE model was intended to shift payments from treating consequences to prevention, first in Medicaid and then in Medicare; before its launch in Medicare, however, the temporary Bridge program was extended with a reduced monthly payment and limited duration.
- Estimates of savings and costs depend on real-world adherence, the proportion of patients on long-term treatment, comorbidities, CKD stages, and the definition of cost offsets.
- Reimbursement uncertainty is a clinical problem: a patient starts treatment and then encounters a barrier to continuation, which may worsen long-term outcomes in patients with CKD, diabetes, and high cardiovascular risk.
- The author of the article recommends documenting the indication and the expected cardiometabolic and kidney benefit, addressing administrative barriers early, monitoring tolerability and metabolic parameters, and explaining treatment to the patient as an investment in the future of the kidneys.
- This is health policy material that may change, so current reimbursement conditions should be verified in primary CMS sources.

3.23 Choline and L-carnitine supplements in diabetic nephropathy: when TMAO can create a vicious circle

Published July 3, 2026 · [online version \(Slovak\)](#)

Abstract. Trimethylamine N-oxide (TMAO) is formed from choline, betaine, and L-carnitine through the action of the gut microbiome and the hepatic enzyme FMO3, and it accumulates in CKD because of reduced excretion. This review of mechanistic, preclinical, and observational data describes the TMAO pathway via the NLRP3 inflammasome and the TGF- β /Smad3 axis leading to inflammation, fibrosis, and endothelial dysfunction. A pilot study showed that six months of L-carnitine supplementation increased TMAO severalfold. Causal evidence on hard kidney endpoints is lacking, but the author recommends caution with these supplements in patients with diabetes who have lower eGFR and albuminuria.

Summary

- TMAO production varies between individuals with the same precursor intake according to the composition of the gut microbiome, which performs the first step of conversion to trimethylamine.

- In preclinical models, TMAO activates the NLRP3 inflammasome with production of IL-1 β and IL-18 and the profibrotic TGF- β /Smad3 axis; chronic dietary exposure to choline and TMAO led to tubulointerstitial fibrosis and declining kidney function in animals.
- TMAO is excreted mainly by glomerular filtration, so its levels are markedly higher in CKD; in patients with diabetes and CKD, they correlate with lower eGFR, albuminuria, and inflammatory markers, which the author describes as a possible vicious circle.
- The pilot study in older women was small and outside the nephrology population, but it shows that a precursor taken as a supplement does translate into higher TMAO production.
- Limitations: correlation does not prove causality; in a prospective cohort with moderate to advanced CKD, TMAO was not an independent predictor of cardiovascular or kidney outcomes after adjustment for kidney function, betaine rather being the independent factor; and large RCTs are lacking.
- The author recommends basing decisions on the CKD stage and albuminuria, the justification for supplementation, and other therapy; in advanced CKD, he considers supplements containing TMAO precursors without clear benefit an unnecessary risk, but he does not restrict the usual diet.

3.24 Tirzepatide and kidney outcomes in the SURPASS program: what they mean for nephrology

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Tirzepatide, a dual GIP/GLP-1 receptor agonist, markedly reduces HbA1c and weight, but its kidney effect remains an open question. This review summarizes a post hoc analysis of SURPASS-4 vs. insulin glargine and a prespecified exploratory analysis of SURPASS-CVOT vs. dulaglutide. In SURPASS-CVOT, after a median of 4.0 years, the kidney composite occurred in 6.0% vs. 7.6% (HR 0.77). Because this is not a primary kidney trial, the author views tirzepatide as a complement to, not a replacement for, RAAS blockade and an SGLT2 inhibitor in patients with type 2 diabetes, obesity, and albuminuria.

Summary

- In SURPASS-4 (mean eGFR 81.3 mL/min/1.73 m², median UACR 15 mg/g), eGFR declined by 1.4 mL/min/1.73 m² per year vs. 3.6 with insulin glargine, a difference of 2.2 mL/min/1.73 m² per year.
- UACR increased by 36.9% with glargine and did not increase with tirzepatide, a difference of approximately 31.9%; the composite of an eGFR decline of $\geq 40\%$, ESKD, kidney death, or macroalbuminuria had an HR of 0.58.
- SURPASS-CVOT enrolled patients with type 2 diabetes and atherosclerotic disease, and its composite comprised persistent macroalbuminuria, an eGFR decline of $\geq 50\%$, kidney failure, or kidney death.
- At low to moderate CKD risk, the benefit derived mainly from fewer cases of new macroalbuminuria, and at high risk from slower eGFR decline; the difference in annual slope was 0.29 mL/min/1.73 m² overall and 0.93 in the high-risk group.
- Limitations: SURPASS-4 was a post hoc analysis and SURPASS-CVOT an exploratory analysis of a cardiovascular trial, not a kidney trial like FLOW; it is unclear what part of the effect is independent of reductions in weight, HbA1c, and blood pressure, and data in advanced CKD are lacking.
- In CKD, gastrointestinal effects and dehydration should be monitored, particularly with diuretics, RAAS blockers, and SGLT2 inhibitors, as should eGFR, UACR, blood pressure, and the risk of hypoglycemia in combination with insulin or a sulfonylurea.

3.25 GLP-1 receptor agonists, “food noise,” and compulsive behavior: what it means for nephrology

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Beyond their metabolic effects, GLP-1 receptor agonists may influence the reward system, craving, and compulsive behavior, which patients describe as a quieting of “food noise.” This review based on Medscape summarizes small RCTs in alcohol use disorder (JAMA Psychiatry 2025, The Lancet 2026), a phase 2a

trial in smoking, an observational cohort of more than 600,000 veterans (BMJ 2026), and an experiment in mice (Nature 2026). The strongest signal concerns alcohol, whereas for nicotine only craving decreased. For nephrology, this represents a possible indirect benefit for cardiorenal risk, not an approved indication.

Summary

- The hypothesis distinguishes liking (pleasure) from wanting (motivational desire); GLP-1 receptor agonists may primarily reduce the intensity of wanting and the salience of food cues, typically of ultraprocessed foods, not necessarily pleasure itself.
- In the RCT in JAMA Psychiatry, low-dose semaglutide reduced laboratory alcohol self-administration and weekly craving, but not all drinking measures; in the RCT in The Lancet in patients with alcohol use disorder and obesity, it reduced heavy drinking days, monthly consumption, and craving after 26 weeks.
- The phase 2a trial in daily smokers did not show a reduction in the number of cigarettes, only a signal of lower craving and weight; the cohort of veterans with type 2 diabetes associated GLP-1 receptor agonists, compared with SGLT2 inhibitors, with a lower risk of new substance use disorders, but these are observational data.
- The mouse study identified GLP-1 receptor neurons in the central amygdala whose activation via dopamine signaling reduces consumption of highly palatable food, although this applies mainly to newer small molecules.
- Some patients describe emotional blunting or reduced motivation, and the evidence is inconsistent; GLP-1 drugs are not an approved treatment for addiction and do not replace psychiatric, addiction, or psychological care.
- In the nephrology outpatient clinic, it is worth asking about changes in the relationship to food, alcohol, and smoking as well as about loss of interest, and in CKD, monitoring for nausea, vomiting, and dehydration, particularly with diuretics, RAAS blockade, and SGLT2 inhibitors.

3.26 Retatrutide outside approval: when an investigational drug outpaces regulation

Published July 10, 2026 · [online version \(Slovak\)](#)

Abstract. Retatrutide, an investigational triple agonist of the GLP-1, GIP, and glucagon receptors, has not yet been approved by any regulatory authority, yet it is being sold online. This review draws on a Medscape report on a CBS News investigation that found more than 120 websites and more than 50 clinics, and on a preprint analyzing Reddit. A physician may therefore encounter a patient self-administering a product of unknown identity, purity, and sterility. The author recommends a nonjudgmental conversation, discontinuation of use, assessment for hypovolemia, AKI, and hypoglycemia, and an offer of approved therapy.

Summary

- According to Eli Lilly, retatrutide is available only in clinical trials, there is no approved dosing or definitive long-term safety profile, and the FDA states that it cannot be used for drug compounding in the United States.
- A product from an unverified source may contain a different substance, an incorrect amount, impurities, or microbial contamination, and the risk is increased by improper reconstitution and storage; the label proves neither identity nor dose.
- The preprint identified 13,589 self-declared users and detected at least one MedDRA symptom in 7,823 (57.6%), such as increased appetite, fatigue, nausea, or insomnia; this represents neither verified incidence nor causality.
- In CKD, nausea, vomiting, and low fluid intake carry a risk of hypovolemia and AKI, particularly with concomitant diuretics, RAAS inhibitors, or SGLT2 inhibitors, as well as hypoglycemia with insulin or a sulfonyleurea and loss of muscle mass in frail patients and patients on dialysis.
- The physician should obtain a history of the source, batch, concentration, reconstitution, and dose, assess vital signs, hydration, urine output, and glycemia, order tests according to the clinical findings, and document the conversation.

- If the patient continues, the physician continues to provide care but neither sets the dose nor titrates the product; with persistent vomiting, oliguria, syncope, severe hypoglycemia, or injection site inflammation, the patient should seek help immediately.

3.27 Semaglutide and the risk of chronic kidney disease in type 2 diabetes: a real-world comparison of GLP-1 receptor agonists

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. The randomized FLOW trial demonstrated the kidney benefit of semaglutide vs. placebo, but a head-to-head comparison of GLP-1 receptor agonists is lacking. An observational study by Neumiller and coauthors in AJKD emulated a target trial using OptumLabs and Medicare data and compared initiation of dulaglutide, exenatide, liraglutide, and semaglutide in adults with type 2 diabetes. The primary kidney composite did not differ; semaglutide was associated with a lower secondary composite including death compared with dulaglutide (HR 0.92) and exenatide (HR 0.88). This is not evidence of superiority within the class.

Summary

- The study included adults aged ≥ 21 years at moderate cardiovascular risk who initiated one of the four drugs from 2019 to 2021; groups were balanced by IPTW with SuperLearner and analyzed by intention to treat.
- The primary composite consisted of new administrative codes for CKD G3 or G4 and kidney failure, without laboratory eGFR and UACR values; the secondary composite added death from any cause.
- The confidence intervals for semaglutide were 0.87 to 0.97 vs. dulaglutide and 0.79 to 0.99 vs. exenatide; all-cause mortality was 19% lower than with dulaglutide (HR 0.81; 95% CI, 0.70 to 0.93), but these are relative differences without absolute risks.
- Limitations include missing HbA1c, weight, and reason for drug choice, residual confounding, unconfirmed adherence, and limited generalizability beyond the United States and the moderate-risk population.
- In FLOW, among 3,533 patients with CKD, semaglutide 1.0 mg weekly reduced the primary composite by 24% (HR 0.76; 95% CI, 0.66 to 0.88), but this is evidence vs. placebo, not vs. other GLP-1 receptor agonists.
- A GLP-1 receptor agonist does not replace blood pressure control, an ACE inhibitor or ARB, an SGLT2 inhibitor, or finerenone; after initiation, eGFR, UACR, blood pressure, volume status, and tolerability should be monitored, and the patient should be instructed on what to do during acute illness.

3.28 Genetic predictors of response to semaglutide and tirzepatide: promise and limits for nephrology practice

Published July 21, 2026 · [online version \(Slovak\)](#)

Abstract. Pharmacogenetic prediction of response to semaglutide and tirzepatide is clinically appealing. A 23andMe genome-wide association study published in *Nature* analyzed questionnaires from 27,885 respondents, and the main GWAS included 15,237 individuals of European ancestry. The rs10305420 variant in the GLP1R gene was associated with greater weight loss, approximately 0.76 kg per allele, and variants in the GLP1R and GIPR regions with nausea and vomiting. The effects are small, the data are mostly self-reported, and the models have not been prospectively validated, so the results do not justify routine testing or genotype-guided treatment adjustment.

Summary

- The population was 82.4% women, with a median age of 52 years and BMI of 35.1 kg/m², and 78.3% were of European ancestry; among 195 people with medical records, the self-reported BMI change was 14.14% vs. 8.43% in the records ($r = 0.57$).

- The T allele of the rs10305420 variant (p.Pro7Leu) was associated with an approximately 0.641 percentage point greater BMI reduction ($P = 2.9 \times 10^{-10}$); fine mapping did not confirm causality, and the association replicated in the All of Us cohort but not in the UK Biobank.
- The signal variant for nausea was rs9357296 (OR 1.36) and for vomiting rs11760106 (OR 1.57) in the GLP1R region; with tirzepatide, vomiting was associated with the GIPR variant rs1800437 (p.Glu354Gln, an odds ratio of approximately 1.83), but not with semaglutide.
- The clinical model explained 21.4% and the extended model with genetics 25% of the variability in BMI change, and the AUCs for nausea and vomiting were 0.654 and 0.680; the ranges of the commercial tool, weight loss of 6% to 20% and risk of 5% to 78%, are not clinically validated.
- Limitations: a selected customer population, self-reporting, nonrandomized treatment, compounded preparations, authors employed by 23andMe, and no kidney outcomes or validation in CKD or dialysis.
- Genotype should not determine drug choice, dose, or titration; in CKD, monitoring of gastrointestinal tolerability, hydration, electrolytes, and kidney function remains key.

3.29 Semaglutide and Wernicke encephalopathy: a rare complication of rapid weight loss

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. During treatment with a GLP-1 receptor agonist, prolonged gastrointestinal symptoms and rapid weight loss can deplete thiamine stores. A systematic review of case reports from July 2026 identified six cases of Wernicke encephalopathy with semaglutide given for obesity; four patients subsequently developed Korsakoff syndrome, and one died. An analysis of Vigibase reports presented at AACE 2026 showed a disproportionality signal (ROR 10.2 for semaglutide and 11.4 for tirzepatide). This is a pharmacovigilance signal, not proof of causality, so early recognition and immediate parenteral thiamine are key.

Summary

- The six cases (mean age 47.2 years) developed after a mean of 4.9 months of semaglutide use; they were preceded by anorexia (5 of 6), weakness (4 of 6), vomiting (3 of 6), and weight loss in all.
- The Vigibase analysis of 18 reports described a triple hit: iatrogenic gastroparesis, persistent vomiting, and weight loss of 3.5 to 13.3 kg per month, with a median onset of 3 to 6 months after treatment initiation or dose escalation.
- The ROR measures reporting disproportionality, not incidence or relative risk; case reports without a control group are subject to publication bias, which explains the difference between Korsakoff syndrome in 4 of 6 patients and cognitive deficit in approximately 20% in the report analysis.
- The similar ROR with tirzepatide supports a nutritional mechanism common to the entire class rather than direct neurotoxicity; thiamine stores (on the order of 25 to 30 mg) are depleted within a few weeks of low intake, even in a person with obesity.
- The diagnosis is clinical, based on the Caine criteria (two of four features); neither a normal thiamine level nor a normal MRI excludes it. Thiamine is given parenterally immediately, before or together with glucose: EFNS 2010 recommends 200 mg three times daily and the review authors 500 mg IV three times daily, although this is not a universal standard; hypomagnesemia should be corrected at the same time.
- For nephrology: thiamine is removed into the dialysate, and confusion is easily attributed to uremia; Wernicke encephalopathy belongs in the differential diagnosis after vomiting or rapid weight loss. The author considers a weight loss exceeding approximately 4 kg per month with nearly zero intake a warning sign.

3.30 Adipose tissue is not a passive reservoir: what its biology tells us about obesity and cardiorenal risk

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. The metabolic and cardiorenal risk of obesity is determined not only by the amount of fat but also by its distribution, storage capacity, and endocrine activity. This review combines established knowledge with new, mostly conference, animal, and cellular data from the International Congress on Obesity 2026: sex-specific transcriptomic profiles of adipose tissue in 41 individuals, communication between adipose tissue and the heart via microRNAs, and the thermogenic effect of genistein. These lines of research are hypothesis-generating and do not change clinical practice. The main message for practice remains that body mass index alone is insufficient, particularly in the nephrology patient.

Summary

- When the storage capacity of adipose tissue is exhausted, adipocyte hypertrophy, hypoxia, inflammation, and fibrosis develop, and lipids are deposited ectopically in the liver, muscle, pancreas, and heart; visceral fat is also a marker of broader metabolic dysfunction.
- Transcriptomic analysis of biopsies from 21 women and 20 men showed that adipose tissue dysfunction in insulin resistance differs by sex and depot; this is a small sample without known control for confounders, and the data require replication.
- Adipose tissue microRNAs (particularly miR-92a and miR-222) altered myocardial pathways in animal and cellular models, but this does not prove a causal relationship with heart disease in humans.
- The reported increase in energy expenditure through thermogenesis of approximately 5% to 10% is compensated by the body; genistein increased UCP1 expression only preclinically, so the author does not recommend it as an obesity treatment.
- BMI does not distinguish fat from muscle mass or visceral fat; the author recommends adding at least waist circumference and assessment of metabolic complications and using the term metabolically healthy obesity with caution.
- In CKD, BMI is distorted by fluid retention, sarcopenia, and amputations; the obesity paradox in patients on dialysis does not prove a protective effect of fat, and during weight reduction, muscle mass should be preserved and malnutrition prevented.

3.31 Retatrutide in type 2 diabetes: marked reduction in HbA1c and body weight, but the cardiorenal benefit is not yet known

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. Retatrutide, a triple GIP, GLP-1, and glucagon receptor agonist, was compared with placebo in the randomized, double-blind, phase 3 TRANSCEND-T2D-1 trial in 537 adults with early type 2 diabetes managed with lifestyle measures alone. Over 40 weeks, the 12 mg dose lowered HbA1c by 1.94 percentage points (placebo-adjusted, 1.12) and body weight by 15.3%. The trial had no active comparator and no kidney or cardiovascular outcomes, so it provides evidence of metabolic efficacy, not organ protection. Retatrutide remains an unapproved investigational drug.

Summary

- Eligible patients had an HbA1c of 7.0% to 9.5% and a BMI of at least 23 kg/m² without pharmacotherapy; mean diabetes duration was 2.5 years, mean BMI was 35.8 kg/m², and 94% of participants completed the trial.
- The placebo-adjusted HbA1c difference was 0.88, 1.04, and 1.12 percentage points for the 4, 9, and 12 mg doses ($P < 0.0001$), with placebo alone lowering HbA1c by 0.81 points; body weight fell by 11.5%, 13.9%, and 15.3% versus 2.6% with placebo and had not yet reached a plateau.
- The ADA press release reports a 16.8% weight loss and an HbA1c reduction of almost 2%, which probably reflects a more favorable estimand; the estimand used should be checked whenever trials are compared.

- Gastrointestinal adverse events predominated and led to discontinuation in 2% to 5% of patients versus 0% with placebo; no severe hypoglycemia occurred with monotherapy, and two deaths in the 4 mg group were judged unrelated.
- Limitations include short follow-up, a selected population, missing body composition data, and funding by Eli Lilly and Company; TRANSCEND-T2D-2 will provide a head-to-head comparison with semaglutide, and TRANSCEND-T2D-3 in patients with impaired kidney function has HbA1c, not kidney events, as its primary endpoint.
- In practice, the author warns of the risk of prerenal AKI with vomiting and diarrhea, particularly in CKD or with diuretics or an SGLT2 inhibitor, and of a spurious rise in creatinine-based eGFR with muscle loss; cystatin C-based eGFR and UACR should also be assessed.

3.32 When the physician tries on the patient's role: can experiential education improve glycemic control in type 2 diabetes?

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. Clinical inertia and the gap between guidelines and practice contribute to poor control of type 2 diabetes. The Chinese open-label, cluster-randomized CREATION trial (205 physicians, 2,017 patients) tested one week of in-person experiential physician training that included playing the patient's role. The proportion of patients with an HbA1c below 7.0% increased by 16.6 percentage points at 6 months and by 17.0 percentage points at 12 months. The effect applies to the entire multicomponent program, not to role-play alone, and the trial assessed neither clinical nor kidney events; patients with an eGFR below 60 mL/min/1.73 m² were not enrolled.

Summary

- The primary endpoint of an HbA1c below 7.0% at 6 months was reached by 58.0% of patients of intensively trained physicians versus 42.9% (adjusted difference, 16.6 percentage points; 95% CI, 7.2 to 25.7), and at 12 months by 60.9% versus 44.6%.
- Other differences were small: 0.3 kg/m² in BMI, 1.4 cm in waist circumference, approximately 0.35 mmol/L in fasting glucose, and 1.5 mm Hg in systolic BP; serious adverse events occurred in 2.6% versus 2.4%.
- According to the author, the mismatch between the large shift in the binary endpoint and the small metabolic changes may be explained by a shift of the distribution around the threshold, postprandial glucose, or intensified pharmacotherapy; without treatment data, this cannot be resolved.
- The main limitations are patient enrollment only after randomization of centers, an open-label design with a possible Hawthorne effect, a control group without comparably intensive training, and missing HbA1c measurements in approximately 19% of patients.
- Patients aged 40 to 65 years with diabetes of up to 10 years' duration were enrolled; patients with an eGFR below 60 mL/min/1.73 m² and with NYHA class III and IV heart failure were excluded, so the results cannot be extrapolated to CKD.
- The author proposes a pilot evaluation in Slovakia with an active control of equal time allocation and a nephrology version of the scenarios: dose adjustment by eGFR, hypoglycemia prevention, sick day rules, prevention of euglycemic ketoacidosis, and management of hyperkalemia.

3.33 Urinary podocalyxin in diabetic nephropathy: remarkable numbers that cannot yet be taken to the bedside

Published August 24, 2026 · [online version \(Slovak\)](#)

Abstract. Albuminuria appears late in diabetic kidney disease, which drives the search for earlier podocyte markers. A cross-sectional single-center study from Dhaka (PLOS One, July 2026) measured urinary podocalyxin in 59 patients with type 2 diabetes and 29 controls. Levels rose with albuminuria, and at a threshold of 2.92 ng/mL, discrimination of nephropathy reached an area under the curve of 0.96. However,

the reference standard was UACR itself, the thresholds were not externally validated, and patients on RAAS blockade were excluded. Nothing changes in practice: screening continues to rely on UACR and eGFR.

Summary

- Podocalyxin rose from 1.16 ± 0.39 ng/mL in normoalbuminuria to 7.28 ± 2.05 ng/mL in macroalbuminuria; the Pearson correlation was 0.79 with UACR and -0.51 with eGFR, but Kendall's τ -b was only 0.61 and -0.36 , suggesting an influence of extreme values.
- At a threshold of 2.92 ng/mL, sensitivity was 86.7% and specificity 100.0%; the adjusted odds ratio of 22.67 per 1 ng/mL has an interval of 2.47 to 208.09, and the authors describe it as exploratory.
- The AUC of 0.96 reflects agreement between two markers from the same sample, not diagnostic validation; the cross-sectional design does not show that podocalyxin is an early biomarker, and the sample of 88 subjects is small.
- A specificity of 29 of 29 has a 95% interval of approximately 88% to 100%, so the PPV of 100% is uncertain; the reported counts yield an NPV of approximately 88%, not 89.9%.
- Before entering practice, the marker would need longitudinal evidence of prediction, external validation of thresholds, added value over UACR and eGFR, data under RAAS blockade, SGLT2 inhibitors, and finerenone, and analytical standardization.

3.34 Serum ketone bodies and the risk of kidney events in type 2 diabetes: what a cohort study shows and what cannot be inferred from it

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. Ketone bodies are being studied as a marker of metabolic status and a possible mediator of the effect of SGLT2 inhibitors. A prospective Korean cohort of 1,392 outpatients with type 2 diabetes not receiving insulin or thiazolidinediones examined the relationship between serum ketone bodies and kidney events. Compared with the lowest tertile, the highest tertile had a lower risk of an eGFR decline of $\geq 40\%$ (aHR, 0.46) and of the composite kidney endpoint (aHR, 0.63). However, more frequent SGLT2 inhibitor treatment may explain the association, so the finding is hypothesis-generating and has no impact on clinical management.

Summary

- Over a median follow-up of 23.2 months, 158 kidney and 64 cardiovascular events occurred; the highest tertile (171.0 to 1,629.6 $\mu\text{mol/L}$) had an aHR of 0.46 for an eGFR decline of $\geq 40\%$, 0.63 for the composite kidney endpoint, and 0.50 for incident CKD.
- Kidney endpoints included new-onset macroalbuminuria, an eGFR decline of $\geq 40\%$, and kidney failure requiring KRT; cardiovascular endpoints trended favorably without statistical significance, which, with 64 events, allows no conclusion about the presence or absence of an effect.
- SGLT2 inhibitors raise ketone bodies and have proven kidney-protective effects, so ketone bodies may merely be a marker of exposure to an effective drug; the abstract does not make clear how this factor was handled, and separating the two would require at least stratification by SGLT2 inhibitor use.
- In the view of the article's author, the study authors' description of ketone bodies as mediators is stronger than the design can support; further limitations are a single measurement of a fluctuating concentration, selection of less advanced diabetes without insulin, short follow-up with possible hemodynamic effects on eGFR decline, and residual confounding.
- Ketone bodies are neither a validated stratification tool nor a treatment target, and the result does not support a ketogenic diet in CKD; the practical message is to start an SGLT2 inhibitor in eligible patients.

3.35 Six SGLT2 inhibitors are not six interchangeable drugs: differences in approved cardiac and kidney indications

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. SGLT2 inhibitors share a mechanism but not their approved indications or the evidence that led to them. A regulatory review compares the six molecules approved in the United States with their status in the EU. In the United States, dapagliflozin and empagliflozin have broad cardiorenal indications regardless of diabetes, canagliflozin has a kidney indication tied to diabetic kidney disease, and sotagliflozin has only a cardiac indication. In the EU and in Slovakia, the choice in heart failure and chronic kidney disease narrows to dapagliflozin and empagliflozin, so selection should be based on the indication being treated.

Summary

- In the United States, dapagliflozin and empagliflozin are indicated for type 2 diabetes, heart failure, and CKD regardless of diabetes; ertugliflozin and bexagliflozin have only a glycemic indication, and sotagliflozin (Inpefa) only a cardiac indication without a glycemic one.
- In the EU, canagliflozin (Invokana) is indicated only for inadequately controlled type 2 diabetes, as is ertugliflozin; sotagliflozin (Zynquista) was withdrawn on March 22, 2022, and bexagliflozin is not approved.
- Canagliflozin has a kidney indication tied to diabetic kidney disease in the United States but not in the EU, although its kidney-protective effect was demonstrated in trials. EU marketing authorization is not the same as reimbursement in Slovakia; categorization should be verified in the ŠÚKL database and the current list of categorized drugs.
- In the CANVAS program, the incidence of amputations with canagliflozin was 6.3 versus 3.4 per 1,000 patient-years (HR 1.97); the FDA removed the 2017 boxed warning on August 26, 2020, but still lists the risk in the warnings.
- When switching products, it should be verified that the new molecule covers the original indication; the review does not compare efficacy, and the absence of an indication does not mean the absence of an effect. Other molecules may be considered for glycemic control without HF or CKD, but the cardiorenal benefit does not automatically extend to them.

3.36 Sequentially or all at once? How to start combined cardiorenal therapy in type 2 diabetes and chronic kidney disease

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. In type 2 diabetes with CKD, four drug classes with cardiorenal benefit are available, and the question is whether to start them sequentially or simultaneously. A critical analysis of the ADA 2026 debate shows that direct evidence comes only from the phase 2 CONFIDENCE trial: the combination of finerenone and empagliflozin reduced albuminuria by 29% more than finerenone and by 32% more than empagliflozin. Arguments from VERIFY, TRIPLE-AXEL, and DCCT/EDIC concern glycemia and are only analogies. The author recommends avoiding unnecessary delay and tailoring the approach to the patient's safety profile.

Summary

- CONFIDENCE randomized patients with an eGFR of 30 to 90 mL/min/1.73 m² and a UACR of 100 to 5,000 mg/g on RAS blockade; the UACR ratio for the combination was 0.71 versus finerenone and 0.68 versus empagliflozin ($p < 0.001$), and the benefit was consistent across KDIGO risk categories.
- Hyperkalemia was more frequent with the combination, but an early eGFR decline of more than 30% was less frequent in higher-risk patients; this was a 180-day trial with a surrogate endpoint and no hard outcomes.
- VERIFY assessed glycemic failure (43.6% vs. 62.1%) in newly diagnosed diabetes, and TRIPLE-AXEL, with 105 patients, also included glimepiride as a comparator, so the advantage in the composite endpoint mainly reflects the choice of comparator.

- According to the author, simultaneous initiation is justified with safe serum potassium, a known eGFR, and a follow-up visit in two to four weeks; a sequential approach is appropriate with borderline serum potassium, unstable comorbidity, or polypharmacy.
- In a sequential approach, RAS blockade and an SGLT2 inhibitor take priority, and a nonsteroidal mineralocorticoid receptor antagonist is added for persistent albuminuria; the early eGFR decline should be explained in advance, and a trial with hard endpoints is still lacking.

3.37 Are GLP-1 drugs now “kidney drugs”? Kidney benefits in the evidence of recent years (and how to translate them into practice)

Published September 2, 2026 · [online version \(Slovak\)](#)

Abstract. GLP-1 receptor agonists are still perceived as diabetes and obesity drugs, but randomized data support their place in kidney protection. This review builds on a commentary by nephrologist Kashif J. Piracha and four scenarios: the FLOW trial with semaglutide 1.0 mg reduced the primary kidney composite by 24% (HR 0.76), the SELECT kidney analysis showed an HR of 0.78 even without diabetes, and tirzepatide lowered UACR in a post hoc analysis of SURMOUNT. In eligible patients with type 2 diabetes and CKD, a GLP-1 receptor agonist belongs in layered kidney protection.

Summary

- FLOW (3,533 patients, median 3.4 years) showed an HR of 0.76 for the primary composite, an eGFR decline slower by 1.16 mL/min/1.73 m² per year, and a 20% reduction in all-cause mortality (HR 0.80).
- In the prespecified kidney analysis of SELECT (semaglutide 2.4 mg, no diabetes), the kidney composite occurred in 1.8% versus 2.2% (HR 0.78), and the eGFR difference at 104 weeks was 0.75 mL/min/1.73 m², rising to 2.19 mL/min/1.73 m² in patients with an eGFR below 60 mL/min/1.73 m².
- In the FLOW subgroup of 550 patients on an SGLT2 inhibitor, the HR was 1.07 (95% CI, 0.69 to 1.67) with no significant interaction (P = 0.109); an incremental effect is therefore unproven because of low statistical power, and no harmful interaction was found.
- In patients with a UACR of ≥ 30 mg/g, tirzepatide reduced placebo-adjusted UACR by approximately 42% (SURMOUNT-1) and 55% (SURMOUNT-2), but this is a biomarker in a post hoc analysis; glycemia alone does not explain the kidney benefit, and the proposed mechanisms have not been definitively proven.
- In practice: consider a GLP-1 receptor agonist early alongside RAS blockade and an SGLT2 inhibitor, counsel patients about nausea (approximately 20% to 30%), reduce basal insulin by approximately 20% when HbA1c is below 8%, and monitor volume status because of the risk of AKI.

3.38 Retatrutide and “pitting the physician against the patient”: barriers in the expanded access process in practice and their relevance for nephrology

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. A commentary by Caroline Messer on Medscape describes how the US expanded access process for unapproved retatrutide can pit the physician against the patient’s expectations. The article separates verifiable facts from opinion: access (NCT07629401) requires a BMI of ≥ 35 kg/m² refractory to approved therapy, two serious complications, and approval at three independent gates, namely the manufacturer, the FDA, and the IRB. A different regime applies in Slovakia under § 46 of Act No. 362/2011 Coll. For the nephrologist, the implication is that until the drug is approved, the patient should be treated with available approved therapy.

Summary

- In phase 2, retatrutide 12 mg reduced body weight by 24.2% versus 2.1% with placebo over 48 weeks, but it has no FDA or EMA approval, and kidney endpoints were not a primary objective.

- In the United States, the physician needs a manufacturer's LOA, submission of FDA Form 3926 or 1571, IRB approval, and informed consent; treatment generally cannot begin until after 30 days, and the FDA cannot compel the manufacturer to supply the drug.
- Entry criteria include age ≥ 18 years, a BMI of ≥ 35 kg/m² despite the highest dose of approved therapy, two serious complications of obesity, inability to participate in a trial, and a discussion of bariatric surgery; even meeting them does not guarantee supply.
- The claim that the process has no one accountable for the whole and allows no appeal is the commentary author's assessment, not the wording of the regulations; expanded access is a regulated route to the authentic drug, whereas retatrutide bought online is its opposite.
- In Slovakia, the Ministry of Health of the Slovak Republic authorizes therapeutic use of an unapproved drug under § 46(4) only when no comparable approved drug is available; because incretin drugs are available, refractory obesity does not automatically establish entitlement to retatrutide.
- For nephrology: patients with CKD are often excluded from trials, no approved retatrutide dosing exists for CKD, and while waiting, approved therapy, including SGLT2 inhibitors and RAAS blockade, should be titrated; a process can be promised, not an outcome.

3.39 Lipedema: multidisciplinary management and the place of surgery in treatment, a practical clinical review

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. Lipedema is a chronic disease of subcutaneous fat, predominantly in women, often mistaken for obesity or edema. This review draws on a conference report from ICO 2026, the US standard of care (Herbst et al., 2021), and the Lipedema World Alliance Delphi consensus (2026). The diagnosis remains clinical, prevalence is uncertain, and treatment is multidisciplinary. Lymph-sparing surgery is reductive, neither cosmetic nor curative, and rests on low-quality evidence (GRADE C). The key point for the nephrologist is that diuretics are not first-line treatment for lipedema.

Summary

- The typical finding is a bilateral, symmetric increase in painful nodular fat on the limbs, sparing the hands and feet, with onset during hormonal changes; the frequently cited prevalence of approximately 11% is not a robust estimate, and in the classic study only 40% to 50% of women reported pain.
- There is no definitive imaging test; pitting edema is not a discriminating sign on its own, and classification (five types and four stages according to Schmeller and Meier-Vollrath) is not uniform.
- Conservative care is the foundation: nutrition, sustainable physical activity, individualized compression, lymphatic therapy, and psychological support; the goals are pain, mobility, and quality of life, not weight loss at any cost.
- Randomized trials with lipedema-specific endpoints are lacking for GLP-1 receptor agonists, so they should be used only for a metabolic indication.
- Surgery should be preceded by conservative treatment (Max Sirota recommended at least six months, the British guideline 6 to 12 months) and evaluation of the venous and lymphatic systems; results come from nonrandomized series, and postoperative compression lasts at least 2 to 3 months.
- Lipedema swelling is not a sign of volume overload, unlike pitting nephrotic and CKD edema; diuretics for lipedema risk hypovolemia and worsening kidney function, and the US standard does not recommend their long-term use.

3.40 "Weight down, steps down": why physical activity may decline with GLP-1 receptor agonists and how to address it in nephrology and cardiometabolic practice

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. GLP-1 receptor agonists reduce body weight independently of exercise, but it is unclear whether patients move more after losing weight. A retrospective before-and-after cohort from the All of Us program

(ENDO 2026 conference abstract, 753 Fitbit wearers) showed a decrease of 560 steps per day and of 5.7 minutes of MVPA per day after treatment initiation; a cross-sectional analysis of 298 adults found that 57.7% did not meet 150 minutes of MVPA per week. These are observational data without evidence of causality. For the nephrologist, the implication is that physical activity should be planned as part of treatment.

Summary

- In the cohort of Maharjan et al., daily steps fell from 5,047 to 4,487 and MVPA from 27.9 to 22.2 min/day ($p < 0.001$); the decline was greater in men and in those with musculoskeletal pain and did not differ by age or severe obesity.
- The design has no control group, did not assess the relationship to the magnitude of weight loss or later trajectories, and Fitbit wearers are a selected group; the proposed mechanisms (NEAT, adverse effects, loss of lean mass) are only hypotheses.
- Three case reports of anhedonia with tirzepatide 15 mg weekly, which improved after dose reduction to 10 mg or less, are a signal for a targeted question, not evidence or guidance for off-label treatment.
- In the randomized S-LITE trial after an initial diet, liraglutide 3.0 mg combined with exercise was superior to exercise alone (-5.4 kg) but not significantly superior to liraglutide alone (-2.7 kg; $p = 0.13$); only the combination improved HbA1c and cardiorespiratory fitness.
- The decline in steps does not call into question the cardiorenal benefit from FLOW (HR 0.76) and SELECT (HR 0.78); in CKD, the average is approximately 4,640 steps per day, and KDIGO 2024 recommends at least 150 minutes of moderate-intensity activity per week.
- The author recommends asking about physical activity at the time of prescribing, measuring baseline steps, combining walking with resistance exercise, distinguishing among barriers, and replacing neither kidney protection with exercise nor exercise with a drug.

3.41 The “gray market” for retatrutide: half the weight loss and more cardiovascular symptoms? Clinical interpretation of a real-world signal and implications for the nephrologist

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. Retatrutide is not approved, yet in the United States it is spreading outside clinical trials. An observational EHR analysis (preprint, Murugadoss et al., 2026) that identified exposure with a language model showed that users of gray-market products lost 7.2% of body weight over 6 to 12 months versus 15.5% in the trial arm, had a transient rise in heart rate, and had a higher burden of cardiovascular (RR 1.56) and neuropsychiatric symptoms (RR 1.95) than users of semaglutide. MACE data were uninformative given the small numbers. This is a preliminary signal; the nephrologist should ask about unapproved peptides and offer approved therapy.

Summary

- Of 983 mentions, exposure was confirmed in 652 patients, and of 531 with a known source, 378 (71.2%) obtained the product outside a trial, mainly through the internet and telemedicine; after manual verification, exposure classification had an accuracy of 99.8%.
- The gray market achieved a weight loss of 7.2%, comparable to tirzepatide (7.7%) and semaglutide (4.6%); a late index date may overstate the difference from the trial, and the data do not reveal the mechanism of attenuation.
- Heart rate rose at month 3 by 4.3 beats/min in the trial arm and by 2.5 beats/min in the gray-market arm and returned to baseline at month 6; it did not change with the comparators.
- The RR of 1.77 for 3-point MACE versus tirzepatide (95% CI, 0.19 to 7.94) belongs to the trial arm, and the gray-market arm trended in the opposite direction; the intervals cross unity, and Medscape’s description of coronary events is incorrect.

- Limitations include unknown composition, purity, and dose, short follow-up, an approximately 1.5-fold higher density of clinical notes without adjustment, and low CKD numbers, so no CKD-specific conclusions can be drawn.
- Retatrutide is not approved in Slovakia; the author recommends asking about unapproved peptides, not adjusting their dose, advising discontinuation, offering approved therapy, monitoring heart rate, hydration, and concomitant medication when use is disclosed, and reporting suspected adverse reactions to ŠÚKL.

3.42 Mounjaro (tirzepatide): a new FDA indication for cardiovascular risk reduction in type 2 diabetes and what it means for practice

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. On August 27, 2026, the FDA expanded the indication of tirzepatide (Mounjaro) to include MACE risk reduction in adults with type 2 diabetes at high risk. The basis is the active-controlled RCT SURPASS-CVOT in 13,299 patients with ASCVD, in which tirzepatide versus dulaglutide 1.5 mg achieved a hazard ratio of 0.92 (95.3% CI, 0.83 to 1.01): noninferiority, not superiority. The primary publication does not report kidney endpoints. The CHMP did not recommend the indication in the EU, so in Slovakia the US label does not entitle prescribers to use tirzepatide for MACE reduction.

Summary

- SURPASS-CVOT was a double-blind, phase 3 RCT with a noninferiority test: subcutaneous tirzepatide once weekly up to 15 mg versus dulaglutide 1.5 mg, mITT 6,586 versus 6,579 patients, mean age 64.1 years, HbA1c 8.4%, and median follow-up 210.1 weeks.
- MACE-3 occurred in 12.2% of patients with tirzepatide and in 13.1% with dulaglutide, HR 0.92 (95.3% CI, 0.83 to 1.01); noninferiority was significant ($P = 0.003$), superiority was not ($P = 0.09$), and the absolute difference was 0.9 percentage points.
- The US label does not require established ASCVD, although the trial enrolled only patients with established ASCVD; it applies to the Mounjaro brand, not Zepbound, and pediatric use from 10 years of age applies only to the glycemic indication.
- Gastrointestinal adverse events were more frequent with tirzepatide; from a nephrology perspective, the class risk of dehydration and AKI with vomiting or diarrhea remains, particularly with a concomitant diuretic, RAS blockade, or an SGLT2 inhibitor.
- The primary publication does not report CKD progression, eGFR, or albuminuria, so the cardiovascular indication cannot be read as a kidney indication; the strongest kidney evidence in the class remains the FLOW trial with semaglutide.
- In June 2026, the CHMP did not recommend the new indication, and the data are to be included only in the product information; the indication and reimbursement framework in Slovakia is determined by ŠÚKL and the current SmPC.
- The author recommends phenotype-based treatment selection, continuing an SGLT2 inhibitor with kidney or cardiac evidence, and, before starting tirzepatide, counseling on gastrointestinal tolerability and the risk of volume depletion.

3.43 HTD1801 in type 2 diabetes: results of the SYMPHONY-2 trial and the limits of the evidence

Published September 5, 2026 · [online version \(Slovak\)](#)

Abstract. HTD1801 is an investigational oral drug, an ionic salt of berberine and ursodeoxycholic acid. The placebo-controlled phase 3 RCT SYMPHONY-2, conducted at 64 centers in China, enrolled 549 patients with type 2 diabetes inadequately controlled on metformin. At 24 weeks, the adjusted difference in HbA1c change was -0.5 percentage points in favor of HTD1801, but diarrhea occurred in 23.8% versus 1.1%. The tri-

al did not assess kidney or cardiovascular outcomes, and the drug is not approved in the EU or the United States, so it is not an established cardiorenal therapy.

Summary

- Randomization 2:1 assigned 365 participants to HTD1801 1,000 mg twice daily and 184 to placebo, all on metformin; entry criteria were an HbA1c of 7.0% to 10.5% and fasting glucose of no more than 13.9 mmol/L.
- HbA1c fell from 8.6% to 7.4% with HTD1801 and from 8.5% to 7.8% with placebo; the adjusted difference of -0.5 percentage points (95% CI, -0.7 to -0.4; $P < 0.0001$), not the 1.2-point decline itself, is the correct estimate of effect.
- Diarrhea was the most frequent adverse event (23.8% vs. 1.1%, three severe cases), and no severe hypoglycemia occurred; diarrhea carries a risk of hypovolemia and worsening kidney function, particularly with diuretics, RAS blockers, or SGLT2 inhibitors.
- The SYMPHONY program enrolled only patients with an eGFR above 60 mL/min/1.73 m², there was no active comparator, all centers were in China, the trial was sponsor-funded, and the open-label 28-week extension had no control group.
- The eGFR changes from the pooled analysis presented at ASN are only a short-term laboratory signal; the kidney question is being addressed by a separate 12-week phase 2 trial, NCT07496177, with 75 participants and UACR as the primary endpoint.
- In March 2026, the NMPA in China accepted the marketing application, which is not marketing authorization; the results justify neither replacing therapy with proven cardiorenal benefit nor self-treatment with berberine or ursodeoxycholic acid.

3.44 Time in target range for HbA1c and systolic blood pressure and kidney outcomes: a post hoc analysis of the ACCORD trial

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. A single HbA1c or BP value does not capture the long-term stability of control. A post hoc observational analysis of the ACCORD trial in 6,542 patients with type 2 diabetes assessed time in target range (TTR) for HbA1c and systolic BP during the first year. Keeping both parameters on target more than 80% of the time was associated with a 29% lower risk of the composite kidney outcome (HR 0.71; 95% CI, 0.61 to 0.83). This is an association, not evidence of causality; the results support monitoring trends, not mechanically pursuing intensive targets.

Summary

- TTR was calculated from at least three clinical measurements of HbA1c and SBP over 12 months; the composite kidney outcome included new albuminuria, an eGFR decline of at least 40%, or kidney failure.
- An HbA1c TTR of 100% had an HR of 0.80 (0.68 to 0.94) and an SBP TTR of 100% an HR of 0.68 (0.58 to 0.80) compared with the lowest tertile; both values above 80% were also associated with 43% lower all-cause mortality and a 33% lower risk of albuminuria.
- No significant association was shown for macroalbuminuria or kidney failure alone because of the small number of events, and no associations were observed in the intensive BP arm; results were similar in the CKD subgroup.
- The analysis is post hoc, and TTR was not randomized, so residual confounding may persist and TTR may reflect overall health status; synergy between the two parameters was not demonstrated by a formal interaction analysis.
- The original ACCORD trial showed higher mortality with an HbA1c target below 6.0%, and an SBP target below 120 mm Hg did not reduce the cardiovascular composite (HR 0.88), while serious adverse events were more frequent (3.3% vs. 1.3%).

- The authors recommend assessing the HbA1c trend over 6 to 12 months, repeated BP measurements, albuminuria, eGFR decline, hypoglycemia, and hypotension, and individualizing targets; RAS blockade, an SGLT2 inhibitor, and, where indicated, finerenone remain the foundation of kidney protection.

3.45 Obesity as a multifactorial disease: a new screening tool and its relevance for nephrology practice

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Obesity is a multifactorial chronic disease, and in nephrology it increases the risk of hypertension, albuminuria, hyperfiltration, and CKD. A brief descriptive piece in Nature Reviews Endocrinology (September 2026) presents the digital tool CheckCausesObesity.com from Erasmus MC, which assesses more than 30 contributing factors in seven domains. Among 62,809 users of the Dutch version, 91.2% reported factors in two or more domains. This is not an original research study; the tool proves neither causality nor improved clinical outcomes, but it can structure the conversation and reduce stigma.

Summary

- The tool assesses lifestyle, social circumstances, psychological factors, medications, endocrine diseases, hypothalamic damage, and genetics; it includes the STOP-Bang screening, NICE criteria, Rotterdam criteria, and AUDIT-C, and the patient completes it at home in approximately 25 minutes.
- Users of the Dutch version had a median age of 53 years, 76.2% were women, and the median BMI was 33.1 kg/m²; the most frequent combination was lifestyle, psychological factors, and medications (39.9%), which describes user reports, not the prevalence of causes.
- In the author's assessment, limitations include selection bias, self-reporting, a risk of false-positive findings, no evidence of improvement in weight, BP, albuminuria, or eGFR, and the authors' declared ties to pharmaceutical companies.
- STOP-Bang has high sensitivity and low specificity, so a positive screen should be confirmed by polygraphy or polysomnography; endocrine and genetic testing should be targeted to the clinical picture, not universal.
- In advanced CKD and on dialysis, BMI is distorted by fluid retention, loss of muscle mass, and malnutrition, so dry weight, waist circumference, nutritional status, and muscle strength should be monitored, and aggressive weight reduction should be avoided in frailty or sarcopenia.
- The author recommends assessing hydration, BP, albuminuria, and the eGFR trend, reviewing weight-promoting medications with attention to kidney safety, and, with GLP-1 receptor agonists, considering kidney function, the risk of dehydration, and reimbursement conditions.

3.46 BMI and hematocrit in type 2 diabetes: what they really tell us about the risk of kidney failure

Published September 11, 2026 · [online version \(Slovak\)](#)

Abstract. The study combined two-sample Mendelian randomization using East Asian data with a retrospective single-center cohort of 875 patients with type 2 diabetes from Shanghai. A higher BMI was associated with a higher risk and a higher hematocrit with a lower risk of kidney failure, and the nomogram achieved an AUC of 0.880. However, the model does not include eGFR or albuminuria, the genetic outcome was not identical to ESRD, and external validation is lacking. Hematocrit therefore cannot be regarded as a kidney-protective treatment target, and the results do not change the KDIGO 2024 recommendations.

Summary

- In the Mendelian randomization, hematocrit had an OR of 0.513 (0.320 to 0.820) and BMI an OR of 1.630 for chronic kidney failure, but the outcome dataset had only 2,117 cases and was not restricted to patients with diabetes; glucose (OR 2.406) relied on a weak instrument with F = 5.65.
- For hematocrit, MR-PRESSO removed the TMPRSS6 variant rs855791, which is linked to iron metabolism, suggesting horizontal pleiotropy; a negative pleiotropy test therefore does not exclude it.

- In the clinical cohort, 140 patients (16%) progressed to ESRD over three years; independent predictors were BMI (OR 1.085 per 1 kg/m²), systolic BP, creatinine, hematocrit (OR 0.940), and albumin, while the abstract erroneously lists diastolic BP.
- A higher hematocrit may reflect SGLT2 inhibitor treatment, which the study does not report, so confounding by indication is possible; the CHOIR RCT showed that a hemoglobin target of 13.5 g/dL versus 11.3 g/dL increased the risk of the cardiovascular composite (HR 1.34).
- The AUC of 0.880 is a measure of discrimination that is probably driven mainly by creatinine; the model lacks a comparison with the Kidney Failure Risk Equation (C statistic 0.841), and the definition of ESRD is not precisely stated.
- The author recommends assessing adiposity and volume status when BMI is higher, investigating the cause of anemia when hematocrit is low, and assessing CKD risk by eGFR, albuminuria, and an externally validated equation in line with KDIGO 2024.

3.47 Intrusive thoughts about food, stigma, and obesity treatment: why tracking body weight is not enough

Published September 11, 2026 · [online version \(Slovak\)](#)

Abstract. Assessing obesity by body weight alone is not enough; intrusive thoughts about food (food noise) and stigma toward patients taking obesity medications are entering the discussion. The sources are three commentaries and conceptual papers from 2026 and a single vignette experiment from 2024 in 357 US adults, in which a woman losing weight with a GLP-1 receptor agonist faced harsher negative judgments. Food noise is not yet a validated measure, and the effect of stigma on treatment discontinuation remains a hypothesis; in patients with CKD, reduced appetite should be assessed together with nutritional status.

Summary

- Umashanker and Mitchell propose characterizing food noise by its frequency, intrusiveness, and impact on behavior; the Food Noise Questionnaire and the Ro Allison Indiana Dhurandhar–Food Noise Inventory do not yet represent a diagnostic standard or a validated threshold for clinically meaningful change.
- GLP-1 receptor agonists may attenuate these thoughts, but the response is not uniform; their persistence does not imply treatment failure or poor adherence, nor is it an automatic reason to increase the dose.
- In the experiment by Post and Persky, negative judgments were mediated by the belief that the woman had taken a shortcut, regardless of body size; the sample was an online US sample, the description was text-based, and the figure was always female.
- Treatment discontinuation can have many causes, including adverse effects, cost, and availability, so a causal relationship between stigma and nonadherence has not been demonstrated.
- In CKD and on dialysis, obesity and malnutrition are not mutually exclusive, weight loss may reflect volume correction, and reduced appetite may be a warning sign in a patient prone to wasting; hydration and kidney function should be monitored with nausea or diarrhea.
- The author recommends distinguishing hunger, meal planning, anxiety, and loss of control in the conversation, asking about experiences of stigma, and judging treatment success also by complications, functional capacity, quality of life, and nutritional status.

3.48 Tirzepatide vs. semaglutide for weight reduction: efficacy, safety, and relevance for nephrology practice

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. A systematic review and meta-analysis by Paccola et al. synthesized 10 head-to-head comparisons of tirzepatide with semaglutide in 41,381 adults with overweight or obesity (3 RCTs and 7 retrospective cohorts, 24 to 72 weeks). Tirzepatide produced 4.28 percentage points greater weight loss and a greater HbA1c reduction, but also a higher incidence of serious adverse events (5.7% vs. 2.9%; RR 1.83). The SAE composite

is not broken down, so this is not evidence of a worse safety profile. In nephrology, the key is managing volume risk during titration.

Summary

- Compared with semaglutide, tirzepatide reduced body weight by 4.28 percentage points (95% CI, -5.28 to -3.28), that is, by 4.43 kg, increased the likelihood of weight loss of at least 10%, 15%, and 20%, and lowered HbA1c by 0.29% ($P = 0.0002$); the drugs did not differ at the 5% threshold.
- SAEs were more frequent with tirzepatide (RR 1.83; $P = 0.007$), while discontinuation because of adverse events (RR 1.28; $P = 0.54$) and overall and gastrointestinal adverse events did not differ significantly; without a breakdown of SAEs, the mechanism of the difference cannot be determined.
- Limitations are a mixed design with only 3 randomized trials, differing populations, doses, and titration, short follow-up, and missing kidney endpoints, so the analysis does not show superiority of tirzepatide for preventing AKI, eGFR decline, or albuminuria.
- With both drugs, nausea, vomiting, and diarrhea can lead to hypovolemia and prerenal AKI, particularly with diuretics, ACE inhibitors, ARBs, NSAIDs, or heart failure; this is a volume risk, not a specific nephrotoxicity of tirzepatide.
- Before initiation or escalation, the author recommends checking creatinine and eGFR, BP, volume status, sodium, and potassium, reviewing diuretics and NSAIDs, and educating the patient about warning symptoms such as repeated vomiting, oliguria, or dizziness.
- With GI symptoms, the dose should not be increased mechanically, sick day management should be individualized according to the drug combination, and the drug should be chosen by its overall profile, tolerability, and kidney risk, not merely by greater weight loss.

3.49 Obesity and cardiometabolic health in kidney disease: from lifestyle to a multicomponent strategy

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. Obesity is an independent factor in the onset and progression of CKD and of cardiovascular complications. A narrative review by Mathew et al. in the Clinical Journal of the American Society of Nephrology (2026) summarizes these links within the cardiovascular-kidney-metabolic (CKM) framework. It describes direct mechanisms, such as glomerular hyperfiltration, and indirect ones through diabetes and hypertension, and emphasizes the importance of visceral and perirenal fat. Management is shifting from lifestyle measures to a multicomponent strategy with incretin therapy and surgery; the review contains no meta-analysis, and its recommendations are expert opinion.

Summary

- The direct pathway includes hyperfiltration with increased intraglomerular pressure and tubular injury, and the indirect pathway includes diabetes and hypertension; adipose tissue produces cytokines and adipokines and activates both the renin-angiotensin-aldosterone system and the sympathetic nervous system.
- Obesity-related glomerulopathy can develop even without diabetes, presents with slowly progressive proteinuria, and may accelerate other nephropathies; CKD is associated mainly with visceral and perirenal fat depots, not total body weight.
- GLP-1 receptor agonists and dual incretin agonists act on metabolism, CKD progression, and cardiovascular risk, while data on triple agonists in kidney disease are limited; SGLT2 inhibitors induce only modest weight loss, and their role is complementary.
- Bariatric surgery can improve hypertension, proteinuria, and diabetes control in CKD, but it brings nutritional deficiencies and a risk of oxalate nephrolithiasis after malabsorptive procedures.
- The obesity paradox in patients on dialysis may be explained by reverse causality, sarcopenia, and selection bias, so weight reduction should be individualized while preserving muscle mass.

- The authors recommend assessing waist circumference, albuminuria, BP, eGFR, sleep apnea, and sarcopenia alongside BMI, using a multidisciplinary model, and communicating without stigma.

3.50 Obesity treatment with GLP-1 receptor agonists: discontinuation, maintenance strategy, and a nephrology perspective

Published September 20, 2026 · [online version \(Slovak\)](#)

Abstract. After obesity treatment with GLP-1 receptor agonists is stopped, a substantial part of the lost weight often returns, and the WHO conditionally recommends long-term use. This review of the evidence as of September 20, 2026, assesses the STEP 1 extension, the randomized SURMOUNT-MAINTAIN trial, the SELECT RCT, and the European product information. After semaglutide was stopped, 11.6 of the 17.3 percentage points of weight loss returned; a tirzepatide maintenance dose of 5 mg preserved a weight loss of 16.6% versus 21.9% with the full dose. Treatment should be long-term with regular reassessment, and in nephrology with attention to eGFR, volume, and nutrition.

Summary

- In STEP 1, body weight fell by 17.3% with semaglutide 2.4 mg over 68 weeks, and after 52 weeks off treatment a net loss of 5.6% remained; this was an exploratory follow-up of 327 participants in whom both the drug and the lifestyle intervention were stopped simultaneously.
- In SURMOUNT-MAINTAIN, 378 participants were randomized after 60 weeks: weight change to week 112 was -21.9% with 10 or 15 mg, -16.6% with 5 mg, and -9.9% with placebo; the trial demonstrated neither equivalence of 5 mg nor preservation of organ protection.
- SELECT, in 17,604 people without diabetes with established cardiovascular disease, reduced MACE with semaglutide to 6.5% versus 8.0% (HR 0.80); the kidney composite of 1.8% versus 2.2% (HR 0.78) also included macroalbuminuria, so it does not represent a 22% reduction in the need for dialysis.
- The European product information for Wegovy does not recommend use at an eGFR < 30 mL/min/1.73 m²; for Mounjaro, the dose is not adjusted in impaired kidney function, but experience in severe impairment and end-stage kidney failure is limited.
- Nausea, vomiting, and diarrhea can lead to dehydration and worsening kidney function; semaglutide is interrupted when pancreatitis is suspected and is not restarted when it is confirmed.
- As a clinical synthesis rather than a validated protocol, the author recommends identifying the reason for the change, assessing eGFR and albuminuria, adapting fluid intake to CKD and heart failure, protecting nutrition and muscle function, and agreeing on a follow-up plan after the treatment change.

3.51 Alogliptin and kidney outcomes: what a comparison of DPP-4 inhibitors shows

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. A target trial emulation using US insurance data (Sklepinski et al., 2026) compared initiation of four DPP-4 inhibitors in 175,468 adults with type 2 diabetes without a code for advanced CKD. Alogliptin was associated with a lower incidence of the composite kidney endpoint, with HRs of 0.50 to 0.63 versus linagliptin, saxagliptin, and sitagliptin. These are observational data without eGFR or UACR, with a small alogliptin group and less convincing supplementary analyses. The study generates a hypothesis, not evidence of kidney protection, and is no reason to change stable therapy.

Summary

- Before weighting, 2,491 patients took alogliptin, 37,438 linagliptin, 17,071 saxagliptin, and 118,468 sitagliptin; inverse probability weighting does not balance unmeasured differences, and the mean age was approximately 60 versus 68 years.
- The three-year incidence of newly recorded CKD stage 3 to 5, kidney failure, or KRT was 3.2% with alogliptin versus 6.2%, 5.0%, and 5.4%; the HR was 0.50 (0.37 to 0.69) versus linagliptin, 0.63 versus saxagliptin, and 0.58 versus sitagliptin.

- In the analysis censoring follow-up at drug discontinuation, the intervals versus sitagliptin and saxagliptin included 1, after the addition of another drug class all three comparisons did, and after the second year the HRs were close to 1.
- The result rests on administrative codes, linagliptin may have been chosen for higher-risk patients because it needs no dose adjustment, and the supplementary materials report acute kidney failure inconsistently; in the CARMELINA RCT, linagliptin did not change kidney outcomes versus placebo (HR 1.04).
- The alogliptin dose is 25 mg daily, 12.5 mg at a CrCl of 30 to 50 mL/min, and 6.25 mg at a CrCl < 30 mL/min or on dialysis; the FDA warns of a possible increase in the risk of heart failure with alogliptin and saxagliptin.
- In type 2 diabetes and CKD, priority goes to an SGLT2 inhibitor at an eGFR of ≥ 20 mL/min/1.73 m² according to KDIGO 2024 and to semaglutide, supported by FLOW; actual measurement of eGFR and albuminuria remains decisive.

3.52 Finerenone in CKD and type 1 diabetes: what FDA approval really means

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. In September 2026, the FDA expanded the indication of finerenone (Kerendia) to adults with CKD associated with type 1 diabetes, to reduce UACR with an expected kidney benefit. The basis is the six-month phase 3 RCT FINE-ONE with 242 participants, in which finerenone reduced UACR by 25% more than placebo (ratio 0.75; 95% CI, 0.65 to 0.87). A reduction in kidney failure or cardiovascular events was not directly demonstrated. Hyperkalemia occurred in 10.1% versus 3.3%; in the EU, this use currently lies outside the approved kidney indication.

Summary

- FINE-ONE enrolled adults with type 1 diabetes, an eGFR of 25 to less than 90 mL/min/1.73 m², a UACR of 200 to less than 5,000 mg/g, potassium of no more than 4.8 mmol/L, and a stable ACE inhibitor or ARB; patients on an SGLT2 inhibitor or a GLP-1 receptor agonist were excluded.
- UACR fell by 34% with finerenone and by 12% with placebo; the placebo-adjusted difference of 25% is a ratio of geometric means, and the US label reports a 22% lower UACR at month 3 and a 28% lower UACR at month 6.
- Over six months, eGFR changed by -5.6 versus -2.7 mL/min/1.73 m² (difference, -2.9) and largely recovered after discontinuation, which supports a hemodynamic component; a long-term eGFR trajectory is lacking.
- Hyperkalemia led to permanent discontinuation in 1.7% of patients treated with finerenone; finerenone is a CYP3A4 substrate, and coadministration with strong CYP3A4 inhibitors is contraindicated in the United States.
- According to the US label, treatment is not started at a potassium level above 5.0 mmol/L, the starting dose is 20 mg at an eGFR of at least 60 and 10 mg at an eGFR of 25 to less than 60 mL/min/1.73 m², initiation is not recommended at an eGFR below 25, and the target dose is 20 mg.
- Potassium is checked after four weeks and after each dose change, and treatment is interrupted at a level above 5.5 mmol/L; in the EU, the indication covers only type 2 diabetes, so the current Slovak SmPC, availability, and reimbursement must be the basis for decisions.

3.53 Hidden hypercortisolism in difficult-to-control type 2 diabetes: what CATALYST showed

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. The US multicenter phase 4 CATALYST study, funded by the manufacturer of mifepristone, tested 1,057 patients with difficult-to-control type 2 diabetes with the overnight 1 mg dexamethasone suppression test. Nonsuppressed cortisol was found in 23.8%, which is not the prevalence of confirmed Cushing's syndrome; an independent meta-analysis reports confirmed syndrome in 2.1% of patients with diabetes. In the

randomized phase, mifepristone lowered HbA1c by 1.32 percentage points versus placebo, at the cost of hypokalemia, a rise in BP, and frequent discontinuations. For the nephrologist, the implications are targeted testing, endocrinological confirmation, and close monitoring of treatment.

Summary

- Cortisol above 1.8 µg/dL (50 nmol/L) after the DST was found in 252 patients (23.8%; 95% CI, 21.3% to 26.5%) and in 36.6% of those taking at least three antihypertensives; the definition relied on a single test without a second confirmatory test, and CT showed an adrenal abnormality in only 34.7%.
- The DST is performed by giving 1 mg of dexamethasone at approximately 23:00 and drawing cortisol at approximately 8:00, with verification of dexamethasone exposure; a positive result requires a second validated test, and ACTH and imaging belong only to the etiologic step.
- In CKD, 24-hour urinary free cortisol may be falsely low; CATALYST excluded patients on dialysis, and the treatment phase also excluded patients with an eGFR < 30 mL/min/1.73 m².
- In the treatment phase (91 mifepristone, 45 placebo, 24 weeks), body weight fell by 5.12 kg versus placebo, but systolic BP rose by 10.1 mm Hg, and 46% versus 18% did not complete treatment.
- Mifepristone does not stop cortisol production, so through mineralocorticoid receptor activation it caused hypokalemia in 29.7%; serious adverse events occurred in 32% versus 5%, and three cases of euglycemic ketoacidosis occurred with an SGLT2 inhibitor.
- The improvement in eGFR in ten case reports is not evidence of kidney protection; the author recommends against universal screening, against sending a positive DST directly to CT or treatment, and, with mifepristone, monitoring potassium, BP, edema, eGFR, UACR, and glycemia.

3.54 DASH4D in type 2 diabetes: better glycemia and blood pressure in a controlled trial

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. DASH4D is a dietary pattern derived from the DASH diet, with fewer carbohydrates, more unsaturated fats, and less potassium, designed for type 2 diabetes. In a randomized crossover feeding trial with 101 adults, compared with a typical American diet, it lowered fructosamine, fasting glucose, and HbA1c by 0.09 percentage points and increased CGM time in range by 5.2 percentage points. With lower sodium, it reduced systolic BP by 4.6 mm Hg. This is short-term evidence of biological efficacy, not prevention of kidney or cardiovascular events.

Summary

- Four isocaloric diets of five weeks each combined DASH4D or the comparison diet with sodium of approximately 1,500 or 3,700 mg per day; all food was supplied by the research team, and body weight was kept stable.
- Fructosamine fell by 5.6 µmol/L, fasting glucose by 4.5 mg/dL, and HbA1c by 0.09 percentage points, a statistically significant but small absolute difference.
- On CGM in 89 participants, mean glucose fell by 11.1 mg/dL, and time in the range of 70 to 180 mg/dL rose by 5.2 percentage points, or approximately 75 minutes per day; time in hypoglycemia did not change.
- The reduction of 4.6 mm Hg in systolic and 2.3 mm Hg in diastolic BP reflects both the dietary change and the sodium change, with sodium reduction explaining the larger part of the effect.
- Limitations are a small single-center sample with a mean age of 67 years and 87% Black participants, short periods, and exclusion of an eGFR < 30 mL/min/1.73 m², potassium ≥ 5.2 mmol/L or < 3.5 mmol/L, and HbA1c > 9%.
- The pattern is suitable as an adjunct to treatment with preserved or mildly to moderately reduced kidney function; in CKD, potassium, sodium, protein, and energy should be adapted to stage, serum potassium, and treatment, and insulin should be reviewed as glycemia improves.

3.55 Switching between GLP-1 receptor agonists: timing, titration, and nephrology safeguards

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. Switching between GLP-1 receptor agonists is common: according to pharmacy data, up to a quarter of patients change drugs after the first year, yet direct switching studies are scarce. This review draws on the recommendations of Almandoz et al. (2020), head-to-head trials, and the authors' experience, and adds nephrology safeguards. Equivalent doses do not exist, timing depends on the reason for the change, and the main kidney risk is volume depletion with gastrointestinal symptoms. At the same time, the randomized FLOW trial showed a 24% lower incidence of the primary endpoint with semaglutide and a favorable safety profile.

Summary

- When the maximum dose is well tolerated, the new drug is started without a gap: dulaglutide 1.5 mg or exenatide 2 mg directly at the maximum dose, and semaglutide via 0.5 mg weekly for 4 weeks; with gastrointestinal intolerance, one waits for symptoms to resolve and starts at the lowest dose.
- During the switch, fasting glucose or HbA1c may rise transiently, which should be explained to the patient in advance; with concomitant insulin or a sulfonylurea, a plan to reduce their dose should be in place.
- The exendin-based analogs exenatide and lixisenatide are eliminated predominantly by the kidneys and are restricted in more advanced renal insufficiency, whereas the human analogs liraglutide, dulaglutide, and semaglutide generally are not; thresholds should be verified in the SmPC.
- According to case reports, most kidney events occurred with gastrointestinal symptoms and volume depletion, particularly with a diuretic and RAAS blockade; patients with an eGFR below 45 mL/min/1.73 m², frail patients, and those after AKI are at increased risk.
- In FLOW, among 3,533 patients with CKD, semaglutide 1.0 mg had an HR of 0.76 (0.66 to 0.88) for the primary endpoint and an HR of 0.71 for cardiovascular death, slowed the annual eGFR slope by 1.16 mL/min/1.73 m², and was associated with fewer serious adverse events (49.6% vs. 53.8%).
- The authors recommend checking eGFR, creatinine, electrolytes, BP, and body weight before the switch, giving the patient a sick day rule, and scheduling follow-up within 2 to 3 months; the switching recommendations are mainly the authors' opinion, not the result of direct trials.

3.56 The 2026 ADA and EASD consensus: type 2 diabetes is no longer a disease of glycemia

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. At the EASD congress in Milan (September and October 2026), a new ADA and EASD consensus on the management of type 2 diabetes was published (Davies et al., Diabetes Care 2026), which broadens the focus from hyperglycemia to holistic risk management of multiple long-term conditions. It recommends earlier use of SGLT2 inhibitors and GLP-1-based therapy, potentially from diagnosis, for organ protection, and their earlier combination in cardiovascular disease, CKD, and heart failure. For the nephrologist, this means earlier kidney protection with safe implementation.

Summary

- The consensus panel included nephrologists Sylvia E. Rosas and Peter Rossing; a healthy lifestyle, including sleep, tobacco avoidance, psychological support, and weight management, remains the foundation of care.
- The shift rests mainly on the FLOW trial in 3,533 patients with type 2 diabetes and CKD, in which semaglutide 1.0 mg reduced the primary endpoint by 24% (HR 0.76), cardiovascular death by 29%, and all-cause mortality by 20%.
- In CKD, metformin remains subject to kidney-related rules based on eGFR, tolerability, and the SmPC, but the order of clinical questions is shifting toward organ protection.

- The initial hemodynamic and reversible eGFR decline after starting an SGLT2 inhibitor should not lead to discontinuation; combination with a GLP-1 receptor agonist, a diuretic, and RAAS blockade requires an agreed sick day rule because of the risk of AKI with volume depletion.
- With insulin or a sulfonylurea, CGM in CKD is primarily a safety tool against hypoglycemia, and the holistic framework also includes metabolic dysfunction-associated steatotic liver disease.
- The author deliberately does not report results of unpublished congress trials, such as RESET for REMISSION; in his view, the main barrier to kidney protection is not a lack of evidence but delayed diagnosis of CKD, absent albuminuria screening, and unclear responsibility for initiating therapy.

CHAPTER 4

Hypertension, lipids, and vascular risk

Blood pressure measurement and targets, resistant hypertension, dyslipidemia, antithrombotic therapy, and pregnancy with kidney risk. Number of articles: 24.

Key messages of the chapter

KDIGO 2021 suggests a systolic BP below 120 mm Hg (grade 2B) under standardized measurement in CKD not on dialysis, yet in real-world practice in 2024 only 21.9% of patients with CKD G3 to G4 achieved it, and a BP of 120 mm Hg or higher was observationally associated with a higher risk of both cardiovascular events (HR 1.19) and kidney failure (HR 1.21) (4.22). A post hoc analysis of ESPRIT showed that the target below 120 mm Hg is achievable even in community centers in 62.5% of patients at the cost of approximately 1.3 additional antihypertensive drugs and 1.5 additional visits per year (4.12). The AHA and ACC guidelines emphasize accurate measurement and recommend sodium below 2,300 mg, ideally approximately 1,500 mg per day (4.8). BP variability is also observationally associated with a worse prognosis, although it mediated only 9.15% of the effect of intensive treatment on white matter lesions (4.18), as is blunted daytime natriuresis with nocturnal hypertension (4.23).

In resistant hypertension, pseudoresistance must first be excluded and primary aldosteronism specifically sought (4.20). In PATHWAY-2, spironolactone lowered home systolic BP by 8.70 mm Hg compared with placebo (4.20). In a phase 2 trial in patients with CKD, the aldosterone synthase inhibitor baxdrostat lowered systolic BP by 8.1 mm Hg and, in an exploratory analysis, albuminuria by 55%, but hyperkalemia occurred in 41% versus 5% (4.3, 4.4). The phase III Bax24 trial showed a decrease in 24-hour systolic BP of 14.0 mm Hg, with potassium above 6.0 mmol/L in 3% of treated patients, but a comparison with spironolactone and data on clinical events are lacking (4.21). Renal denervation lowers BP by approximately 5 to 10 mm Hg, about one third of patients do not respond, and the procedure complements optimized pharmacotherapy rather than replacing it (4.6).

In lipidology, the ACC/AHA 2026 guidelines introduce the PREVENT calculator, the return of LDL cholesterol targets, an LDL below 1.4 mmol/L in secondary prevention, and CKD stages 3 and 4 as a stand-alone indication for a statin (4.2, 4.10). A nephrology commentary adds a non-HDL cholesterol target below 2.2 mmol/L and does not support routine new statin initiation in maintenance hemodialysis (4.14). Critics point to conflicts of interest in 12 of 33 authors and the absence of an independent systematic review (4.1), while European guidelines remain decisive in Slovak practice (4.14). ESC and EAS 2025 retain the targets from 2019; lowering LDL cholesterol by 1 mmol/L reduces major vascular events by approximately one fifth, and decisions should distinguish CKD not on dialysis, dialysis, and transplantation (4.17). Lipoprotein(a) should be measured at least once in adulthood, and in CKD its acquired elevation should also be considered (4.19).

Antithrombotic prevention in advanced CKD has negative randomized evidence: in the TRACK trial, rivaroxaban 2.5 mg twice daily did not reduce cardiovascular events (HR 1.09) and increased major bleeding (HR 1.51) (4.7). Factor XI inhibitors hold promise according to the OCEANIC-STROKE trial, but the review is based on an expert meeting (4.11). In the CRIC cohort, a conventionally normal ABI of 0.9 to 1.0 carried an almost threefold risk of peripheral artery disease (HR 2.76), so in CKD the ABI should be read as a continuous variable (4.24).

Pregnancy is a window into future risk. Preeclampsia arises from placental hypoperfusion and systemic endothelial dysfunction (4.5). In a Danish cohort, women with preeclampsia and significant proteinuria had a 10-year risk of CKD of 5.1% versus 0.4% (4.9), and in a Stockholm cohort, the risk of

laboratory signs of CKD was more than doubled (HR 2.43); at an albuminuria threshold of 30 mg/g, the number of detected events rose approximately 4.5-fold (4.15). The randomized CHAP trial showed that treating mild chronic hypertension to a target below 140/90 mm Hg reduced the composite adverse outcome from 37.0% to 30.2% (4.16). CKD occurs in approximately 3% to 4% of pregnancies and requires preconception counseling (4.13).

This yields a framework for the outpatient clinic: measure BP in a standardized way and add ABPM when in doubt (4.8, 4.23); in CKD, aim for a low target with protocol-based titration and safety monitoring (4.22, 4.12); in resistance, investigate the aldosterone axis, and with new drugs, monitor hyperkalemia closely (4.20, 4.3). Individualize lipid therapy according to absolute risk and dialysis status (4.1, 4.17), do not use prophylactic rivaroxaban in advanced CKD (4.7), and follow women after preeclampsia long term with BP and UACR (4.15, 4.16).

4.1 The new dyslipidemia guidelines: why they need to be read critically

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. The new ACC/AHA guideline for the management of dyslipidemia replaces the Pooled Cohort Equations with the PREVENT calculator, lowers the 10-year risk thresholds for statin therapy, expands coronary artery calcium scoring and lipoprotein(a) measurement, and reinstates LDL cholesterol targets. This is a commentary by family physician Dr. Kenny Lin assessing the quality of the guideline development process. He criticizes the conflicts of interest of 12 of the 33 authors, the nonparticipation of the AAFP and ACP, the absence of an independent systematic review, and the marginal incorporation of patient preferences. In practice, clinicians should know the guideline but decide individually according to absolute benefit.

Summary

- The ACC/AHA guideline replaces the Pooled Cohort Equations with the PREVENT calculator, lowers the 10-year risk thresholds for starting a statin, expands the use of coronary artery calcium scoring, recommends more frequent measurement of lipoprotein(a), and reintroduces LDL cholesterol targets with regular lipid profile checks. The result may be a larger number of candidates for lipid-lowering therapy.
- Of the 33 authors, 12 had declared relevant financial conflicts of interest, while the panel chair and vice chair had none; Lin considers this still acceptable, although not ideal.
- A more serious weakness is the nonparticipation of the American Academy of Family Physicians and the American College of Physicians, which had methodological reservations, even though they represent the frontline physicians who deliver prevention.
- The guideline is not based on independent systematic reviews of predefined clinical questions and does not publish search strategies or inclusion criteria, unlike the 2022 US Preventive Services Task Force recommendation and the 2025 VA/DoD guideline.
- The panel had a single patient representative. Lin cites a study in which, at a 10-year risk of 10%, more than 40% of US adults would decline a statin even with a hypothetical reduction of absolute risk to zero, whereas the real benefit of a statin is approximately 2.5 percentage points.
- The author recommends knowing the guideline but not adopting it mechanically: treatment decisions should combine individual risk, comorbidities, expected absolute benefit, adverse effects, and patient preferences.

4.2 The 2026 dyslipidemia management guideline: what it brings to clinical practice

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. The 2026 multisociety ACC/AHA/AACVPR/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA guideline for the management of dyslipidemia, published in *Circulation*, focuses on the assessment, treatment, and monitoring of lipid disorders to reduce the risk of atherosclerotic cardiovascular disease. This

professional guideline promotes comprehensive risk assessment with the PREVENT calculator, greater emphasis on non-HDL cholesterol, apoB, and measurement of Lp(a) at least once in adulthood, and the return of LDL cholesterol targets. Statins remain the foundation of treatment, with escalation to ezetimibe and PCSK9 inhibitors. In practice, the key is to decide according to overall risk and together with the patient.

Summary

- Treatment decisions should be based on the overall risk profile, including age, BP, smoking, diabetes, chronic kidney disease, prior cardiovascular disease, family history, and atherogenic lipoprotein levels.
- Greater weight is given to non-HDL cholesterol and apoB, which may better capture risk when LDL is acceptable, and to lipoprotein(a), a genetically determined risk factor for atherosclerosis and aortic stenosis, to be measured at least once in adulthood.
- The summary table lists indicative LDL targets derived from ESC/EAS 2019 and ACC/AHA: secondary prevention below 1.4 to 1.8 mmol/L; high risk, including CKD with proteinuria, a reduction of 50% or more or below 1.8 to 2.6 mmol/L; intermediate risk (10-year PREVENT risk 7.5% to 20%) below 2.6 mmol/L; and low risk below 3.0 mmol/L; exact values should be verified in the guideline text.
- LDL reduction: lifestyle 5% to 15%, moderate-intensity statin (atorvastatin 10 to 20 mg, rosuvastatin 5 to 10 mg) 30% to 49%, high-intensity statin (atorvastatin 40 to 80 mg, rosuvastatin 20 to 40 mg) at least 50%, ezetimibe 10 mg/day a further 15% to 25%, and a PCSK9 inhibitor (evolocumab, alirocumab) a further 50% to 60%.
- In diabetes, CKD, and familial hypercholesterolemia, risk should be assessed particularly rigorously, and shared decision-making applies at intermediate risk; no HDL-raising drug has shown clinical benefit.
- After starting or changing treatment, the lipid profile should be checked to detect poor adherence, adverse effects, or the need for combination therapy.

4.3 A new pill lowers difficult-to-control blood pressure and may protect the kidneys

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. Patients with chronic kidney disease and uncontrolled hypertension have a high cardiovascular and kidney risk. A randomized, double-blind, placebo-controlled phase 2 trial published in the Journal of the American Society of Nephrology tested the aldosterone synthase inhibitor baxdrostat in 195 patients. At 26 weeks, systolic BP fell by 8.1 mm Hg more than with placebo, and albuminuria by 55%. Hyperkalemia occurred in 41% versus 5%. This is a promising signal of kidney protection, not evidence of prevention of kidney failure; phase 3 trials are expected to provide confirmation.

Summary

- The phase 2 trial enrolled 195 patients with CKD and uncontrolled hypertension, most on an ACE inhibitor or an ARB at the maximum tolerated dose; baseline systolic BP was approximately 151 mm Hg, mean eGFR 44 mL/min/1.73 m², UACR approximately 714 mg/g, and 80% had type 2 diabetes.
- At 26 weeks, baxdrostat reduced systolic BP by 8.1 mm Hg more than placebo and urinary albumin by 55% compared with placebo.
- Hyperkalemia occurred in 41% of patients on baxdrostat versus 5% on placebo, mostly of mild to moderate severity; no deaths or unexpected adverse events were reported.
- The reduction in albuminuria suggests a possible kidney-protective effect beyond hemodynamics, but it is not yet evidence that the drug prevents kidney failure; this is to be tested in large phase 3 trials.
- According to the author, this is a further signal that CKD treatment will rely not only on conventional RAAS blockade but also on targeting other hormonal and inflammatory pathways.

4.4 New treatment for resistant hypertension in CKD: baxdrostat lowers both blood pressure and albuminuria

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. Resistant hypertension in chronic kidney disease remains a common problem despite RAAS blockade. A randomized, double-blind, placebo-controlled phase 2 trial in the *Journal of the American Society of Nephrology* assessed the aldosterone synthase inhibitor baxdrostat in 195 patients with CKD and uncontrolled hypertension. At 26 weeks, it reduced systolic BP by 8.1 mm Hg more than placebo and, in an exploratory analysis, albuminuria by 55%. The main risk was hyperkalemia (41% vs. 5%). The results do not change current practice; phase 3 trials with hard clinical endpoints will be decisive.

Summary

- The trial enrolled 195 patients, of whom 192 were randomized to low-dose baxdrostat, a higher dose, or placebo in addition to an ACE inhibitor or an ARB at the maximum tolerated dose; at baseline, systolic BP was approximately 151 mm Hg, eGFR 44 mL/min/1.73 m², UACR approximately 714 mg/g, and 80% had type 2 diabetes.
- At 26 weeks, the reduction in systolic BP was 8.1 mm Hg greater than with placebo, and in an exploratory analysis urinary albumin was 55% lower, suggesting a potential kidney-protective effect.
- Hyperkalemia occurred in 41% of patients on baxdrostat versus 5% on placebo, mostly of mild to moderate severity; serious adverse events occurred in 9% versus 3%, and no deaths were reported.
- These are phase 2 results with a biological and clinical signal, not evidence of protection against kidney failure; phase 3 trials are under way, including a combination with dapagliflozin.
- Baxdrostat is not approved by the FDA for any use, and the trial was funded by AstraZeneca, which the author recommends taking into account in interpretation.
- Nothing changes in current practice; if the drug is introduced, the author notes that potassium, eGFR, and drug interactions will need careful monitoring, particularly with concomitant ACE inhibitor or ARB therapy and in diabetes.

4.5 Preeclampsia and the kidney: why damage occurs and what it means for practice

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. Preeclampsia combines new-onset hypertension with organ dysfunction, including the kidneys, and with adverse effects on the fetus. A review in *Nature Reviews Nephrology* explains the kidney manifestations with a two-stage model: inadequate placental perfusion leads to the release of signals, an imbalance of angiogenic and antiangiogenic factors, and systemic endothelial dysfunction. In the kidneys, this manifests as glomerular endothelial injury with proteinuria and impaired filtration. The article reports no quantitative results; in practical terms, it emphasizes monitoring the trend of kidney parameters and long-term postpartum follow-up because of the higher cardiovascular and kidney risk.

Summary

- According to the two-stage framework, defective placentation and reduced perfusion lead to the release of placental signals, an imbalance between proangiogenic factors and antiangiogenic influences, and whole-body endothelial dysfunction, to which the kidney, with its rich microcirculation, is particularly sensitive.
- In the kidneys, a state close to glomerular endothelial injury develops, clinically presenting with proteinuria, declining function ranging from mild changes to acute deterioration, and altered perfusion; this is a network of signals with varying timing and severity.
- Proteinuria does not automatically mean kidney failure; acute deterioration depends on the severity of endothelial dysfunction, the extent of vascular changes, renal reserve, comorbidities, and the timing of intervention.

- When preeclampsia is suspected, the trend of creatinine and eGFR, proteinuria, and fluid balance should be monitored, and secondary factors, such as hypovolemia with vomiting, infections, medication timing, and cardiovascular instability, should not be underestimated.
- Because the mechanisms run in parallel, there is no single targeted therapy, and clinical management is guided by reducing the acute risk to both mother and fetus, not by correcting a single biomarker. After delivery, the author recommends planning long-term follow-up because of the higher cardiovascular and kidney risk.

4.6 Renal denervation in resistant hypertension: a promising method, but not a shortcut in treatment

Published June 5, 2026 · [online version \(Slovak\)](#)

Abstract. Resistant hypertension carries a high risk of myocardial infarction, stroke, heart failure, and CKD. Based on an expert discussion in the Medscape InDiscussion podcast, the article describes the place of renal denervation, a catheter-based procedure that reduces sympathetic activity around the renal arteries. The mean reduction in systolic BP is modest, approximately 5 to 10 mm Hg, and about a third of patients do not respond adequately; complications are rare, with renal artery stenosis reported in under 0.2%. The method complements a confirmed diagnosis, exclusion of secondary causes, and optimized pharmacotherapy; it does not replace them.

Summary

- According to the 2025 ACC/AHA and other societies' guideline, resistant hypertension is BP above target despite three antihypertensives with complementary mechanisms at maximally tolerated doses, including a diuretic, or controlled BP on four or more drugs; pseudoresistance due to measurement error, poor adherence, or white coat effect must first be excluded.
- Before the procedure, home or ambulatory monitoring, optimization of treatment including a diuretic, and exclusion of secondary causes are required, particularly primary aldosteronism, renal artery stenosis, CKD, obstructive sleep apnea, and endocrine diseases.
- Candidates are selected by a multidisciplinary team, and significant stenoses, fibromuscular dysplasia, aneurysms, or a recent renal artery stent are exclusions. There is no absolute eGFR threshold, but trials included few patients with an eGFR below approximately 40 mL/min/1.73 m², so caution is needed in advanced CKD.
- The procedure is performed through the femoral artery with radiofrequency or ultrasound energy, takes up to approximately one hour, and its effect develops gradually; response is assessed after 3 to 6 months, or according to some experts only after one year.
- Most patients continue antihypertensive treatment; complications are mainly related to arterial access, artery dissection or perforation is very rare, and excessive worsening of kidney function is not observed.
- In trials, patients were followed up at 1, 3, and 6 months, at one year, and annually thereafter; renal artery duplex ultrasound may be considered at approximately 6 months. The author recommends clearly defining who manages the adjustment of antihypertensives.

4.7 Low-dose rivaroxaban in advanced CKD provided no cardiovascular benefit

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Patients with advanced CKD have high cardiovascular and bleeding risks and were often excluded from cardiology trials. The randomized TRACK trial, published in JAMA, compared rivaroxaban 2.5 mg twice daily with placebo in 1,458 patients with an eGFR ≤ 29 mL/min/1.73 m² or on dialysis. After a median of 1.7 years, it did not reduce major cardiovascular events (22.6% vs. 20.7%; HR 1.09) and increased major bleeding

(8.8% vs. 6.0%; HR 1.51). The results do not support preventive use, but they do not apply to clear indications for anticoagulation.

Summary

- TRACK enrolled adults with an eGFR ≤ 29 mL/min/1.73 m² or on dialysis with increased cardiovascular risk (coronary heart disease, peripheral artery disease, diabetes, nonhemorrhagic nonlacunar stroke, or age 65 years or older); the primary endpoint included cardiovascular death, nonfatal myocardial infarction, stroke, and a peripheral artery disease event.
- After a median of 1.7 years, the primary endpoint occurred in 22.6% of patients on rivaroxaban and in 20.7% on placebo (HR 1.09; 95% CI, 0.87 to 1.36; P = 0.46); all-cause mortality was 25.6% versus 23.0%.
- Major bleeding was more frequent with rivaroxaban (8.8% vs. 6.0%; HR 1.51; 95% CI, 1.02 to 2.22; P = 0.04), approximately 1.7 additional events per 100 person-years; in patients aged 65 years and older it was 10% versus 5.6%, and in younger patients 7.3% versus 6.4%.
- The benefit from COMPASS and VOYAGER PAD (rivaroxaban with aspirin) cannot be extrapolated, because in advanced CKD the pathophysiology of atherothrombosis and bleeding differs, and rivaroxaban is partly eliminated by the kidneys.
- Limitations: early termination with enrollment of 1,458 instead of the planned 1,900 patients at 90 centers in 12 countries, permanent treatment discontinuation in almost a third of patients on rivaroxaban, and baseline aspirin use in only about half.
- The result does not apply to clear indications such as atrial fibrillation or venous thromboembolism; decisions should be individualized according to absolute ischemic and bleeding risk, age, dialysis status, and concomitant antiplatelet therapy.

4.8 New hypertension guidelines: less improvisation, more accurate measurement, and practical decisions

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. The new American Heart Association and American College of Cardiology hypertension guideline and its practical interpretation emphasize accurate BP measurement before diagnosis or treatment escalation. This is a guideline with commentary for outpatient practice. Cuffless devices, with a sensitivity of 41%, are not yet adequate; optimal alcohol intake is zero, and sodium should be below 2,300 mg, ideally approximately 1,500 mg per day. In stage 1 hypertension, a 10-year PREVENT risk above 7.5% also informs the decision, and in stage 2, early combination therapy is preferred.

Summary

- During measurement, the patient should sit with the back supported, feet on the floor, legs uncrossed, and the arm supported at heart level, not immediately after coffee, smoking, a meal, or exertion; the first triage reading tends to be higher than a repeat reading after several minutes of rest.
- Cuffless devices, including the technology in the Apple Watch, have a sensitivity of 41% for detecting high BP, and treatment should not be adjusted based on them; validated cuff measurement supplemented by home or 24-hour ambulatory monitoring is needed.
- The relationship between alcohol and BP is linear, and zero intake is optimal; sodium should be below 2,300 mg per day, ideally approximately 1,500 mg, with a large share coming from processed foods.
- Higher dietary potassium intake may lower BP, but in CKD, with RAAS inhibitor or mineralocorticoid receptor antagonist therapy, or after hyperkalemia, the author does not recommend it across the board without checking laboratory values.
- In stage 1 hypertension (systolic BP 130 to 139 mm Hg or diastolic BP 80 to 89 mm Hg), established cardiovascular disease, diabetes, or a 10-year PREVENT risk above 7.5% justifies earlier pharmacotherapy; otherwise, lifestyle changes and reassessment come first.
- In stage 2 hypertension (above 140/90 mm Hg), dual combinations are preferred, or a triple combination of an angiotensin receptor blocker, a calcium channel blocker, and a thiazide or thiazide-like diuretic,

ideally in a single pill. Before changing treatment, measurement, adherence, and BP-raising drugs, such as nonsteroidal anti-inflammatory drugs, should be checked.

4.9 Proteinuria in preeclampsia may signal a higher long-term risk of hypertension and CKD

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Preeclampsia may signal future cardiometabolic and kidney risk. A Danish population-based cohort study from the Danish Medical Birth Register included 286,078 women with pregnancies from 1998 to 2018 and a median follow-up of 6.4 years. Women with preeclampsia and moderate or severe proteinuria had a 10-year risk of hypertension of 16.0% versus 3.6% and of CKD of 5.1% versus 0.4% compared with women without preeclampsia; an association with cardiovascular disease was not confirmed. These observational data support structured postpartum monitoring of BP and kidney parameters.

Summary

- Proteinuria was classified by the highest value between week 20 of pregnancy and week 6 postpartum: none or mild with a UACR below 30 mg/g or a negative dipstick, and moderate or severe with at least 30 mg/g or a positive dipstick.
- Preeclampsia occurred in 3.3% of pregnant women, of whom 58.3% had no or mild and 41.7% moderate or severe proteinuria.
- With moderate or severe proteinuria, the 10-year risk of hypertension was 16.0% versus 3.6% and the 15-year risk 22.6% versus 7.2% in women without preeclampsia; the 10-year risk of CKD was 5.1% versus 0.4%, with a similar difference at 15 years.
- A clear association between the degree of proteinuria and cardiovascular disease was not confirmed, which, according to the author, does not diminish the importance of preeclampsia as an overall risk marker.
- Limitations: inconsistent recording of dipstick tests in Danish databases and identification of CKD from clinical laboratory data, so asymptomatic cases may have been missed; this is a strong observational signal, not evidence of causality.
- The postpartum plan should include checks of BP, creatinine and eGFR, albuminuria or proteinuria, and other risk factors, and should specify who follows the patient; with persistent hypertension, albuminuria, or reduced eGFR, nephrology or internal medicine follow-up is warranted.

4.10 New dyslipidemia guidelines: PREVENT replaces the old calculator, and LDL targets are lower

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. The 2026 ACC/AHA/ADA dyslipidemia guideline replaces the PCE calculator, based on data from approximately 25,000 people from 1960 to 1990, with the PREVENT calculator, derived from more than 3 million people, which estimates both 10-year and 30-year risk. Because PREVENT yields an estimate roughly half as high, treatment thresholds were lowered. Risk categories with LDL targets are introduced, and in secondary prevention, an LDL below 55 mg/dL (1.4 mmol/L) becomes the key value. Importantly for nephrology, CKD stage 3 and 4 remains an independent indication for a statin.

Summary

- Primary prevention by 10-year PREVENT risk: below 3% is low (moderate-intensity statin with LDL 160 to 189 mg/dL or a 30-year risk of at least 10% at age 30 to 50 years), 3% to 5% borderline, 5% to 10% intermediate with an LDL target below 100 mg/dL (2.6 mmol/L), and above 10% high with a target of 70 mg/dL (1.8 mmol/L) or less, possibly with ezetimibe.
- Without risk calculation, a statin is indicated in adults aged 40 to 75 years with diabetes, CKD stage 3 or 4, HIV infection, or LDL above 190 mg/dL (4.9 mmol/L); in diabetes with multiple risk factors, the LDL target is below 70 mg/dL.

- Lp(a) should be measured at least once in a lifetime, with a value above 125 being a risk factor; apoB refines risk in metabolic syndrome, hypertriglyceridemia, and type 2 diabetes.
- The coronary artery calcium score (men over 40, women over 45 years) helps when there is uncertainty: with CAC 0, a statin is usually not needed; with CAC 1 to 99, the LDL target is below 100 mg/dL; with 100 to 999, below 70 mg/dL; and with CAC above 1,000, a target below 55 mg/dL may be considered.
- In secondary prevention, most patients with ASCVD fall into the very high risk category with an LDL target below 55 mg/dL (1.4 mmol/L), so a high-intensity statin is often not enough, and ezetimibe, PCSK9 inhibitors, bempedoic acid, or inclisiran should be added early.
- The author recommends first determining risk correctly, choosing an appropriate treatment intensity, and checking whether the target is reached.

4.11 New approaches in antithrombotic therapy: from “good” and “bad” clots to safer prevention

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. Antithrombotic therapy balances clot prevention against bleeding risk. This review is based on an expert meeting on secondary stroke prevention without atrial fibrillation and on combining anticoagulation with antiplatelet therapy after coronary interventions. The greatest expectations are attached to factor XI inhibitors: the phase 3 OCEANIC-STROKE trial with asundexian showed a marked reduction in recurrences without an excessive increase in bleeding, including intracranial hemorrhage, and results with milvexian are awaited. In practice, timely de-escalation of combination therapy and coordination across specialties are key.

Summary

- In noncardioembolic stroke, antiplatelet therapy is the foundation, and in atrial fibrillation, anticoagulation; earlier trials of anticoagulants in noncardioembolic events ran into an excessively high bleeding risk.
- Factor XI is presented as a target more closely linked to pathological thrombosis and less to vessel wall repair; OCEANIC-STROKE with asundexian showed a marked reduction in recurrences without excessive bleeding, and results are awaited from LIBREXIA-STROKE and LIBREXIA-AF with milvexian, the latter with apixaban as the comparator.
- For atrial fibrillation with a stent after NSTEMI, a de-escalation framework was presented: approximately 1 month of triple therapy (an anticoagulant, acetylsalicylic acid, and a P2Y12 inhibitor, most often clopidogrel), then dual therapy for approximately 5 months, and at approximately 6 months or later often a DOAC alone.
- In practice, de-escalation is often missed because of fragmented care, when no specialist is willing to take responsibility for shortening antiplatelet therapy.
- With gastrointestinal bleeding, rivaroxaban was temporarily interrupted and resumed after the bleeding stopped; hidden causes, such as malignancy, should also be sought.
- Neither ablation nor left atrial appendage occlusion eliminates residual risk in every patient, and a safer anticoagulant could shift the threshold of benefit from anticoagulation. Some of the latest data were not available at the time this material was prepared.

4.12 Implementation strategies for intensive blood pressure control: lessons for everyday clinical practice (with a focus on nephrology)

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. In practice, intensive blood pressure control runs up against concerns about the J-curve and organizational barriers. A commentary on a post hoc analysis of the ESPRIT trial shows that a systolic target below 120 mm Hg is achievable even in community primary care centers: 62.5% of patients in the intensive arm reached it at the cost of approximately 1.3 additional antihypertensive drugs and 1.5 additional visits

per year. Patients with a longer history of hypertension and uncontrolled blood pressure needed more time. The key elements are protocol-driven titration, centralized monitoring, and, in CKD, rigorous safety surveillance.

Summary

- ESPRIT compared a systolic blood pressure target below 120 mm Hg with a standard target below 140 mm Hg in patients with hypertension and high cardiovascular risk; according to the source, it was the only trial adequately powered for a mean systolic blood pressure below 120 mm Hg.
- The target was reached by 62.5% of patients in the intensive arm, who required approximately 1.3 more antihypertensive drugs and 1.5 more visits per year than those in the standard arm.
- Patients with a longer duration of hypertension and uncontrolled blood pressure at baseline were less likely to achieve sustained intensive control and required more time and more visits.
- Implementation in community centers relied on standardized titration protocols, centrally managed blood pressure monitoring, and clinical decision support; this is a post hoc analysis, not a new guideline.
- According to the source, the J-curve exists in theory, but a protocol-driven process with monitoring of tolerability widens the room for safe low targets; in CKD, the author recommends titrating with an emphasis on tolerability and monitoring for hypoperfusion, kidney function, and electrolytes.
- For the outpatient clinic, the author recommends a clear titration algorithm, a follow-up schedule, a uniform measurement method including home measurements, and realistic expectations combined with targeted work on adherence.

4.13 Kidney disease in pregnancy: early risk assessment and multidisciplinary care

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Pregnancy in kidney disease is a dynamic state in which the physiological burden of gestation overlaps with CKD, hypertension, proteinuria, and the risk of preeclampsia. A review in the Journal of Nephrology reports that CKD occurs in approximately 3 to 4% of pregnancies and increases the risk of preeclampsia, preterm birth, fetal growth restriction, and progression of kidney dysfunction. It emphasizes bridging care between nephrology and maternal-fetal medicine. In practice, preconception counseling, timely treatment adjustment, and a shared monitoring plan that includes the postpartum period are essential.

Summary

- CKD occurs in approximately 3 to 4% of pregnancies, and the risk depends on CKD stage, baseline proteinuria, blood pressure control, underlying diagnosis, diabetes, and previous pregnancy complications; fertility declines as kidney function decreases.
- The highest risk is seen in patients with advanced CKD, significant proteinuria, uncontrolled hypertension, active glomerular disease, lupus nephritis, or diabetic kidney disease; in active glomerulonephritis or lupus nephritis, pregnancy should preferably be postponed until remission.
- Standard eGFR equations are unreliable in pregnancy, so serum creatinine and its trajectory, proteinuria, and blood pressure are assessed against the prepregnancy baseline; even a mildly elevated creatinine level may be significant.
- In the first trimester, causes of AKI include hyperemesis gravidarum, dehydration, and sepsis, and later preeclampsia, HELLP syndrome, thrombotic microangiopathies, hemorrhage, and obstruction; in CKD, preeclampsia is suggested mainly by a change from baseline, and angiogenic biomarkers may help.
- Antihypertensive drugs, RAAS blockers, some immunosuppressants, and statins should be reviewed before conception; an untreated relapse or severe hypertension may be more dangerous than appropriately chosen treatment.
- Pregnancy in kidney failure requires more frequent or longer dialysis and management in an experienced center. After delivery, blood pressure, creatinine, proteinuria, and treatment safety during breastfeeding should be checked, and the nephrologist should actively ask about reproductive plans.

4.14 Dyslipidemia in chronic kidney disease: what in the 2026 ACC/AHA guideline changes nephrology practice

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. In Nephrology Dialysis Transplantation, Van Craenenbroeck and coauthors comment on the 2026 ACC/AHA dyslipidemia guideline from a nephrologist's perspective. The guideline reintroduces explicit LDL-C and non-HDL-C targets, introduces the PREVENT model, which incorporates eGFR, and formulates three recommendations for CKD stage G3 or higher. In clinical ASCVD, the targets are LDL-C below 1.4 mmol/L, non-HDL-C below 2.2 mmol/L, and a reduction of at least 50%. Routine new initiation of a statin in patients on maintenance hemodialysis is not supported. In Slovak practice, European and national guidelines remain decisive.

Summary

- At ages 40 to 75 years with CKD stage G3 or higher and LDL-C of 1.8 to 4.9 mmol/L, a moderate-intensity statin or its combination with ezetimibe is recommended (I, B-R); a low calculator result is not a reason for delay, but the recommendation should not be mechanically extrapolated outside this age and LDL-C range.
- In CKD stage G3 or higher with clinical ASCVD, a high-intensity statin is indicated, with the possible addition of ezetimibe or a PCSK9 antibody, targeting LDL-C below 1.4 mmol/L, non-HDL-C below 2.2 mmol/L, and a reduction of at least 50% (I, B-R).
- PREVENT mandatorily includes eGFR and optionally UACR, but it was validated in the US population and does not replace SCORE2 or SCORE2-OP in European practice.
- Ezetimibe in CKD is supported by the SHARP trial, data on PCSK9 antibodies at an eGFR below 20 mL/min/1.73 m² are limited, and outcome trials of inclisiran are still ongoing; in severe CKD, the rosuvastatin dose may need to be limited.
- Neither the 4D and AURORA trials nor the dialysis subgroup of SHARP demonstrated a benefit of new statin initiation after the start of dialysis, but established therapy may be continued individually in patients on hemodialysis (IIb, C-LD). After transplantation, KDIGO suggests a statin with screening for interactions, particularly with cyclosporine.
- Lp(a) above 125 nmol/L corresponds to an approximately 1.4-fold and above 250 nmol/L to an at least twofold risk of ASCVD; the coronary artery calcium score should not override a clear indication in CKD, and blood pressure, diabetes, and albuminuria should be treated concurrently.

4.15 Kidney risk after preeclampsia: the hidden burden and missed opportunities for early detection

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. A Stockholm population-based cohort study in JASN (171,693 pregnancies in primiparous women from 2006 to 2020, preeclampsia in 6%) showed a more than twofold risk of laboratory signs of CKD after preeclampsia, with an HR of 2.43 for the composite of albuminuria above 300 mg/g and eGFR below 60 mL/min/1.73 m². The absolute risk was low (0.6%), but subsequent correspondence showed that it is underestimated owing to the low testing rate and the strict albuminuria threshold; at a threshold of 30 mg/g, the number of events rose approximately 4.5-fold. Postpartum follow-up should therefore be based on UACR.

Summary

- Over a median of 7 years, the incidence per 1,000 person-years in women with versus without preeclampsia was 1.53 versus 0.57 for albuminuria above 300 mg/g, 0.52 versus 0.18 for eGFR below 60 mL/min/1.73 m², and 2.00 versus 0.73 for the composite; the HRs were 2.53, 2.18, and 2.43.
- In absolute terms, albuminuria occurred in 0.5%, eGFR decline in 0.1%, and the composite in 0.6% of pregnancies, but in the first year after delivery only 20% of women had creatinine measured, 10% had albuminuria assessed, and 5% had both tests.

- The 300 mg/g threshold (KDIGO A3) misses category A2; at a threshold above 30 mg/g, the number of events rose from 113 to 428 after preeclampsia and from 662 to 3,025 without it.
- Albuminuria precedes eGFR decline because glomerular endotheliosis and nearly irreversible podocyte loss damage the filtration barrier before nephron mass is lost; creatinine-based follow-up detects fewer cases and later.
- Detection bias underestimates absolute risk in both groups, but more intensive follow-up after preeclampsia may overestimate the risk ratio; further limitations include the observational design, inclusion of primiparous women only, short follow-up, and a single Swedish region.
- The author recommends measuring UACR, not creatinine alone, assessing it from the 30 mg/g threshold, and recording preeclampsia as a permanent risk factor; the 2021 ISSHP recommendations envisage a checkup approximately three months after delivery.

4.16 Hypertension in pregnancy and after delivery: why the postpartum visit must not be the last

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. A review by Line Malha and Phyllis August in JASN emphasizes that hypertensive disorders of pregnancy, which affect approximately 9% of pregnancies in the United States, do not end with delivery: cardio-renal damage does not fully resolve, and blood pressure peaks on the third to sixth day after delivery. The CHAP trial showed that treating mild chronic hypertension to a target below 140/90 mm Hg reduced the composite adverse outcome from 37.0% to 30.2% without an increase in small-for-gestational-age newborns. The postpartum visit should mark the beginning of long-term nephrology follow-up.

Summary

- Preeclampsia is initiated by placental sFlt-1, which binds vascular endothelial growth factor and placental growth factor and induces widespread endothelial dysfunction; delivery treats the acute phase, not its consequences. CKD is among the strongest risk factors for preeclampsia.
- CHAP randomized 2,408 women with mild chronic hypertension before week 23; active treatment reduced the composite endpoint from 37.0% to 30.2% (adjusted relative risk, 0.82; 95% CI, 0.74 to 0.92).
- In pregnancy, labetalol and extended-release nifedipine are first-line agents, with methyldopa as an alternative; ACE inhibitors, ARBs, and direct renin inhibitors are contraindicated, and mineralocorticoid receptor antagonists and SGLT2 inhibitors should also be avoided, starting before conception.
- Blood pressure of 160/110 mm Hg or higher requires treatment within 30 to 60 minutes; magnesium sulfate accumulates when filtration is reduced, so the dose should be reduced and reflexes, urine output, and serum magnesium monitored.
- In CKD, low-dose aspirin prophylaxis started between weeks 12 and 16 is indicated. During breastfeeding, labetalol, nifedipine, and the ACE inhibitors enalapril and captopril are suitable, whereas atenolol is less suitable.
- The nephrologist should ensure a blood pressure check between the third and sixth day after delivery, assess kidney function and albuminuria, and record preeclampsia as a permanent risk factor. This is a review, and US data cannot be automatically transferred to European settings.

4.17 Dyslipidemia in 2026: cardiovascular risk assessment, LDL cholesterol targets, and treatment in chronic kidney disease

Published August 15, 2026 · [online version \(Slovak\)](#)

Abstract. The October 2025 update of the ESC and EAS guidelines retains the 2019 LDL cholesterol targets, refines risk assessment with age-specific SCORE2 thresholds, introduces early combination therapy after acute coronary syndrome, and expands drug options. The review recalls that lowering LDL cholesterol by approximately 1 mmol/L reduces major vascular events by about one fifth in relative terms. In chronic kidney disease, lipid-lowering therapy is important before dialysis, but new statin initiation during mainten-

ance dialysis has no convincing benefit. Decision-making should distinguish nondialysis CKD, dialysis, and transplantation.

Summary

- The ESC/EAS LDL targets remain unchanged: below 1.4 mmol/L for very high risk and below 1.8 mmol/L for high risk, both with a reduction of at least 50%, below 2.6 mmol/L for moderate risk, and below 3.0 mmol/L for low risk; SCORE2 has age-specific thresholds and is not used in CKD.
- Lp(a) should be measured at least once in a lifetime, with a risk threshold above 50 mg/dL (approximately 105 to 125 nmol/L); its targeted lowering does not yet have outcome evidence, as Lp(a)HORIZON with pelacarsen had not published results by mid-August 2026.
- After acute coronary syndrome, combination of a statin with ezetimibe should be considered during hospitalization, with lipids checked after 4 to 6 weeks. In CLEAR Outcomes, bempedoic acid reduced the composite endpoint by 13% in relative terms without an effect on mortality, and for inclisiran the results of ORION-4 and VICTORION-2P are still lacking.
- KDIGO recommends a statin or its combination with ezetimibe for most adults aged 50 years or older with an eGFR below 60 mL/min/1.73 m² who are not on dialysis and have not undergone transplantation; because its 2013 recommendation has not been updated, the author advises combining it with the ESC/EAS target-based approach. A normal LDL level in CKD does not rule out an indication.
- The 4D and AURORA trials did not demonstrate a benefit of statin initiation in patients on dialysis, and in SHARP the evidence in the dialysis subgroup was less clear; therapy may be continued when dialysis is started, but routine new initiation is not recommended.
- Atorvastatin requires no adjustment for eGFR, the rosuvastatin dose should be limited in severe kidney impairment, and gemfibrozil should not be combined with a statin; after transplantation, a statin is generally recommended with screening for interactions.

4.18 Visit-to-visit systolic blood pressure variability and faster progression of white matter lesions: implications for nephrology practice

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. White matter lesions are a marker of cerebral small vessel disease, and whether blood pressure stability matters in addition to its mean level has remained unanswered. A post hoc pooled analysis of individual data from 952 participants in the SPRINT MIND and ACCORD MIND substudies showed that higher visit-to-visit systolic blood pressure variability is independently associated with faster lesion progression. Intensive blood pressure control slowed progression, but variability mediated only 9.15% of this effect. For nephrology practice, this supports an emphasis on stable blood pressure rather than on variability as a separate therapeutic target.

Summary

- A total of 952 participants with a mean age of 64.8 years and a median of 12 blood pressure measurements were analyzed; variability was calculated from month 3 onward using the VIM metric, which is independent of the mean blood pressure level.
- The median volume of abnormal white matter increased from 1.69 mL to 2.58 mL, that is, by 0.43 mL per year; the highest variability tertile had progression faster by 0.160 mL per year than the lowest, corresponding to roughly one third of the typical annual increase.
- Higher variability was associated with progression ($\beta = 0.017$; 95% CI, 0.001 to 0.033), intensive blood pressure control slowed progression ($\beta = -0.270$; 95% CI, -0.393 to -0.146), and the mediating effect of variability was 0.014, that is, 9.15% of the total effect.
- Lower confidence interval limits just above zero indicate statistically significant but fragile findings; this is a post hoc analysis in which treatment intensity, not variability, was randomized, and the outcome was an imaging surrogate rather than stroke or dementia.

- The cohort was selected for high cardiovascular risk, not for eGFR, so generalizability to CKD is uncertain, and residual confounding or reverse causation cannot be ruled out.
- In CKD, the author recommends gradual titration of antihypertensive drugs, verification of adherence, home or ambulatory monitoring, management of volume status and orthostatic hypotension, and, in patients on dialysis, particular attention to intradialytic and interdialytic blood pressure fluctuations.

4.19 Lipoprotein(a): a cardiovascular risk factor we often miss in primary care

Published August 22, 2026 · [online version \(Slovak\)](#)

Abstract. Lipoprotein(a) is a predominantly genetically determined causal risk factor for ASCVD and calcific aortic stenosis that a standard lipid profile does not detect. A review based on Medscape and the EAS, ESC/EAS 2025, ACC/AHA 2026, and NLA recommendations supports a one-time measurement of Lp(a) in every adult. According to ACC/AHA 2026, a level ≥ 50 mg/dL or ≥ 125 nmol/L is associated with an approximately 1.4-fold risk of ASCVD. The main practical consequence is more rigorous control of LDL cholesterol and other risk factors; in CKD, acquired elevation should also be considered.

Summary

- Lp(a) is measured at least once in adulthood, does not require fasting, and, when markedly elevated, cascade testing of first-degree relatives is worthwhile; LPA genotyping is not routinely required.
- Units of mg/dL and nmol/L must not be converted with a fixed factor; pragmatic ranges are low below 30 mg/dL or 75 nmol/L, a gray zone of 30 to 50 mg/dL or 75 to 125 nmol/L, and high from 50 mg/dL or 125 nmol/L, whereas ESC/EAS 2025 uses a threshold of ≥ 105 nmol/L.
- According to ACC/AHA 2026, a level ≥ 250 nmol/L or ≥ 100 mg/dL is associated with an approximately twofold risk of ASCVD; risk increases continuously, and its absolute significance depends on the overall risk profile.
- Statins generally do not lower Lp(a), PCSK9 antibodies lower it by approximately 20 to 30%, and pelacarsen and olpasiran lowered it by 70 to 98% in earlier studies, but the outcome trials Lp(a)HORIZON and OCEAN(a)-Outcomes had no results as of August 22, 2026.
- High Lp(a) mainly changes the intensity of prevention: reaching the LDL-C target, adding ezetimibe or a PCSK9 inhibitor early, and treating blood pressure, glycemia, and smoking; it is not an indication for routine aspirin or screening echocardiography.
- In CKD, Lp(a) may rise as kidney function declines and in nephrotic syndrome and peritoneal dialysis with protein losses; the elevation may resolve after remission or transplantation, so the author recommends considering a recheck after remission and not regarding Lp(a) alone as a reason to start a statin or apheresis in patients on dialysis.

4.20 Uncontrolled and resistant hypertension: diagnosis, the aldosterone axis, and new treatment options

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. Uncontrolled, apparent resistant, and true resistant hypertension are not synonyms, and distinguishing them determines both workup and treatment. The review bases its approach on the ESC 2024, AHA/ACC 2025, and Endocrine Society 2025 guidelines and summarizes trials of aldosterone synthase inhibitors. In PATHWAY-2, spironolactone lowered home systolic blood pressure by 8.70 mm Hg compared with placebo, and in BaxHTN, baxdrostat lowered it by 8.7 to 9.8 mm Hg. Before using the new class, pseudo-resistance should be excluded, primary aldosteronism should be actively sought, and in CKD, potassium should be closely monitored.

Summary

- According to ESC 2024, resistant hypertension is defined as blood pressure of at least 140/90 mm Hg despite maximal doses of at least three drugs, including a renin-angiotensin system blocker, a calcium

channel blocker, and a diuretic, confirmed by out-of-office measurement; at an eGFR below 30 mL/min/1.73 m², a titrated loop diuretic should be used.

- Apparent resistance should be excluded by verifying measurement technique, home or 24-hour monitoring, nonjudgmental assessment of adherence, and a search for substances that raise blood pressure, such as nonsteroidal anti-inflammatory drugs, calcineurin inhibitors, or erythropoiesis-stimulating agents.
- Endocrine Society 2025 conditionally suggests screening for primary aldosteronism with the ARR in all patients with hypertension; the ARR is a screening tool, not a diagnosis, and adrenalectomy is the standard for the lateralized form.
- According to ESC, spironolactone is considered the preferred add-on drug at an eGFR of at least 30 mL/min/1.73 m² and a potassium level of at most 4.5 mmol/L, with frequent monitoring.
- Lorundrostat 50 mg lowered systolic blood pressure by 9.1 mm Hg; in FigHTN in CKD, baxdrostat lowered blood pressure by 8.1 mm Hg, but hyperkalemia occurred in 41% versus 5% with placebo, and vicadrostat reduced albuminuria by approximately 22 to 39%.
- The FDA approved baxdrostat (BAXFENDY) on May 15, 2026; an effect on cardiovascular events or kidney outcomes has not been demonstrated, data are lacking at an eGFR below 45 mL/min/1.73 m², and US approval does not mean availability in Slovakia.

4.21 Baxdrostat markedly lowers 24-hour blood pressure in resistant hypertension: results of the Bax24 trial

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. Inappropriate aldosterone production contributes to resistant hypertension even in the absence of classic primary aldosteronism. The randomized, double-blind, phase III Bax24 trial evaluated the selective aldosterone synthase inhibitor baxdrostat at 2 mg daily for 12 weeks in 217 patients taking at least three antihypertensive drugs, including a diuretic. Mean 24-hour ambulatory systolic blood pressure fell by 14.0 mm Hg more than with placebo. Potassium above 6.0 mmol/L occurred in 3% of treated patients. A comparison with spironolactone and data on clinical events are lacking.

Summary

- The trial was conducted at 79 centers in 22 countries; after a two-week placebo run-in, patients with a 24-hour systolic blood pressure of at least 130 mm Hg were randomized, and 199 patients were excluded during this run-in phase.
- Blood pressure fell by 16.6 mm Hg with baxdrostat and by 2.6 mm Hg with placebo, a difference of 14.0 mm Hg (95% CI, -17.2 to -10.8; $p < 0.0001$); according to the authors, the effect was consistent across subgroups.
- An adverse event occurred in 52% versus 37% of patients, and confirmed potassium above 6.0 mmol/L in 3 patients receiving baxdrostat versus none receiving placebo; in practice, the risk may be higher because the trial required preserved kidney function and normal serum potassium.
- The primary outcome was available in only 184 of 217 patients (approximately 85%) without imputation of missing data, follow-up lasted 12 weeks, and the trial was funded by AstraZeneca.
- The trial did not demonstrate a reduction in mortality or cardiovascular events, kidney protection, safety in advanced CKD, or superiority over spironolactone, which remains the drug with an extensive evidence base.
- Before targeted treatment, the author recommends confirming hypertension by ambulatory or home measurement, objectively assessing adherence, optimizing diuretics and sodium intake, screening for primary aldosteronism and sleep apnea, and considering the risk of hyperkalemia.

4.22 A systolic blood pressure target below 120 mm Hg in chronic kidney disease has seen only limited uptake in practice

Published September 11, 2026 · [online version \(Slovak\)](#)

Abstract. KDIGO 2021 suggests a systolic blood pressure below 120 mm Hg, measured in a standardized manner, in CKD not on dialysis (grade 2B). An observational analysis of electronic records from the Mass General Brigham network found that in 2024 only 21.9% of patients with CKD stage G3 to G4 had a routine mean blood pressure below 120 mm Hg, compared with 18.3% in 2020. A blood pressure of 120 mm Hg or higher was associated with a higher risk of cardiovascular events (HR 1.19) and kidney failure (HR 1.21). However, nonstandardized measurement does not allow assessment of true adherence to the guideline.

Summary

- Annual cross-sections included 45,651 to 52,855 patients aged 18 to 79 years; the increase of 3.6 percentage points compared with 2020 became significant only from 2023, and the mean number of antihypertensive drug classes changed only from 1.53 to 1.55.
- At a threshold of 130 mm Hg, the proportion of controlled patients rose from 44.7% to 50.4%, so inadequate control cannot be explained by the strict KDIGO target alone.
- In a longitudinal cohort of 18,996 patients with a median follow-up of 6.9 years, a blood pressure of 120 mm Hg or higher was associated with an HR of 1.19 (95% CI, 1.11 to 1.27) for cardiovascular events and an HR of 1.21 (95% CI, 1.08 to 1.35) for kidney failure, with a lower incidence of hypotension (HR 0.80).
- These are observational data with routine, nonstandardized measurement, a cohort enrolled before 2021, missing albuminuria, and exclusion of persons aged 80 years or older, so they do not prove that lowering blood pressure below 120 mm Hg causally reduces risk.
- In the SPRINT CKD subgroup (2,646 persons), the composite cardiovascular outcome did not reach significance (HR 0.81; 95% CI, 0.63 to 1.05), mortality decreased (HR 0.72), and kidney protection was not demonstrated; ESC 2024 recommends 120 to 129 mm Hg and ACC/AHA 2017 less than 130 mm Hg.
- Before intensification, the author recommends verifying standardized measurement, using home or 24-hour monitoring, assessing the orthostatic response and albuminuria, optimizing renin-angiotensin system blockade and SGLT2 inhibitor therapy, and individualizing the target according to frailty and tolerability.

4.23 Diurnal sodium excretion and nocturnal hypertension: salt timing as a clinical problem

Published September 14, 2026 · [online version \(Slovak\)](#)

Abstract. Loss of the nocturnal blood pressure dip and nocturnal hypertension increase both cardiovascular and kidney risk, and renal sodium excretion has its own diurnal rhythm with daytime predominance. A review in Nephron (2026) and supporting cohort studies show that blunted daytime natriuresis is associated with higher nocturnal blood pressure and impaired dipping; in CKD, the differences between quartiles were on the order of a few mm Hg. This is an association, not proven causality. In practice, it strengthens the role of ABPM and, in selected patients, timed urine collection for sodium.

Summary

- According to ESH 2023, nocturnal hypertension is defined as a mean nighttime blood pressure $\geq 120/70$ mm Hg, normal dipping is a decline of approximately 10 to 20%, and non-dipping is a decline below 10%; the ABPM report should state the thresholds and time windows used.
- With blunted daytime natriuresis, sodium excretion shifts to the night, and higher nocturnal blood pressure may be part of compensatory pressure natriuresis; cohorts (Bankir 2008, Del Giorgio 2020, Liu 2024, Zhang 2020, Liu 2026) link a low day-to-night sodium excretion ratio with nocturnal blood pressure, target organ damage, and CKD progression.

- Twenty-four-hour natriuresis alone does not reveal the phenotype, and a disrupted rhythm may be both a mechanism and a marker of eGFR decline, autonomic dysfunction, or sleep apnea; randomized data on chronotherapeutic interventions are lacking.
- Reliable assessment requires ABPM with a sleep diary, high-quality split urine collection in the same time windows, and consideration of diuretics, SGLT2 inhibitors, nocturia, and sleep, so it is appropriate on a targeted rather than universal basis.
- With high nocturnal blood pressure and a low sodium ratio, the author recommends controlling salt intake, individualized timing of salt and antihypertensive doses, and a search for OSA, volume overload, and contributing drugs; he does not recommend universal evening dosing of all medications.

4.24 Peripheral artery disease in CKD: why an ABI threshold below 0.9 is not enough

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. Peripheral artery disease is common in CKD and often asymptomatic for a long time, and medial calcification makes the ankle-brachial index unreliable. The article combines a systematic review in The Lancet Global Health (global prevalence of 6.58%), the CRIC cohort of 3,627 patients with CKD, and a meta-analysis of 13,378 patients. In CRIC, a conventionally normal ABI of 0.9 to 1.0 carried a nearly threefold risk of PAD (HR 2.76), and the risk also rose at an ABI ≥ 1.4 . In CKD, the ABI should be read as a continuous variable, and a normal result should not be regarded as excluding disease.

Summary

- In 2023, PAD affected 316.54 million persons aged 25 years or older, and prevalence rose from 4.31% at ages 25 to 29 years to 22.09% at ages 90 to 99 years.
- In CRIC, the lowest risk was seen at an ABI of 1.0 to less than 1.4; at an ABI below 0.9, the HR for PAD was 5.78, at an ABI of 0.9 to 1.0 the HR was 2.76 for PAD and 1.85 for myocardial infarction, that is, higher than at an ABI below 0.9, and at an ABI ≥ 1.4 the HR for PAD was 4.85.
- In a meta-analysis (Crit Pathw Cardiol 2025), a high ABI ≥ 1.3 was associated with cardiovascular mortality with an HR of 2.77 and a low ABI with an HR of 2.23, with higher risk in patients on hemodialysis.
- There is no randomized evidence that ABI screening in CKD improves outcomes, and the thresholds below 1.0 and ≥ 1.4 are not formally adopted; these are prognostic associations.
- The author recommends targeted screening in diabetes, smoking, cardiovascular disease, claudication, a foot wound, frailty, and before transplantation or vascular access creation; at an ABI ≥ 1.4 , the toe-brachial index with a threshold below 0.70, Doppler waveform analysis, and duplex ultrasonography should be added.
- An abnormal ABI is in itself an indication for intensive cardiovascular prevention, including smoking cessation and walking exercise training; in CKD, contrast studies should be weighed with the vascular team, and a nonhealing wound or rest pain requires urgent consultation.

CHAPTER 5

Glomerular and immune-mediated kidney diseases

IgA nephropathy, membranous nephropathy, lupus nephritis, ANCA-associated vasculitis, and complement-mediated diseases. New targeted therapies and their place according to current guidelines. Number of articles: 31.

Key messages of the chapter

The treatment of glomerular diseases is moving from nonselective corticosteroid therapy to drugs that target specific steps of pathogenesis (5.1, 5.5). In IgA nephropathy, the framework is the four-hit model: production of Gd-IgA1, autoantibodies, immune complexes, and their mesangial deposition with inflammation, followed by targeting of the APRIL and BAFF axis, complement, mucosal IgA production, and the endothelin pathway (5.3, 5.28). The acceptance of proteinuria as a surrogate endpoint has enabled accelerated approval (5.5). KDIGO 2025 sets a proteinuria target below 0.5 g/day, ideally below 0.3 g/day, and an eGFR decline of less than approximately 1 mL/min per year (5.4, 5.9). A post hoc analysis of the PROTECT trial showed an eGFR decline of 0.7 versus 4.2 mL/min/1.73 m² per year with complete remission, but this is a prognostic association, and proteinuria later rose again in 35% of patients (5.21).

The evidence for new drugs is so far predominantly surrogate. The strongest data on kidney function came from the final analysis of the APPLAUSE-IgAN RCT: iptacopan slowed the annualized eGFR decline to 3.10 versus 6.12 mL/min/1.73 m² and reduced the composite kidney endpoint (HR 0.57), at the cost of serious infections in 6.7% versus 2.1% of patients (5.22, 5.25). Atacicept received FDA accelerated approval based on a 42% reduction in UPCR versus placebo (5.17); the final ORIGIN 3 analysis with an HR of 0.24 for progression is so far a company announcement, not peer reviewed, with few measurements at week 104 (5.29). In the TELIGAN interim analysis, telitacicept reduced UPCR by 58.9% versus 8.8% (5.14), and sibeprenlimab by 51.2% (5.28). In PROTECT, sparsentan reduced proteinuria by approximately 40% and modestly slowed the chronic eGFR decline, but the composite kidney endpoint did not differ (5.26). A network meta-analysis did not demonstrate the superiority of any regimen (5.2), and KDIGO explicitly warns against ranking drugs by numbers from different trials (5.28). KDIGO 2025 prefers a 9-month course of Nefecon, but KDOQI points to a possible loss of response after its completion (5.9). Data on combining sparsentan with an SGLT2 inhibitor are short term and surrogate (5.24), and the Spanish cohort is uncontrolled (5.26).

Beyond IgA nephropathy, precision therapy is advancing similarly. In FSGS, the FDA approved sparsentan based on the DUPLEX trial, although the difference in eGFR slope was not significant; genetic testing is indicated, since up to 15% of adults carry a causal mutation (5.7). In a phase 2 RCT, a TRPC6 inhibitor achieved a response in 35% versus 7% with placebo in a small sample (5.13). In C3 glomerulopathy, phase 3 RCTs of iptacopan and pegcetacoplan showed a consistent decrease in proteinuria, while long-term kidney outcomes are uncertain (5.12). The KDIGO Controversies Conference confirmed anti-CD20 therapy as first-line treatment in membranous nephropathy and the limited benefit of rituximab in IgA nephropathy, and it emphasized that the right depth of depletion, not the maximal intensity of intervention, is what matters (5.16, 5.27). Retrospective signals, such as persistent hematuria in podocytopathies (HR 3.55) and anti-PLA2R titer for thrombosis risk, are hypotheses to be verified, not decision parameters (5.18, 5.19). Even the baseline eGFR at biopsy can bias observational studies (5.20).

In systemic diseases, the ACR 2025, EULAR 2025, and KDIGO 2024 guidelines for lupus nephritis agree on hydroxychloroquine, rapid glucocorticoid tapering, and a proteinuria target below 0.5 g/day, but differ in their preference for triple therapy (5.8). Treatment should not be delayed because of the biopsy, which should be performed as soon as possible; the benefit of early response comes from observational cohorts (5.30). In ANCA-associated vasculitis, the EMA recommended revoking the marketing authorization of avacopan because of data integrity problems in the ADVOCATE trial (5.15), and the other recommendations rest on unchallenged RCTs: the reduced glucocorticoid regimen according to PEXIVAS, plasma exchange without proven benefit, and rituximab maintenance (5.31). In connective tissue diseases, an organ-oriented risk assessment, not the disease label itself, is decisive (5.10, 5.11).

This yields a stepwise framework for the outpatient clinic. The diagnosis should rest on a biopsy with MEST-C scoring and risk stratification (5.4, 5.9). Kidney protection remains the foundation: RAS blockade, SGLT2 inhibitors, sodium below 2 g/day, and BP up to 120/70 mm Hg (5.9). Proteinuria should be measured repeatedly, and a value sustained over time should be documented before changing therapy (5.21); urinary sediment should be verified by microscopy (5.18). A targeted drug is added on an individual basis, with vaccination before complement inhibition or B cell-targeted therapy (5.22, 5.25, 5.17), and bearing in mind that atacicept is not authorized in the EU (5.17). Patients also need guidance on information from social media, where the popularity of a video does not guarantee its quality (5.6).

5.1 A golden age for IgA nephropathy and rare glomerulopathies

Published April 25, 2026 · [online version \(Slovak\)](#)

Abstract. Treatment of IgA nephropathy and rare glomerulopathies long relied on nonselective immunosuppression with corticosteroids, which carries marked adverse effects. A brief commentary describes the shift toward precision medicine with drugs that act directly on pathogenesis. It is an opinion piece without original data. According to the author, the dual endothelin and angiotensin receptor antagonist sparsentan, the complement inhibitor iptacopan, and blockade of the APRIL/BAFF pathways with sibeprenlimab are gradually transforming IgAN into a condition with a more manageable long-term prognosis.

Summary

- Drugs targeting pathogenesis are replacing nonselective corticosteroid therapy.
- Examples include sparsentan, iptacopan, and sibeprenlimab.
- The author anticipates a better long-term prognosis for IgAN.

5.2 Comparative effectiveness of treatments for IgA nephropathy: a network meta-analysis

Published May 8, 2026 · [online version \(Slovak\)](#)

Abstract. In IgA nephropathy, direct comparisons of conventional immunosuppression with new targeted drugs are lacking. A Bayesian network meta-analysis of 17 randomized trials published in April 2026 compared methylprednisolone, mycophenolate mofetil, tacrolimus, Nefecon, iptacopan, and sibeprenlimab with placebo or supportive care. Nefecon had the most favorable point estimate for eGFR slope, but the intervals included no effect; methylprednisolone reduced UPCR the most. The authors emphasize the exploratory nature of the findings, which can support individualized decision-making but cannot prove superiority.

Summary

- The analysis included trials up to March 2025 and assessed eGFR slope, UPCR, and serious adverse events; Nefecon had the highest SUCRA ranking for eGFR slope, but without demonstrated superiority.

- Methylprednisolone reduced proteinuria the most, and iptacopan and sibeprenlimab also produced reductions; iptacopan had numerically fewer serious adverse events, but the signal is uncertain, and the effect did not depend on whether eGFR was below or above 60 mL/min/1.73 m².
- The comparison network was sparse, regimens were heterogeneous, and follow-up was short, so head-to-head trials with hard kidney endpoints are needed.

5.3 IgA nephropathy: the role of APRIL in the four-hit model of pathogenesis

Published May 11, 2026 · [online version \(Slovak\)](#)

Abstract. IgA nephropathy is now understood as a complex immunologic process rather than passive deposition of IgA in the mesangium. A review based on the ReachMD educational program explains the four-hit model of pathogenesis: production of Gd-IgA1, autoantibody formation, immune complex formation, and their mesangial deposition with inflammation. The cytokine APRIL promotes B-cell survival and Gd-IgA1 production and therefore represents a therapeutic target at the beginning of the cascade. In practice, management rests on three pillars: diagnosis including biopsy, risk stratification, and individualized treatment on top of a kidney-protective foundation.

Summary

- APRIL promotes the differentiation of B cells and plasma cells and the production of pathologic Gd-IgA1 at the beginning of the cascade, so targeting it may intervene higher up in the pathogenic chain.
- The course is heterogeneous, ranging from incidental microscopic hematuria to proteinuria, hypertension, and reduced eGFR, so stratification by proteinuria, blood pressure, eGFR trajectory, and histology is decisive.
- Blood pressure control, RAAS blockade, and proteinuria reduction remain the foundation, now supplemented by targeted strategies directed at the APRIL/BAFF axis, complement, mucosal IgA production, and the endothelin pathway.

5.4 IgA nephropathy in the era of new therapeutic options: from supportive care to targeted therapy

Published May 11, 2026 · [online version \(Slovak\)](#)

Abstract. Treatment of IgA nephropathy is shifting from general kidney protection to targeted therapy that acts on the mechanisms of the four-hit model. This expert summary draws on the KDIGO guideline and the PROTECT, ALIGN, NefIgArd, APPLAUSE-IgAN, DAPA-CKD, and EMPA-KIDNEY trials. KDIGO sets a goal of proteinuria below 0.5 g/day at a minimum, ideally below 0.3 g/day. Supportive care, including SGLT2 inhibitors, remains the foundation, new drugs complement it, and the choice should be individualized according to the risk of progression, histology, and comorbidities.

Summary

- Proteinuria is the main therapeutic target, a marker of activity, and a predictor of kidney function decline; in patients at risk of progression, the goal is less than 0.5 g/day, ideally less than 0.3 g/day.
- The foundation is early kidney protection with ACE inhibitors or ARBs, blood pressure control, and SGLT2 inhibitors, whose effect in proteinuric phenotypes, including IgAN, is supported by DAPA-CKD and EMPA-KIDNEY.
- In the PROTECT trial, sparsentan reduced proteinuria more than irbesartan, and in ALIGN, atrasentan reduced it compared with placebo; fluid retention should be monitored with atrasentan.
- Targeted-release budesonide acts on Peyer's patches in the distal ileum and in NefIgArd reduced proteinuria with a more favorable eGFR trajectory; in APPLAUSE-IgAN, iptacopan significantly reduced proteinuria and requires prevention of infections with encapsulated bacteria, including vaccination.
- According to available data, the anti-APRIL antibody sibeprenlimab significantly reduces proteinuria, and further molecules targeting the APRIL/BAFF axes, atacicept and povetacicept, are in development.

- Diagnosis relies on biopsy and MEST-C scoring, and treatment should be chosen according to proteinuria, eGFR trajectory, blood pressure, age, comorbidities, pregnancy planning, availability, and patient preferences.

5.5 IgA nephropathy: new insights bring a wave of targeted therapies

Published May 13, 2026 · [online version \(Slovak\)](#)

Abstract. IgA nephropathy is the most common primary glomerulonephritis and an important cause of kidney failure. A review describes the four-step pathogenesis and the arrival of targeted drugs, made possible by the acceptance of proteinuria as a surrogate endpoint for accelerated approval by the FDA and EMA. Sparsentan, atrasentan, targeted-release budesonide, iptacopan in APPLAUSE-IgAN, and sibeprenlimab in ENVISION significantly reduce proteinuria, although the text does not give figures. In practice, early identification of high-risk patients by biopsy and the International IgAN Prediction Tool score is essential.

Summary

- Pathogenesis involves the production of Gd-IgA1, antibodies mainly of the IgG class, circulating immune complexes, and their mesangial deposition with inflammation and fibrosis.
- Proteinuria is an accepted surrogate endpoint; a drug that significantly reduces it within a year will probably slow kidney function decline in the long term.
- In APPLAUSE-IgAN, iptacopan reduced proteinuria, and the data suggest eGFR stabilization; in ENVISION, sibeprenlimab produced a marked and sustained reduction in both proteinuria and Gd-IgA1.
- The author anticipates that the new standard will be a combination of drugs with different mechanisms according to the dominant pathway, which has not yet been validated.

5.6 TikTok and IgA nephropathy: short videos can help, but the quality of information is uneven

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. Patients with IgA nephropathy increasingly obtain information from short videos on social media. A cross-sectional study evaluated 186 videos from the TikTok and Bilibili platforms using the GQS, mDISCERN, JAMA, and VIQI tools. Overall, the videos provided only moderate-quality information, with better results for videos by health professionals and in monologue format. Engagement correlated with GQS and VIQI but not with reliability according to mDISCERN and JAMA. For nephrology, this is a call to actively create accurate professional content and to explain to patients that popularity is no guarantee of quality.

Summary

- A total of 186 videos retrieved with the keyword IgA nephropathy were analyzed and rated with the GQS, mDISCERN, JAMA benchmark, and VIQI tools; videos by health professionals had significantly higher reliability and educational quality than videos by ordinary users.
- TikTok had better visual quality according to VIQI and more likes, comments, and shares; engagement correlated moderately to strongly with GQS and VIQI but not with mDISCERN and JAMA, so a popular video is not necessarily medically reliable.
- The videos focused mainly on treatment and prognosis and less on the causes of the disease and the financial burden of long-term care, which may lead to an oversimplified understanding of a complex disease.
- The quality of information depends on the author, the handling of sources, and whether the video clearly distinguishes facts from personal experience; IgAN treatment must remain individualized according to proteinuria, kidney function, and the risk of progression.
- The author recommends greater involvement of health professionals in creating factual and transparent videos and explaining to patients that a video should prompt a conversation with a nephrologist, not serve as a treatment recommendation.

5.7 Overview of the latest research: FSGS and diabetic kidney disease (2025 to 2026)

Published June 6, 2026 · [online version \(Slovak\)](#)

Abstract. The years 2025 to 2026 brought fundamental advances in both FSGS and diabetic kidney disease. The review summarizes RCTs, registry studies, and observational studies: on April 13, 2026, the FDA approved sparsentan as the first drug specifically for FSGS without nephrotic syndrome on the basis of the DUPLEX trial, in which it reduced proteinuria by 46% versus 30% with irbesartan. In the FLOW trial, semaglutide reduced the risk of the primary endpoint by 24%. According to the author, the combination of an SGLT2 inhibitor, a GLP-1 receptor agonist, and finerenone is becoming the standard in DKD.

Summary

- In DUPLEX (n = 371), the difference in eGFR slope in the overall population was not significant; KDIGO recognizes 4 subtypes of FSGS, immunosuppression achieves remission in fewer than 25% of patients, and genetic testing is indicated regardless of age because up to 15% of adults carry a causal mutation.
- Anti-nephrin autoantibodies are found in approximately 10% of patients with FSGS, positive patients respond well to rituximab, and titers correlate with response; inaxaplin and baricitinib for APOL1, BI 764198, and other molecules are in development, mostly in phase 2.
- In FLOW, semaglutide had a consistent benefit across CKD stages, and the FDA approved it for CKD in type 2 diabetes in January 2025; SELECT showed a 22% reduction in serious kidney events in obesity or overweight.
- In Danish registry data, the 5-year risk of CKD was 6.7% with SGLT2 inhibitors and 8.2% with GLP-1 receptor agonists, with less frequent AKI with SGLT2 inhibitors; the classes are regarded as complementary.
- In an observational target trial emulation (n = 2,268), finerenone compared with spironolactone had an aHR of 0.31 for mortality, 0.74 for MACE, and 0.47 for MAKE, with hyperkalemia in 17.2% versus 26.4%; these are real-world data, not an RCT.
- The author regards sparsentan as the drug of first choice in FSGS without nephrotic syndrome and, in DKD, recommends triple therapy on a background of an ACE inhibitor or ARB, which is supported by the ADA Standards of Care 2025.

5.8 Comparison of the three major lupus nephritis guidelines: ACR 2025, EULAR 2025, and KDIGO 2024

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. The ACR 2025, EULAR 2025, and KDIGO 2024 guidelines for lupus nephritis draw on a shared evidence base but differ in practical decision-making. A comparative analysis shows agreement on hydroxychloroquine, rapid glucocorticoid tapering, and complete renal response defined as proteinuria below 0.5 g/day. ACR prefers initial triple therapy, EULAR lists MMF with obinutuzumab as a first-line option, and KDIGO also accepts dual therapy. These differences explain the ongoing debate over whether triple therapy should be the standard for all or whether treatment should be tailored to risk.

Summary

- All three documents recommend biopsy for proteinuria above 0.5 g/day or declining kidney function; ACR prefers assessment every 6 to 12 months, EULAR and KDIGO individualized monitoring, and KDIGO emphasizes compartment-based histologic assessment.
- In proliferative LN, methylprednisolone pulses of 250 to 1,000 mg daily for at most 3 days are used, with tapering to 5 mg/day or less within 4 to 6 months; rituximab is not first-line therapy in any guideline because of the negative results of the LUNAR trial.
- ACR is the most prescriptive: with proteinuria of 3 g/day or more, it prefers CNI-based regimens, and with extrarenal involvement or reduced kidney function, belimumab, whereas EULAR points out the post hoc origin of these profile-based conclusions.

- KDIGO and ACR expect a complete response within 6 to 12 months, whereas EULAR uses an interim target of proteinuria below 0.7 g/day at month 12.
- Robust data are lacking for class V: EULAR treats it as a proliferative form, ACR recommends triple therapy for proteinuria above 1 g/day, and KDIGO dual therapy; classes I and II remain a gray zone, with the concept of lupus podocytopathy according to KDIGO.
- All three documents consistently recommend kidney protection with a target blood pressure below approximately 120 to 130/80 mm Hg, RAAS blockade, and selective consideration of SGLT2 inhibitors.

5.9 IgA nephropathy according to KDIGO 2025 and the KDOQI US Commentary: from biopsy to more targeted treatment

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. KDIGO 2025 shifts the management of IgA nephropathy toward early identification of high-risk patients and treatment targeting disease mechanisms, and the KDOQI US Commentary addresses its implementation in practice. This is a guideline with a commentary. The goals are an eGFR decline of less than approximately 1 mL/min/year and proteinuria below 0.5 g/day at a minimum, ideally below 0.3 g/day. The preferred specific treatment is a 9-month course of Nefecon or, when it is unavailable, a reduced dose of systemic glucocorticoids with prophylaxis. Diagnosis requires biopsy.

Summary

- Biopsy should be considered in adults with proteinuria ≥ 0.5 g/day if IgAN is a possibility; after confirmation, secondary causes and the MEST-C score according to the Oxford classification are assessed, and no validated biomarkers exist.
- Prediction tools serve for risk stratification and shared decision-making, not for drug selection, and are not recommended in atypical phenotypes such as nephrotic syndrome or rapidly progressive glomerulonephritis.
- Supportive measures include sodium below 2 g/day, cessation of smoking and vaping, blood pressure $\leq 120/70$ mm Hg, RAS blockade, consideration of an SGLT2 inhibitor, and cardiovascular prevention.
- For Nefecon, KDOQI cautions that the response may not persist after discontinuation, data on repeated courses are limited, and serious adverse events were more frequent in the trial; a reverse dip in eGFR may occur at the start.
- An alternative is methylprednisolone 0.4 mg/kg/day, at most 32 mg/day, for 2 months, tapered by 4 mg/day each month, for a total of 6 to 9 months, with *Pneumocystis jirovecii* prophylaxis, antiviral prophylaxis in HBsAg carriers, gastroprotection, and bone protection.
- Increased caution with corticosteroids is needed at an eGFR < 30 mL/min/1.73 m² and in diabetes, obesity, latent infections, peptic ulceration, psychiatric illness, osteoporosis, and cataract.

5.10 Connective tissue diseases in practice: diagnostic and therapeutic challenges in systemic autoimmune diseases

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. Connective tissue diseases are systemic autoimmune diseases with heterogeneous, often overlapping presentations and a risk of rapid organ damage. This clinical review builds on the Kollagenosen 2026 event of the RheumaLive series and materials from the Rheuma-Liga; it presents no original data or figures. Its main message is that no single test establishes the diagnosis; instead, an organ-oriented approach combines clinical findings, urinalysis, muscle enzymes, imaging, and autoantibodies. Treatment should be guided by the severity of organ involvement and should balance efficacy against safety.

Summary

- The author recommends mapping three areas: systemic symptoms with inflammatory markers, organ warning signs from the kidneys, muscles, lungs, skin, and mucous membranes, and the pattern of symptoms by domain, for example the myositis spectrum in muscle weakness.

- ANA are often present in connective tissue diseases, and specific antibodies, such as anti-dsDNA in active SLE or anti-Ro and anti-La in Sjögren's syndrome, help distinguish subgroups; however, positivity without a corresponding phenotype calls for caution.
- Nephritis in SLE can be detected by urinalysis, and ankle swelling and foamy urine are often the first warning signs; in muscle weakness, muscle enzymes, ultrasonography, MRI, EMG, and, if needed, biopsy are added.
- The author describes the therapeutic challenges as his own interpretation, without a record of the lectures: in organ-threatening forms, treatment should be started according to severity, prevention of infections and toxicity monitoring should be planned in advance, and the aim should be to maintain remission.
- This is a group of diseases in which diagnosis is often delayed by seemingly unrelated involvement of the skin, joints, lungs, kidneys, muscles, or nervous system.
- In active organ involvement, multidisciplinary decision-making with a rheumatologist, nephrologist, pulmonologist, or neurologist is appropriate; in unclear cases, the time course, repeated organ assessment, and a clear plan in case of deterioration are decisive.

5.11 Connective tissue diseases: a nephrologist's practical clinical view of diagnosis, organ risk, and treatment decisions

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. In connective tissue diseases, diagnosis is not a single-step process, and organ risk may precede a clear clinical picture. This review by a nephrologist builds on the Kollagenosen 2026 event and materials from the Rheuma-Liga and presents no original data. The essence is assessment of organ risk rather than merely naming the disease: a combination of clinical findings, organ function, targeted laboratory and imaging tests, and autoantibodies interpreted in context. For practice, the author offers a five-step reasoning algorithm with an emphasis on early detection of nephritis and myositis.

Summary

- A connective tissue disease can manifest in various ways and change over time, so the correct diagnosis is not the goal but a means of determining the degree of urgency.
- Diagnosis relies on a targeted history and examination, laboratory tests including function of the affected organs, ultrasonography or radiography, and biopsy if it would change the diagnosis or treatment.
- In SLE, nephritis may be signaled by urinary findings, ankle swelling, and foamy urine due to high protein content; the author stresses that urine findings and edema should not be underestimated because nephritis may determine the prognosis.
- In muscle weakness, polymyositis and dermatomyositis should be considered, muscle enzymes, ultrasonography, MRI, and EMG should be combined, and muscle biopsy may provide definitive proof of inflammation.
- ANA and specific antibodies, for example anti-dsDNA in active SLE or anti-Ro and anti-La in Sjögren's syndrome, are signposts, neither a definitive diagnosis nor an exclusion.
- The author's algorithm: map the phenotype, assess the organs, add targeted laboratory tests and imaging, interpret antibodies in context, and intensify the workup when uncertainty would change management; treatment intensity is determined by active organ threat and the risk of progression.

5.12 Complement 3 glomerulopathy (C3G): standard of care, nonspecific therapy, and targeted complement inhibition

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. In C3 glomerulopathy, randomized trials were lacking, and KDIGO 2021 relies mainly on expert consensus and retrospective cohorts. The review summarizes supportive care, immunosuppression, and targeted complement inhibition, including phase 3 RCTs. In the APPEAR-C3G trial, iptacopan reduced 24-

hour UPCR by 35.1% compared with placebo, and in VALIANT, pegcetacoplan reduced it by 67.3%, a difference of 68.3% compared with placebo. Drugs blocking the alternative pathway produce a more consistent reduction in proteinuria, but long-term kidney outcomes and patient selection remain uncertain.

Summary

- KDIGO 2021 recommends an ACEi or ARB for all patients, statins, and, for proteinuria > 1 g/day with hematuria or declining function for at least 6 months, MMF with glucocorticoids; according to preliminary analyses, SGLT2 inhibitors may reduce proteinuria.
- The efficacy of nonspecific therapy is variable in retrospective cohorts, and post-transplant recurrence persists despite MMF in baseline immunosuppression, which supports the need for targeted treatment.
- Eculizumab is considered in progression without response to other treatment; in pooled data from 122 patients, it was associated with graft failure in 33% versus 42% with plasmapheresis and 81% with rituximab, with large variability. Avacopan did not meet the primary endpoint in the ACCOLADE trial.
- In APPEAR-C3G (74 patients, iptacopan 200 mg twice daily), eGFR changed by +1.3 versus -0.9 mL/min/1.73 m² at 6 months, serum C3 rose by 185%, and the effect on proteinuria appeared within approximately 2 weeks; 1 serious pneumococcal infection was recorded.
- Pegcetacoplan is FDA approved for C3G and IC-MPGN; in VALIANT (96 patients, 26 weeks), eGFR changed by -1.6 versus -7.9 mL/min (nominal P = 0.0322), and 3 serious infections were not attributed to encapsulated bacteria.
- In practice, treatment should start with supportive care, MMF with corticosteroids should be added according to severity, and targeted complement therapy should be considered in progression; criteria for the subgroup with a clear benefit from terminal blockade have not been reliably defined.

5.13 TRPC6 inhibition in focal segmental glomerulosclerosis (FSGS): what a randomized phase 2 trial shows and why it is a significant step toward precision nephrology

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. Excessive activity of the TRPC6 ion channel in podocytes leads, through calcineurin/NFAT signaling, to loss of the filtration barrier and proteinuria. A randomized, placebo-controlled phase 2 trial tested the oral TRPC6 inhibitor BI 764198 at doses of 20, 40, and 80 mg for 12 weeks in 62 patients with biopsy-proven FSGS. A UPCR reduction of at least 25% was achieved by 35% of treated patients versus 7% with placebo and by all 4 treated patients with a pathogenic TRPC6 variant. This is the first RCT in FSGS with a drug targeting a causal gene.

Summary

- Patients with primary FSGS without evidence of a secondary cause or with genetic FSGS due to a TRPC6 variant were enrolled and randomized in a 1:1:1:1 ratio.
- The placebo-adjusted change in UPCR at week 12 was 27% for all treated patients; the 20-mg dose had the most pronounced effect, with a 44% response rate and a UPCR reduction of approximately 40%, which the authors interpret as a possible plateau.
- In the TRPC6 variant subgroup, 4 of 4 treated patients and 0 of 3 receiving placebo responded, but statistical power is limited by the small numbers.
- Adverse events were comparable to placebo, most commonly headache, peripheral edema, and hypertension, and only 3 patients discontinued treatment because of adverse events.
- Limitations include the small sample, follow-up of only 12 weeks without data on long-term kidney survival, and unclear selection of responders without a TRPC6 variant; direct comparison with sparsentan is methodologically problematic.
- The authors anticipate greater efficacy with selection based on biomarkers of TRPC6 activation, which is being tested with the NEPTUNE network; phase 3 trials will determine its place in practice.

5.14 Telitacicept in IgA nephropathy: a phase 3 interim analysis (TELIGAN) shows a marked reduction in proteinuria with a higher incidence of adverse events

Published June 22, 2026 · [online version \(Slovak\)](#)

Abstract. Telitacicept is a fusion protein that neutralizes both BAFF and APRIL, pathways of B-cell dysregulation involved in the pathogenesis of IgA nephropathy. An interim analysis of the randomized, double-blind, placebo-controlled phase 3 TELIGAN trial (NEJM 2026) included 318 adults with proteinuria of at least 1.0 g/day despite supportive care. At 39 weeks, telitacicept 240 mg subcutaneously once weekly reduced 24-hour UPCR by 58.9% versus 8.8% with placebo. Adverse events were more frequent, and serious adverse events less frequent. For practice, this is a promising but still interim and surrogate result.

Summary

- The TELIGAN trial (NCT05799287, funded by RemeGen) randomized 318 adults with biopsy-proven IgA nephropathy and persistent proteinuria of at least 1.0 g/day in a 1:1 ratio to telitacicept 240 mg subcutaneously once weekly or placebo; the primary endpoint was the change in 24-hour UPCR at week 39.
- UPCR fell by 58.9% with telitacicept and by 8.8% with placebo; the relative difference, expressed as the ratio of geometric means, was -55.0% (95% CI, -61.3 to -47.6; $p < 0.001$) in favor of active treatment.
- eGFR changed by -1.0% (95% CI, -3.2 to 1.2) with telitacicept versus -7.7% (95% CI, -9.9 to -5.4) with placebo; the author cautions that at the interim analysis this is only an encouraging signal, not proof of long-term kidney protection.
- Adverse events of any type occurred in 89.3% of patients receiving telitacicept versus 78.6% receiving placebo, and serious adverse events in 2.5% versus 8.2%; no unexpected safety findings were reported.
- Limitations include the interim nature of the analysis, in which the effect size may still change, and the surrogate primary endpoint; with immunomodulation of BAFF and APRIL, an increase in related risks is biologically expected.
- The reduction in proteinuria is large and clinically relevant, but a decision on the benefit-risk balance in practice will require confirmation of the durability of the effect and long-term kidney safety in the ongoing part of the trial.

5.15 New data cast doubt on efficacy: EMA has recommended revoking the authorization of Tavneos (avacopan) in the EU, and what it means for ANCA-associated vasculitis and nephrology

Published June 26, 2026 · [online version \(Slovak\)](#)

Abstract. The European Medicines Agency (EMA) has recommended that the European Commission revoke the authorization of Tavneos (avacopan) in the EU. This is a regulatory decision: serious data integrity problems and breaches of good clinical practice were identified in the pivotal ADVOCATE trial (331 patients with GPA or MPA, 52 weeks), so the benefits can no longer be shown to outweigh the risks. Treatment should not be started in new patients, and patients already treated should be switched to an alternative with liver monitoring. This matters for nephrology because GPA and MPA are among the most common causes of rapidly progressive kidney failure.

Summary

- Avacopan, a C5a receptor antagonist, was used in adults with severe active GPA or MPA in combination with other immunosuppressants, typically rituximab or cyclophosphamide, with the aim of reducing the corticosteroid burden.
- In the ADVOCATE trial, 331 patients received avacopan or placebo for 52 weeks, and both arms also received corticosteroids for 20 weeks; avacopan was judged at least comparable to high-dose corticosteroids in achieving remission and superior in sustaining it long term.
- After re-examination, EMA concluded that the data submitted during the assessment were incorrect and misleading and therefore no longer constitute reliable evidence of efficacy; the final, legally binding decision for member states will be issued by the European Commission.

- EMA recommends not starting treatment in new patients and switching those already treated to a suitable alternative therapy.
- The drug is associated with an increased risk of drug-induced liver injury and the rare vanishing bile duct syndrome (VBDS); liver function should be monitored until treatment is permanently discontinued, and the drug should be stopped immediately if VBDS is suspected.
- The author recommends preparing local protocols for a rapid change of therapy, coordinating the approach within a multidisciplinary team, and individually reassessing the trade-off between vasculitis control and steroid toxicity; patients should be told that this is not a fluctuation in efficacy but serious doubt about the evidence.

5.16 B-cell-targeted therapies in immune-mediated kidney diseases: what the KDIGO Controversies Conference shows

Published June 26, 2026 · [online version \(Slovak\)](#)

Abstract. Therapies that deplete or modulate B cells are used in several glomerular diseases, but their efficacy and safety differ markedly across diagnoses. The KDIGO Controversies Conference (Panama City, June 2025) reviewed the available evidence and knowledge gaps; this is a consensus conference report, not new clinical data. Anti-CD20 therapy is first-line in membranous nephropathy, has limited efficacy in IgA nephropathy, where BAFF and APRIL inhibition is more promising, and in lupus nephritis there is room for CAR T cells. Validation of response biomarkers remains the key need.

Summary

- In IgA nephropathy, rituximab has limited efficacy, whereas BAFF and APRIL inhibition and plasma cell-targeted approaches yield a better signal; activating mechanisms may return after treatment ends, so whether to maintain suppression longer in selected phenotypes remains an open question.
- In membranous nephropathy, anti-CD20 antibodies have moved to first line, and monotherapy leads in many cases to at least partial remission within approximately 18 months; obinutuzumab is used mainly in refractory cases, but a clear advantage has not been demonstrated.
- Rituximab effectively prevents relapses of steroid-dependent nephrotic syndrome, particularly in children, but the benefit may be transient; in ANCA-associated vasculitis, its efficacy is established in both induction and maintenance therapy.
- In lupus nephritis, obinutuzumab and CAR T-cell therapy, which introduces the concept of longer remission, are more promising; however, the notion of cure is controversial and requires precise immunologic and clinical criteria.
- Conventional anti-CD20 regimens have a more favorable safety profile, whereas CAR T cells carry a risk of serious adverse events, including cytokine release syndrome, and require strict patient selection.
- The main research gaps are biomarker validation, objective timing of escalation, liquid kidney biopsy, and determining when inflammation transitions to fibrosis; according to the report, decisions should be based on the diagnosis, phenotype, expected biology of response, and safety capacities.

5.17 Atacicept in primary IgA nephropathy: what FDA accelerated approval means

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. On July 7, 2026, the FDA granted accelerated approval to Trutakna (atacicept-vymj), a TACI-Fc fusion protein that inhibits both BAFF and APRIL, to reduce proteinuria in adults with primary IgA nephropathy at risk of progression. The basis is a prespecified analysis of the phase 3 ORIGIN 3 RCT in 203 patients: at 36 weeks, atacicept 150 mg subcutaneously once weekly reduced UPCR by 42% compared with

placebo. A long-term slowing of eGFR decline has not yet been demonstrated. The drug is not authorized in the EU or in Slovakia.

Summary

- ORIGIN 3 (NCT04716231) enrolled adults with biopsy-proven primary IgA nephropathy, an eGFR of at least 30 mL/min/1.73 m², and a UPCR of at least 1 g/g or proteinuria of at least 1 g per 24 hours on stable, maximally tolerated renin-angiotensin system blockade; mean eGFR was 65 mL/min/1.73 m², and 53% were taking an SGLT2 inhibitor.
- UPCR fell by 46% with atacicept (95% CI, 38 to 53%) and by 7% with placebo, and the relative reduction compared with placebo was 42% (29 to 52%; $p < 0.0001$); this is a ratio of geometric means, not a difference in percentage points.
- Gd-IgA1 fell by approximately 68%, and IgA, IgG, and IgM also decreased, but this biomarker was not tested with control for multiplicity and does not prove prevention of dialysis or transplantation.
- In the safety analysis of 428 patients, infections occurred in 32% versus 28% and injection-site reactions in 30% versus 5%, mostly mild; median exposure was only 34 and 31 weeks.
- Before treatment, active infection should be ruled out and vaccinations completed, and live vaccines are not recommended within 30 days before the first dose or during treatment; combination with other immunomodulatory drugs, including corticosteroids, has not been evaluated.
- Atacicept does not replace foundational kidney-protective therapy; the confirmatory assessment of annualized eGFR change at 52 and 104 weeks is ongoing, and in November 2024 EMA granted only orphan drug designation, which does not constitute marketing authorization.

5.18 Persistent microscopic hematuria in podocytopathies: a prognostic signal, not a therapeutic target

Published August 4, 2026 · [online version \(Slovak\)](#)

Abstract. In podocytopathies, assessment relies mainly on proteinuria and eGFR, whereas hematuria is often regarded as an incidental finding. A retrospective cohort of 236 adults with biopsy-proven membranous nephropathy, minimal change disease, or FSGS published in Nephrology showed that persistent microscopic hematuria within three months after biopsy was, after multivariable adjustment, associated with kidney failure requiring chronic kidney replacement therapy (HR 3.55). This is an association, not proof of causality. The finding supports systematic monitoring of the urinary sediment, not escalation of immunosuppression.

Summary

- The cohort included 114 patients with membranous nephropathy, 76 with minimal change disease, and 46 with FSGS; persistent hematuria was defined as at least five red blood cells per high-power field in at least three consecutive urine samples within three months after biopsy.
- Persistent hematuria was present in 73 patients (31%), who had higher proteinuria (5.3 versus 3.2 g/g) and more frequent hypertension (55% versus 36%); over a median follow-up of 96 months, 32% of them reached kidney failure versus 9% of patients without hematuria.
- After adjustment for diagnosis, baseline eGFR, proteinuria, Charlson comorbidity index, and histologic chronicity score, the HR was 3.55 (95% CI, 1.78 to 7.05); the direction of the association is convincing, but the precise effect size is uncertain.
- The result is supported by the work of Marchel and coauthors from the NEPTUNE and CureGN cohorts (1,516 children and adults), in which hematuria was associated with a composite kidney outcome (HR 1.31) and a lower rate of proteinuria remission (HR 0.80).
- Limitations include the retrospective design, only 38 events, a heterogeneous cohort, and residual bias; a mechanism through tubular toxicity of hemoglobin and heme remains a hypothesis.
- The author recommends confirming a positive dipstick with microscopy of fresh sediment, monitoring persistence together with proteinuria, eGFR, and blood pressure, reconsidering alternative causes of

hematuria, and not regarding it as a separate treatment target or a threshold for escalating immunosuppression.

5.19 Anti-PLA2R and thrombosis risk in membranous nephropathy: the antibody as a possible complement to albumin

Published August 24, 2026 · [online version \(Slovak\)](#)

Abstract. Thromboprophylaxis in membranous nephropathy is currently guided by the depth of hypoalbuminemia and the bleeding risk. A retrospective cohort of 67 patients with serum albumin below 2.5 g/dL published in August 2026 found that, in the subgroup with anti-PLA2R-positive disease without anticoagulant prophylaxis, a higher antibody level was associated with a higher risk of thrombotic complications (adjusted hazard ratio, 1.51 per 10 RU/mL). However, the model rests on at most ten events and does not account for immunosuppression. This is a signal for prospective validation, not a new decision-making parameter.

Summary

- The starting point is an inception cohort of 898 patients with a venous thromboembolic event in 65 persons (7.2%) and a Markov model according to which the benefit-to-risk ratio of prophylaxis rises from approximately 4.5:1 at albumin below 3 g/dL to 13.1:1 below 2 g/dL at low bleeding risk; KDIGO 2021 is based on this reasoning.
- In the new cohort, the median UPCr was 11.9 g/g and the median nadir albumin 1.9 g/dL; 11 persons (16%) had a thrombotic event, with an incidence of 0.11 per person-year without prophylaxis, 0.10 with antiplatelet therapy alone, and 0.02 with anticoagulation.
- The adjusted hazard ratio of 1.51 per 10 RU/mL (95% CI, 1.10 to 2.09; $p = 0.012$) after adjustment for nadir albumin and age cannot be extrapolated to usual titers in the hundreds of RU/mL because linearity of the relationship has not been demonstrated.
- The lower incidence with anticoagulation is not evidence of its efficacy, since prophylaxis was at the physician's discretion, a statistical comparison is lacking, and bleeding complications were not quantified.
- The abstract does not report the number of anti-PLA2R-positive patients, the types of events, the duration of follow-up, or the result in antibody-negative disease, and both sites belong to a single Boston tertiary system, which limits generalizability.
- According to the author, decisions on thromboprophylaxis continue to rest on albumin and bleeding risk; a high titer with severe hypoalbuminemia warrants attention, a lower threshold for imaging, and patient education, particularly in the first two years, when most events occur.

5.20 Baseline eGFR at kidney biopsy: how its selection and imputation can bias the results of observational studies

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. Baseline eGFR at kidney biopsy is a key variable in observational studies of glomerular diseases, but it is often defined as the first value within a six-month window or imputed. A methodological analysis of a population-based cohort of 2,874 patients from British Columbia compared these approaches with a reference value obtained within 15 days of biopsy. Bias increased with the width of the time window, whereas a baseline eGFR predicted by a mixed-effects model from repeated measurements improved both the accuracy and the precision of imputation. For research, baseline eGFR should be understood as an estimate with its own uncertainty.

Summary

- The reference standard was the closest eGFR within 15 days of biopsy; the comparators were the closest observed measurement in a widening window and single and multiple imputation with or without a baseline eGFR predicted by a mixed-effects model, with mean absolute error as the measure of bias.

- Bias of the closest observed measurement increased with distance from biopsy; in single imputation, the predicted value had consistently higher accuracy, and in multiple imputation it improved both accuracy and precision.
- Misclassification of baseline function, that is, measurement error in an explanatory variable, silently biases associations and progression thresholds in subsequent analyses and cannot be corrected by adding further variables to the model.
- Limitations include a single population-based cohort, a reference that was defined rather than measured with an exogenous marker, and availability of the abstract only; better imputation of one variable does not automatically yield a more accurate prognostic model.
- When reading observational studies, attention should be paid to the window width, the proportion of missing values, the imputation method, and reporting of its uncertainty; in registries with repeated creatinine measurements, using them is an accessible improvement that requires no new data collection.

5.21 Complete remission of proteinuria in IgA nephropathy is associated with slower decline in glomerular filtration rate

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. A post hoc analysis of the randomized phase III PROTECT trial (404 patients with IgA nephropathy, sparsentan versus irbesartan) stratified patients by achievement of complete remission of proteinuria below 0.3 g/day. Patients with remission by week 110 (21%) had a total eGFR slope of -0.7 versus -4.2 mL/min/1.73 m² per year and the composite kidney endpoint in 1% versus 14%. Remission was more frequent with sparsentan. This is a strong prognostic association, not proof of causality, and it supports the goal of a sustained reduction in proteinuria.

Summary

- Complete remission by week 36 was achieved by 43 patients (11%, of whom 81% were on sparsentan) and by week 110 by 85 patients (21%, of whom 73% were on sparsentan); at week 110, UPCR had changed by -73% with remission versus -3% without it.
- The absolute change in eGFR by week 110 was -4.0 versus -8.6 mL/min/1.73 m², and the chronic slope was -0.4 (-1.4 to 0.6) versus -4.2 mL/min/1.73 m² per year; with increasing residual proteinuria, eGFR decline accelerated continuously rather than at a threshold.
- The composite endpoint (eGFR decline of at least 40%, kidney failure, or death) occurred in one patient with remission versus 43 patients without it, but a single event does not provide a precise estimate of the clinical effect.
- Patients with remission had a more favorable baseline profile, blood pressure approximately 4 mm Hg lower during treatment, and more hypotensive events (16% versus 10%); the analysis carries a risk of residual bias, reverse causation, and immortal time bias.
- In 30 of 85 patients (35%), proteinuria later rose above 0.75 g/day after a median of 24.2 weeks, so a single value below 0.3 g/day does not mean sustained remission.
- The results are consistent with the KDIGO goal of maintaining proteinuria below 0.5 g/day and, if possible, below 0.3 g/day; the author recommends monitoring proteinuria repeatedly and documenting a value sustained over time before changing or stopping treatment.

5.22 Iptacopan slows kidney function decline in IgA nephropathy: final 24-month results of the APPLAUSE-IgAN trial

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. The final 24-month analysis of the randomized, double-blind phase III APPLAUSE-IgAN trial evaluated iptacopan, an oral selective inhibitor of factor B of the alternative complement pathway, in 477 adults with IgA nephropathy and proteinuria of at least 1 g/g despite supportive care. Iptacopan slowed the annualized eGFR decline to -3.10 versus -6.12 mL/min/1.73 m² per year with placebo and reduced the incidence

of the composite kidney endpoint (21.4% versus 33.5%; HR 0.57). Serious infections were three times more frequent. This is the first evidence of an effect of complement inhibition on loss of kidney function.

Summary

- Patients with an eGFR of at least 30 mL/min/1.73 m² received iptacopan 200 mg orally twice daily or placebo on top of the maximally tolerated dose of an ACE inhibitor or ARB; only about one fifth of patients were taking an SGLT2 inhibitor at baseline.
- The difference in eGFR slope was 3.02 mL/min/1.73 m² per year (95% CI, 2.02 to 4.01; $p < 0.001$), that is, roughly half of the annual loss; the nine-month analysis showed a 38.3% reduction in UPCR compared with placebo.
- The composite endpoint (eGFR decline of at least 30%, eGFR below 15 mL/min/1.73 m², dialysis, transplantation, or death from kidney failure) represented an absolute difference of 12.1 percentage points and an NNT of approximately 8 to 9 over two years; however, this is not a 43% reduction in the risk of kidney failure.
- Overall and serious adverse events were similar (87.0% versus 89.1%; 12.2% versus 11.7%), but serious infections occurred in 6.7% versus 2.1% of patients; there were no deaths.
- Before treatment, vaccination against *Neisseria meningitidis* and *Streptococcus pneumoniae*, and against *Haemophilus influenzae* type b according to local rules, is required, and the patient must know the symptoms of serious infection.
- Limitations include a selected population, two-year follow-up, low use of SGLT2 inhibitors, lack of head-to-head comparison with other targeted drugs, funding by Novartis, and the absence of a biomarker for patient selection; iptacopan is an add-on, not a replacement, therapy.

5.23 Transverse coaxial biopsy of the native kidney: higher specimen yield, but insufficient evidence to change practice

Published September 11, 2026 · [online version \(Slovak\)](#)

Abstract. A single-center retrospective study of 184 adults from China (2019 to 2025) compared three techniques of ultrasound-guided native kidney biopsy. The transverse coaxial technique achieved the highest mean number of glomeruli (26.90 versus 22.93 and 20.87), a 100% success rate according to the 10-glomerulus threshold, and the shortest sampling time. However, the technique was chosen by the operator, the publication contains four internal inconsistencies, including a reversed direction of the difference in macroscopic hematuria, and the only randomized trial of transverse orientation did not confirm a difference in yield. The technique is promising, not a new standard.

Summary

- The transverse coaxial technique (47 patients) had a success rate of 100% versus 89.1% with the longitudinal noncoaxial technique (92 patients; $p = 0.016$), and the difference from the longitudinal coaxial technique (45 patients) was not significant ($p = 0.489$); sampling time was 1.45 versus 2.22 and 2.61 minutes.
- Perirenal hematoma occurred in 10.6%, 24.4%, and 40.2% and macroscopic hematuria in 14.9%, 42.2%, and 40.2%; the longitudinal coaxial technique did not differ from the noncoaxial technique in any complication, so the coaxial system alone did not demonstrate a benefit.
- According to the tables, macroscopic hematuria was less frequent with the transverse technique than with the longitudinal coaxial technique, but both the abstract and the ReachMD portal state the opposite; the author recommends giving priority to the tabulated counts.
- Hematuria is an order of magnitude more frequent than in the meta-analysis by Corapi et al. (3.5%), probably because of a different definition; no major complication occurred, but the upper limit of the 95% CI for 47 patients reaches approximately 7.6%.

- Causal inference is precluded by operator choice of technique, unequal kidney selection, the seven-year period, and the declining proportion of 16 G needles (74.5%, 68.9%, and 59.8%), which, however, explains only about one glomerulus.
- The randomized trial by Akkakrisee et al. (70 patients) found no difference in the number of glomeruli, and the success rate of 85.7% versus 62.9% did not reach significance ($p = 0.056$).
- The 10-glomerulus threshold cannot be equated with diagnostic adequacy; a center should decide on adopting the technique according to its experience, procedure volume, and availability of embolization, and the next step should be a four-arm multicenter RCT.

5.24 Sparsentan with an SGLT2 inhibitor in IgA nephropathy: SPARTACUS and PROTECT OLE

Published September 14, 2026 · [online version \(Slovak\)](#)

Abstract. Both sparsentan and SGLT2 inhibitors reduce proteinuria in IgA nephropathy, but their combination has been little studied. Ayoub et al. (Nephrology Dialysis Transplantation 2026) summarized two clinical scenarios; the text is based on the published abstract. In the open-label phase 2 SPARTACUS trial (48 patients), switching from a RASi to sparsentan on a stable SGLT2 inhibitor led to a 56% reduction in UACR by week 24. In the randomized PROTECT OLE substudy (63 patients), adding an SGLT2 inhibitor to sparsentan improved the change in UPCR (ratio 0.75). These are surrogate and short-term data.

Summary

- SPARTACUS (NCT05856760) enrolled 48 patients with a mean age of 48.9 years, of whom 39 completed treatment; UACR fell by week 24 by an LS mean of -56% (95% CI, -66% to -3%), with a markedly asymmetric confidence interval.
- The trial was open-label and had no control arm with continued RASi, so it supports the hypothesis of a benefit of sparsentan on an SGLT2 inhibitor but does not prove superiority over an optimally titrated RASi with an SGLT2 inhibitor; its endpoint was UACR, not UPCR.
- In the PROTECT OLE substudy (NCT03762850), 63 patients on sparsentan were randomized 1:1 to the addition of an SGLT2 inhibitor or continued monotherapy; at week 12, UPCR changed by -11.1% versus +17.7%, and the between-group ratio was 0.75 (95% CI, 0.58 to 0.98; $P < 0.05$).
- This represents a modest further reduction in proteinuria with a small number of patients and a short randomized horizon; 54 patients completed 24 weeks of combination therapy.
- According to the abstract, the combination was well tolerated without unexpected safety signals, but data on hyperkalemia, eGFR decline, hepatotoxicity, fluid retention, and hypotension are lacking.
- The data do not provide evidence of slower eGFR decline or of effects on dialysis or cardiovascular events; the author regards the combination as a reasonable route to further proteinuria reduction, with attention to reimbursement, contraindications, monitoring, and a possible need for targeted immunomodulation.

5.25 Complement in nephrology: diagnosis, infection risk, and targeted inhibition in IgA nephropathy

Published September 14, 2026 · [online version \(Slovak\)](#)

Abstract. The FDA has already approved complement inhibitors for five kidney indications, including IgA nephropathy. The 2026 Core Curriculum review in the American Journal of Kidney Diseases (Thurman, Poppelelaars), summarized from the abstract and secondary sources, emphasizes distinguishing systemic from local activity, managing infection risk, and targeting the right pathway. In IgA nephropathy, the alternative

pathway is central: iptacopan reduced proteinuria by 38.3% and slowed eGFR decline at 24 months, whereas the lectin pathway strategy with narsoplimab failed in phase III. Vaccination is mandatory before inhibition.

Summary

- Normal serum C3 and C4 do not exclude local activation in the glomerulus, activation fragments are not a validated decision-making parameter, and CH50 and AH50 serve mainly to monitor inhibition; with C5 blockade, the target is often considered to be a CH50 close to zero or below approximately 10% of normal.
- In IgA nephropathy, C3 is found in more than 90% of biopsies and C4d in about one third of patients (30 to 45%), where it is associated with a worse prognosis; deletion of CFHR1 and CFHR3 is associated with a lower risk of the disease, which supports the dominance of the alternative pathway.
- Iptacopan reduced UPCR at 9 months by 38.3% (95% CI, 26.0 to 48.6), which formed the basis of the FDA accelerated approval in August 2024; the final analysis showed an eGFR slope of -3.10 versus -6.12 mL/min/1.73 m² per year and serious infections in 6.7% versus 2.1%.
- Narsoplimab did not meet the primary endpoint in the phase III ARTEMIS-IGAN trial, which, according to the review, does not refute the lectin hypothesis because the trial did not select patients by C4d or other biomarkers.
- Vaccination against meningococcus (A, C, W, Y, and B), pneumococcus, and Hib should ideally be completed at least 2 weeks before treatment; otherwise, antibiotic prophylaxis is needed. Vaccination does not eliminate the risk, as 14 of 16 eculizumab recipients with invasive meningococcal infection had received at least one vaccine dose.
- The pre-inhibition checklist includes confirmation of complement mediation, exclusion of active infection, a monitoring plan, and assessment of irreversible damage; randomized data on combinations with budesonide, sibeprenlimab, or sparsentan are still lacking.

5.26 Sparsentan and rapid, sustained reduction of proteinuria in IgA nephropathy in real-world practice

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. Sparsentan, a dual endothelin type A receptor and AT1 receptor antagonist, is considered in IgA nephropathy with persistent proteinuria. A multicenter retrospective Spanish cohort of 41 patients (Clinical Kidney Journal) with a median eGFR of 44 mL/min/1.73 m², of whom 95% were taking an SGLT2 inhibitor, showed a median proteinuria reduction of 44.1% at 3 months and a reduction of more than 50% in 51% of patients, with the effect persisting up to 12 months. These are uncontrolled observational data that do not demonstrate a causal effect or prevention of kidney failure.

Summary

- Median proteinuria fell from 1,163 to 772 mg/24 h at 3 months; the median reduction was 46.9% at 6 months and 38.9% at 12 months, although the abstract does not report the number of patients at each time point.
- According to the abstract, kidney function remained stable, and the authors observed no clinically significant hyperkalemia or serious adverse events; however, 12 months and an uncontrolled design are insufficient to determine the eGFR slope.
- In the PROTECT RCT (406 patients), proteinuria at 110 weeks was approximately 40% lower with sparsentan than with irbesartan, and the chronic eGFR slope was -2.7 versus -3.8 mL/min/1.73 m² per year ($P = 0.037$), but the composite kidney endpoint did not differ significantly (relative risk, 0.7; 95% CI, 0.4 to 1.2).
- A 2024 meta-analysis (884 patients with IgAN or FSGS) confirmed mainly the antiproteinuric effect and more frequent complete remission (relative risk, 2.57), with no significant difference in eGFR and more frequent hypotension (relative risk, 2.02).

- Limitations of the Spanish cohort include the retrospective design, lack of a control group, possible selection bias with longer follow-up, and a surrogate endpoint.
- After initiation, blood pressure, volume status, creatinine or eGFR, potassium, and liver tests should be monitored on a scheduled basis, and in hypotension or acute worsening of kidney function, concomitant therapy and hydration should be reassessed.

5.27 Targeting B cells in immune-mediated kidney diseases: conclusions of the KDIGO conference and why “the stronger, the better” does not apply

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. The KDIGO Controversies Conference (Panama, June 2025; Kidney International 2026) reviewed B-cell-targeted therapy in immune-mediated kidney diseases. Its output is a consensus and a set of research priorities, not a graded guideline. Efficacy differs substantially across diagnoses: anti-CD20 therapy is first-line in membranous nephropathy and of limited benefit in IgA nephropathy, whereas BAFF and APRIL inhibitors and anti-CD38 antibodies reduce proteinuria and slow eGFR decline. The appropriate depth of depletion, not maximal intensity of intervention, is what matters, and validated biomarkers are still lacking.

Summary

- CD20 is not expressed on plasma cells, so CD20 depletion may spare long-lived plasma cells as a source of antibodies; this explains part of the differences across diagnoses and distinguishes it from targeting BAFF and APRIL, CD38, or CD19.
- In membranous nephropathy, most patients achieve at least partial remission within 18 months; CAR T-cell therapy is considered here only for selected severe cases, because the disease is usually not rapidly progressive and the population tends to be older.
- In steroid-dependent nephrotic syndrome, rituximab prevents relapses, particularly in children, but the benefit is transient and relapses usually follow B-cell reconstitution.
- In lupus nephritis, obinutuzumab and CAR T cells have yielded promising results, with the potential for longer periods free of disease activity and even of treatment; in ANCA-associated glomerulonephritis, rituximab has proved effective for both induction and maintenance.
- Conventional anti-CD20 therapy has a favorable safety profile; its risks include infections, cytopenias, hypogammaglobulinemia, impaired vaccine response, and, rarely, progressive multifocal leukoencephalopathy, while CAR T-cell therapy adds CRS and ICANS.
- Immunoglobulins, blood count, and vaccination status should be assessed before treatment, but treatment should not be delayed for a vaccination window when an organ is threatened; circulating B-cell count is only a surrogate marker, and criteria for stopping treatment are lacking.

5.28 Complement, B cells, and the endothelin system in IgA nephropathy: where new therapies fit according to the KDIGO commentary

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. The pace of approvals in IgA nephropathy has outrun the KDIGO 2025 guideline, so the KDIGO work group issued a brief commentary placing the new drugs within a four-hit pathogenetic model. It is an interim position based on prespecified interim analyses of proteinuria. Atrasentan reduced uPCR by 36.1 percentage points versus placebo, iptacopan by 38.3%, sibeprenlimab by 51.2%, and atacicept by 41.8 percentage points. KDIGO explicitly cautions against ranking the drugs by these figures and does not yet endorse combinations of the new therapies.

Summary

- Hits 1 to 3 describe the formation of Gd-IgA1, the generation of immune complexes, and their mesangial deposition, and hit 4 describes inflammatory and complement-mediated injury; each drug is assigned by its dominant presumed effect, and the order does not imply a treatment hierarchy.

- Atrasentan, a selective ET-A receptor antagonist, reduced uPCR at week 36 by 38.1% versus 3.1% with placebo in the ALIGN trial (270 patients in the interim analysis); fluid retention was reported in 11.2% versus 8.2% without treatment discontinuation, and KDIGO places it in a position similar to sparsentan.
- Iptacopan reduced uPCR at month 9 by 38.3% in the APPLAUSE-IgAN trial without an increased risk of infection, but vaccination against encapsulated bacteria is mandatory according to the prescribing information because of its mechanism of action.
- Sibeprenlimab, a selective APRIL inhibitor, reduced uPCR by 51.2% versus placebo and Gd-IgA1 levels by 67.1% in the VISIONARY trial (320 patients in the interim analysis); atacicept reduced uPCR by 45.7% versus 6.8% in the ORIGIN 3 trial (203 patients).
- The trials differ in time point, population, and expression of effect, so the figures cannot be compared; proteinuria is only a surrogate endpoint, and eGFR slope data are still awaited for three of the four molecules.
- The new drugs are added to optimized supportive therapy, the choice should be individualized according to mechanism, risk of progression, availability, and cost, and volume status should be monitored with atrasentan; the commentary does not address maintenance therapy, treatment duration, or switching between therapies.

5.29 Trutakna in IgA nephropathy: the final ORIGIN 3 analysis and its implications for practice

Published September 20, 2026 · [online version \(Slovak\)](#)

Abstract. The final analysis of the phase 3 ORIGIN 3 RCT, announced by the sponsor on September 15, 2026, extends the data on atacicept (Trutakna), a BAFF and APRIL inhibitor, in primary IgA nephropathy. In 428 patients, the annualized eGFR slope through week 104 was -0.6 versus -5.6 mL/min/1.73 m² with placebo, and the composite progression endpoint occurred in 11 versus 38 events (HR 0.24). These are company data, not yet peer reviewed, and no new FDA decision exists. The results support a disease-modifying effect beyond proteinuria.

Summary

- The peer-reviewed interim analysis of 203 patients showed a reduction in UPCR at week 36 of 45.7% versus 6.8%, a relative reduction versus placebo of approximately 42%; this underpins the FDA accelerated approval of July 7, 2026.
- The change in eGFR at week 52 was -0.1 versus -5.7 mL/min/1.73 m² (difference, 5.6; 95% CI, 3.7 to 7.5), and the annualized rate of change through week 104 was -0.6 versus -5.6 mL/min/1.73 m² (difference, 5.0; 95% CI, 3.6 to 6.5), with $p < 0.0001$.
- The composite endpoint comprised death, transplantation, dialysis lasting at least 30 days, eGFR below 15 mL/min/1.73 m², or an eGFR decline of at least 30%; the HR of 0.24 does not mean reduced mortality, and dialysis, transplantation, or death together accounted for 0 versus 8 events.
- At week 104, eGFR measurements were available for only 50 patients on atacicept and 37 on placebo out of 214 in each arm, so the label “two-year results” does not mean that all patients were followed for two years.
- Adverse events occurred in 75% versus 78% and infections in 51% versus 52% of patients, but injection-site reactions occurred in 25% versus 7%; vaccinations should be completed before treatment, and live vaccines are not recommended within 30 days before initiation or during treatment.
- The manufacturer planned to file for full approval in the fourth quarter of 2026; a head-to-head comparison with other targeted drugs is lacking, and the FDA decision does not establish marketing authorization or reimbursement in Slovakia.

5.30 Lupus nephritis: treat now, biopsy as soon as possible

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. In lupus nephritis (LN), whether to perform a biopsy before treatment or to start treatment immediately has long been debated. A review of the ACR, EULAR, and KDIGO guidelines, cohort studies, and a panel discussion at the Brazilian rheumatology congress supports a compromise: perform a biopsy whenever possible, but do not delay treatment for it. Prognosis is determined by early response; in a Hong Kong cohort, achieving mCRR was associated with an 86% reduction in the adjusted risk of kidney failure. In practice, this means starting treatment concurrently with an urgent biopsy, combination immunosuppression, and hydroxychloroquine for every patient without a contraindication.

Summary

- In the Hong Kong cohort, 43.8% of patients achieved mCRR, and their kidney survival was 92.2% versus 71.7% in nonresponders; in a Toronto cohort of 374 patients, only 25.7% had a favorable monophasic course.
- Suspected LN (proteinuria ≥ 0.5 g/g creatinine, active urinary sediment, or unexplained eGFR decline in SLE) warrants concurrent initiation of treatment and urgent biopsy; both ACR and KDIGO allow pulse glucocorticoids before histologic confirmation.
- Conventional markers cannot replace histology: active disease occurs even with proteinuria below 500 to 1,000 mg/day, and in a pediatric cohort, 11 of 18 children in complete response had persistent proliferative nephritis; new markers (e.g., ICAM2, with an AUC of 0.92) currently serve mainly for monitoring.
- In a retrospective cohort of 370 patients, repeat biopsy in 122 patients revealed histologic transformation in 68% and led to a change in treatment in 82.8%; in a Spanish cohort, it revealed a diagnosis incompatible with active LN in 15.8%, but the retrospective design does not demonstrate a benefit for long-term survival.
- According to RCTs, combinations of two immunosuppressants (belimumab with MMF or cyclophosphamide, voclosporin with MMF, obinutuzumab with MMF) increase kidney response; ACR recommends treatment for 3 to 5 years, guidelines differ in the intensity of the initial regimen, and some authors reject routine triple therapy.
- The author recommends repeat biopsy for persistent proteinuria after ≥ 6 months of adequate treatment and for relapse or refractory disease, glucocorticoids at the lowest possible dose, and hydroxychloroquine with ophthalmologic monitoring, although the evidence for HCQ in LN is predominantly observational.

5.31 ANCA-associated vasculitis with kidney involvement: which treatment standards still apply in 2026

Published September 29, 2026 · [online version \(Slovak\)](#)

Abstract. ANCA-associated vasculitides are among the most common causes of rapidly progressive glomerulonephritis, and after the retraction of the ADVOCATE trial and the revocation of the European marketing authorization of avacopan on August 4, 2026, the KDIGO 2024 and EULAR 2022 guidelines had to be reassessed. An expert summary of the guidelines and primary studies shows that the remaining recommendations rest on unchallenged trials. Central is the reduced-dose glucocorticoid regimen from PEXIVAS, which was noninferior to the standard regimen (death or kidney failure, 27.9% vs. 25.5%) and associated with fewer infections. In practice, induction with rituximab or cyclophosphamide, selective plasma exchange, and maintenance therapy with rituximab remain the standard.

Summary

- Avacopan is no longer an option in the EU: on June 29, 2026, NEJM retracted the ADVOCATE trial because of post hoc readjudication of the primary endpoint in nine patients, and on August 4, 2026, the European Commission revoked the marketing authorization of Tavneos.

- Induction relies on glucocorticoids with rituximab or cyclophosphamide (remission in 64% vs. 53% in the RAVE trial); with serum creatinine > 354 $\mu\text{mol/L}$, the evidence for rituximab alone is limited, and cyclophosphamide (orally 2 mg/kg/day, reduced for age and GFR, or IV 15 mg/kg) or its combination with rituximab should be considered.
- PEXIVAS (704 patients) confirmed the noninferiority of the reduced-dose glucocorticoid regimen, with an incidence rate ratio for serious infections of 0.69; LoVAS showed fewer serious infections with rituximab (7.2% vs. 20.0%) but excluded severe glomerulonephritis; the target is approximately 5 mg of prednisolone daily by 4 to 5 months.
- Plasma exchange provided no benefit in PEXIVAS (hazard ratio, 0.86; 95% CI, 0.65 to 1.13); KDIGO recommends considering it for serum creatinine > 300 $\mu\text{mol/L}$, dialysis requirement, rapidly rising creatinine, and hypoxemic alveolar hemorrhage, and adding it in anti-GBM overlap; anti-GBM antibodies should also be tested in RPGN.
- Maintenance therapy should preferably use rituximab (RITAZAREM: hazard ratio for relapse, 0.41 versus azathioprine) for 18 months to 4 years; methotrexate should not be used with GFR < 60 mL/min/1.73 m².
- In a patient on permanent dialysis without extrarenal manifestations, stopping immunosuppression after 3 months should be considered, and transplantation is deferred until remission has lasted at least 6 months; in EGPA, mepolizumab or benralizumab may be considered.

CHAPTER 6

Acute kidney injury, critical care, and toxicology

Acute kidney injury, perioperative and intensive care, nephrotoxicity of drugs and environmental agents, and clinical toxicology. Number of articles: 17.

Key messages of the chapter

The common denominator of this chapter is a shift from reflexive interventions to indication-driven decision-making. A review of the randomized trials ELAIN, AKIKI, IDEAL-ICU, STARRT-AKI, and AKIKI-2 shows that immediate initiation of kidney replacement therapy without an urgent indication generally does not improve outcomes and in STARRT-AKI led to more hypotension and hypophosphatemia, whereas the very delayed strategy in AKIKI-2 suggested higher 60-day mortality (6.4). An RCT in JAMA with 221 patients points in the same direction: dialysis only when metabolic or clinical criteria were met led to recovery of kidney function in 64% versus 50% of patients and to fewer episodes of hypotension, although the difference was not significant in the adjusted analysis (6.1). The evidence therefore supports standardized triggers and a time-limited trial of therapy with agreed goals (6.4), not an automatic dialysis schedule (6.1).

The second theme is the difference between predicting risk and modifying it. In the single-center ESTOP-AKI RCT, a machine learning algorithm identified at-risk patients, but early nephrology consultation did not affect the maximal change in creatinine, and the recommendations were implemented in only 41% of cases (6.10). In MERCURI-2, dapagliflozin before cardiac surgery reduced AKI from 52% to 28%, but most of the difference probably consisted of oliguric episodes, which the drug affects, and neither mortality nor hard outcomes changed (6.11). A warning in the opposite direction is the VICTORY RCT, in which high-dose vitamin C in burns increased both 28-day mortality (15.0% versus 7.6%) and the need for KRT (6.6). Positive and negative randomized results thus lead to the same conclusion: an intervention should be judged by clinical outcomes, not by surrogate markers (6.6, 6.11).

Observational data add prognostic context but do not prove causality. In an Ontario cohort of 531,090 ICU patients, both 90-day mortality and dialysis dependence rose with declining eGFR; in stage G5 not on dialysis, dialysis dependence reached 83.8% of survivors after KRT initiation (6.12). A Dutch cohort showed AKI in 35.2% of patients treated with flucloxacillin versus 20.2% with ce-fazolin, consistent with the randomized SNAP and CloCeBa trials (6.15). A large study after COVID-19 describes a higher risk of AKI and CKD progression than after influenza, especially in the first six months (6.2).

The toxicology section of this chapter emphasizes a targeted history instead of blanket screening. Lead, cadmium, and mercury have the best documented nephrotoxicity, whereas for PFAS, bisphenol A, and microplastics, causality is uncertain (6.9, 6.7). With uranium in well water, chemical toxicity to the proximal tubule predominates, and the association below the limit of 30 µg/L comes from observational studies (6.7); a Taiwanese case-control study on MMP and metals has no clinical application (6.13). Therapeutic plasma exchange reduced microplastics only at higher burden, and this is mechanistic evidence without clinical benefit (6.5). A case report of DRESS after allopurinol with granulomatous interstitial nephritis (6.3), a review of ketamine-associated uropathy (6.16), and an analysis of fly agaric products, in which AKI should rather prompt a search for another substance (6.17), are reminders that only a targeted question reveals the etiology. Ketoacidosis can occur with

normal glycemia, particularly during SGLT2 inhibitor therapy, and β -hydroxybutyrate should be measured in blood (6.8).

This yields a simple framework for outpatient and consultation practice. A rise in creatinine should first be confirmed, pseudo-AKI distinguished, and prerenal, drug-induced, obstructive, parenchymal, and vascular causes systematically sought, with ultrasound when obstruction is suspected (6.14). In older adults, diuretics, RAS blockade, and NSAIDs should be reviewed, and treatment interruption should be considered temporary (6.14). With flucloxacillin, creatinine should be monitored from the first day, and cefazolin should be considered in suitable patients with MSSA (6.15). SGLT2 inhibitors are withheld before planned surgery according to current rules, and dapagliflozin should not be started for AKI prevention outside a trial (6.11, 6.8). After COVID-19, monitoring of UACR and kidney function is advisable (6.2), and CKD stage alone should not be a reason to refuse intensive care (6.12).

6.1 A more conservative dialysis strategy in AKI may support recovery of kidney function

Published May 11, 2026 · [online version \(Slovak\)](#)

Abstract. In hospitalized patients with dialysis-requiring AKI, it is unclear how intensively to dialyze during the period of potential recovery. A multicenter RCT published in JAMA (221 patients, four US centers) compared dialysis only when metabolic or clinical criteria were met with scheduled dialysis three times weekly. Recovery of kidney function at discharge was achieved in 64% vs. 50% of patients, but the difference was not statistically significant in the adjusted analysis. An indication-guided approach may be preferable in stable patients, but larger trials are needed.

Summary

- The conservative strategy led to recovery of kidney function in 64% of patients versus 50% (absolute difference, 13.8%), but the significance seen in the unadjusted analysis was not confirmed in the prespecified adjusted analysis, so the effect size remains uncertain.
- Patients in the conservative group underwent a median of 1.8 vs. 3.1 dialysis sessions per week, had a median of 21 vs. 5 consecutive dialysis-free days through day 28, and had 69 vs. 97 episodes of dialysis-associated hypotension.
- Randomization took place a median of 9 days after KRT initiation, and recovery was defined as survival without dialysis dependence with at least 14 consecutive dialysis-free days.
- In hemodynamically stable patients, automatic dialysis three times weekly may not be optimal; an indication-guided approach is reasonable to consider, but the results should be interpreted with caution.

6.2 COVID-19 and kidney health: what impact the infection may have and whom to monitor

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. A study of more than 3 million patients in Communications Medicine showed that, compared with influenza, COVID-19 carries a higher risk of AKI, new or worsening CKD, and progression to kidney failure. Written as a question-and-answer interview with a nephrologist, that is, observational data supplemented by expert opinion, the article describes risk groups, the timing of complications, and monitoring. AKI occurs mainly in the first six months after infection. In practice, UACR testing, monitoring of kidney function, and timely nephrology evaluation of patients after COVID-19 are important.

Summary

- The risk is highest after severe COVID-19 but is not limited to it; in the data, AKI is reported mainly in the first six months after SARS-CoV-2 infection, and CKD often remains asymptomatic for a long time.

- The spectrum ranges from acute tubular injury to inflammatory and immune-mediated glomerulopathies; the infection often acts as a trigger in predisposed individuals, for example in APOL1 variants and COVID-19-associated nephropathy.
- Manifestations may appear early or over weeks to months, and some patients recover fully once immune activity subsides.
- Monitoring of UACR, blood pressure, diabetes, and subtle persistent decline in kidney function is recommended, together with ascertainment of family history of kidney disease and, in selected cases, consideration of genetic evaluation.
- Beyond RAAS inhibitors, SGLT2 inhibitors, GLP-1 receptor agonists, and MRAs, knowing the cause of CKD is crucial; with unclear etiology and significant proteinuria without an evident diabetic origin, the threshold for biopsy should be lower.
- According to the article, current variants damage the kidneys less than earlier ones, but given the high number of infections, even a small increase in individual risk represents a real burden, so long-term continuous monitoring is important.

6.3 DRESS syndrome after allopurinol: the diagnostic trap of granulomatous interstitial nephritis with pancreatitis

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. DRESS syndrome is a rare, potentially life-threatening delayed hypersensitivity reaction frequently triggered by allopurinol, and kidney involvement most often takes the form of acute interstitial nephritis (AIN). This case report describes a 71-year-old man who developed DRESS three weeks after starting allopurinol, with granulomatous AIN, acute pancreatitis, neurologic manifestations, and HHV-6 reactivation. With rising creatinine, the diagnosis was confirmed by biopsy. Drug withdrawal and high-dose corticosteroids led to near-complete recovery of kidney function. DRESS should be considered in multifactorial AKI with eosinophilia, and biopsy should not be delayed.

Summary

- The patient had fever, a diffuse morbilliform rash without mucosal involvement, altered consciousness, AKI, marked eosinophilia, hepatocellular injury, and very high lipase with acute pancreatitis without an evident biliary cause; infectious and autoimmune workups did not explain his condition.
- Creatinine continued to rise even after allopurinol was stopped, so the nephrologists performed a biopsy, which showed granulomatous interstitial nephritis with an eosinophilic infiltrate, tubulitis, and edema; the granulomatous variant is less typical and obscures the diagnosis even further.
- HHV-6 reactivation is reported in approximately six of ten patients with DRESS and is associated with a more severe and prolonged course and a higher risk of multiorgan failure.
- After allopurinol withdrawal and high-dose corticosteroids, the mental status, rash, and eosinophilia resolved and kidney function improved; in severe forms, however, recovery may remain incomplete.
- DRESS should be considered in a patient with fever, rash, eosinophilia, liver injury, and AKI 2 to 6 weeks after starting a high-risk drug; in worsening AKI of unclear etiology, biopsy can expedite appropriate treatment.
- The risk of hypersensitivity is higher in CKD, so it is recommended to start allopurinol at a low dose (e.g., up to 100 mg daily, even lower in CKD), titrate slowly, and consider screening for the HLA-B*58:01 allele in high-risk ethnic groups; this is a single case report.

6.4 When to start kidney replacement therapy (KRT) in acute kidney injury (AKI)

Published June 17, 2026 · [online version \(Slovak\)](#)

Abstract. The timing of kidney replacement therapy (KRT) in severe AKI without urgent complications remains controversial. A review of the randomized trials ELAIN, AKIKI, IDEAL-ICU, STARRT-AKI, and AKIKI-2 reaches a consistent conclusion: immediate initiation of KRT without an urgent indication generally does

not improve important outcomes and, in STARRT-AKI, led to more adverse events. Excessive delay, however, suggested higher 60-day mortality in AKIKI-2. In practice, KRT should be started for urgent indications, triggers should be standardized, and decisions should be made in the context of goals of care.

Summary

- KRT removes fluid and corrects electrolyte, metabolic, and acid-base disturbances, but it does not restore intrinsic kidney functions such as erythropoietin production or vitamin D activation.
- Urgent indications include refractory severe metabolic acidosis and hyperkalemia, pulmonary edema, uremic complications, fluid overload with organ dysfunction, and intoxication with a dialyzable substance; for relative indications, neither creatinine nor urine output reliably identifies patients who will need KRT.
- The furosemide stress test assesses preserved tubular response and, in pilot data, identified patients at higher risk of requiring KRT, but biomarkers are unsuitable as the sole basis for decisions because of differing cutoff values and confounding by comorbidities.
- Only the single-center ELAIN trial showed lower 90-day mortality with KRT within 8 hours (with a fragility index limitation); AKIKI, IDEAL-ICU, and STARRT-AKI showed no difference in mortality, and STARRT-AKI had more KRT-related hypotension and hypophosphatemia in the accelerated arm.
- In AKIKI-2, in persistent stage 3 AKI, the number of KRT-free days did not differ significantly, but pre-specified analyses suggested higher 60-day mortality with a markedly delayed strategy.
- The author recommends standardizing triggers for starting KRT (pH, hyperkalemia, intoxication, severe edema, oliguria, uremic symptoms) and, under uncertainty or with a low likelihood of improvement, considering a time-limited trial of KRT with agreed goals and a review date.

6.5 Can we remove microplastics from the body? What the first clinical study of therapeutic plasma exchange suggests

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. The presence of microplastics in blood raises the question of whether their removal is clinically meaningful. A study summarized by Medscape analyzed 174 therapeutic plasma exchange (TPE) procedures in 114 patients, with PlasticTox test measurements before and after the procedure. With a higher baseline burden (approximately $\geq 20/100 \mu\text{L}$), microplastics decreased markedly, in some cases by more than half, whereas at low values the effect was inconsistent and even included increases. This is proof of mechanism without demonstrated clinical benefit, so TPE should not be a routine treatment for microplastics.

Summary

- The aim was to determine whether microplastics partition into the plasma fraction, and are thus theoretically removable, and whether TPE leads to a measurable reduction in circulating particles.
- Whereas the reduction was marked at a higher burden, it was inconsistent at lower baseline values, and at the lowest values a significant increase was even observed, which expert commentators attribute to possible contamination during blood warming and from single-use plastic components.
- A reduction in blood levels may not reflect tissue burden, the follow-up measurement at 30 days did not allow a robust conclusion because of patient attrition, and the PlasticTox test does not capture the full spectrum of nanoplastics, which may be more biologically active.
- TPE is an invasive procedure with risks related to vascular access, hemodynamic stress, and plasma replacement, so it should be judged by clinical endpoints, not merely by a reduction in laboratory markers.
- For nephrology, the topic is mainly conceptual: the result is not a recommendation for routine removal of microplastics, not even in patients on dialysis; further research should improve the analytics, and the planned STOMP initiative (2026) aims to bring both measurement and removal strategies to a reproducible level.

6.6 VICTORY: high-dose intravenous vitamin C does not improve outcomes in severe burns and may be harmful, with signals important for nephrology

Published June 22, 2026 · [online version \(Slovak\)](#)

Abstract. High-dose intravenous vitamin C has been studied as a modulator of inflammation in severe burns, although it is metabolized to oxalate and may burden the kidneys. The international, double-blind, placebo-controlled phase 3 VICTORY RCT (JAMA 2026, 238 patients) did not improve the primary composite outcome (40.8% vs. 29.7%; RR 1.28) and was stopped early. Mortality at 28 days was higher (15.0% vs. 7.6%; RR 1.96), and AKI and KRT initiation were more frequent. Routine use outside clinical trials is therefore difficult to justify.

Summary

- Adults with deep burns covering at least 20% of body surface area and requiring skin grafts were enrolled and received vitamin C 50 mg/kg IV every 6 hours for 96 hours or placebo; the primary outcome was death or persistent organ dysfunction at day 28.
- The primary outcome was worse with vitamin C (adjusted RR 1.28; 95% CI, 0.99 to 1.65; $p = 0.06$), the trial crossed the futility or harm threshold, and its early termination reduces the precision of the estimates.
- Both 28-day mortality (15.0% vs. 7.6%; RR 1.96; $p = 0.001$) and in-hospital mortality (23.3% vs. 16.1%; RR 1.44; $p = 0.03$) were higher, and there were fewer days free of persistent organ dysfunction (median, 12.5 vs. 19.5).
- Among nephrologic signals, AKI (15.0% vs. 11.0%) and KRT initiation (10.8% vs. 5.9%) were more frequent; no new oxalate stones or severe hemolysis were reported, but the trial was not powered to detect rare events.
- If vitamin C is nevertheless used, the author recommends monitoring creatinine, urine output, need for KRT, blood glucose, electrolytes, acid-base balance, and signs of hemolysis, and individually assessing the risk of oxalate load.
- Strategies with a proven effect on perfusion, AKI prevention, and rational timing of KRT take precedence, and the signal of higher mortality should be communicated across disciplines.

6.7 Uranium and the kidneys: chemical nephrotoxicity, radiation injury, and the risks of environmental metals

Published July 11, 2026 · [online version \(Slovak\)](#)

Abstract. Uranium is both a radioactive actinide and a toxic metal, and with ordinary environmental exposure, particularly from contaminated well water, its chemical toxicity matters more for the kidneys than its radioactivity. This review synthesizes experimental, epidemiologic, and regulatory data, including Slovak, Czech, and European maps. The target of injury is the proximal tubule; observational studies from 2025 suggested an association even below the limit of 30 $\mu\text{g/L}$ but do not prove causality. The nephrologist should specifically evaluate patients with a compatible tubular phenotype and a verified exposure pathway rather than screen all patients with CKD.

Summary

- Soluble uranium enters proximal tubular cells partly via the NaPi-IIa cotransporter and damages mainly the S2 and S3 segments; acute high exposure leads to tubular proteinuria, incomplete or complete Fanconi syndrome, and AKI.
- With chronic exposure, nonspecific tubular markers (β_2 -microglobulin, α_1 -microglobulin, retinol-binding protein) change before eGFR; elevated urinary uranium proves exposure, not kidney injury.
- A 2024 meta-analysis of 36 studies found no higher mortality from kidney disease, but in the California Teachers Study (88,185 women), the risk of CKD at 10 to less than 15 $\mu\text{g/L}$ was approximately one third higher than below 2 $\mu\text{g/L}$; these, however, are observational data.

- The Slovak limit under Decree No. 91/2023 Coll. and the provisional WHO value are both 30 µg/L, and the Czech limit is 15 µg/L; exceeding the limit is not proof of injury, boiling does not remove uranium, and reverse osmosis or anion exchange is usually effective.
- Radiation nephropathy develops with a whole-kidney dose of about 7 to 8 Gy acutely or 18 to 20 Gy fractionated, an order of magnitude higher than with environmental exposure; neither radium nor radon is a convincingly proven cause of CKD.
- Cadmium, lead, and inorganic mercury have the best-documented nephrotoxicity, and with contaminated water, other metals should also be considered, as their mixtures may act synergistically.
- Targeted evaluation is warranted with a Fanconi phenotype, proteinuria markedly exceeding albuminuria, unexplained AKI after exposure, or CKD in a well user in a high-risk area; residence in such an area alone or a hair test has a low pretest probability.

6.8 Ketoacidosis in nephrology practice: from starvation to euglycemic diabetic ketoacidosis

Published July 24, 2026 · [online version \(Slovak\)](#)

Abstract. Ketoacidosis occurs not only in diabetes but also with starvation, excessive alcohol intake, pregnancy, very low-carbohydrate diets, and SGLT2 inhibitor therapy. This review focused on nephrology practice summarizes the pathophysiology, diagnosis according to the 2024 international consensus, and pitfalls in interpreting serum potassium and acid-base balance. DKA requires diabetes or hyperglycemia, ketonemia, and metabolic acidosis, and a β -hydroxybutyrate level of at least 3.0 mmol/L has high diagnostic accuracy. Normal blood glucose does not exclude ketoacidosis, and treatment must be substantially individualized in patients on dialysis.

Summary

- The dominant ketone body is β -hydroxybutyrate, which the urine nitroprusside test does not detect, so it should be measured directly in blood for both diagnosis and monitoring, and a negative urine test should not outweigh the clinical picture.
- In euglycemic DKA, blood glucose is usually below 11.1 mmol/L, and risk factors include SGLT2 inhibitor therapy with omitted insulin, starvation, infection, dehydration, alcohol, or a ketogenic diet; before elective surgery, the drug is generally stopped 3 to 4 days in advance.
- Serum potassium is often normal or elevated in DKA despite a total body deficit; according to the 2024 consensus, insulin is withheld when serum potassium is below 3.5 mmol/L, and replacement is started when it falls below approximately 5.0 to 5.3 mmol/L.
- The normal anion gap is approximately 8 to 12 mmol/L, and in hypoalbuminemia about 2.5 mmol/L should be added for every 10 g/L decrease in albumin; persistent hyperchloremic acidosis after treatment is not automatically persistent DKA.
- Alcoholic ketoacidosis is treated with volume, thiamine before glucose, glucose, and electrolytes, but not routinely with insulin; markedly elevated lactate should prompt a search for sepsis, hypoperfusion, or intoxication.
- In CKD, renal compensation of acidosis is reduced and hypovolemia leads to prerenal AKI; in patients on hemodialysis, standard volume resuscitation risks pulmonary edema and hyperkalemia, and dialysis should not replace insulin.

6.9 Environmental toxins and kidney injury: what the nephrologist should specifically ask about

Published July 25, 2026 · [online version \(Slovak\)](#)

Abstract. The kidneys are vulnerable to environmental toxins because of their high blood flow, filtration, and tubular concentration of substances. A narrative review by Strasma, Jayasundara, and Anand in the *American Journal of Kidney Diseases* offers a practical framework for recognizing environmentally induced kidney

disease, although it is not a systematic review. Strong evidence exists for lead, cadmium, mercury, and some acute intoxications, whereas causality is uncertain for PFAS, bisphenol A, microplastics, and pesticides. An occupational and environmental history should be part of the evaluation of unexplained tubulopathy, proteinuria, or a disproportionately rapid decline in eGFR.

Summary

- Exposure to lead, cadmium, mercury, arsenic, industrial chemicals, pesticides, and some cosmetic products can lead to AKI, proximal tubulopathy, Fanconi syndrome, glomerular disease, nephrolithiasis, or chronic tubulointerstitial nephritis.
- For many modern contaminants, such as PFAS, bisphenol A, microplastics, melamine, and hair-straightening products, the available data are predominantly experimental or cross-sectional and do not confirm a causal relationship.
- Mechanisms include accumulation in the proximal tubule, oxidative stress, transporter dysfunction, endothelial injury, immune-mediated glomerular injury, and crystal precipitation; chronic exposure leads to tubular atrophy and fibrosis with a relatively bland sediment.
- Proximal tubular injury is signaled by low-molecular-weight proteinuria, normoglycemic glycosuria, phosphaturia, uricosuria, aminoaciduria, and proximal renal tubular acidosis.

6.10 ESTOP-AKI: the algorithm recognized the risk, but early nephrology consultation did not improve outcomes

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. Machine learning models predict AKI with an area under the ROC curve of 0.85 to 0.86 approximately 27 to 39 hours before creatinine doubles, but it is unclear whether the time gained improves outcomes. The single-center ESTOP-AKI RCT randomized 180 at-risk hospitalized patients to early nephrology consultation or standard care. The maximum change in creatinine over seven days did not differ ($P = 0.30$), although AKI developed in 38.9% of the cohort, and recommendations were followed in only 41% of cases. An alert without a prioritized and implemented intervention does not protect the kidneys.

Summary

- The primary endpoint did not differ (creatinine increase of 0.04 mg/dL with consultation vs. decrease of 0.03 mg/dL), and differences in stage 2 or higher AKI (19% vs. 13%), 90-day mortality (14.8% vs. 18.7%), and rehospitalizations (34.1% vs. 44.4%) were also not significant.
- Nephrologist recommendations were implemented in 41% of cases with early consultation versus 68% in the control group, in which the nephrologist was called for a more evident problem; the trial thus tested the offer of early care rather than its efficacy.
- The author interprets the result to mean that the model predicts risk, not a treatable mechanism: the same score can arise from hypovolemia, congestion, sepsis, drug toxicity, or obstruction, and consultation does not change most risk factors.
- Limitations include a single site, a small unblinded sample, a heterogeneous intervention, a short-term creatinine endpoint, and insufficient power to assess mortality, need for dialysis, or long-term kidney function.
- The foundation of prevention remains cause-directed hemodynamic stabilization, accurate volume assessment, medication review and dose adjustment, rational use of contrast, and monitoring of creatinine and urine output trends; future systems should deliver a brief, prioritized prompt with verification of its implementation.

6.11 Dapagliflozin before cardiac surgery reduced the incidence of acute kidney injury. Changing perioperative practice is still premature

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. AKI is among the most common complications of cardiac surgery, and proven pharmacologic prevention is lacking. The multicenter, double-blind MERCURI-2 RCT (JAMA 2026, 784 patients in the Netherlands) administered four doses of dapagliflozin 10 mg around elective cardiac surgery and reduced AKI within 7 days from 52% to 28% (RR 0.54). Most of the difference, however, probably consisted of cases defined by oliguria, which the drug affects, and neither mortality nor hard outcomes changed. Routine initiation of dapagliflozin or abandoning perioperative withholding of SGLT2 inhibitors is not yet justified.

Summary

- The absolute difference of 24 percentage points corresponds to an approximate NNT of 4.2, but the incidence of AKI on placebo was almost 2.4 times the anticipated 22%, suggesting capture of very mild, mainly oliguric episodes.
- According to the author's rough calculation, the urine output criterion accounts for approximately four fifths of the difference, and with assessment by creatinine alone, the rates would be 22% vs. 27%, a reduction of roughly one fifth.
- Mortality, length of hospital and ICU stay, and major cardiorenal events within 30 days did not change; the finding of fewer stage 2 or 3 AKI cases (19 vs. 50 patients) is not yet verified.
- Ketoacidosis occurred in one patient on dapagliflozin (upper bound of the 95% CI about 1.4%), and patients with type 1 diabetes, a history of DKA, eGFR below 20 mL/min/1.73 m², or urgent surgery, as well as chronic users of SGLT2 inhibitors, were excluded.
- The FDA recommends stopping dapagliflozin, canagliflozin, and empagliflozin at least 3 days and ertugliflozin 4 days before elective surgery, even in patients without diabetes; the SPAQI 2026 statement advocates an individualized approach, and in Slovakia, the summary of product characteristics and the local protocol are decisive.
- The author recommends not starting dapagliflozin for AKI prevention outside a trial, resuming the drug only once the patient is stable with adequate intake, monitoring blood ketones in urgent surgery, and not regarding preserved urine output as proof of kidney protection.

6.12 Severity of chronic kidney disease and prognosis of patients admitted to the intensive care unit

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. A population-based retrospective cohort study from the Canadian province of Ontario, including 531,090 adults admitted to the ICU from 2008 to 2021, assessed how prognosis changes with baseline eGFR. As eGFR declined, 90-day mortality, the need for acute KRT, length of stay, and dialysis dependence all increased; the adjusted odds ratio for death reached 2.32 in nondialysis stage G5. These are observational data without proof of causality. The results help in estimating prognosis and communicating with the patient and family, not in deciding alone to limit treatment.

Summary

- Chronic kidney disease, defined as eGFR below 60 mL/min/1.73 m² or maintenance dialysis, was present in one quarter of the 531,090 patients admitted to the ICU.
- The adjusted odds ratio for 90-day mortality rose from 1.14 in G3a through 1.38 in G3b and 1.95 in G4 to 2.32 in nondialysis G5, and it was 1.92 with maintenance dialysis. When the outcome is common, the odds ratio overestimates the relative risk.
- The odds ratio for KRT initiation in the ICU reached 1.79 in G3a, 3.02 in G3b, 6.71 in G4, and 32.00 in nondialysis G5, partly reflecting lower renal reserve. The hazard ratio for discharge decreased from 0.94 in G3a to 0.65 in nondialysis G5.

- Among patients surviving to day 90 after KRT initiation, 7.2% of those without CKD, 22.5% in G3b, 50.3% in G4, and 83.8% in previously nondialysis G5 remained dialysis dependent; this is a survival-conditioned selection, and later recovery is possible.
- The lower mortality of patients on dialysis compared with nondialysis G5 probably reflects selection, frailty, conservative management, and established nephrology care, not a protective effect of dialysis.
- Limitations include CKD classification based on a single outpatient creatinine up to one year before admission, missing albuminuria, undetected AKI at admission, and residual confounding.
- The author recommends assessing the eGFR trajectory, reversibility, potassium, vascular access, frailty, and patient preferences at admission. CKD stage alone should not be a reason to refuse ICU admission, and when dialysis is started in a patient with G5, it is honest to discuss the low likelihood of returning to dialysis-free management.

6.13 Matrix metalloproteinases, toxic metals, and chronic kidney disease: a strong association without evidence of causality

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. A Taiwanese case-control study of 215 patients with CKD and 389 controls from two hospitals in Taipei examined the relationship between plasma matrix metalloproteinases MMP-2 and MMP-9, toxic elements, and chronic kidney disease. The adjusted odds ratio for CKD was 12.45 for MMP-2 and 3.77 for MMP-9, and the association was stronger with higher arsenic, cadmium, or lead and lower selenium. Because of the cross-sectional design, neither the direction of the relationship nor causality can be determined. MMPs remain research biomarkers with no application in clinical practice.

Summary

- Patients with CKD had eGFR below 60 mL/min/1.73 m² for more than three months by the MDRD equation and did not require KRT; controls had eGFR above 60 mL/min/1.73 m², but without albuminuria data, some of them may have been misclassified.
- The adjusted odds ratio was 12.45 (95% CI, 6.04 to 25.66) for MMP-2 and 3.77 (2.14 to 6.66) for MMP-9; these are odds of having CKD, not a twelvefold risk of developing it, and the wide interval indicates an imprecise estimate. Both MMPs correlated inversely with eGFR.
- Patients with CKD had higher urinary arsenic, higher blood cadmium and lead, and lower plasma selenium. The authors described additive interactions of MMP-2 with arsenic, cadmium, lead, MMP-9, and low selenium and a multiplicative interaction with lead, but statistical interaction does not prove biological synergy.
- Polymorphisms in the MMP2 and MMP9 genes were not associated with CKD, which is more consistent with the notion that elevated MMPs are an epiphenomenon of the disease.
- The main limitations are the inability to establish temporal sequence, reverse causation with declining filtration, a single measurement, hospital-based controls, preanalytic variability of MMP-9, multiple testing, and residual confounding.
- Measurement of MMP-2 or MMP-9 cannot be recommended for screening, determining the cause, or estimating the prognosis of CKD. Metals and metalloids should be tested selectively according to the exposure history, and low selenium is not an indication for unsupervised supplementation.

6.14 Sudden rise in creatinine in an older patient with hypertension: causes, diagnosis, and clinical approach

Published September 8, 2026 · [online version \(Slovak\)](#)

Abstract. This clinical review discusses a rise in serum creatinine from 90 to 160 µmol/L in a patient of about 80 years with hypertension. If it occurred within seven days, it represents 1.78 times the baseline value, that is, KDIGO stage 1 AKI; with an unknown time course, the value alone does not prove AKI. Drawing on the KDIGO and NICE NG148 guidelines, the text emphasizes a systematic search for prerenal, drug-

related, obstructive, parenchymal, and vascular causes instead of automatically attributing the change to hypertension or age.

Summary

- By KDIGO criteria, AKI means a creatinine rise of at least $26.5 \mu\text{mol/L}$ within 48 hours, an increase to at least 1.5 times baseline within seven days, or urine output below 0.5 mL/kg/h for at least six hours; stage 3 includes at least a threefold increase, a rise to at least $353.6 \mu\text{mol/L}$, or initiation of KRT.
- First, a true and acute change should be confirmed by repeat testing, and pseudo-AKI due to trimethoprim, cimetidine, cobicistat, or dolutegravir should be considered. Normal urine output does not exclude AKI.
- In older adults, volume depletion, a diuretic, an ACE inhibitor or ARB, and NSAIDs frequently coincide; KDIGO 2024 recommends checking blood pressure, creatinine, and potassium within two to four weeks after initiation or dose increase and searching for a cause if creatinine rises by more than 30%, and any interruption should be temporary.
- Obstruction should be actively excluded by ultrasound bladder volume measurement and renal ultrasonography; NICE recommends ultrasound within 24 hours when the cause is unexplained or obstruction is a risk and within six hours when pyonephrosis is suspected. Hematuria or red cell casts warrant consideration of glomerulonephritis.
- FeNa below 1% is most useful in oliguric patients without CKD and without diuretics, and FeUrea had only moderate accuracy with low certainty of evidence in a 2026 systematic review.
- Fluids should not be given merely because of elevated creatinine, diuretics do not treat AKI, and blanket withdrawal of all antihypertensives is not safe. Anuria, severe hyperkalemia, acidosis, pulmonary edema, or uremic complications require urgent management, and according to NICE, a nephrologist should be contacted within 24 hours at the latest for stage 3 AKI, an unclear cause, or CKD G4 to G5.

6.15 Flucloxacillin in Staphylococcus aureus bacteremia: more acute kidney injury than with cefazolin

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. A retrospective multicenter cohort study from three Dutch hospitals compared the incidence, timing, severity, and recovery of AKI in 1,408 patients with *Staphylococcus aureus* bacteremia treated with flucloxacillin or cefazolin. AKI developed in 35.2% of patients on flucloxacillin and in 20.2% on cefazolin, with an adjusted odds ratio of 2.37 (95% CI, 1.30 to 4.55). The result is consistent with the randomized trials SNAP and CloCeBa. Most AKI developed within seven days, so kidney function should be monitored from day one, and cefazolin should be considered in suitable patients with MSSA.

Summary

- Of the 1,408 patients, 1,324 received flucloxacillin and 84 cefazolin; AKI by KDIGO creatinine criteria developed in 466 (35.2%) versus 17 (20.2%), $P = 0.004$. The AKI rate was 1.66 versus 0.84 per 100 patient-days.
- In the randomized SNAP trial, cefazolin had an AKI rate of 13.9% versus 19.6% with antistaphylococcal penicillin, with noninferior 90-day mortality, and in the CloCeBa trial, 1% versus 12% with cloxacillin.
- The median onset of AKI was 1.3 days, and 80.3% of episodes occurred within seven days, which does not match typical tubulointerstitial nephritis after seven to ten days and suggests the coexistence of several mechanisms, including sepsis-associated AKI.
- Recovery within 30 days was achieved by 81.7% of patients with stage 1 AKI, 64.5% with stage 2, and 50.5% with stage 3. The high dose of 12 g daily and the lower dose of 6 to 8 g daily had almost the same incidence of AKI (35.2% versus 35.7%), but dosing was not randomized.
- Limitations include the observational design with confounding by indication, the small cefazolin group, estimated baseline creatinine in 456 patients, nonuse of the urine output criterion, 212 patients receiving both drugs, and heterogeneity across centers.

- With flucloxacillin, the authors recommend measuring creatinine before treatment and checking it from the first days, monitoring urine output, limiting other nephrotoxic drugs, and following kidney function after AKI even after discharge. The choice of antibiotic should take into account the site of infection, isolate susceptibility, and allergies; caution with cefazolin applies to CNS infection.

6.16 Ketamine-associated uropathy in children and adolescents: diagnosis and management

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. This review, synthesizing retrospective cohorts, case reports, mechanistic experiments, and consensus recommendations, addresses ketamine-associated uropathy in children and adolescents. It comprises injury to the bladder and upper urinary tract with repeated nontherapeutic ketamine use; in the original cohort of 59 patients, 51% had hydronephrosis. The key is to consider the diagnosis, ask without stigmatization, detect hydronephrosis and declining kidney function early, and secure lasting abstinence, because the efficacy of individual interventions has not been confirmed by randomized trials.

Summary

- KAU should be considered with unexplained urinary frequency, urgency, dysuria, pelvic pain, and hematuria with repeatedly negative cultures, reduced bladder capacity, or hydronephrosis. In a cohort of 59 selected patients, 30 (51%) had hydronephrosis and eight had elevated creatinine, but this is not a population prevalence.
- The presumed mechanism is injury to the urothelium by ketamine and its metabolites, with fibrosis and loss of bladder compliance, as supported by experimental data whose applicability to children is uncertain. The biliary tract and liver may also be affected.
- Diagnosis includes a direct, nonjudgmental history of use, urine sediment and culture, creatinine with eGFR, electrolytes, liver tests, and ultrasonography as first-line imaging; a negative toxicology screen does not exclude KAU, and cystoscopy or urodynamics are the urologist's domain.
- Oliguria, rising creatinine, hyperkalemia, fever with a systemic response, or bilateral hydronephrosis require urgent urologic and nephrologic assessment with prompt decompression of the urinary tract.
- Complete abstinence with addiction support is the cornerstone of treatment; pentosan polysulfate, hydrodistension, or botulinum toxin have been described only in small series, and NSAIDs should not be given automatically given the risk of AKI. Reconstructive surgery is considered only for severe refractory disease.

6.17 Fly agaric in “gummies”: toxicological risks and their true nephrological dimension

Published September 25, 2026 · [online version \(Slovak\)](#)

Abstract. This review of toxicological data on fly agaric (*Amanita muscaria*), sold as gummies and other supplements, separates two questions: whether the mushroom damages the kidneys and what the patient actually ingested. The largest clinical series of 34 cases recorded not a single case of acute kidney injury, and the FDA memorandum does not mention kidney involvement. Laboratory analyses, however, have repeatedly shown that product contents do not match the label; for example, seven of eight contained psilocin. AKI after mushroom ingestion should therefore prompt a search for another mushroom species or another substance.

Summary

- The active substances are muscimol (a GABA A receptor agonist), ibotenic acid (an NMDA receptor agonist), and muscarine; muscarine content measured in 2025 was 0.004% to 0.043%. Symptoms begin within 30 minutes to 2 hours, with alternating depression and excitation and no uniform toxidrome.
- In a series of 34 cases from 2002 to 2016, 25 were symptomatic and 5 were intubated, with no AKI and no deaths. The only report of kidney failure with rhabdomyolysis is a conference abstract, the lowest level of evidence, and AKI in this poisoning is probably secondary.

- Latency is decisive: nephrotoxic white *Amanita* species (*A. smithiana*, *A. echinocephala*, and others) cause reversible AKI with interstitial nephritis after 10 to 12 hours, orellanine from the genus *Cortinarius* causes often irreversible injury after 1 to 2 weeks, *Tricholoma equestre* causes rhabdomyolysis, and amatoxins cause liver injury.
- In an analysis of eight commercial products, two declared extracts contained no ibotenic acid or muscimol but did contain psilocin; the outbreak linked to Diamond Shrooms bars involved 180 cases, 73 hospitalizations, and 2 deaths. In 2024, the FDA designated these substances as unapproved food additives.
- The history should cover mushroom products, mushroom foraging during the preceding two to three weeks, latency, and preservation of any remnants; routine toxicology screening does not detect these substances. With seizures, creatine kinase and myoglobin should be monitored.
- No antidote exists, and treatment is supportive; KRT is indicated according to standard nephrologic criteria, not for exposure alone. For AKI after mushroom ingestion, the author recommends consultation with the National Toxicological Information Center and mycological identification.

CHAPTER 7

Hemodialysis and extracorporeal therapies

Dose and technique of hemodialysis and hemodiafiltration, vascular access, volume management, complications, quality of life, and the organization of dialysis care. Number of articles: 37.

Key messages of the chapter

The strongest evidence in this chapter concerns convection dose. In the CONVINCe RCT with 1,360 patients, 17.3% died with high-volume postdilution online hemodiafiltration versus 21.9% with high-flux hemodialysis (7.16, 7.27). Guidelines of the Italian Society of Nephrology based on the GRADE methodology therefore recommend OL-HDF to reduce cardiovascular mortality, and an individual participant data meta-analysis showed a graded relationship between convection volume and the risk of death (7.27). The EuCliD registry cohort of incident patients describes a similar association, which, however, diminishes in a more rigorous analysis and does not prove causality (7.31). MCO dialyzers improve laboratory clearance of middle molecules without evidence of an effect on survival (7.16). The benefit is thus tied to an actually delivered volume of more than approximately 23 L, which is limited by vascular access, blood flow, and treatment time (7.16, 7.27).

The second axis is volume management, where observational and pilot data predominate. Dry weight is a dynamic target that is best determined multimodally: clinical assessment and BP trends form the basis, and bioimpedance, the relative blood volume curve, and POCUS complement them (7.8). Machine learning models were least accurate precisely in the sickest patients with malnutrition (7.9), and a model with relative blood volume predicted hypotension in new patients only weakly, without external validation (7.32). A small crossover RCT of low-sodium dialysate did not demonstrate an overall decrease in tissue sodium (7.3). A traffic-light educational diary improved adherence to recommendations in a small RCT (7.15). According to a meta-analysis of observational studies, 55% of patients have muscle cramps, which should be managed mainly by adjusting dry weight and ultrafiltration, not with quinine (7.20). In a small prospective study, disequilibrium syndrome occurred in 22.9% of incident patients and was associated with intradialytic hypertension, not with predialysis urea (7.11).

Vascular access and safety form the third area. A Korean cohort supports creating an AVF at least three months in advance (7.14), and a negative surgical recommendation is a prognostic marker that should prompt consideration of both PD and palliative care (7.34). A pragmatic RCT showed that an EDTA lock reduces mainly catheter occlusions, not infections (7.4); the SWAM technique is promising but lacks comparative data (7.10). In bloodstream infections, the catheter remains the main modifiable risk, and empirical therapy is guided by the local antibiogram (7.23). Cardiac arrest in the dialysis unit requires a written protocol, including aspiration of the lock solution (7.24). Incremental hemodialysis has no kidney-protective effect proven in RCTs (7.36).

Symptoms and frailty are often overlooked. Patients do not report pruritus on their own, and antihistamines often do not help (7.28), whereas difelikefalin improved itch in the KALM RCT despite a high placebo response (7.18). Frailty is common even under 65 years of age (7.26), and the nurse-assessed CFS scale predicts mortality (7.33); CT myosteatosis does not replace functional assessment (7.22). Olfactory dysfunction (7.35), blood group A (7.17), the relationship of GLP-1 receptor agonists to ESA consumption (7.29), and predictors of tenapanor dose (7.30) are observational or post hoc signals without treatment implications. In the pharmacotherapy of obesity, dosing must be adjusted for dialysis (7.6), and IV L-carnitine does not completely eliminate TMAO production (7.7). Wearable sensors have so far been validated mainly for the hemoglobin trend (7.25).

System-level data add context. The ERA Registry shows faster growth in prevalence than in incidence in Central and Eastern Europe and a declining prevalence of PD in Slovakia (7.13); a comparison of Europe and the United States reveals sex differences (7.5), and a US analysis links insurance coverage with an optimal start of treatment (7.2). Decision-making about modality is hampered by the reluctance of a quarter of patients to discuss values, which trust in the nephrologist mitigates (7.19).

This yields a framework for the outpatient clinic: plan the modality and AVF in good time through shared decision-making with involvement of the care partner (7.1, 7.14, 7.37); in suitable patients, prefer high-volume OL-HDF with measurement of the delivered volume (7.27); reassess dry weight multimodally (7.8); actively ask about itch, cramps, and frailty (7.20, 7.28, 7.33); and in frail older adults, offer palliative tools including conservative management and deprescribing (7.21). The carbon footprint can be reduced first by measures that do not affect treatment (7.12).

7.1 Trends in dialysis care: access, home modalities, and the patient at the center

Published May 13, 2026 · [online version \(Slovak\)](#)

Abstract. Based on an interview by Joel Topf with Ann Hinckley, a former dialysis nurse and advanced practice nephrology provider, at the NKF Spring Clinical Meetings 2026, this text summarizes current trends in dialysis care. It is not a research study but an expert commentary without quantitative data. The main message is a shift from a uniform model toward individualized treatment with shared decision-making, better access, and expansion of home modalities. In practice, this calls for timely predialysis education and team-based care with a strong role for nurses.

Summary

- Access to dialysis care involves not only the distance to the center but also access to medications, home therapy, education, vascular access, transplant evaluation, and a multidisciplinary team.
- Peritoneal dialysis and home hemodialysis can offer greater flexibility and autonomy, but their expansion depends on timely education, predialysis care, trained staff, family support, and continuous follow-up; not every eligible patient is given a real choice.
- Dialysis nurses, nurse practitioners, and physician assistants play a key role in detecting changes in clinical status, supporting adherence, and educating patients, because quality of care is often determined by continuity and communication.
- A patient with end-stage kidney failure comes with a family, a job, concerns, transportation options, and comorbidities, so treatment should be grounded in this context and based on education and shared decision-making.
- Preparation for KRT begins in the outpatient clinic months to years before the first dialysis session, with vascular access planning, discussion of transplantation, and modality selection; combined eGFR and UACR testing can improve early detection of CKD and identify patients for nephrology follow-up.

7.2 Medicaid expansion and “optimal starts” in incident kidney failure: results of an observational cohort analysis (2008 to 2019)

Published May 26, 2026 · [online version \(Slovak\)](#)

Abstract. An observational interrupted time-series analysis of the US Renal Data System from 2008 to 2019 assessed whether the 2014 Medicaid expansion was associated with the proportion of optimal starts of kidney failure treatment, that is, preemptive transplantation, home dialysis, or hemodialysis with arteriovenous access. Among adults aged 26 to 64 years, expansion states had an adjusted proportion of optimal starts ap-

proximately 3.9% higher by 2019, driven mainly by home dialysis. This is an association without proof of causality, suggesting the importance of insurance coverage for KRT preparation.

Summary

- Data from 323,807 patients in Medicaid expansion states and 271,725 patients in nonexpansion states were analyzed; trends were similar before 2014 and diverged thereafter, and the change in trend was statistically significant ($P = 0.02$).
- Most of the difference was explained by home dialysis, whose share grew 0.29% per year faster in expansion states. Preemptive transplantation was consistently higher in expansion states and hemodialysis with arteriovenous access increased, but neither showed a break in trend after 2014.
- Medicaid coverage in expansion states rose 1.97% per year faster, and the uninsured share fell 1.04% per year faster. Nephrology care before treatment initiation was improving but did not accelerate after 2014, and a separate analysis of incidence did not support an explanation based on growth in the number of treated patients.
- Limitations include unmeasured state-level confounders, concurrent interventions outside Medicaid, lack of clinical outcomes after treatment initiation, and restriction to ages 26 to 64 years.
- The results signal that access barriers and insurance coverage may affect the quality of treatment entry, particularly the availability of home dialysis; confirmation requires studies with more rigorous causal identification.

7.3 Low-sodium dialysate and tissue sodium: what a randomized crossover study showed

Published May 26, 2026 · [online version \(Slovak\)](#)

Abstract. A randomized crossover study in patients on maintenance hemodialysis compared dialysate sodium of 132 mEq/L and 138 mEq/L over four-week periods, with tissue sodium measured by ^{23}Na MRI. Of 28 randomized participants, 23 provided complete data. Overall differences in skin and muscle sodium were not statistically significant, but after adjustment, low-sodium dialysate was associated with a decrease in skin sodium, and a higher ultrafiltration rate with lower sodium in both compartments. Larger studies are needed to establish clinical relevance.

Summary

- Each period lasted 4 weeks, and tissue sodium was measured at its end by ^{23}Na MRI in skin and muscle. Compared with 138 mEq/L dialysate, the estimated difference with low-sodium dialysate was -0.8 in muscle ($p = 0.27$) and -1.5 in skin ($p = 0.09$), with no statistically significant overall difference, although both compartments tended to decrease.
- After adjustment for baseline tissue sodium and ultrafiltration rate, low-sodium dialysate was associated with a decrease in sodium in skin but not in muscle. A higher ultrafiltration rate was significantly associated with lower sodium in both skin and muscle.
- hsCRP and prealbumin levels did not differ significantly between periods, so the dialysate change produced no apparent systemic effect, and no serious study-related adverse event was reported with either regimen.
- Limitations are the small number of analyzed participants (23 with data from both phases) and the lack of clinical endpoints, so the authors call for larger studies. The results shift attention from the dialysate itself to the overall treatment prescription, including ultrafiltration.

7.4 EDTA lock solutions: can they improve central venous catheter patency?

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. A pragmatic, multicenter, cluster-randomized, triple-blind crossover trial published in JAMA compared a 4% tetrasodium EDTA lock solution with saline, or with 4% citrate for hemodialysis lines, in 1,468 ICU patients in six Canadian hospitals. The combined incidence of catheter infection and occlusion de-

creased from 19.9 to 13.1 events per 1,000 catheter-days (incidence rate ratio, 0.68). The benefit derived mainly from fewer occlusions, not from a reduction in infections. EDTA is a promising strategy for maintaining central venous catheter patency.

Summary

- Units used blinded syringes with 2.5 mL of lock solution, each strategy for 3.5 months with a 1-month washout interval; the mean patient age was 60 years, and all patients had at least one unused lumen.
- The primary composite outcome comprised catheter-related bloodstream infection, occlusion requiring alteplase, and catheter removal for occlusion; the incidence rate ratio was 0.68 (95% CI, 0.47 to 0.96; $P = 0.03$), corresponding to a relative reduction of approximately one third.
- The effect was driven mainly by a reduction in occlusions requiring alteplase, whereas the difference in catheter-related bloodstream infections was not statistically significant, so EDTA cannot be presented as proven infection prevention.
- Reducing occlusions matters in practice, because a blocked lumen means alteplase administration, treatment delay, catheter manipulation, and a risk of premature removal. No relevant safety signal was reported. EDTA acts as a cation chelator with anticoagulant and antibiofilm effects in the catheter lumen.
- Before routine adoption, the author recommends considering the cost and availability of the solution, the local incidence of occlusions, compatibility with the catheter type, standardization of administration, and staff training.

7.5 Sex differences in dialysis and transplantation: Europe vs. the US according to the ERA Registry and USRDS

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. A registry analysis published in *Nephrology Dialysis Transplantation* compared 2022 data from the ERA Registry and USRDS on KRT incidence and prevalence, transplantation rates, and mortality, separately for women and men. KRT incidence was 388.7 per million population in the US and 146.2 in Europe, with a larger difference in women (3.2-fold versus 2.4-fold in men). The relationship between sex and mortality differed between regions. The results are an epidemiologic signal, not a causal explanation, and support systematic monitoring of outcomes in both sexes.

Summary

- Women accounted for 35% of patients starting KRT in Europe and 41% in the US. From 2013 to 2022, incidence among women in Europe was stable (+0.1% per year) while it increased among men (+1.1% per year), whereas in the US it increased similarly in both sexes.
- KRT prevalence as of December 31, 2022, was 2,444.2 per million population in the US and 1,218.6 in Europe, 2.2-fold higher in the US among women and 1.9-fold higher among men; women accounted for 38% and 41% of patients on KRT, respectively.
- The transplantation rate was 79.1 per million population in the US versus 45.4 in Europe, and the proportion of women among recipients was similar (37% and 39%).
- Mortality on KRT was 145.0 per 1,000 patient-years in the US and 100.5 in Europe; in Europe, women had lower mortality than men (93.7 versus 104.6), whereas in the US it was higher (148.9 versus 142.2).
- Intercontinental comparison is affected by differences in definitions and coding, transplantation policy, comorbidity management, and timing of KRT, so this is an epidemiologic signal without identification of a mechanism.
- The author recommends timely KRT planning in the outpatient clinic, thorough management of comorbidities, and systematic monitoring of outcomes in both sexes.

7.6 Pharmacological treatment of obesity in patients with advanced CKD and on dialysis: practical dosing, dialysis-specific considerations, and safety

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. This practical review, based on full-text material, summarizes the pharmacotherapy of obesity in advanced CKD and on dialysis with respect to elimination, removal by dialysis, titration, and safety. The key finding is that approximately 50% of topiramate is removed by dialysis, so it is dosed after dialysis, whereas naltrexone is not removed by dialysis. GLP-1 receptor agonists and tirzepatide are eliminated by proteolysis independently of the kidneys, but data in patients on dialysis are limited. In practice, the drug should be chosen individually according to GFR, dialysis regimen, and risks.

Summary

- Naltrexone is metabolized in the liver, excreted by the kidneys, and not removed by dialysis; the maintenance dose is 16 mg twice daily with 180 mg of bupropion, and the patient must be opioid-free for 7 to 10 days before initiation.
- Topiramate is 60 to 70% excreted in urine, its clearance is approximately halved at GFR below 30 mL/min, and dialysis removes approximately 50% of the dose, so it is given after dialysis; risks include paresthesia, cognitive difficulties, metabolic acidosis, glaucoma, and kidney stones.
- Bupropion is metabolized via CYP2B6 to active metabolites excreted by the kidneys, so at GFR below 30 mL/min/1.73 m², it is started low (IR 75 mg daily, ER 150 mg every other day or daily); seizure disorders and severe electrolyte disturbances are warnings.
- Approximately 97% of orlistat is excreted in feces, it requires no dose adjustment in ESKD, and it is given at 60 to 120 mg three times daily, but steatorrhea, malabsorption of fat-soluble vitamins, and the risk of oxalate nephropathy should be kept in mind.
- Liraglutide is titrated to 3.0 mg daily and semaglutide to 2.4 mg weekly, with lower doses in dialysis, and in ESKD, tirzepatide is recommended to be started at 2.5 mg weekly with titration to a maximum of 15 mg; medullary thyroid carcinoma and MEN2 are contraindications.
- For tirzepatide and other GLP-1 receptor agonists, an evidence gap remains in dialysis, and tirzepatide has no ESKD-specific approval, so careful titration with monitoring of gastrointestinal tolerance is needed.

7.7 Does injectable L-carnitine prevent TMAO formation? What the data really say

Published July 4, 2026 · [online version \(Slovak\)](#)

Abstract. This analysis of mechanistic and clinical data assesses whether intravenous L-carnitine after hemodialysis prevents TMAO formation. The conversion of carnitine to trimethylamine is obligately microbial, as confirmed by Koeth et al. (Nature Medicine 2013), so parenteral administration markedly attenuates but does not abolish TMAO formation. In patients on dialysis, TMAO levels are determined mainly by absent kidney elimination, and a benefit of TMAO reduction for hard cardiovascular outcomes has not been demonstrated. When correcting carnitine deficiency, IV administration is more rational than oral.

Summary

- The gut microbiota converts L-carnitine to trimethylamine, which the hepatic enzyme FMO3 oxidizes to TMAO; in germ-free mice and in humans after antibiotics, virtually no TMAO is formed from carnitine. Oral supplementation in patients on hemodialysis normalizes carnitine but substantially increases TMAO.
- The intravenous route does not completely eliminate TMAO formation, because carnitine undergoes enterohepatic circulation and part of it is excreted into the gut. In patients on dialysis, the level of TMAO as a uremic retention solute is determined mainly by absent kidney elimination, with rebound between sessions, which the route of carnitine administration does not fundamentally change.

- The association of TMAO with atherosclerosis and MACE is mechanistically supported, but these are associative data without interventional evidence that lowering TMAO improves hard cardiovascular outcomes.
- Carnitine deficiency is common in patients on hemodialysis and is associated with ESA-resistant anemia, intradialytic hypotension, cramps, and inflammation, but evidence of a benefit from supplementation is mixed.
- The reported dosing is 10 to 20 mg/kg IV after each HD session, commonly 1 g IV after HD, or 500 to 1,000 mg daily orally; a high oral dose of 3 g daily increased triacylglycerols and platelet aggregation in a small study.

7.8 Determining dry weight (EDW) in hemodialysis: clinical assessment, BCM, BVM, and POCUS

Published July 6, 2026 · [online version \(Slovak\)](#)

Abstract. This expert review discusses the determination of estimated dry weight (EDW) in hemodialysis as a dynamic therapeutic target rather than the output of a single formula. It compares clinical assessment, the cardiothoracic ratio, bioimpedance spectroscopy (BCM), relative blood volume monitoring, POCUS of the inferior vena cava and lungs, biomarkers, and machine learning models. The key conclusion favors a multimodal approach in which the clinical picture and blood pressure trends form the foundation and objective methods complement it. The goals are euvolemia, better blood pressure control, and fewer intradialytic complications.

Summary

- An EDW set too high leads to chronic hypervolemia, hypertension, left ventricular hypertrophy, and higher cardiovascular risk, whereas one set too low leads to intradialytic hypotension, cramps, and a risk of myocardial or cerebral ischemia.
- A cardiothoracic ratio below 48% combined with normotension without antihypertensives is a historical criterion affected by cardiac pathology and radiographic technique. BCM quantifies overhydration and reveals silent hypervolemia, but amputations, implants, or extreme BMI complicate interpretation.
- The relative blood volume curve is valuable for trends: a steep decline with hypotension signals poor tolerance of ultrafiltration but does not determine absolute dry weight. A dilated inferior vena cava with low collapsibility supports congestion, and a strategy guided by lung ultrasound B-lines reduced ambulatory blood pressure in a randomized trial.
- NT-proBNP, albumin, and CRP are nonspecific, and machine learning models such as Random Forest or XGBoost have only a supporting role without external validation.
- The author recommends first distinguishing water from a change in tissue mass, assessing both hypervolemia and hypovolemia, changing the target in small steps together with adjustment of salt intake and HD duration, verifying home blood pressure, and reassessing EDW after hospitalization, infection, or a change in urine output.

7.9 Artificial intelligence in determining dry weight in hemodialysis patients

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Abstract. A retrospective single-center study from Chungnam National University Hospital, published in PLOS One, assessed in 1,672 patients on hemodialysis whether machine learning models predict clinically appropriate dry weight better than bioimpedance spectroscopy. In the entire cohort, accuracy was below 40%, but when the difference between the bioimpedance and clinical values was below 1 kg, XGBoost

achieved approximately 83%. Large discrepancies were associated with extracellular overhydration and signs of malnutrition. The algorithm can complement, not replace, clinical decision-making.

Summary

- The authors compared BIS-derived dry weight (DW BIS) with the clinically appropriate value (DW CP) and termed their difference Gap DW; it was below 1 kg in 972 patients, 1 to 2 kg in 303, and at least 2 kg in 384.
- XGBoost, LightGBM, and random forest models were trained on 80% and tested on 20% of the data in 20 repetitions; in the entire cohort, their accuracy was below 40%, with an error in fluid excess of up to approximately 1.5 kg, and with Gap DW below 2 kg, approximately 72%.
- With increasing Gap DW, hemoglobin, total protein, albumin, creatinine, phosphorus, potassium, and fat mass index decreased, while extracellular water and the ECW/ICW ratio increased, indicating overhydration with altered body composition and malnutrition.
- In malnutrition, extracellular water may be increased despite insufficient intravascular volume, and aggressive weight reduction guided by BIS then leads to hypotension and cramps.
- Limitations include the retrospective single-center design, measurements by a single operator, lack of external validation, and no demonstrated effect on mortality or hospitalizations; the models were least accurate precisely in the clinically most severe patients.
- The author recommends determining dry weight by combining blood pressure, intradialytic symptoms, edema, interdialytic weight gains, nutritional parameters, residual urine output, and BIS or lung ultrasound results.

7.10 The SWAM technique for hemodialysis access thrombosis: a new option for mechanical thrombectomy

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. A technical description with initial clinical experience, published in Hemodialysis International, presents the SWAM technique (Separator With Aspirator for Mechanical Thrombectomy) for mechanical thrombectomy of hemodialysis access. It combines a large-bore aspiration catheter with a separator that fragments the thrombus and keeps the system patent. A favorable effect was observed mainly with acute thrombus and only partially with chronic thrombus. This is not a randomized study, so SWAM should be regarded as a promising option for access salvage, not as a new standard.

Summary

- The technique was used for thrombosis of arteriovenous fistulas, arteriovenous grafts, and central venous catheters, with the aim of removing thrombus more effectively and shortening procedural time; a shorter procedure may mean less contrast medium and lower radiation exposure, but this needs to be confirmed in comparative studies.
- Chronic thrombus is firmer and often associated with stenosis or fibrosis, so it may require a combination of techniques or surgical management; the sooner thrombosis is treated, the greater the chance of salvaging the access.
- Thrombus removal alone is insufficient unless a hemodynamically significant stenosis is treated at the same time, in grafts most often by balloon angioplasty. With a catheter, a fibrin sheath, malposition, central stenosis, and infection must be distinguished.
- Data are lacking on long-term patency, reinterventions, complications, costs, and comparison with other mechanical or pharmacomechanical techniques; prospective studies with patency through 12 months are needed.
- The nephrologist should promptly recognize warning signs of dysfunction, such as reduced flows, higher venous pressures, prolonged bleeding, or a weakened thrill, and expedite ultrasound or interventional evaluation in collaboration with the interventional radiologist and vascular surgeon.

7.11 Dialysis disequilibrium syndrome: an underdiagnosed complication at the start of hemodialysis

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. Dialysis disequilibrium syndrome (DDS) is traditionally explained by a rapid fall in urea, but prospective data in adults have been limited. A prospective, observational, single-center study from France followed 48 patients during their first four hemodialysis sessions using a standardized symptom score. DDS occurred in 22.9% of patients, predialysis urea was not a predictor, and intradialytic hypertension was an independent risk factor (OR 4.43). When starting hemodialysis, therefore, not only urea but also blood pressure, pH, hydration, and neurologic symptoms should be monitored before and after the session.

Summary

- The study enrolled 48 patients starting hemodialysis (70.8% men, median age 67 years); DDS was defined as any increase in the severity score of 0 to 3 after dialysis compared with before (Δ score ≥ 1).
- DDS occurred in 22.9% of patients, and two had more than one episode, so with systematic screening, it is not a rare complication.
- In univariate analysis, DDS was associated with centrally acting drugs (45.5% vs. 8.1%), overhydration (45.4% vs. 13.5%), lower predialysis pH (7.28 vs. 7.37), and higher chloride (100 vs. 94 mmol/L); predialysis urea was not significant ($P = 0.15$).
- In multivariate analysis, intradialytic hypertension was an independent risk factor (OR 4.43; 95% CI, 1.38 to 18.15), occurring in 78.6% of sessions with DDS versus 40.0% of sessions without DDS.
- Limitations are the single-center design, the small sample, and a sensitive definition that may also capture nonspecific symptoms; the results are a signal, not a definitive risk model, and the authors call for multicenter validation.
- In high-risk patients, the author recommends not starting with aggressive dialysis, increasing the dose gradually, individualizing ultrafiltration, monitoring pH, hydration, and blood pressure, and documenting neurologic symptoms before and after the session.
- When DDS is suspected, dialysis intensity should be reduced, interruption and neurologic evaluation should be considered for more severe manifestations, and other causes, such as hypoglycemia, hypertensive encephalopathy, hemorrhage, hyponatremia, or sepsis, should be excluded.

7.12 The carbon footprint of hemodialysis: measurement, practical measures, and the limits of emission reduction

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. Hemodialysis consumes large amounts of energy, water, transportation, and single-use materials. A pilot uncontrolled study in Nephrology Dialysis Transplantation evaluated the DGfN web-based calculator in five German hemodialysis centers; it was neither an RCT nor a full LCA. The mean operational carbon footprint was 3.72 ± 0.44 t CO_{2e} per patient per year and decreased by 9.1% between 2021 and 2023, mainly in the energy, heating, and water category. The decrease cannot be causally attributed to individual measures; clinical changes, such as a reduction in Qd, require individualization and verification of the delivered dose.

Summary

- The calculator covered transportation, consumables and waste, electricity, heating, and water, including RO, and other operations; it did not include manufacture of machines, drugs, buildings, or transport of materials, so it is an operational inventory with a limited boundary.
- Consumables and waste accounted for approximately 40% of the footprint; the total decreased from 3.91 to 3.56 t CO_{2e} per patient per year ($p = 0.04$), and the energy, heating, and water category by 20.5% ($p = 0.01$).

- Reducing Qd from 500 to 350 mL/min in 50 to 80% of selected patients reduced RO water consumption by 14%; the decrease in the electricity and water footprint was not significant ($p = 0.07$), and a four-week laboratory comparison in one center provided no data on Kt/V or long-term safety.
- Photovoltaics reduced the footprint of the electricity component in one center by 22%, a change in refreshments reduced the catering footprint by 53% in modeling (approximately 2.6 to 3% of the total), and group transportation made only a small difference.
- The scenario of a 38.7% reduction is a hypothesis assembled from models, including a 40% share of APD and incremental HD, not a measured result, and the savings may overlap when summed.
- The author recommends defining the calculation boundary, collecting data for at least 12 months, starting with measures that do not affect treatment, and governing any clinical change in Qd, temperature, or frequency by a protocol with monitoring of dose, electrolytes, and symptoms; modality choice remains a medical decision.

7.13 Kidney replacement therapy in Central and Eastern Europe: incidence and prevalence trends in the ERA Registry

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. This retrospective observational analysis of the ERA Registry describes trends in kidney replacement therapy in 19 countries of Central and Eastern Europe from 2010 to 2019, extended to 2021. Unadjusted KRT incidence increased by an average of 1.5% per year, similar to Western Europe, but prevalence increased by 5.1% per year, approximately twice as fast. Differences between countries were large, and rates were not standardized for age or sex. In Slovakia, the incidence of dialysis initiation was stable and PD prevalence declined, and the most practical conclusion is the need for a complete registry.

Summary

- In Central and Eastern Europe, KRT incidence increased from 106.3 to 119.6 pmp (AAPC +1.5%) and prevalence from 426.2 to 651.2 pmp (AAPC +5.1%); in Western Europe, prevalence increased by 2.6% per year.
- Incidence in 2019 ranged from 39.9 pmp in Ukraine to 283.8 pmp in Cyprus, an approximately sevenfold difference, and prevalence from 244.1 pmp to 1,413.3 pmp in Greece.
- In the region, hemodialysis accounted for approximately 66% of prevalent KRT, a functioning transplant for 29%, and PD for approximately 5%; growth in the number of patients with a functioning graft contributed most to the rise in prevalence, but the analysis does not explain the mechanism.
- In Slovakia, the incidence of dialysis initiation was statistically stable (AAPC -1.2%), hemodialysis prevalence increased (+2.0%), and PD prevalence declined (-3.3%); total KRT figures are not comparable, because transplant recipients were missing in 2010, 2011, and 2021.
- The study had no data on mortality, comorbidities, or patients receiving conservative care, registry completeness varied, and neither incidence nor prevalence is a measure of health system quality.
- The author recommends a complete Slovak registry including preemptive transplantation and conservative care, publication of both unadjusted and standardized rates, modality-neutral education, an audit of barriers to PD, and timely referral of suitable candidates for transplant evaluation.

7.14 Timing of vascular access before hemodialysis: an AVF needs more lead time than an AVG

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. Late vascular access leads to starting hemodialysis via a catheter, whereas a procedure performed too early burdens the patient unnecessarily. A retrospective nationwide Korean cohort of 24,713 patients who started hemodialysis in 2013 assessed the lead time of AVF and AVG creation. For AVF, creation less than three months before dialysis was associated with worse outcomes, and an interval of less than one

month with an OR of 14.03 for catheter use. For AVG, unfavorable associations were concentrated in the last month. The result supports a timely, individualized ESKD Life-Plan rather than a fixed calendar.

Summary

- The cohort comprised 20,894 patients with an AVF and 3,819 with an AVG from the Korean National Health Insurance Service database; the reference interval was 6 to less than 9 months.
- An AVF created less than one month before dialysis had an OR of 14.03 for catheter use, an HR of 2.64 for loss of primary patency, an HR of 4.56 for loss of secondary patency, and an HR of 1.67 for mortality; an interval of 1 to 3 months was also unfavorable (OR 1.85; mortality HR 1.25).
- An AVG created less than one month in advance had an OR of 8.45 for catheter use and an HR of 1.43 for mortality, and with an interval of 1 to 3 months, only loss of primary patency was higher.
- These are observational data without absolute risks; a late procedure may be a marker of rapid progression or late referral, and results from 2013 may not apply to Slovak conditions.
- The study did not determine the time of first cannulation; the European guideline specifies AVF cannulation from four weeks, not before two weeks, and at least two weeks for a standard AVG.
- The author recommends creating an AVF at least approximately three months in advance when dialysis is likely to be needed, preserving veins, checking maturation after approximately 4 to 6 weeks, and basing the plan on the eGFR trend and the risk of kidney failure according to KDIGO 2024 (two-year risk above 40%).

7.15 A “traffic light” diary in hemodialysis: better short-term fluid management in a small RCT

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. Fluid management in maintenance hemodialysis balances volume overload against intradialytic hypotension. A small 12-week RCT of 80 patients compared a traffic-light diary, combining self-monitoring, feedback, and education, with usual education. The intervention increased adherence to recommendations from 75% to 95% and improved knowledge, attitude, and practice scores; IDWG, UFR, blood pressure, and the incidence of hypotension and cramps were more favorable, but the abstract does not report effect sizes. It is a promising educational adjunct, not a universal protocol.

Summary

- Patients were randomized 40 per group; the control group received routine treatment and usual education, and the intervention group additionally used the traffic-light diary for 12 weeks.
- The intervention group had higher scores for knowledge (9 vs. 7 points), attitudes (9.2 vs. 7.0), and practical habits (22.9 vs. 18.7 points) and adherence to recommendations of 95% versus 75%.
- IDWG and UFR were lower, blood pressure control was better, and hypotension and cramps during dialysis were less frequent ($p < 0.05$), but the abstract does not report values, event counts, or confidence intervals.
- Limitations include the small sample, short follow-up, unblinded intervention, lack of a description of the diary's color thresholds, multiple comparisons, and absence of data on hospitalizations or mortality; it has not been shown that the colors themselves caused the effect.
- The author recommends preparing a local diary with the patient according to target weight, residual urine output, and dialysis regimen and recording weight, symptoms, sodium, and home blood pressure; the diary must not lead to unilateral changes in target weight, medications, or ultrafiltration and does not replace clinical and objective assessment of volume status.

7.16 Beyond diffusion: online hemodiafiltration and MCO dialyzers in the treatment of kidney failure

Published July 25, 2026 · [online version \(Slovak\)](#)

Abstract. Conventional hemodialysis poorly removes middle molecules such as β_2 -microglobulin or free light chains. This review compares online hemodiafiltration and MCO dialyzers. Randomized evidence exists for high-volume postdilution OL-HDF: in the CONVINCe trial of 1,360 patients, 17.3% died versus 21.9% with high-flux hemodialysis (HR 0.77). MCO dialyzers mainly improve laboratory clearance, and survival evidence is lacking. Modality selection depends on vascular access, achievable convection volume, and the patient's condition.

Summary

- The clinical benefit of OL-HDF is associated with a convection volume of at least approximately 23 liters per session, a practical target derived from trials rather than a biological threshold; it generally requires a blood flow above 350 mL/min, and extending the session by 30 minutes adds about 2 to 3 liters.
- In the CONVINCe trial, the absolute difference in mortality was approximately 4.6 percentage points over a median of 30 months; the trial was open-label and included patients able to achieve high volumes, which may have favored them.
- It has not been adequately demonstrated that low-volume HDF also confers benefit, and the dose-survival relationship in observational analyses may not be causal.
- MCO dialyzers (HDx) remove larger middle molecules more effectively without substitution fluid, but they have not shown reductions in mortality or hospitalizations or improvement in quality of life, and they may slightly lower serum albumin.
- Kt/V reflects only urea removal; with OL-HDF, convection volume, time, effective blood flow, filtration fraction, and transmembrane pressure should also be recorded.
- The author recommends preferring OL-HDF when ultrapure water and access with adequate flow are available, considering an MCO dialyzer when OL-HDF is unavailable or with a catheter or low flows, and monitoring albumin and nutritional status with MCO.

7.17 Blood group A and lower mortality on hemodialysis: an intriguing association with no implications for current treatment

Published August 2, 2026 · [online version \(Slovak\)](#)

Abstract. In the general population, blood group O is the one associated with lower thrombotic risk. However, the Japanese prospective multicenter Osaka Dialysis Complication Study cohort of 1,671 patients on hemodialysis found that blood group A, compared with group O, was associated with lower all-cause mortality (HR 0.780) and lower cardiovascular mortality. This is an observational association without an explained mechanism, and adding blood group improved the prognostic model only minimally. The finding generates a research hypothesis and does not change prevention or treatment of patients on hemodialysis.

Summary

- Follow-up ran from 2012 to 2017 with a median of 1,826 days; 464 patients died, 278 of cardiovascular causes, and sudden death accounted for 123 of the cardiovascular deaths.
- Group A versus O had an HR of 0.780 (95% CI, 0.619 to 0.981) for all-cause mortality, which does not mean a 22% reduction in absolute mortality; groups B and AB showed no significant association, and no association was found for noncardiovascular mortality.
- An exploratory comparison of A versus non-A for cardiovascular mortality showed an HR of 0.733, but it was neither prespecified nor adjusted for multiplicity.
- Adding blood group raised Harrell's C-statistic only from 0.749 to 0.751, so its clinical prognostic utility is minimal.

- Limitations include the observational design, a prevalent cohort with possible survivor selection, single baseline measurements, no VWF or factor VIII measurements, no genotyping, and limited generalizability to European populations.
- Blood group should not alter cardiovascular prevention, antithrombotic therapy, dialysis dose, vascular access, or transplant evaluation; modifiable risk factors remain decisive.

7.18 The neuroimmune architecture of uremic pruritus: a four-node model and its therapeutic implications

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. CKD-associated pruritus (CKD-aP) is common and burdensome, yet its treatment remains largely empirical. In a DOPPS analysis, 51% of patients on hemodialysis had moderate to severe itch at least once, and persistent itch was associated with higher mortality (HR 1.29). A Brazilian review in the Journal of Nephrology groups the mechanisms into four neuroimmune nodes that explain the failure of antihistamines and the efficacy of gabapentinoids and difelikefalin. This is a hypothesis-generating framework, not validated endotyping, so node-directed treatment still requires prospective validation.

Summary

- In DOPPS, among 7,976 patients, 22% had persistent itch, which was associated with an HR of 1.17 for hospitalization and 1.48 for cardiovascular events; causal interpretation should be cautious, as itch may also be a marker of poorer overall condition.
- Node I is the cutaneous neuroimmune synapse, with xerosis as the only fully modifiable element; node II is neural hyperexcitability (the target of gabapentinoids); node III is μ and κ opioid receptor imbalance; and node IV is amplification via substance P and NK-1, which remains largely hypothetical.
- In a pooled analysis of KALM-1 and KALM-2, difelikefalin 0.5 $\mu\text{g/kg}$ IV after each hemodialysis session reduced itch by at least 3 points in 51.1% of patients versus 35.2% with placebo, although the placebo response was high.
- On dialysis, gabapentin and pregabalin are given at markedly reduced doses after the session because of accumulation and the risk of sedation and falls; antihistamines act mainly as sedatives and increase the risk of confusion in older patients.
- The author recommends actively asking about and measuring itch with the WI-NRS, excluding other causes, using emollients consistently, optimizing dialysis, and, for persistent itch, considering a low-dose gabapentinoid or difelikefalin, while not overlooking sleep and depression.

7.19 When the patient does not want to talk about their values: a hidden barrier to shared decision-making about dialysis

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. Shared decision-making about the kidney replacement therapy modality assumes that the patient articulates their values. A Japanese cross-sectional study with a convergent mixed-methods design included 474 patients with advanced CKD who had already chosen a modality. Unwillingness to share personal values was reported by 24.2% of patients, and they rated their involvement 11.39 points lower on the CollaboRATE questionnaire. Trust in the nephrologist was protective, whereas depressive symptoms increased unwillingness. A patient's silence should be read as information about the relationship or their psychological state, not as nonadherence.

Summary

- Minimally unwilling patients had CollaboRATE scores 5.12 points lower and very or somewhat unwilling patients 11.39 points lower, indicating a graded relationship.
- Trust in the primary nephrologist reduced unwillingness, whereas depressive symptoms increased it; willingness to discuss values is thus a property of the relationship, and silence or apathy may be a sign of untreated depression rather than the patient's stance.

- Qualitative analysis identified five themes: differing sensitivity to privacy, a preference for physician-led decisions, personality and psychological adaptation, prioritizing health over privacy, and trust that overcomes the barrier.
- Unwillingness had measurable consequences for process quality, because a patient who does not disclose their values leaves the physician to fill the gap with their own assumptions, so the decision remains formally correct but is in fact one-sided.
- These are cross-sectional associations without causality, based on self-reported variables, excluding patients who never reached a decision, and set in Japanese culture, so the 24.2% figure cannot be directly transferred to Slovakia.
- Before modality decisions, the author recommends screening for depressive symptoms, building trust continuously, explicitly asking about the preferred level of involvement, spreading the decision over several visits, and respecting the patient's right not to discuss private matters.

7.20 Skeletal muscle cramps on dialysis: a global prevalence of 55% and what to do about it in practice

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. Muscle cramps are among the most troublesome symptoms of dialysis, yet they are not systematically assessed. The first global meta-analysis of 94 observational and nonrandomized studies with 32,223 patients from 36 countries found a pooled prevalence of 55% for cramps and 33% for intradialytic cramps. However, prediction intervals of 20% to 86% and 4% to 87% indicate substantial heterogeneity, which likely reflects differing definitions and ascertainment methods. Cramps should be actively sought and managed primarily by adjusting dry weight and ultrafiltration, not with drugs.

Summary

- Prevalence was higher with hemodialysis than with PD, in incident than in prevalent patients, and with a four-week than with a one-week recall period; 53% of studies had a moderate risk of bias.
- Intradialytic cramps are associated mainly with rapid or excessive ultrafiltration and falling below true dry weight, intradialytic hypotension, rapid sodium shifts, and hypomagnesemia, hypocalcemia, or hypokalemia; cramps outside dialysis have a broader differential diagnosis, including neuropathy, diuretics, and statins.
- The narrow 95% CI of 50% to 59% describes the precision of the mean, whereas the wide prediction interval means that prevalence in one's own center cannot be inferred from the meta-analysis.
- Limitations include the observational nature, which cannot address causes or treatment, inclusion of English-language publications only, exclusion of hospitalized patients, and exclusively self-reported cramps.
- The author recommends actively asking about cramps during and outside dialysis, reassessing dry weight, lowering the ultrafiltration rate by extending or adding sessions and restricting salt, checking electrolytes including magnesium, and considering dialysate cooling in the presence of hypotension.
- Quinine should not be used because of the 2010 FDA warning about serious hematologic reactions; evidence for levocarnitine, vitamin E, and gabapentin is weak, and drug therapy should follow only after the dialysis regimen has been adjusted.

7.21 Palliative care in routine nephrology: low-threshold tools for frail and older patients

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. The number of patients reaching end-stage kidney disease at an advanced age and with frailty is growing, yet palliative principles are insufficiently implemented in practice. A narrative review in Nephrology Dialysis Transplantation covers prognostication, goals-of-care conversations, conservative management, individualized dialysis, deprescribing, and dialysis withdrawal. In a Dutch cohort, no survival difference between dialysis and conservative management was confirmed in patients aged 80 years and older (2.1

vs. 1.4 years; $p = 0.08$). Nephrologists can use these tools at a low threshold even without specialized palliative training.

Summary

- The surprise question about death within twelve months has only moderate accuracy and serves as a trigger for a goals-of-care conversation, not as a prognosis to be communicated to the patient.
- In a Dutch cohort of patients aged 70 years and older, median survival was 3.1 years with KRT versus 1.5 years with conservative management, and only 1.8 versus 1.0 years with a Davies score of 3 or higher; a British study showed a similar number of days out of hospital and a higher likelihood of dying at home or in a hospice (OR 4.15).
- A palliative approach can also be applied on dialysis through incremental initiation, symptom-guided prescription, realistic access choice including a catheter, and a time-limited trial.
- Deprescribing should regularly reassess statins, phosphate binders, hemoglobin targets, antihypertensives, and dietary restrictions.
- Dialysis withdrawal accounts for roughly one fifth of deaths; the plan should include anticipatory prescriptions for symptom control, the place of care, informing the family, and the course of action if the patient deteriorates.
- Limitations include the narrative nature of the review, the Central European context, and observational data on conservative management with marked selection bias.

7.22 Myosteatorsis in hemodialysis: CT measures of muscle quality are promising but do not yet replace functional assessment

Published August 15, 2026 · [online version \(Slovak\)](#)

Abstract. In hemodialysis, muscle weakness may result from impaired muscle quality, not only from muscle loss. A Japanese single-center cross-sectional study of 114 patients evaluated CT measures of myosteatorsis at the L3 level and their relationship to handgrip strength. Mean muscle density and three new indices were all independently associated with handgrip strength, with a virtually identical C-statistic of approximately 0.81. Without external validation, these measures cannot be introduced as standalone diagnostic tests; their greatest practical value lies in opportunistic assessment of existing CT scans followed by functional testing.

Summary

- Mean muscle density was 34.4 ± 9.0 HU, and the new indices correlated strongly with it ($r = 0.987$ for the proportion of normal-density muscle), which stems mainly from their shared mathematical construction rather than from independent biological information.
- Standardized regression coefficients for handgrip strength were 0.305 for density, 0.297 for the proportion of normal-density muscle, and -0.321 for the proportion of intermuscular fat; the value of -310 in the abstract is probably an error and most likely means -0.310 .
- After adjustment for age and sex, all measures had a C-statistic of 0.809 to 0.810 with no significant difference; two new indices held up in models of low-strength risk, which, with 114 patients, is only a fragile hypothesis.
- Limitations include the cross-sectional design, small sample, lack of external validation, the influence of CT protocol, contrast, segmentation, and hydration on density, and the absence of data on falls, hospitalizations, or mortality.
- CT does not assess strength or performance, so it cannot by itself diagnose sarcopenia according to EWGSOP2, and because of radiation exposure it should not be ordered as routine screening.
- The author recommends following a suspicious finding with standardized handgrip dynamometry, performance tests, nutritional assessment, and treatment of reversible causes; the interventions are resistance training and individualized nutrition, and no specific drug exists.

7.23 Bloodstream infections in hemodialysis: incidence is falling, but the microbiological spectrum may be shifting

Published August 15, 2026 · [online version \(Slovak\)](#)

Abstract. Bloodstream infections are among the most serious complications of hemodialysis. A retrospective single-center Irish cohort of 1,248 patients with 522 episodes from 2009 to 2023 used interrupted time series analysis. The proportion of staphylococci fell from 60% to 34.8% and the proportion of gram-negative bacilli rose from 18.5% to 43.9%, but an absolute increase was not demonstrated. After a new unit opened, the incidence of *S. aureus* fell (IRR 0.42), although this is a temporal association. The central venous catheter remains the main modifiable risk.

Summary

- The relative rise in gram-negative pathogens does not mean an absolute increase; the modeled incidence rate ratio for them after the new unit opened was 0.49 ($P = 0.048$), suggesting a decrease of border-line significance.
- A central venous catheter increased the risk of infection, and male sex was associated with an IRR of 1.43; the age threshold of over 34 years has little clinical value in the dialysis population.
- Limitations include the retrospective single-center design without randomization, possible changes in definitions and diagnostics over 15 years, missing data on trends in catheter use, and concurrent organizational changes, so causality cannot be established.
- When infection is suspected, blood cultures should be drawn before antibiotics, empirical therapy should cover MRSA and relevant gram-negative bacteria according to the local antibiogram, dosing should be adjusted for hemodialysis, therapy should be narrowed, and catheter removal should be strongly considered in sepsis, persistent bacteremia, or infection with *S. aureus*, *P. aeruginosa*, or fungi.
- Prevention rests on minimizing catheters, hand hygiene, aseptic handling, alcoholic chlorhexidine, hub disinfection, patient education, and audit with feedback.
- The author recommends monitoring incidence by access type, pathogen, and susceptibility, and not using the Irish results to guide empirical therapy in Slovakia, where the local antibiogram and de-escalation are decisive.

7.24 Cardiac arrest during hemodialysis: the extracorporeal circuit, the catheter lock, and the silent burden on dialysis nurses

Published August 23, 2026 · [online version \(Slovak\)](#)

Abstract. Cardiac arrest in the dialysis unit requires managing the extracorporeal circuit, the vascular access, and the machine in addition to resuscitation. This text fact-checks a commentary by Malaysian nursing educators against the European Resuscitation Council (ERC) 2021 and 2025 guidelines. The key correction is that blood in the circuit (approximately 150 to 300 mL) is returned to the patient together with a fluid bolus and not discarded, but without delaying compressions. Hyperkalemia is important but not proven to be the most common cause. In practice, a written protocol with assigned team roles is needed.

Summary

- The ERC calls for stopping dialysis, returning the blood volume with a fluid bolus, disconnecting the patient from the machine unless it is certified as defibrillation-proof, and leaving the vascular access in place for drug administration; a separate trained person operates the machine, otherwise compressions and defibrillation take priority.
- Before a tunneled catheter is used, the lock solution must be aspirated: at heparin concentrations of 1,000 to 10,000 U/mL and volumes of 1.3 to 2.0 mL per lumen, this amounts to 1,300 to 20,000 units; the same principle applies to a 4% citrate lock solution.
- Implantable loop recorder monitoring in 66 patients captured 1,678 arrhythmic events, including 1,461 bradyarrhythmias, 14 asystoles, and a single sustained ventricular tachycardia; dialysate potassium below 2 mmol/L, low dialysate calcium, and higher ultrafiltration volume also increase the risk of arrest.

- In a cohort of 32,065 patients, mortality after the long interdialytic interval was 22.1 versus 18.0 per 100 patient-years.
- For hyperkalemic arrest, the ERC specifies 10 mL of 10% calcium chloride (or 30 mL of 10% calcium gluconate), 10 units of insulin with 25 g of glucose, and 50 mmol of sodium bicarbonate; dialysis is the definitive treatment, and salbutamol and oral potassium binders have no role during arrest.
- The claim that the kidneys cannot eliminate resuscitation drugs is inaccurate, as epinephrine is degraded enzymatically and amiodarone in the liver; after return of spontaneous circulation, the cause, electrolytes, blood glucose, and continuation of elimination therapy are decisive.
- The author recommends a written protocol, rehearsed catheter handling, an on-site blood gas analyzer, review of the dialysis prescription, an agreed transfer route, and a structured post-event debriefing.

7.25 Wearable sensors in nephrology and dialysis: what already has validation data and what is still in development

Published August 23, 2026 · [online version \(Slovak\)](#)

Abstract. Wearable sensors promise early detection of deterioration, but in patients on dialysis, data from the general population translate poorly because of vascular changes, arteriovenous access, and rapid volume and electrolyte shifts. The review sorts applications by level of evidence. The most concrete is a prospective validation of a patch hemoglobin sensor in 116 patients on hemodialysis, with negligible bias but limits of agreement of approximately ± 2 g/dL. Arrhythmia detection and potassium measurement remain unvalidated. In practice, a device should be introduced only with verified accuracy and a clear decision pathway.

Summary

- A patch sensor over the fistula in 116 patients at four sites showed a hemoglobin bias of -0.04 g/dL, 95% limits of agreement of -2.00 to $+1.86$ g/dL, and $r = 0.64$; it is suitable for trend monitoring, not for changing the dose of an erythropoiesis-stimulating agent without laboratory confirmation.
- Limitations of the validation include a single session, exclusion of hemoglobinopathies, native fistulas only, and unclear performance across skin types.
- Published studies of noninvasive arrhythmia detection in kidney failure are lacking, and the clinical follow-through is unclear, since the indication for anticoagulation in atrial fibrillation in patients on dialysis is controversial.
- A deep learning model detected hyperkalemia from two ECG leads with an area under the curve of 0.883, but this is screening performance on a standard recording, not a validated wearable sensor.
- Bioimpedance is more sensitive than weight alone for guiding ultrafiltration, but it is a point-in-time assessment without convincing long-term outcomes; remote monitoring of home HD and PD has the strongest, mostly observational data.
- Prototypes of a wearable artificial kidney (AWAK, Carry Life, WEAKID, ViWAK) are in early stages; a wearable monitor is realistic, wearable dialysis is not.
- Before implementation, the author recommends verifying the clinical question, validation in the dialysis population, limits of agreement, responsibility for responding to alerts, the false alarm rate, data protection, and regulatory status; the risks include alert fatigue and the digital divide.

7.26 Frailty in dialysis patients younger than 65 years: what a cross-sectional study from Tbilisi adds

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Abstract. In nephrology, frailty is regarded as an issue of older patients. A single-center cross-sectional study from Tbilisi (Cureus, 2026) used the FRAIL scale to assess 50 patients on hemodialysis aged 18 to 64 years, with a mean age of 54 years. Of these, 20% were frail and 44% prefrail, so only 36% were robust. The regression findings are unstable because of the small sample, and the inverse association of smoking with

frailty is almost certainly an artifact. The main message supports routine frailty screening without an age cutoff.

Summary

- A meta-analysis reports a pooled prevalence of frailty on hemodialysis of 34.3% (95% CI, 24.5 to 44.1), with a range of 6% to 82%, and 56.0% in patients younger than 55 years; this gradient reflects heterogeneity of instruments, not greater frailty in younger patients.
- In a cohort of 2,275 incident patients on dialysis, frailty was associated with mortality (HR 2.24; 95% CI, 1.60 to 3.15) and with hospitalization or death (HR 1.90); screening is justified by these prognostic data, not by this study.
- The regression in ten frail patients reports only p values without effect sizes; associations with serum phosphate ($p = 0.04$) and dialysis vintage ($p = 0.016$) were significant in only one comparison, and the class 1 obesity subgroup comprised seven patients.
- The author attributes the inverse association of smoking with frailty to reverse causality, survivor selection bias, or chance, so smoking cannot be interpreted as a protective factor; FRAIL is a questionnaire, not a performance test.
- The author recommends screening at entry into the program, at least once a year, and after hospitalization or a fall; a positive result should trigger nutritional assessment, an exercise intervention including intradialytic exercise, management of anemia and acidosis, and medication review, and should inform transplantation and goals-of-care decisions.

7.27 Online hemodiafiltration as a dose-dependent therapy: recommendations of the Italian Society of Nephrology

Published August 23, 2026 · [online version \(Slovak\)](#)

Abstract. Recommendations of an international panel of the Italian Society of Nephrology (2026) used the GRADE methodology to evaluate randomized evidence on online postdilution hemodiafiltration versus high-flux hemodialysis. They recommend it to reduce cardiovascular mortality (HR 0.78) and suggest high-volume HDF to reduce all-cause mortality (HR 0.84; 95% CI, 0.74 to 0.95; moderate certainty). The panel views OL-HDF as a dose-dependent therapy: the benefit depends on the convective volume actually delivered. In practice, it is essential to monitor delivered volume, vascular access, blood flow, and treatment time.

Summary

- ESHOL (906 patients) showed an HR for all-cause mortality of 0.70, whereas cardiovascular mortality (HR 0.67) did not reach significance; CONVINCe (1,360 patients, target of at least 23 L per session, mean achieved 25.3 L) showed an HR of 0.77 and mortality of 17.3% versus 21.9%.
- An individual participant data meta-analysis of five trials with 4,153 patients showed an HR of 0.84, mortality of 23.3% versus 27.0%, a graded relationship between convective volume and risk of death, and no heterogeneity of effect across subgroups.
- Absence of heterogeneity does not prove that no unsuitable subgroup exists, because CONVINCe enrolled only candidates able to achieve a volume of at least 23 L; benefit has not been demonstrated for the predilution mode or for quality of life (low certainty).
- The achievable volume is limited by the filtration fraction and depends on blood flow and time; it is constrained by vascular access, shortened sessions, hematocrit, anticoagulation, and membrane surface area, and safety requires ultrapure fluid and ultrafilters.
- The authors recommend measuring delivered rather than prescribed volume at every session, tracking the proportion of sessions above target, not shortening treatment time, addressing vascular access as part of the dose, and reassessing settings after changes in hematocrit or anticoagulation.
- Patients should be informed soberly: a mortality benefit at an adequate dose has been demonstrated, not an improvement in how they feel during treatment.

7.28 Itch (pruritus) on dialysis: why reported prevalence varies twofold and what patients will not tell you unprompted

Published August 24, 2026 · [online version \(Slovak\)](#)

Abstract. CKD-associated pruritus is highly burdensome for patients on dialysis, yet it is rarely mentioned in the clinic. A Turkish study with a mixed quantitative and qualitative design assessed 294 patients on hemodialysis at three centers in Ankara using the 5-D itch scale. Moderate to severe itch was present in 16.3% and at least mild itch in 34.4%; however, the figure depends on the instrument and threshold, and other cohorts report 23.5% to 40%. More valuable is the finding that patients do not report the symptom and nearly half of those treated derive no benefit from treatment. In practice, ask actively and regularly.

Summary

- The threshold for moderate pruritus (5-D ≥ 12 , in practice 11.5 in this study) does not come from the original publication of the scale but from a single-center Taiwanese study of 409 patients; the scale was developed to measure change, not for screening, and the no-pruritus category (score 5 to 8) also includes mild itch.
- Exclusion of psychiatric, dermatologic, and other comorbidities produced a cleaner sample and probably underestimates the symptom burden.
- Antipruritic treatment was used by 212 patients (72.1%), most often oral antihistamines (143); 47.2% of users reported no effect, 36.8% a short-term effect, and only 16.0% a longer-lasting effect.
- Interviews with 48 patients with moderate to severe itch revealed the burden of pruritus, coping strategies, communication barriers with busy staff, and a mismatch between expected and actual treatment effects.
- The association of prior education with more severe itch (odds ratio 3.145) reflects reverse causality; the model is underpowered and lacks adjustment for Kt/V, phosphate, parathyroid hormone, or hemoglobin, and modern therapy does not feature in the study.
- The author recommends asking at the monthly review, using a single instrument, not stopping at antihistamines, checking for xerosis, dialysis dose, calcium-phosphate balance, parathyroid hormone, and anemia, and making the question a fixed part of the review structure.

7.29 GLP-1 receptor agonists, inflammation, and anemia in patients on hemodialysis: what real-world evidence shows and how to translate it into practice

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. The effect of GLP-1 receptor agonists on inflammation and anemia in hemodialysis has not been prospectively tested. A retrospective cohort from the Fresenius Kidney Care network (Kidney Medicine, 2026) compared 2,468 pairs of treated patients and nonusers after propensity score matching over 12 months. Treated patients had a lower neutrophil-to-lymphocyte ratio (difference -0.67), lower cumulative ESA use, and a lower erythropoietin resistance index, but hemoglobin did not differ. This is a hypothesis-generating association; a GLP-1 receptor agonist is not a treatment for anemia, and the KDIGO 2026 algorithm continues to apply.

Summary

- Of 76,011 patients on hemodialysis, 2,683 (3.5%) started a GLP-1 receptor agonist, mainly semaglutide (48.1%), dulaglutide (29.2%), and tirzepatide (14.3%); 215 patients with a BMI of 42.4 kg/m^2 could not be matched, so the conclusions do not apply to the most severe obesity.
- At month 12: NLR 3.97 versus 4.64, albumin $+0.06 \text{ g/dL}$, ferritin $+40.50 \text{ ng/mL}$, cumulative ESA dose $-123.9 \text{ } \mu\text{g}$, ERI 3.31 versus 3.63; hemoglobin 10.90 versus 10.84 g/dL, with a confidence interval including zero.
- Ferritin exceeded 800 ng/mL in both groups, a range in which routine iron is usually withheld in HD, and as an acute phase reactant it does not indicate better iron stores; hs-CRP was not available, and NLR is only a surrogate marker.

- Limitations: the per-protocol analysis excluding deaths, transplantations, and treatment discontinuation introduces survivor and healthy user bias, there is no active comparator, and the analysis was internally supported by Fresenius Medical Care.
- The author recommends interpreting the data to mean that, when a GLP-1 receptor agonist is already indicated, inflammatory markers and ESA use may fall concurrently; they are no reason to start treatment for anemia, which is managed according to KDIGO 2026.
- During ongoing treatment on dialysis, safety takes priority: gastrointestinal tolerability, protein and energy intake, loss of muscle mass, volume status, and blood glucose; the study cannot be read as a FLOW trial for dialysis.

7.30 Which dialysis patients need a higher dose of tenapanor? The role of bone resorption and intestinal transit

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. Tenapanor reduces paracellular phosphate absorption by inhibiting intestinal NHE3. A pooled post hoc analysis of three Japanese phase 3 trials (PLOS ONE, 2026) in 212 patients on dialysis sought baseline characteristics associated with the need for a higher dose after titration. Higher TRACP-5b concentration (OR 1.121 per 100 mU/dL) and harder stools on the Bristol scale (OR 0.667 per grade) were independently confirmed. These are exploratory signals, not validated predictors. Tenapanor is not available in the EU, including Slovakia, but the concept of a bone phosphate reservoir also applies to binders.

Summary

- The Japanese regimen starts at 5 mg twice daily with titration to 10, 20, or 30 mg, whereas in the US it is given at a fixed 30 mg twice daily as add-on therapy; the analysis included 127 patients on the lower and 85 on the higher dose (mean 8.0 vs. 26.7 mg twice daily).
- The decrease in serum phosphate did not differ between groups ($p = 0.456$), which, because of confounding by indication, does not mean equal efficacy of the doses but rather effective titration; the baseline serum phosphate of 7.39 mg/dL is an artifact of the design after binder washout.
- The association with TRACP-5b is of borderline significance (95% CI, 1.005 to 1.249); biologically, it may reflect phosphate release from bone, which holds approximately 85% of total body phosphorus, but the study did not measure this directly.
- The Bristol scale finding probably reflects better tolerance of the osmotic effect rather than pharmacological need; diarrhea was common in both groups (71.4% and 67.1%), and stool frequency was not a predictor.
- Limitations: pooling of three different trials, of monotherapy with add-on therapy, and of HD, HDF, and PD modalities, only eight weeks of follow-up, a risk of circular inference, and funding by Kyowa Kirin; clinical outcomes were not assessed.
- When the response to gut-acting therapy is inadequate, the author recommends assessing phosphate intake including additives, adherence, dialysis dose, treatment of secondary hyperparathyroidism, and bone turnover, and asking specifically about stool consistency.

7.31 High-volume hemodiafiltration and mortality in incident dialysis patients: a convincing association, not proof of causality

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. Randomized evidence for the benefit of high-volume hemodiafiltration concerns prevalent patients. A retrospective international cohort from the EuCliD registry (18,515 incident patients, 2019 to 2022, mean convective volume 24.9 L) showed, after IPTW weighting, lower all-cause mortality with HDF (HR 0.80; 95% CI, 0.75 to 0.86) and lower cardiovascular mortality (HR 0.71). In a stricter analysis, the effect

shrank to an HR of 0.88. This is an observational association close to the CONVINCe result, not proof of causality. In practice, HDF should be considered early in suitable patients.

Summary

- Crude mortality was 11.7 versus 15.6 deaths per 100 person-years; after adjustment for age and vascular access, the HR was 0.82, and in an analysis approximating intention to treat it was 0.88, so the stricter the method, the smaller the estimated effect.
- Cardiovascular mortality was 4.1 versus 6.7 per 100 person-years, and the cause of death was known for 93.4% of deaths; in patients with diabetes, the cardiovascular outcome did not reach significance.
- The 28% figure from the Fresenius Medical Care press release comes from a different study (JASN, HR 0.72) in a prevalent population with exposure in at least 90% of sessions, where the absolute difference after two years was only 1.7 percentage points; for incident patients, the figure is 20%.
- Before weighting, the groups differed (fistula 40% vs. 28%, age 62.0 vs. 64.5 years), classification based on 75% of sessions in the first year carries a risk of immortal time bias, and unstable patients and those with Kt/V below 1.2 were excluded.
- The data come from a single provider network with authors affiliated with Fresenius Medical Care; independent replication and a randomized trial in incident patients are needed.
- The author recommends considering postdilution HDF when ultrapure fluid, functioning vascular access, adequate treatment time, and a safe volume above 23 L are available, but not at the cost of instability or shortened treatment; timely fistula creation is a prerequisite.

7.32 Continuous relative blood volume monitoring may predict intradialytic hypotension

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. Intradialytic hypotension is among the most common complications of hemodialysis. A French prospective multicenter observational study in 56 high-risk patients (459 sessions) evaluated continuous relative blood volume (RBV) monitoring. RBV alone predicted hypotension poorly in new patients (AUC 0.62), whereas a model combining RBV, its change, systolic BP, its decline, and heart rate achieved an AUC of 0.85 to 0.86. This is internal validation without evidence of clinical benefit. RBV should not be the sole signal for automatically reducing ultrafiltration.

Summary

- Hypotension (systolic BP below 90 mm Hg and a decrease of at least 20 mm Hg) occurred in 29.7% of sessions in a population deliberately enriched for recurrent instability; each percentage point decrease in RBV increased the odds of concurrent hypotension (OR 1.05).
- A dynamic RBV threshold, around 90% after 150 minutes, was associated with concurrent hypotension (OR 2.37), while the association was weaker for prediction over the following 10 to 60 minutes.
- In session-level validation, RBV alone had an AUC of 0.77, but in patient-level validation only 0.62, which quantifies leakage of information about known patients; XGBoost was not superior to the mixed model, and part of the performance came from an already ongoing BP decline.
- In high-risk clusters, prescribed dry weight was approximately 0.9 kg lower than the bioimpedance estimate, although bioimpedance is not an absolute standard.
- Limitations: 56 patients, clusters and threshold derived from the same dataset without external validation, technical confounders affecting RBV, and two coauthors employed by Fresenius Medical Care.
- The authors caution that reducing ultrafiltration alone leads to chronic fluid overload; it is advisable to reassess target weight, sodium intake, dialysis duration and frequency, dialysate temperature, and anti-hypertensive medication.

7.33 Nurse-assessed frailty predicts mortality on dialysis: a multicenter validation of the Clinical Frailty Scale

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Routine frailty assessment on dialysis is constrained by time and staffing. A prospective multicenter cohort from six centers in northeastern France (Kidney Medicine) validated the Clinical Frailty Scale (CFS), assessed by dialysis nurses, in 792 prevalent patients. Each CFS point was associated with a subdistribution hazard ratio for death of 1.33, independent of the comorbidity-based REIN Predictive Score, and at 24 months 45.87% of frail patients had died versus 9.78% of fit patients. Discrimination of the two tools was comparable. The CFS is a screening and prognostic tool, not a basis for mechanical decisions.

Summary

- Cumulative mortality at 12 and 24 months was 3.56% and 9.78% in fit patients (CFS 1 to 3), 10.49% and 21.30% in vulnerable patients (4 to 5), and 27.27% and 45.87% in frail patients (6 to 9); transplantation was treated as a competing event.
- In a joint model, both the CFS (sHR 1.33; 95% CI, 1.19 to 1.49) and the RPS (sHR 1.05) were independently associated with mortality; the C-statistic at 12 months was 0.73 versus 0.69 with overlapping intervals, indicating comparable, not superior, performance.
- Agreement between two nurses in 509 patients was moderate for three categories ($\kappa = 0.60$), so broader categories rather than individual points are more suitable for decision-making.
- The CFS threshold of 6 is exploratory, and the authors call for external validation; it should trigger geriatric and nutritional assessment, fall risk assessment, medication review, and a goals-of-care conversation.
- Limitations: a prevalent cohort with survivor bias, a single region, a one-time assessment not standardized in timing, and a purely associational design; the RPS was used outside its original derivation context.
- The authors recommend rating the usual status before acute illness, involving the family, training assessors, repeating the assessment when the patient's condition changes, and not using the score as the sole reason to refuse treatment or transplantation.

7.34 When arteriovenous vascular access is not possible: a negative surgical recommendation as a prognostic marker

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Not every patient with advanced CKD can have an arteriovenous fistula or graft created. A retrospective descriptive cohort from São João Hospital in Porto (Hemodialysis International) followed 40 of 647 patients (6.2%) for whom the multidisciplinary team did not recommend arteriovenous access. Among predialysis patients who started hemodialysis, 90% had an unplanned start, and all-cause mortality reached 42.5%. A negative surgical recommendation is a prognostic marker, not a proven cause of death. Practice requires early planning, including PD, conservative care, and palliative care.

Summary

- The reason for the negative recommendation was exhaustion of the vascular bed in 23 patients (57.5%) and contraindication due to comorbidities in 17 (42.5%); half were in CKD stage G4 to G5 and half were already on kidney replacement therapy.
- Of 20 predialysis patients, 11 started kidney replacement therapy (ten HD, one PD), and nine of the ten started hemodialysis unplanned, most often because of acute cardiorenal or coronary syndrome; 90% of patients already on dialysis remained on a catheter long term.
- Mortality was 35% before dialysis and 50% on kidney replacement therapy, but with 20 patients per group the difference cannot be considered significant or causal.
- Limitations: 40 patients, no control group, median follow-up not reported, retrospective design, and pooling of two distinct clinical situations.

- Only one of 11 patients chose peritoneal dialysis, so its potential may be underused; the author recommends also considering assisted or urgent-start PD and, in line with KDOQI, planning according to an individual ESKD Life-Plan.
- The authors emphasize earlier identification of these patients, structured modality assessment, and timely integration of palliative care; repeated procedures should have a clear expected benefit.

7.35 Olfactory dysfunction in hemodialysis patients: a common but overlooked marker of overall morbidity

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Olfactory dysfunction is among the least monitored complications of CKD. A prospective observational study from Ankara (Journal of Nephrology) compared odor identification using the Sniffin' Sticks test in 100 patients on hemodialysis and 52 healthy controls. Dysfunction was present in 91% of patients versus 44% of controls, with mean scores of 60.6% versus 76.1%. Better smell was associated with a higher cognitive score and albumin, not with Kt/V. The dysfunction appears to be a marker of overall morbidity. In practice, consider it in cases of poor appetite, weight loss, or cognitive change, because patients usually do not notice it.

Summary

- Normosmia was present in 9% of patients versus 55.8% of controls, and anosmia in 17% versus 3.8%; the category thresholds are descriptive, and only identification, which also depends on cognition, was measured, so the between-group difference is more credible than the absolute prevalence.
- In the multivariable model, cognition ($\beta = 0.42$) and albumin ($\beta = 0.29$) were associated with better smell; neither the direction of the relationship nor causality is known, and albumin is not a pure marker of nutrition.
- Only 60 patients attended follow-up after one year, no mean decline was demonstrated, and Kt/V was unrelated to the trajectory; dropout of sicker patients introduces selection bias.
- Fewer than 25% of people are aware of their olfactory dysfunction before testing; severe dysfunction calls for advice on smoke and gas detectors and checking food, but an olfactory test is not dementia screening.
- A systematic review rates olfactory training as the best supported intervention (moderate certainty, 19 patients): after 12 weeks, identification improved from 15.3 to 18.8 points; curcumin did not improve the objective score, and intranasal theophylline has very low certainty of evidence.
- When dysfunction is suspected, the author recommends distinguishing it from taste disorder, considering infection including COVID-19, nasal disease, medications, smoking, cognition, and nutrition, not giving zinc without proven deficiency, and referring for further evaluation in case of sudden or unilateral symptoms.

7.36 Incremental hemodialysis and residual kidney function: what the evidence shows

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. Incremental hemodialysis adjusts the dose to residual kidney function (RKF) and increases it as RKF declines. A narrative review evaluates observational cohorts, randomized trials, and meta-analyses. Observational studies suggest slower RKF loss, but randomized trials have not confirmed this: in the Fernández Lucas RCT, the difference in the slope of residual GFR decline was $-0.09 \text{ mL/min/1.73 m}^2$ per month ($p = 0.230$). Meta-analyses showed no difference in mortality. A twice-weekly regimen is an option for carefully

selected patients with regular RKF measurement and readiness to increase the dose promptly, but a causal kidney-protective effect has not yet been proven.

Summary

- Obi et al. (351 vs. 8,068 matched patients) found 16% better preserved KRU and 15% higher urine volume, but a higher risk of death with $KRU \leq 3 \text{ mL/min/1.73 m}^2$ (HR 1.61; 95% CI, 1.07 to 2.44); the analysis included only patients who survived the first year.
- The Vilar pilot RCT (55 patients) found no difference in the decline of urea clearance over 12 months; serious adverse events were less frequent (incidence rate ratio 0.47), and the incremental group had lower bicarbonate.
- The Fernández Lucas RCT (150 patients, postdilution HDF) showed no difference in residual GFR decline, anuria, hospitalizations, or mortality, but it was not designed for noninferiority; the number of sessions per year was 69.1 versus 122.6.
- The meta-analyses by Caton (mortality HR 0.99) and Takkavatakarn (OR 0.87; lower odds of hospitalization, OR 0.44, and of RKF loss, OR 0.31) rest mainly on observational data, and their datasets overlap.
- A 25% decrease in KRU within three months was associated with a baseline $GFR < 7 \text{ mL/min/1.73 m}^2$ (OR 25.41; 95% CI, 1.22 to 530.27), which, given the small sample, is a highly uncertain estimate, not a validated threshold.
- For regimens other than three sessions per week, KDOQI 2015 suggests a target standard Kt/V of 2.3 per week and a minimum of 2.1, including ultrafiltration and residual function; RKF should be measured periodically by timed urine collection.
- The author recommends increasing frequency promptly in case of inadequate clearance, hyperkalemia, acidosis, or fluid overload; results of the VA IncHVets trial (NCT05465044) are not yet available.

7.37 Living with hemodialysis: advice from older patients and their care partners

Published September 27, 2026 · [online version \(Slovak\)](#)

Abstract. Older patients starting hemodialysis often need help from a close person, a care partner. A descriptive qualitative study by DePasquale et al. at Duke Health analyzed semistructured interviews with 33 patients aged 65 years or older in their first year of in-center hemodialysis and 36 care partners. The result is seven themes of advice concerning attitude, self-care, commitment, relationship, information, and planning. The study does not test the efficacy of an intervention, but in practice it supports stepwise education of the dyad, checking understanding, and attention to the care partner's burden.

Summary

- Participants advised patients to accept the new reality with a positive but realistic attitude and to look after their overall health, including activity, medications, and the regimen agreed with the team.
- They advised care partners to prepare for a significant commitment and not to set aside their own needs for long periods, because an exhausted carer has less capacity to provide safe support.
- Shared advice emphasizes a collaborative relationship with agreed roles, gathering information both before and after starting dialysis, and planning transport, medications, meals, and family obligations around the dialysis schedule.
- These are qualitative data from a single US health system; the study does not demonstrate effects on mortality, hospitalizations, or complications and does not compare patients with and without a care partner.
- As a clinical interpretation, the author recommends spreading education over time, including the care partner with the patient's consent, checking understanding with the teach-back method, and asking about the partner's health, sleep, and burden.
- A simple shared plan is useful, with the dialysis and transport schedule, a medication list, contacts, vascular access instructions, and the course of action for warning signs, while the fluid and dietary regimen remains individualized.

CHAPTER 8

Peritoneal and home dialysis

Peritoneal dialysis and home therapies: patient selection, prevention and treatment of infections, solute transport, and program sustainability. Number of articles: 13.

Key messages of the chapter

The chapter on peritoneal and home dialysis rests predominantly on observational data, expert opinion, and case reports; randomized evidence is rare. Its common message is that the success of home therapy is determined more by a functioning system than by the choice of modality itself. The Barcelona program doubled the number of patients on PD thanks to a shared infrastructure for training and monitoring and to nurses who support patients both in the center and at home; however, this is an institutional report without clinical outcomes (8.2). The KDIGO consensus report on home hemodialysis in East Asia emphasizes funding, regulation, the team, and safety backup, and regional referral centers are proposed for smaller countries (8.9).

Randomized data on more intensive home hemodialysis are mixed. The in-center FHN Daily trial showed a benefit of more frequent dialysis, but the home-based FHN Nocturnal and ACTIVE trials did not improve their primary endpoints, and long-term follow-up of FHN Nocturnal revealed an uncertain but serious signal of higher mortality (8.9). Registry comparisons are burdened by selection bias (8.9). Similarly, a Catalan cohort of patients aged 80 years and older found no difference in survival by initial modality after adjustment; risk emerged only with the transfer from PD to HD, which is rather a marker of deterioration (8.12). A Taiwanese cohort linked icodextrin in heart failure with lower mortality (HR 0.16), but an effect size of this magnitude cannot be read as a treatment effect because of marked selection bias (8.5). A mechanistic model of buffer transport does not demonstrate the superiority of any solution (8.6).

Patient selection and preparation form the second axis. A Polish cohort showed that after systematic predialysis education, 62.7% of preselected patients chose PD, but without a control group, a causal effect of education has not been proven; a lower likelihood of choosing PD was associated with age, diabetes, obesity, and the need for assistance (8.7). A predictive suitability model from Alberta has only moderate discrimination and lacks external validation; the main loss occurs between recognition of suitability and initiation of therapy, since only 22.3% of suitable candidates started PD within six months (8.8). Both studies share the recommendation to consider assisted PD and not to confuse unpreparedness with unsuitability (8.7, 8.8). In sarcopenia, the modified creatinine index is only a screening and short-term prognostic marker, and its change over time does not reflect muscle mass (8.10).

According to expert opinion, the prevention and treatment of infections is a continuous team process with root cause analysis of every episode and an audit of the unit's own data (8.1). In relapsing peritonitis, taurolidine may be an adjunct within a narrow window for catheter salvage, but the evidence comes only from case reports and small series, and it must not delay indicated catheter removal (8.3). A Slovak case report recalls *Mycobacterium fortuitum* in refractory culture-negative peritonitis, which according to ISPD requires prolonged culture and simultaneous catheter removal with combination antibiotic therapy (8.13). A media story of improvised home hemodialysis illustrates that the survival of an individual is not proof of safety, since water and solution quality and safety alarms cannot be ensured without a managed program (8.11). A green nephrology questionnaire study revealed that one fifth of patients also recycle bags, which are medical waste, and that most prefer in-person visits to telemedicine (8.4).

This yields a practical framework for the outpatient clinic. Modality education should begin as patients approach stage G4 and should offer all options, including conservative care, while monitoring concordance between the preferred and the initiated modality (8.7). Modality should not be selected by age, and with PD, possible technique failure should be planned for in advance (8.12). Icodextrin should be considered individually in inadequate ultrafiltration, with a glucose-specific glucometer (8.5). Every peritonitis episode should be analyzed; with repeatedly negative cultures, microbiology should be extended, and decisions about the catheter should be made promptly (8.1, 8.13). Strength and performance should be measured directly (8.10), and the patient's financial and transportation burden should be actively assessed (8.11).

8.1 Core interventions in peritoneal dialysis: infection prevention as a shared team responsibility

Published June 9, 2026 · [online version \(Slovak\)](#)

Abstract. In peritoneal dialysis, infectious complications, particularly peritonitis, exit-site infection, and tunnel infection, lead to hospitalizations, technique failure, and transfer to hemodialysis. An overview of an American Society of Nephrology podcast with Dr. Jeffrey Perl presents no new study data but summarizes expert recommendations. The main message is that infection prevention is a continuous team process built on education, aseptic technique, trend monitoring, and root cause analysis of every episode. In practice, this calls for standardized procedures and audit of one's own data.

Summary

- Gaps in practice arise mainly from inconsistent procedures for education, exchange technique checks, and infection documentation, not only from a lack of knowledge.
- If an infection is merely treated without root cause analysis, the risk of repeating the same error increases; the program should track the frequency, type, and pathogens of infections, their link to specific procedures, and treatment outcomes.
- A center's own data should serve as a quality improvement tool, for example to change the education process, improve hygiene practices, strengthen home support for the patient, or adjust the follow-up system.
- Prevention requires a coordinated team that includes the nephrologist, dialysis nurse, surgeon or interventionalist, microbiologist, and the patient and family; the patient should know what they are doing, why, and when to seek help.
- This is expert opinion from a discussion, not results of an interventional study, so quantified effects of individual measures are lacking.
- For practice, the podcast recommends standardized procedures, systematic infection audit, thorough education and regular retraining of patients, high-quality catheter insertion and care, and sharing of experience between centers, because every step from patient selection to home exchanges affects infection risk.

8.2 One hundred patients on home dialysis: a model of training and home treatment monitoring

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. Hospital Clínic Barcelona announced that it had reached 100 patients on home dialysis, 80 on peritoneal dialysis and 20 on home hemodialysis. This is an institutional report, not a clinical study. The number of patients on PD doubled from 40 to 80 since December 2023, and 61 people started home treatment in 2025. The hospital attributes the growth to a shared infrastructure for training and monitoring of both

modalities and to nurses who support patients both in the center and at home. The model shows the importance of organizational pillars in developing home dialysis.

Summary

- Home HD grew from 14 to 20 patients (by 43%), 61 people started home treatment in 2025 (50 PD, 11 home HD), and another 42 have been added so far in 2026.
- New premises on Carrer Manso serve both training and monitoring of both modalities, which enables peer learning among patients and reduces initial anxiety.
- The nursing team leads training both in the center and at home, so support is a single continuum rather than separate phases; this is a program description without evaluation of clinical outcomes.

8.3 Adjuvant taurolidine for relapsing peritonitis in peritoneal dialysis: catheter salvage or a dead end?

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Relapsing peritonitis in peritoneal dialysis often signals a persistent focus, particularly catheter biofilm. An educational article from the Lessons for the clinical nephrologist series in the Journal of Nephrology discusses adjuvant taurolidine in carefully selected patients. Taurolidine also acts on microorganisms within biofilm, but the evidence comes from case reports, small series, and extrapolation from catheter-related infections, not from randomized trials. It may be an adjunct within a narrow window for technique salvage, but never a substitute for antibiotics or a reason to delay indicated catheter removal.

Summary

- According to the ISPD, relapsing peritonitis is a recurrence within 4 weeks after treatment with the same organism or one sterile episode, recurrent peritonitis involves a different organism, repeat peritonitis means the same organism after more than 4 weeks, and refractory peritonitis fails to clear, typically after 5 days of treatment.
- Bacteria in biofilm have reduced antibiotic susceptibility, so the infection may reactivate after treatment ends; the ISPD therefore recommends timely consideration of catheter removal or replacement in relapsing, recurrent, or repeat peritonitis, and removal in refractory peritonitis.
- Taurolidine as a lock solution may be considered for relapse with the same organism, suspected biofilm, clinical response to antibiotics, and a substantial burden of catheter removal, for example in patients in rural and remote areas.
- Contraindications to delaying catheter removal are refractory or fungal peritonitis, severe sepsis, tunnel involvement, and clinical failure of conservative management.
- Use of taurolidine has not been standardized by large randomized trials; it is an option, not a standard for every patient.
- The author recommends the following approach: confirm relapse by comparing microbiology, assess the catheter and tunnel, check exchange technique, start targeted antibiotics, decide on catheter salvage or removal, and plan retraining and an audit of episodes.

8.4 A sustainable future for peritoneal dialysis: patients, recycling, and green nephrology

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Peritoneal dialysis reduces patient travel but generates large quantities of solutions, plastic bags, and packaging and requires home delivery of supplies. An observational two-center questionnaire study in the Journal of Nephrology surveyed 90 patients on PD from a green nephrology perspective. Recycling of at least one PD-related material was reported by 84.4% of patients, but 21.3% also recycled solution bags or

drainage bags, which are medical waste. In practice, this calls for clear education on waste sorting, stable supply logistics, and a hybrid follow-up model.

Summary

- The sample comprised 90 patients with a mean age of 54.4 ± 14.3 years (64.4% men); a 17-item questionnaire assessed recycling, supply logistics, and perceptions of telemedicine.
- The most commonly recycled materials were paper (83.3%) and plastic packaging (66.7%); recycling behavior was unrelated to age, place of residence, or PD modality, so waste education should be directed at all patients.
- Monthly deliveries were received by 85.6% of patients, and 88.9% were satisfied with delivery; the author emphasizes that predictable delivery has clinical value and that logistics changes must respect household space and continuity of treatment.
- In-person visits were preferred over telemedicine by 84.4% of patients, and younger patients were more open to digital forms of information; an in-person visit allows assessment of the catheter exit site, exchange technique, hydration, and signs of infection.
- Limitations are the observational design, 90 patients from two centers, and self-reported rather than verified behavior, so transfer to other countries and waste management systems is not automatic.
- The author recommends that centers provide a local overview of what counts as recyclable packaging and what is medical waste, how to store supplies, and whom to contact about a faulty delivery, and that they introduce changes with patient feedback.

8.5 Icodextrin in peritoneal dialysis and heart failure: how to read the new observational data

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. In patients on peritoneal dialysis with heart failure, volume control is key, and icodextrin maintains more stable ultrafiltration during the long dwell. A retrospective cohort study from the Chang Gung Research Database (Taiwan, 1,800 patients) in Nephrology Dialysis Transplantation found that icodextrin use was associated with lower all-cause mortality (adjusted HR 0.16) and fewer MACE (HR 0.68). These are observational data with a high risk of bias, so the HR cannot be read as the magnitude of a treatment effect. The results support individualized consideration of icodextrin, not its blanket prescription.

Summary

- All-cause mortality was 3.2 versus 14.0 events per 100 patient-years (HR 0.16; 95% CI, 0.13 to 0.20), cardiovascular mortality 1.0 versus 3.4 (HR 0.20), sudden death 2.3 versus 11.0 (HR 0.15), and MACE 9.9 versus 11.6 (HR 0.68); target trial emulation yielded similar estimates.
- Hospitalizations for heart failure (HR 1.06), myocardial infarction (HR 0.94), and ischemic stroke (HR 1.39) did not differ significantly, so the difference in MACE was driven mainly by cardiovascular mortality and sudden death.
- Icodextrin was also associated with a lower risk of transfer to hemodialysis (HR 0.52), EPS (HR 0.40), and transplantation (HR 0.22), which the author attributes to patient selection and unmeasured factors rather than to an effect of the solution.
- The groups differed in diabetes, hypertension, albumin, hemoglobin, and the period of PD initiation; Taiwanese reimbursement tied icodextrin to HbA1c above 7% or high transport status, most patients had an ejection fraction above 60%, and the mechanism through volume status was not measured.
- The author recommends considering icodextrin for inadequate ultrafiltration during the long dwell, volume overload, high transport status, or a need to limit glucose exposure, with an individualized prescription alongside standard heart failure therapy.
- Maltose can falsely elevate blood glucose readings on GDH-PQQ, GDO, and some GDH-FAD glucometers and mask hypoglycemia, so a glucose-specific method should be used; interference may persist for up to two weeks after discontinuation.

8.6 Buffer transport and acidosis correction in peritoneal dialysis: the integrated 3PM/ABE model

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. Acid-base balance in peritoneal dialysis is determined not only by dialysate mass balance but also by lactate metabolism and pulmonary ventilation. A mechanistic modeling study in Scientific Reports combined the three-pore transport model (3PM) with a whole-body acid-base model (ABE) and calibrated it on data from six patients during 4-hour dwells with Physioneal (B/L) and Dianeal PD4 solutions. The model reproduced the data with an RMSE of 6.6% and 6.4% and estimated similar transport parameters despite different buffer kinetics. It demonstrates neither clinical superiority nor equivalence of either solution.

Summary

- Each patient underwent two dwells with 2 liters of 3.86% glucose; at 3 minutes, lactate was 15.2 versus 38.1 mmol/L and bicarbonate 25.4 versus 3.4 mmol/L with B/L and PD4, respectively, and absorption did not differ significantly (3.66 vs. 2.79 mL/min).
- The model estimated a transfer of 65.7 ± 10.8 mmol of bicarbonate from blood to dialysate with PD4 and 23.1 ± 6.8 mmol with B/L, without accounting for reabsorption, so this is not a measured systemic loss.
- Plasma bicarbonate remained nearly constant, but the model reproduced this only by assuming instantaneous 1 : 1 conversion of lactate to bicarbonate, which was not measured.
- A significantly higher PS with B/L was found only for chloride (11.35 vs. 7.74 mL/min) and lactate (19.80 vs. 15.43 mL/min), while values for glucose and creatinine were similar.
- Minute ventilation was derived rather than measured, so the model is not a validated tool for patients with chronic obstructive pulmonary disease or respiratory failure.
- Limitations are six patients with no low transporters, calibration and evaluation on the same data without external validation, a fixed perfusion of 55 mL/min, and no clinical outcomes, Kt/V, or long-term data.
- The author concludes that the study supports the physiological plausibility of integrated modeling, but the choice of solution in liver or lung disease should be guided by clinical data and individual patient assessment.

8.7 Predialysis education and the choice of peritoneal dialysis: what a Polish cohort showed

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. Timely and balanced predialysis education should enable patients with advanced CKD to make an informed modality choice. A retrospective single-center study from Warsaw evaluated 118 preselected patients with CKD G4 to G5 after an individual 90- to 120-minute meeting with physicians and nurses. Peritoneal dialysis was chosen by 62.7% of patients, and 51.7% of the entire sample started it. Without a control group, the study does not prove a causal effect of education, but it shows the feasibility of a systematic modality discussion and the importance of assisted PD for functionally limited patients.

Summary

- No barriers to PD were present in 63.5% of patients, 14.4% had relative barriers, 16.1% required assisted PD, and 6.0% were excluded; thus 94% of referred patients had no absolute barrier, which, because of preselection, cannot be extrapolated to the whole population.
- Of 74 patients preferring PD, 61 (82.4%) started it, four underwent preemptive transplantation, and in four the plan was not implemented because of complications.
- In univariable models, older age (OR 0.967 per year), diabetes (OR 0.353), higher BMI (OR 0.884), abdominal surgery scars (OR 0.423), and need for assistance (OR 0.264) were associated with lower odds of choosing PD; these are associations without control for confounding.
- Of 19 patients requiring assistance, only six chose PD, because help could come only from family; the ISPD regards assisted PD as a legitimate model.

- Limitations are the lack of a control group, a single center, no assessment of knowledge or quality of life, and an accepted manuscript with minor discrepancies in subgroups.
- Following KDIGO, the author recommends starting education as CKD approaches G4, offering transplantation, PD, in-center and home HD, and conservative care, and monitoring concordance between the preferred and the actually started modality instead of a mandatory PD-first approach.

8.8 Predicting suitability for peritoneal dialysis: a model that needs validation and clear incremental value

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. Predictive models are intended to help select candidates for peritoneal dialysis. A commentary on a letter to the editor in Peritoneal Dialysis International discusses the logistic regression model of Yang et al. from the province of Alberta (598 patients), which achieved a sensitivity of 91.3%, an accuracy of 0.68, and an area under the ROC curve of 0.69 to 0.71. Discrimination is only moderate, and external validation, calibration, and evidence of incremental value are lacking. Moreover, the model predicts the physician's decision, not the patient's outcome. The key bottleneck is that only 22.3% of candidates deemed suitable started PD within six months.

Summary

- Older age, higher body mass index, dialysis initiation in an intensive care unit, and polycystic kidney disease were associated with a lower likelihood of being deemed suitable, that is, factors clinicians consider routinely.
- With an accuracy of 0.68 and a sensitivity of 91.3%, specificity must be low, so the model frequently labels unsuitable patients as suitable; specificity was not explicitly reported.
- According to the TRIPOD statement, calibration should also be assessed, and the value for decision-making should be shown by decision curve analysis according to Vickers and Elkin, which is missing.
- The author cautions that a model trained on physicians' decisions may formalize existing biases, such as underestimation of older patients; it would be clinically meaningful to predict technique failure, peritonitis, transfer to hemodialysis, or quality of life.
- Of 391 patients deemed suitable, only 22.3% started PD within six months, so the main loss occurs between decision and treatment because of waiting for a catheter, training capacity, urgent-start HD, or lack of support.
- The author recommends using the model to identify rather than exclude candidates, not mistaking unreadiness for unsuitability, considering assisted PD, comparing predictions with actual outcomes locally, and respecting patient preferences.

8.9 Home hemodialysis: greater patient autonomy is not enough without a functioning system

Published August 6, 2026 · [online version \(Slovak\)](#)

Abstract. Home hemodialysis gives patients greater autonomy, but it is underused in East Asia. A KDIGO consensus report in Kidney International Reports from a workshop in Kawagoe proposes an implementation framework for Hong Kong, Japan, South Korea, and Taiwan; it is not a clinical guideline. The article supplements it with randomized data: FHN Daily (in-center) showed a benefit of more frequent dialysis (HR 0.61), but home nocturnal FHN Nocturnal and ACTIVE did not. A causal effect on survival has not been proven, and program development requires funding, a team, training, and safety backup.

Summary

- FHN Daily (245 patients, six times weekly in-center) reduced the risk of death or increase in left ventricular mass (HR 0.61; 95% CI, 0.46 to 0.82) but increased vascular access interventions (HR 1.71).

- FHN Nocturnal (87 patients, nocturnal home HD) showed no effect on the primary outcomes and improved only serum phosphate and hypertension; in long-term follow-up (median 3.7 years), there were 14 versus 5 deaths (HR 3.88; 95% CI, 1.27 to 11.79), an uncertain but serious signal given the small sample.
- ACTIVE (200 patients, ≥ 24 hours per week) did not change quality of life on the EQ-5D (difference 0.04; $p = 0.29$) but improved serum phosphate, serum potassium, and hemoglobin.
- A registry comparison of 1,873 patients on daily home HD with 9,365 matched patients showed an HR for death of 0.87, but these are observational data subject to selection bias; KDIGO 2023 considers outcomes across modalities largely similar.
- Barriers include the culture of in-center dialysis, hospital infrastructure, regulation, reimbursement, limited staff experience, and patient concerns; risks include needle dislodgement, air embolism, and family burden.
- The report proposes three phases: establishing reimbursement and regulatory conditions, launching a pilot program with a multidisciplinary team, and expanding it after evaluating adverse events, hospitalizations, technique survival, and costs.
- For smaller European countries, the author proposes regional reference centers with centralized training, technical support, round-the-clock assistance, and an outcomes registry, while modality choice should remain informed.

8.10 Sarcopenia in peritoneal dialysis: why the modified creatinine index is not enough

Published August 15, 2026 · [online version \(Slovak\)](#)

Abstract. Sarcopenia in peritoneal dialysis is associated with falls, hospitalizations, and mortality, but its assessment is challenging. The review discusses two studies from Prince of Wales Hospital in Hong Kong on the modified creatinine index. At a single time point, the index correlated with muscle mass ($r = 0.684$ with creatinine kinetics, $r = 0.641$ with bioimpedance spectroscopy) and predicted 12-month mortality, but its change over time did not reflect muscle mass. The index is a screening and short-term prognostic marker, not a diagnostic test for sarcopenia or a tool for monitoring the effect of an intervention.

Summary

- According to EWGSOP2, probable sarcopenia is based on low muscle strength, confirmation requires low muscle quantity or quality, and reduced physical performance indicates severe sarcopenia; EWGSOP2 and AWGS thresholds are not interchangeable.
- Canaud's equation, derived from hemodialysis, uses age, sex, predialysis creatinine, and Kt/V , with total weekly Kt/V substituted in PD; validation in 605 incident patients confirmed its single-time-point validity.
- In serial follow-up of 351 patients every six months over two years, the systematic bias was small (+4.9% and -7.0%), but the limits of agreement exceeded 60 percentage points, and the change in the index was unrelated to survival, hospitalizations, or peritonitis.
- The main confounder is residual kidney function, followed by peritoneal transport, dialysis prescription, incomplete urine and dialysate collection, meat intake, and an unstable state during peritonitis or hospitalization.
- DXA and bioimpedance overestimate muscle in overhydration, albumin is not a standalone measure of nutrition, and sarcopenia, malnutrition, and protein-energy wasting are not synonyms.
- The author recommends the following approach: identify at-risk patients, measure handgrip strength or perform the chair stand test, assess muscle mass and performance (SPPB, Timed Up and Go), and look for reversible causes, including acidosis, inflammation, and inadequate dialysis.
- Treatment includes resistance and aerobic training with attention to hernia risk, individualized nutrition without a blanket high-protein diet, and correction of acidosis; PD-specific evidence is limited, and pharmacotherapy for sarcopenia has no standard role.

8.11 Improvised home hemodialysis: why one person's survival is not evidence of safety

Published August 23, 2026 · [online version \(Slovak\)](#)

Abstract. The story of a Chinese patient who built his own hemodialysis setup and survived on it for thirteen years is circulating again on social media. It is a media report from January 2013 without clinical documentation, not a new event. The article explains factually why one person's survival is not evidence of safety: approximately 120 liters of fluid pass the membrane per session, and improvisation cannot verify its microbiological purity or composition. In practice, the key is to detect financial and transport burdens early and to offer supervised home programs.

Summary

- According to ISO 23500 standards, the limits for dialysis water are < 100 CFU/mL and < 0.25 EU/mL, for standard dialysis fluid < 0.5 EU/mL, and for ultrapure fluid < 0.1 CFU/mL and < 0.03 EU/mL; the frequently cited threshold of 2 EU/mL is an outdated US standard.
- Water treatment must also remove chemical contaminants; historically, failures led to encephalopathy from aluminum, hemolysis from chloramines, and poisoning from fluoride.
- Dialysis fluid requires precise sodium (usually 135 to 140 mmol/L), potassium (2 to 3 mmol/L), calcium (1.25 to 1.50 mmol/L), and bicarbonate (30 to 35 mmol/L); errors lead within minutes to hemolysis, cerebral edema, or arrhythmia.
- Safety is ensured by an air detector, a blood leak detector, temperature control (risk of hemolysis above approximately 42 °C), conductivity monitoring, volumetrically controlled ultrafiltration, pressure alarms, and validated disinfection, none of which an improvised setup has.
- The story lacks data on fluid composition, microbiology, laboratory results, and infections, and it reflects survivorship selection; reuse of single-use sets without a validated procedure makes it impossible to demonstrate sterility.
- The author recommends actively asking about financial and transport burdens, treating missed sessions as a warning sign, offering PD or home HD in a supervised program, and addressing the social situation as part of the treatment plan.

8.12 Dialysis modality and survival in patients aged 80 years and older: the course of treatment matters more than the initial choice

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Data on survival by dialysis modality in patients aged 80 years and older are scarce. A population-based retrospective cohort study from the Catalan Renal Registry in the Clinical Kidney Journal included 4,205 patients who started dialysis from 2000 to 2022, with modality transfer modeled as a time-dependent variable. After adjustment, the initial modality was not associated with a difference in five-year survival, but transfer from PD to HD was (HR 1.46; 95% CI, 1.02 to 2.08). Transfer is more likely a marker of deterioration; the results support individualized choice and timely planning of modality change.

Summary

- Of the cohort, 3,966 patients (94.3%) remained on HD, 176 (4.2%) were treated exclusively with PD, and 63 (1.5%) transferred from PD to HD; 1 : 3 propensity score matching included 637 patients on HD and 213 on PD.
- PD without transfer was not significantly associated with mortality (HR 1.19; 95% CI, 0.95 to 1.48), whereas higher risk was associated with age (HR 1.06 per year) and initiation from 2000 to 2007 (HR 1.37).
- The apparently worse survival with PD in the matched Kaplan-Meier analysis ($p = 0.008$) disappeared after modeling transfer as a time-dependent variable, which prevents immortal time bias.
- Reasons for transfer were loss of membrane function (30.4%), severe intercurrent illness (30.4%), peritoneal infection (23.2%), and patient preference (16.1%), consistent with confounding by indication; the E-values were 1.72 and 2.02.

- Limitations are the observational design without data on frailty, cognition, and nutrition, only 63 patients after transfer, unmodeled PD duration before transfer and competing events, and missing center-level data.
- The authors recommend not choosing modality by age, planning in advance for possible technique failure on PD including vascular access, monitoring technique sustainability, and regarding unplanned transfer as a high-risk moment rather than a reason to postpone it.
- The study did not assess quality of life or compare dialysis with conservative care, which the author considers an appropriate alternative for markedly frail patients according to their wishes.

8.13 Refractory culture-negative peritonitis in peritoneal dialysis: think of *Mycobacterium fortuitum*

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Peritonitis in peritoneal dialysis with repeatedly negative cultures and no response to empirical therapy is a diagnostic challenge. A Slovak case report from Košice and Martin, published in the Journal of Nephrology, draws attention to *Mycobacterium fortuitum*, a rapidly growing nontuberculous mycobacterium that forms biofilm on the catheter. The article supplements it with ISPD recommendations: diagnosis is delayed by six to thirty days, and complete cure was achieved in only 14.8% of episodes in an observational series. Repeatedly negative cultures should prompt extended microbiology and an early decision about the catheter.

Summary

- According to the ISPD, peritonitis is diagnosed when two of three criteria are met: clinical features, more than 100 leukocytes/ μL in dialysate after a dwell of at least two hours with more than 50% polymorphonuclear cells, and a positive culture; the culture-negative form meets the first two.
- Most episodes of NTM peritonitis are caused by *M. fortuitum* and *M. chelonae*; on Gram stain they may be mistaken for diphtheroids, so published series report a median diagnostic delay of six to thirty days.
- If cultures remain negative after three to five days, the ISPD recommends repeating the cell count with differential, adding fungal and mycobacterial cultures, and extending incubation to seven days when NTM is suspected; negative acid-fast microscopy does not exclude the diagnosis.
- Refractory peritonitis is failure of the dialysate to clear after five days of appropriate treatment; the 2022 update allows further observation when leukocytes are falling (grade 2C), and the five-day threshold is arbitrary.
- The ISPD recommends treating NTM peritonitis with antibiotics combined with catheter removal (grade 2D); expert consensus specifies two antibiotics according to susceptibility for at least six weeks. Macrolides carry a risk of inducible resistance (*erm* genes), so monotherapy is not appropriate.
- The case report is intended for clinical teaching and is not a controlled study; the public record has no abstract, so the conclusions rest on ISPD recommendations rather than on case details.
- In practice: consult a microbiologist, examine the catheter exit site and tunnel, search for a surgical cause, do not delay catheter removal, and discuss possible transfer to hemodialysis with the patient early.

CHAPTER 9

Kidney transplantation

Living donation, immunology, long-term outcomes, and new directions in kidney transplantation, from biomarkers of immunosuppression to xenotransplantation. Number of articles: 14.

Key messages of the chapter

Kidney transplantation remains the preferred method of kidney replacement therapy, and this chapter shows that early planning brings the greatest benefit. Preemptive transplantation is associated with lower mortality and better graft survival; in the French registry, 10-year graft survival was 80% versus 61%, and KDIGO recommends it with grade 1A; however, the evidence is observational, and part of the benefit may be explained by lead-time bias and the selection of younger and healthier patients (9.12). Transplantation at an eGFR above 10 to 20 mL/min/1.73 m² brings no additional benefit, and education should begin at an eGFR below 30 mL/min/1.73 m² (9.12). Retransplantation follows the same logic: a qualitative Canadian study identified sensitization, comorbidities, a history of nonadherence, and delayed referral as the main barriers, so retransplantation should be prepared before definitive graft failure (9.13). Nor should obesity be an automatic barrier, since at a BMI up to 40 kg/m², transplantation reduces mortality compared with remaining on the waiting list by more than 60% (9.4).

The second axis of this chapter is individualized immunosuppression. The randomized phase II TTVguideIT trial in 260 low-risk recipients met noninferiority of tacrolimus dosing guided by TTV load (35% versus 38%), but a statistically significant reduction in infections was not demonstrated, and the result applies only to the first year (9.1). A retrospective Irish cohort with 33 grafts functioning for 40 years or more supports maintaining low individualized immunosuppression, and in it a living donor was associated with better ultra-long-term survival (OR 3.51); however, the results of a historical cohort cannot be mechanically transferred to the present (9.5). Belatacept is an established alternative for selected EBV-seropositive recipients, while tegoprobart remains experimental (9.14).

Long-term immunosuppression carries an oncologic and infectious toll. The risk of cancer is approximately 2- to 4-fold higher than in the general population, skin cancers are the most common malignancy, and PTLT threatens mainly EBV-seronegative recipients (9.3). In the Irish cohort, approximately 67% of long-term survivors had nonmelanoma skin cancers (9.5). Vaccination should ideally be timed before transplantation, and after it, without live vaccines; reviews differ on intervals, with influenza cited from 4 weeks and pneumococcus and RSV from 6 months, while the Core Curriculum recommends waiting at least 3 months (9.2, 9.3). A meta-analysis of observational studies links persistent hyperparathyroidism after transplantation with higher mortality (HR 1.74) and graft failure (HR 2.16), but this is a risk marker, not evidence that treatment improves outcomes (9.6).

The chapter consistently separates the proven from the promising. Hypothermic machine perfusion has randomized support, with a reduction in delayed graft function (OR 0.57), whereas 1-hour normothermic perfusion did not reduce DGF in an RCT, and the C1 esterase inhibitor yielded only a pilot signal (9.14). At the level of hypotheses or single cases are regulatory T cells, which in small phase 1 and 1/2 studies did not demonstrate complete withdrawal of immunosuppression (9.9), teclistamab in extreme HLA sensitization in a single patient (9.10), and xenotransplantation, in which a functioning graft was accompanied by Banff 2A rejection and the patient died on day 52 (9.8).

In living donors, neutral or uncertain results predominate. In the randomized AL-DON trial, allopurinol lowered uric acid but did not change left ventricular mass or BP, so it cannot be recommen-

ded to donors for cardiovascular prevention (9.11). The observational LETO study showed a more frequent decline in eGFR below 45 mL/min/1.73 m² with the high-risk APOL1 genotype (RR 2.31), but after adjustment for baseline eGFR, the association lost significance; genotyping should serve stratification and informed consent, not automatic exclusion (9.7).

A practical framework for the outpatient clinic: refer candidates early, ideally before dialysis, and do not regard age or obesity as an automatic contraindication (9.4, 9.12, 9.14); with a failing graft, update HLA typing and search for living donors (9.13); regularly reassess immunosuppression according to immunologic, infectious, and oncologic risk (9.5); ensure an annual dermatologic examination and photoprotection and check vaccination status (9.2, 9.3); monitor PTH, calcium, phosphate, and 25-OH vitamin D (9.6); and in donors, check BP, BMI, eGFR, and albuminuria at least once a year (9.7).

9.1 Biomarker-guided reduction of immunosuppression after kidney transplantation: TTV as a measure of immune activity

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. Tacrolimus dosing by target trough levels may not accurately reflect the degree of immunosuppression in an individual patient. The randomized phase 2 TTVguideIT trial from 13 European centers tested dosing guided by Torque teno virus (TTV) load in 260 stable low-risk kidney recipients enrolled approximately 4 months after transplantation. The composite endpoint (infections, rejection, graft loss, death) occurred in 35% versus 38%, meeting noninferiority with lower tacrolimus doses. The approach is promising but so far applies only to the first year and to low-risk patients.

Summary

- The primary composite endpoint occurred in 35% of patients in the TTV-guided group and 38% in the control group, so the trial met its noninferiority objective; rejection on protocol biopsies at 12 months was comparable.
- Patients in the TTV group had lower tacrolimus levels and lower daily doses, but a statistically significant reduction in infections was not demonstrated.
- The result applies to stable patients with low immunological and infectious risk in the first year after transplantation; benefit in later phases is being tested in a follow-up trial in France from the second year onward.
- According to the author, this is a clear direction, not a finished algorithm; before implementation, the appropriate risk profile, the period of use, and the course of action upon transition to higher immunological or infectious risk must be defined.

9.2 Vaccination in chronic kidney disease (CKD) and after kidney transplantation: a practical overview of vaccines and timing

Published June 17, 2026 · [online version \(Slovak\)](#)

Abstract. Adults with CKD and kidney recipients are at higher risk of serious infections because of an impaired immune response, immunosuppression, and frequent healthcare contacts. The article summarizes the review Vaccinations to Prevent Infections in Adult Individuals With CKD and After Kidney Transplantation, covering indications, timing, and vaccine types. Timing after transplantation is key: influenza from 4 weeks, pneumococcal and RSV vaccines from 6 months, and no live vaccines. The result is two practical outpatient tables that should be adapted to local and product-specific recommendations.

Summary

- Influenza: everyone annually; after transplantation from 4 weeks, no live vaccine, standard-dose vaccine up to 60 years and high-dose vaccine from 60 years.
- Pneumococcal: in CKD, everyone aged 19 years and older, after transplantation from 6 months; unvaccinated patients receive 1 dose of PCV, and those previously given PPSV23 receive 1 booster dose of PCV

more than 1 year later. RSV as a single dose with no preference of vaccine type: in CKD stage 3+ from 75 years, on dialysis and after transplantation from 60 years (after transplantation from 6 months).

- COVID-19: annual revaccination with a single dose of an mRNA vaccine targeting the current variant, with an interval of 6 months mentioned after transplantation. Hepatitis B in CKD stages 3b to 5: 3 to 4 injections of a standard or double dose depending on the product, with serologic monitoring and a booster dose; no specific recommendation after transplantation.
- Herpes zoster: 2 doses of recombinant vaccine 2 to 6 months apart; age thresholds in CKD differ (ACIP from 50, RKI from 60, NHS from 65 years), and in immunosuppression and after transplantation from 19 years.
- Meningococcal (MenB or MenACWY) only with specific risk, such as asplenia, complement inhibition, or travel, with consideration of antibiotic prophylaxis; Tdap/Td for everyone every 5 to 10 years, HPV 3 doses at months 0, 2, and 6.
- For the outpatient clinic, the author recommends the following approach: determine the CKD stage or immunosuppression status, select the appropriate table, check vaccination history (pneumococcal, hepatitis B, zoster), and give missing vaccines according to the required time intervals.

9.3 Malignancy and kidney transplantation: what the nephrologist should know (Core Curriculum 2026)

Published June 17, 2026 · [online version \(Slovak\)](#)

Abstract. Long-term immunosuppression increases the risk of nonmelanoma skin cancers, EBV-associated lymphoproliferative disorders, and some solid tumors in kidney recipients. The AJKD Core Curriculum 2026 review summarizes screening, prevention, and management of malignancy in candidates and recipients. Cancer risk is approximately 2 to 4 times higher than in the general population, and tumors are the second most common cause of death. Screening should be individualized by risk and life expectancy, vaccination should be timed before immunosuppression, and PTLD should be considered early.

Summary

- Screening in candidates aims to detect occult malignancy before immunosuppression and relies mainly on general population schedules; the benefit of early detection must be balanced against competing risks, overdiagnosis, and overtreatment. More frequent cervical screening is recommended, and ultrasound or urine cytology for RCC and urothelial cancer in high-risk patients.
- Vaccination should ideally be completed before transplantation; afterward, the review recommends waiting at least 3 months, and live vaccines should be avoided during immunosuppression.
- Skin cancers are the most common malignancy: UV protection (SPF 30+) and an annual full-body dermatologic examination are recommended. Azathioprine accelerates carcinogenesis of nonmelanoma tumors; with recurrences, immunosuppression is reduced or the patient is converted to an mTOR inhibitor, taking into account a possibly higher cardiovascular risk.
- PTLD is a risk mainly for EBV-seronegative recipients, and viremia typically develops between months 2 and 4; EBV PCR is recommended every 7 to 14 days in the first month, then every 28 days until 6 months, and then every 3 to 6 months until 1 year.
- Diagnosis of PTLD includes a complete blood count, LDH, CT, and above all biopsy with EBER detection. The first step is minimization of immunosuppression, but in monomorphic and aggressive forms it does not replace oncologic therapy; the benefit of mTOR inhibitors has not been clinically demonstrated.
- In practice, involve a hematologist or oncologist urgently in the presence of red flags such as unexplained fever, lymphadenopathy, B symptoms, or declining performance status.

9.4 Obesity in nephrology: screening, management, and impact on dialysis and transplantation (a practical perspective based on the AJKD Core Curriculum)

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. Obesity is the most common risk factor for kidney disease and affects the course of CKD, dialysis, and access to transplantation. A review based on the AJKD Core Curriculum summarizes mechanisms of injury, screening by BMI and central adiposity, behavioral treatment, ketogenic diets, peritoneal dialysis, and transplantation. The key finding is that with a BMI up to 40 kg/m², transplantation reduces mortality by more than 60% compared with remaining on the waiting list, and KDIGO 2020 does not recommend excluding patients on the basis of obesity alone. Weight loss is indicated in overweight patients with comorbidities without further measurements.

Summary

- Obesity damages the kidneys through altered adipokines with sympathetic and RAAS activation, glomerular hyperfiltration and barotrauma, and ectopic lipid deposition; abdominal adiposity is more consistently associated with mortality than overall adiposity.
- Screening: overweight is a BMI of 25 to 29.9, obesity from 30 kg/m² (classes I to III), with thresholds of 23.0 and 27.5 in Asians; central adiposity is indicated by a waist circumference above 88 cm in women and above 102 cm in men or a waist-to-hip ratio above 0.85 and 0.9, respectively. In overweight with comorbidities, weight loss is recommended without DXA.
- A behavioral program should last at least 6 months: a deficit of 500 to 750 kcal/day, protein 0.8 g/kg ideal body weight/day, fiber around 24 g/day, and physical activity of 30 to 60 minutes at least 5 days a week. In LOOK AHEAD, weight loss was 8% versus 0.7%, but slowing of progression in established CKD was not demonstrated.
- Ketogenic diets are generally not recommended in advanced CKD because of insufficient safety evidence and the risk of higher LDL and kidney stones; the 12-week data in ADPKD are short term.
- In PD, sustainable exit-site hygiene is decisive, and with an abdominal pannus an upper abdominal or presternal catheter is considered; Kt/V may falsely underestimate adequacy in obesity, so clinical judgment, bioimpedance, or kinetic modeling are helpful.
- With a BMI of 40 kg/m² or higher, the risk of delayed graft function, rejection, and wound complications rises, but the survival benefit of transplantation persists. Weight loss should be discussed with the patient's consent, without stigmatizing language, and in relation to a specific goal.

9.5 Forty years with a functioning kidney transplant: what ultra-long-term survivors show

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Most transplant cohorts follow graft survival for no more than ten years. A retrospective analysis of the Irish national registry in the Journal of Nephrology evaluated 428 transplantations in 394 patients performed from 1970 to 1983, with follow-up to 2023. Thirty-three grafts (7.7%) survived forty years or more, and living donor transplantation was associated with better ultra-long-term survival (OR 3.51). As many as 67% of long-term survivors had nonmelanoma skin cancers. Success over decades requires low, individualized immunosuppression and lifelong comprehensive follow-up.

Summary

- Of 428 transplanted kidneys, 33 grafts (7.7%) survived 40 years or more, and 25 were functioning at the time of analysis; median creatinine was 107 µmol/L (range, 66 to 322 µmol/L). Longitudinal data were available for 98.9% of the cohort.
- In multivariable analysis, a living donor was associated with better ultra-long-term graft survival (OR 3.51; 95% CI, 1.17 to 11.1; *p* = 0.027), whereas donor and recipient age and sex were not significant. Some grafts from deceased donors also survived long term.

- Graft function was excellent in patients maintained on low-level immunosuppression; the authors emphasize regular reassessment of the regimen according to immunological, infectious, and oncologic risk.
- Nonmelanoma skin cancers affected 22 of 33 patients (approximately 67%), so dermatologic screening and photoprotection are an essential part of care.
- This is a retrospective historical cohort with different immunosuppression, surgery, and prophylaxis, so the results cannot be mechanically applied to current patients.
- In practice, a long-term recipient needs monitoring of eGFR and proteinuria, blood pressure, immunosuppressant toxicity, infections, skin and other cancers, metabolic and bone complications, drug interactions, and frailty.

9.6 Persistent hyperparathyroidism after kidney transplantation as a risk marker for mortality and graft failure

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. Secondary hyperparathyroidism persists in some patients even after successful kidney transplantation. A systematic review and meta-analysis in *Clinical Kidney Journal* (23 observational studies, 19 included in meta-analyses, 26,266 recipients) assessed its relationship with mortality and graft failure. Persistent posttransplant hyperparathyroidism was consistently associated with higher mortality (HR 1.74) and graft failure (HR 2.16), most markedly in the hypercalcemic form. The prognostic significance of pretransplant hyperparathyroidism was uncertain. It is a risk marker that calls for systematic CKD-MBD monitoring, not evidence of causality.

Summary

- Persistent hyperparathyroidism after transplantation was associated with all-cause mortality (HR 1.74), overall graft failure (HR 2.16), and death-censored graft failure (HR 1.92); the results held in the Hartung-Knapp sensitivity analysis.
- The risk of graft failure showed a gradient: HR 1.66 for the normocalcemic form and HR 2.67 for the hypercalcemic form, so even normocalcemia may not be benign.
- Pretransplant hyperparathyroidism was not associated with mortality (HR 0.80); its association with death-censored graft failure (HR 1.42) lost significance after sensitivity adjustment.
- Limitations: observational studies with residual confounding and inconsistent PTH definitions and thresholds; continuous PTH analyses were heterogeneous. The results do not show that treatment improves graft survival, nor do they define a target PTH level.
- Cinacalcet lowers calcium and PTH, but its effect on hard outcomes remains unresolved; parathyroidectomy is an option for the refractory hypercalcemic form, with a risk of hypocalcemia and hungry bone syndrome.
- The author recommends regular posttransplant monitoring of PTH, calcium, phosphate, and 25-OH vitamin D, always interpreting PTH in the context of graft function, and consulting the transplant center or an endocrinologist if the disorder persists.

9.7 APOL1 genotyping in living kidney donation: more precise stratification, not automatic exclusion

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. High-risk APOL1 variants are more common in people of recent sub-Saharan African ancestry, and their significance for living kidney donors is uncertain. The retrospective LETO cohort study in *JAMA Internal Medicine* followed 653 US donors for a median of 18.5 years after donation. Donors with two risk alleles had an eGFR below 45 mL/min/1.73 m² in 14.7% versus 6.4% (RR 2.31), but after adjustment for pre-

donation eGFR the association lost significance (RR 1.91). Genotyping should serve more precise stratification and informed consent, not automatic exclusion.

Summary

- A high-risk genotype (G1/G1, G1/G2, G2/G2) was present in 68 of 445 donors in the Black category (15.3%); the primary outcome of eGFR below 45 mL/min/1.73 m² occurred in 14.7% versus 6.4%, an absolute difference of 8.3 percentage points (RR 2.31; 95% CI, 1.16 to 4.61).
- After adjustment for predonation eGFR, the RR fell to 1.91 (95% CI, 0.90 to 4.03; *p* = 0.09); for eGFR below 60 mL/min/1.73 m², the association persisted (RR 1.50), and the genotype was also associated with severe albuminuria.
- Limitations: observational design, a single measurement without confirmation of chronicity, no control group of nondonors, selection bias, and only three cases of kidney failure, which precludes quantifying the risk of kidney failure.
- The LETO authors propose testing all US candidates in the Black category, but this is their interpretation; KDIGO 2017 states that testing may be offered, and the KDIGO 2025 consensus does not recommend routine population screening.
- Race, ancestry, and genotype are not synonymous; testing should be voluntary, with qualified genetic counseling before and after, and with the right to decline.
- The genotype should be interpreted together with age, eGFR, UACR, blood pressure, and metabolic and family risk; every donor should have BP, BMI, eGFR, and albuminuria checked at least once a year.

9.8 Kidney xenotransplantation from a gene-edited pig: a functioning graft does not yet mean immunologic quiescence

Published July 24, 2026 · [online version \(Slovak\)](#)

Abstract. Kidney xenotransplantation from a gene-edited pig could help alleviate the organ shortage in the future. An editorial in AJKD discusses a multiomic analysis from Nature Medicine in the first living recipient, a 62-year-old man who received a kidney from a miniature pig with 69 genomic edits. The xenograft provided clinically meaningful function, but despite intensive immunosuppression, T-cell-mediated Banff 2A rejection developed on day 8, and innate immune activation persisted. The patient died on day 52 of a probable arrhythmia. This is a single case, and xenotransplantation remains experimental.

Summary

- The edits included knockout of GGTA1, CMAH, and B4GALNT2, seven human transgenes, and inactivation of porcine endogenous retroviruses; induction comprised antithymocyte globulin, rituximab, tegoprubart (anti-CD154, investigational), and ravulizumab.
- Banff 2A rejection on day 8 improved after methylprednisolone, ATG, tocilizumab, and pegcetacoplan; however, the Banff classification has not been validated for xenografts.
- Despite depletion of circulating lymphocytes, CD8⁺ T cells persisted in regional lymph nodes, along with activation of NK cells, monocytes, and macrophages; a biopsy on day 34 showed interstitial fibrosis and tubular atrophy in approximately 30% of the cortex.
- Circulating antibodies against porcine cells were not prominent, but glomerular deposits of IgG, IgA, C3, and C5b-9 with albuminuria suggest humoral injury not detected by crossmatch.
- Autopsy showed coronary atherosclerosis and left ventricular fibrosis without acute infarction; the contribution of systemic inflammation to death is only a hypothesis.
- This is a single patient without a control group, so the risk of rejection, long-term survival, and zoonotic risk cannot be estimated; eGFR equations may be inaccurate for monitoring a xenograft, and peripheral lymphocyte counts do not reflect immune activity in the graft.

9.9 Regulatory T cells after kidney transplantation: a path to immunologic tolerance without lifelong immunosuppression?

Published July 25, 2026 · [online version \(Slovak\)](#)

Abstract. Lifelong immunosuppression after kidney transplantation carries risks of infection, malignancy, and nephrotoxicity and does not prevent chronic graft injury. A strategic review in *Frontiers in Science*, whose authors have commercial ties to Treg therapies, discusses regulatory T cells as a path to targeted tolerance. The first small phase 1 and 1/2 trials, including the ONE Study, confirmed feasibility and acceptable short-term safety, and some allowed immunosuppression to be reduced. However, they did not demonstrate complete withdrawal of immunosuppression, which would not be safe outside clinical trials.

Summary

- Treg cells are CD4⁺ lymphocytes with high expression of CD25 and FOXP3; they suppress immune responses through IL-10 and TGF- β , CTLA-4, IL-2 consumption, adenosine (CD39, CD73), and bystander suppression.
- Products in development include polyclonal autologous, donor-reactive, TCR-modified, and CAR-Treg products (for example, against HLA-A2), universal off-the-shelf products, and in vivo expansion with low-dose IL-2; CAR-Tregs remain largely experimental.
- Clinical trials were small, mostly nonrandomized, with different products, concomitant immunosuppression, and short follow-up, so they have not confirmed routine and complete withdrawal of immunosuppression.
- Efficacy should be assessed by GFR, albuminuria, protocol biopsies, DSA, donor-derived cell-free DNA, and molecular markers; a rise in circulating Tregs is not sufficient.
- Risks include loss of the regulatory phenotype under inflammation (IL-6, IL-12, IL-1 β) with production of interferon gamma or IL-17, suppression of antiviral and antitumor immunity, and complex, costly manufacturing.
- The success of Orca-T in hematopoietic cell transplantation is not evidence of tolerance after kidney transplantation; a realistic goal is safe reduction of immunosuppressant doses with stable graft function and histology, which must be confirmed by randomized trials.

9.10 Teclistamab before kidney transplantation: a first clinical signal in extreme HLA sensitization

Published August 4, 2026 · [online version \(Slovak\)](#)

Abstract. Preformed anti-HLA antibodies can keep an extremely sensitized patient on dialysis for years. A letter in the *New England Journal of Medicine* describes a 37-year-old patient with two previous transplants who had waited more than 12 years and who, after 31 weeks of treatment with teclistamab, a BCMA $\times$ CD3 bispecific antibody, achieved a decline in anti-HLA antibodies, an expanded donor pool, and successful transplantation with a functioning graft. This is a proof of biological concept in a single patient, off label, and with substantial safety risks. Routine use is not yet justified.

Summary

- The patient had been on the waiting list for more than 12 years because of broad HLA sensitization; after 31 weeks of treatment, selected anti-HLA antibodies declined enough for the team to reassess unacceptable antigens and find a suitable organ, and at short-term assessment the patient had a functioning graft without dialysis.
- Teclistamab destroys BCMA-positive plasmablasts and plasma cells, thereby limiting the production of new antibodies, but it does not immediately remove circulating IgG, and memory B cells may persist.
- A high cPRA means few compatible donors; a cPRA of 0% does not mean zero compatibility, as the source text erroneously stated. A decrease in MFI is not a universal titer; DSA and virtual and physical crossmatches are decisive.

- In the EU, teclistamab is approved for relapsed and refractory multiple myeloma; its risks include cytokine release syndrome, hypogammaglobulinemia, cytopenias, infections, and neurotoxicity, compounded by transplant immunosuppression.
- There is no control group, no estimate of the response rate, optimal dose, or interval to transplantation, and no long-term graft follow-up; CAR T cells targeting CD19 and BCMA are a different intervention.
- The author recommends first using allocation priority, the acceptable mismatch program, paired exchange, and standard desensitization, and using teclistamab at most within an approved clinical trial with predefined goals and stopping rules.

9.11 Allopurinol in living kidney donors lowered uric acid but did not improve cardiac structure or blood pressure

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. Serum uric acid rises after donor nephrectomy, and observational data link uric acid to higher left ventricular mass. The single-center, randomized, double-blind, placebo-controlled phase 2b AL-DON trial from Oslo assigned 71 healthy living donors to allopurinol 300 mg daily or placebo for 9 months. Uric acid decreased significantly, but left ventricular mass on magnetic resonance imaging did not change (-0.25 g vs. $+1.04$ g; $p = 0.25$), nor did blood pressure or insulin sensitivity. Allopurinol therefore cannot be recommended to donors solely for cardiovascular prevention.

Summary

- Uric acid decreased in the allopurinol group ($p < 0.001$), left ventricular mass did not differ (a difference of approximately 1.3 g with a standard deviation of about 4.4 g; $p = 0.25$), and neither blood pressure ($p = 0.98$) nor insulin sensitivity ($p = 0.62$) changed.
- Adverse events occurred in 18 participants receiving allopurinol and 10 receiving placebo; all three serious events were in the active arm. The percentages in the abstract do not match the arm allocation, so the absolute counts are more reliable.
- Earlier positive studies (Kao 2011, Rekhraj 2013, Szwejkowski 2013) enrolled patients with left ventricular hypertrophy and used double the dose (600 mg); healthy donors had no room for regression.
- Limitations: a small single-center sample, 9 months of treatment, a highly selected population, a surrogate endpoint, and unpublished albuminuria and eGFR data; the result does not refute a role of urate or the standard indications for allopurinol.
- The result is consistent with the negative CKD-FIX and PERL trials; elevated urate may be a marker rather than a cause of injury.
- In practice, urate-lowering therapy in donors should be indicated only for gout or uric acid stones, and follow-up should be based on blood pressure, eGFR, albuminuria, weight, and metabolic risks.

9.12 Preemptive transplantation: the optimal form of kidney replacement

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. According to a review based on the work of Prof. Tomáš Reischig (Postgraduální nefrologie, 2026), preemptive kidney transplantation before initiation of any dialysis is associated with lower mortality and better graft survival. The evidence comes from observational registries and two meta-analyses; in the French registry, 10-year graft survival was 80% versus 61%. KDIGO recommends it with grade 1A at an eGFR below 10 mL/min/1.73 m². The benefit may be partly overestimated because of lead-time bias and selection, and early education is the key to a higher proportion of preemptive transplants.

Summary

- Time on dialysis before transplantation is a risk factor for mortality and graft failure with both living and deceased donors; in the French registry of more than 20,000 patients, the difference was significant even with dialysis lasting less than 6 months.

- Randomization is not ethically feasible; an Australian analysis correcting for lead-time bias found no significant survival difference compared with very short dialysis, and preemptively transplanted patients are younger and healthier.
- Proposed mechanisms include prevention of dialysis-associated myocardial ischemia and fibrosis and a lower burden of uremia and inflammation; a lower incidence of rejection has not been consistently confirmed.
- Transplantation at an eGFR above 10 to 20 mL/min/1.73 m² provides no additional benefit, and the IDEAL trial showed no advantage of early initiation of hemodialysis.
- The proportion of preemptive transplants is low: in the US, about 9% of deceased donor and 33% of living donor transplants, and in Europe, 19% of all transplants; University Hospital Pilsen (FN Plzeň) achieves more than 25% through education.
- Education is recommended to begin at an eGFR below 30, and the KRT modality should be chosen at an eGFR below 20 mL/min/1.73 m²; in a small RCT, enhanced nurse-led education increased the number of preemptive transplants.

9.13 After kidney graft failure: barriers to retransplantation through the eyes of patients, caregivers, and physicians

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Graft failure is among the main causes of dialysis initiation, and retransplantation confers a significant survival benefit compared with remaining on the waiting list. A Canadian qualitative study in CJASN, based on interviews with 23 patients, 23 caregivers, and 11 transplant nephrologists, identified four main barriers to retransplantation: sensitization, comorbidities, a history of nonadherence, and delayed referral. With living donation, psychological and structural barriers are added. A commentary by a patient who has undergone three transplants adds the human dimension. Planning retransplantation before definitive graft failure is essential.

Summary

- Main barriers: high sensitization with a shortage of compatible donors, comorbidities leading to ineligibility, a history of nonadherence, and delayed referral with poor coordination; with living donation, a preference for a deceased donor, a sense of an exhausted donor pool, psychological burden, and structural barriers.
- This is a qualitative design with purposive sampling that does not establish the prevalence or impact of the barriers; the patient commentary is a personal account, not clinical evidence, and transferability from Canada is limited.
- Sensitization can be counteracted by avoiding unnecessary transfusions, updating the antibody profile, and considering paired exchange or acceptable mismatch programs early; compatibility is determined by the ABO system, not by the Rh factor.
- Nonadherence should be viewed as a solvable problem, with identification of its causes and a support plan, not as a permanent contraindication; active malignancy requires individual oncologic assessment.
- Continuation of immunosuppression after graft failure should be decided individually, since both withdrawal and transplant nephrectomy are associated with sensitization; transplant nephrectomy is not performed routinely.
- In practice, when graft function declines progressively, the author recommends updating HLA typing, identifying living donors, preparing vascular access, offering psychological support, and designating a responsible coordinator.

9.14 Kidney transplantation: what is standard, what is innovation, and what is still experimental

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. Not every technical innovation in kidney transplantation improves clinical outcomes. This review maps the level of evidence for organ preservation and immunosuppression. Hypothermic machine perfusion has randomized support (DGF in 70 versus 89 recipients; OR 0.57), 1-hour normothermic perfusion did not reduce DGF in a UK RCT (60.7% vs. 58.5%), and a local C1-esterase inhibitor yielded only a pilot signal. Belatacept is an established alternative for selected patients, whereas tegoprobart remains experimental. The review warns against confusing feasibility or a mechanistic signal with proven benefit.

Summary

- In a 2009 paired study (336 donors, 672 recipients), DGF developed in 70 recipients after machine perfusion and in 89 after cold storage (OR 0.57; $p = 0.01$); 1-year graft survival was 94% versus 90%.
- In 277 kidneys from donors after circulatory death, 1 hour of NMP did not reduce DGF (OR 1.13; 95% CI, 0.69 to 1.84); NMP was feasible and safe, but other protocols have not been validated.
- A pilot study using 500 units of C1-esterase inhibitor in 20 kidneys did not reduce DGF, but median eGFR was higher at 6 months (55 vs. 39) and at 30 months (54 vs. 43 mL/min/1.73 m²); this is an exploratory signal, and the authors have patent ties.
- In the 7-year BENEFIT follow-up, belatacept was associated with higher eGFR and a lower risk of death or graft loss, but the comparator was cyclosporine; it is indicated only for EBV-seropositive recipients.
- The phase 2 BESTOW trial of tegoprobart in 127 recipients ended in September 2025 without published results, so the drug remains experimental.
- In practice, candidates should be referred early, ideally before dialysis, age should not be considered an automatic contraindication, and immunologic, infectious, metabolic, and behavioral risks should be monitored together.

CHAPTER 10

Anemia, mineral metabolism, electrolytes, and acid-base balance

Anemia of chronic kidney disease, iron metabolism, potassium, vitamin D, and other disorders of the internal milieu. Number of articles: 13.

Key messages of the chapter

Anemia of chronic kidney disease (CKD) is entering a new era with the KDIGO 2026 guideline, which defines anemia as Hb below 130 g/L in men and below 120 g/L in women and emphasizes the correction of reversible causes before starting an ESA (10.2). The most common and least expensive correctable cause is disordered iron homeostasis, yet all recommendations in the iron chapter carry only grade 2D (10.13). Initiation thresholds differ by dialysis status: in hemodialysis, ferritin up to 500 ng/mL and TSAT up to 30% with a preference for the IV route; outside hemodialysis, including PD, ferritin below 100 ng/mL and TSAT below 40%, or ferritin of 100 to 299 ng/mL and TSAT below 25% (10.2, 10.8, 10.13). Proactive IV iron in hemodialysis has randomized support from PIVOTAL (HR 0.85), and ferric carboxymaltose in CKD not on dialysis from FIND-CKD (HR 0.65); however, the safety limits of ferritin above 700 ng/mL or TSAT of 40% or higher have limited support outside hemodialysis (10.8, 10.13).

In the choice of erythropoiesis stimulation, formal guidelines clearly agree: an ESA remains the first choice over HIF-PHI, and maintenance Hb should usually be below 115 g/L because of the cardiovascular and thromboembolic risk with higher targets (10.2, 10.8, 10.9). A brief commentary without figures or studies nevertheless describes HIF-PH inhibitors becoming established as an oral alternative (10.1). In ESA hyporesponsiveness, a nonrandomized cohort linked roxadustat with more frequent achievement of Hb of at least 10 g/dL (OR 4.44), but it did not reduce Hb variability, and the result cannot be read as a causal effect (10.9). Hyporesponsiveness is therefore mainly a signal to search for iron deficiency, blood loss, inflammation, and inadequate dialysis (10.9).

The safety of IV iron shows a tension between guidelines and observational data. KDIGO 2026 and KDOQI recommend withholding iron during systemic infection (10.8, 10.13), but a retrospective TriNetX cohort found no worsening of infection and showed better hemoglobin recovery and a lower transfusion burden; these are associations without available specific figures (10.4). Ferric carboxymaltose has the highest incidence of FGF23-mediated hypophosphatemia, so phosphate should be measured before repeated doses, and ERBP does not recommend it in recent transplant recipients (10.8, 10.13). A shortened 60-minute iron dextran protocol demonstrated feasibility, but with one case of anaphylactic shock; this is not evidence of safety equivalence (10.12).

In this chapter, hyperkalemia is primarily a problem of undertreatment. In the CARE-HK registry, RAAS inhibitors or MRAs were not adequately used in about one third of patients (10.5), and in a Spanish prospective observational study, hyperkalemia recurred in 43.8% of patients, with MRAs discontinued in 21.82% of users versus 6.80% for ACE inhibitors (10.11). There is agreement that hyperkalemia alone should not lead to discontinuation of RAAS blockade and that the solution is a potassium binder, which the randomized AMBER and DIAMOND trials support (10.5, 10.6, 10.11). Patiromer and SZC differ in sodium load, and the signal of worsening heart failure with SZC comes from smaller RCTs and pooled analyses (10.5). Dietary education should be individualized, without universal lists of forbidden foods, with an emphasis on portion size, preparing potatoes by boiling with a change of water, and treating constipation (10.7).

Uncertainty prevails for other disorders. The rate of correction of chronic hyponatremia remains controversial: observational Canadian data report osmotic demyelination syndrome in only approximately 0.05% of cases, whereas preliminary PRONATREOUS data at serum sodium up to 105 mmol/L report 4.5% confirmed cases (10.3). Vitamin D is not proven kidney protection, since the randomized VITAL-DKD trial did not confirm preservation of eGFR, and calcitriol should not be used routinely in CKD not on dialysis (10.10).

A practical framework for the outpatient clinic: in anemia, check the blood count, reticulocytes, ferritin, and TSAT, treat iron deficiency first, and only then consider an ESA through shared decision-making (10.2, 10.8); in hyperkalemia, exclude pseudohyperkalemia, correct acidosis, volume status, and constipation, add a binder, and reduce the dose only as a last resort (10.6, 10.11); in chronic profound hyponatremia in high-risk patients, keep the correction within 8 mmol/L per 24 hours with frequent sodium monitoring (10.3); and decide on vitamin D according to trends in calcium, phosphate, and PTH (10.10).

10.1 Innovations in dialysis and anemia management

Published April 25, 2026 · [online version \(Slovak\)](#)

Abstract. A brief review highlights two areas of change in nephrology practice. In dialysis, the key development is the spring 2026 recommendations for incremental peritoneal dialysis, which is gentler, prolongs preservation of residual kidney function, and improves quality of life. In anemia of CKD, according to the author, HIF-PH inhibitors such as roxadustat and daprodustat are becoming established; they activate the hypoxia response without injectable ESAs. This is a short commentary without reported figures or studies.

Summary

- According to the spring 2026 recommendations, incremental PD is gentler and protects residual function.
- HIF-PH inhibitors (roxadustat, daprodustat) allow oral treatment of anemia without ESAs.
- The benefits are patient comfort and a lower logistical burden on staff.

10.2 Anemia in chronic kidney disease according to KDIGO: a brief summary of the recommendations

Published May 14, 2026 · [online version \(Slovak\)](#)

Abstract. Anemia of chronic kidney disease affects fatigue, exercise tolerance, quality of life, cardiovascular risk, and transfusion requirements. The review summarizes the KDIGO 2026 clinical practice guideline, which updates the 2012 document. Key thresholds: anemia at Hb below 130 g/L in men and below 120 g/L in women; iron at ferritin below 100 ng/mL and TSAT below 40% outside hemodialysis; and a target Hb during ESA or HIF-PHI therapy usually below 115 g/L. In practice, treatment should be individualized and correctable causes eliminated before an ESA is started.

Summary

- The main mechanism of anemia of CKD is inadequate erythropoietin production, compounded by iron deficiency, chronic inflammation, shortened red cell survival, and blood loss.
- Screening includes hemoglobin, ferritin, TSAT, and a complete blood count and, depending on the situation, vitamin B12, folate, inflammation, and blood loss; testing is performed at referral to a nephrologist, when symptoms occur, and periodically.
- Anemia is defined as Hb below 130 g/L in men and below 120 g/L in women. Outside hemodialysis (including PD), iron is started at ferritin below 100 ng/mL and TSAT below 40%, or at ferritin from 100 to below 300 ng/mL and TSAT below 25%; in hemodialysis, at TSAT up to approximately 30% and ferritin up to approximately 500 ng/mL.

- Routine iron is withheld or discontinued at ferritin of 700 ng/mL or higher or TSAT of 40% or higher; outside HD, Hb, ferritin, and TSAT are checked approximately every 3 months. In patients on hemodialysis, intravenous iron is often preferred.
- In dialysis, an ESA is considered at an Hb of approximately 90 to 100 g/L; outside dialysis, the decision is individual and usually not automatic at an Hb of 100 g/L or higher. Maintenance Hb during ESA or HIF-PHI therapy should usually be below 115 g/L, and routinely targeting an Hb of approximately 130 g/L or higher should be avoided because of cardiovascular and thromboembolic risk.
- HIF-PHIs are an option depending on the patient profile and availability. Transfusions are reserved for severe or symptomatic anemia and carry a risk of sensitization in transplant candidates; the goal is symptom improvement, not normalization of Hb at any cost.

10.3 Fast or slow correction of chronic hyponatremia? A controversy returning to clinical practice

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. How fast chronic hyponatremia should be corrected was debated by two nephrologists at the National Kidney Foundation 2026 Spring Clinical Meetings. The proponent of faster correction cites a Canadian observational analysis of more than 22,000 hospitalizations in which osmotic demyelination syndrome occurred in only approximately 0.05% of cases. The opponent points to preliminary PRONATREOUS data: at serum sodium up to 105 mmol/L, 4.5% of patients had confirmed ODS and another 14.7% had suspected ODS. In practice, individualization, conservative limits in high-risk patients, and rigorous monitoring remain the rule.

Summary

- Current guidelines for chronic hyponatremia with high ODS risk (serum sodium up to approximately 105 mmol/L, alcohol use disorder, malnutrition, advanced liver disease, hypokalemia, hypophosphatemia) specify correction of no more than 8 mmol/L in 24 hours, and 10 to 12 mmol/L per day at average risk.
- Juan Carlos Q. Velez argues that ODS is very rare in recent observational data (approximately 0.05% of more than 22,000 hospitalizations in a 2023 Canadian analysis) and that faster correction was associated in another observational study with a lower risk of 90-day mortality or delayed neurologic events.
- Helbert Rondon Berrios emphasizes that the 8 mmol/L limit applies to chronic hyponatremia at high risk, not to the acute form or to most patients with serum sodium above 120 mmol/L; at sodium up to 110 mmol/L, the incidence of ODS rose markedly.
- In preliminary PRONATREOUS data (serum sodium up to 105 mmol/L), 4.5% of patients had imaging-confirmed ODS and another 14.7% had suspected ODS, and all had correction faster than 8 mmol/L per day.
- Limitations of the evidence: these are observational analyses that often do not distinguish acute from chronic hyponatremia, lack systematic brain imaging, and are subject to confounding by indication.
- The author recommends acting quickly in acute symptomatic hyponatremia, proceeding cautiously in chronic profound hyponatremia, actively monitoring sodium because overcorrection is common, and maintaining conservative limits in high-risk patients until prospective data are available.

10.4 IV iron during acute infection: real-world data versus the fear of “worsening the infection”

Published May 22, 2026 · [online version \(Slovak\)](#)

Abstract. The rule of not giving intravenous iron during acute infection stems from the theoretical concern that iron promotes pathogen growth. A retrospective cohort study from the TriNetX network (2000 to June 2025), published in *Blood*, evaluated hospitalized adults with iron deficiency anemia and acute infection. IV iron was not associated with worsening of infection and was associated with better hemoglobin recovery, a

lower transfusion burden without prolonged hospitalization, and better survival; the available materials do not report specific figures. Systematically postponing IV iron may therefore not be justified.

Summary

- This is a retrospective real-world cohort study from the TriNetX network (2000 to June 2025) in hospitalized adults with iron deficiency anemia during acute infection; outcomes included infection-related and hard clinical outcomes, length of stay, transfusions, and hemoglobin recovery.
- IV iron was not associated with worsening of infection and was associated with better hemoglobin recovery, fewer transfusion days, no prolongation of hospitalization, and, in the broader assessment, better survival.
- The available materials do not report specific percentages or absolute differences; the author recommends verifying the figures in the primary publication.
- These are associations, not causality: patients treated with IV iron may have differed in access to treatment, anemia severity, local practices, or timing of the intervention.
- In nephrology care, iron deficiency often coincides with infection; anemia impairs tolerance of infection and recovery and increases transfusion requirements, and oral iron does not work quickly for everyone. For nephrology practice (CKD, hemodialysis, acute hospitalizations), the author recommends reconsidering local habits but always confirming true iron deficiency and excluding another dominant cause of anemia, such as bone marrow suppression.

10.5 Case-based: how to manage hyperkalemia in CKD and heart failure without reducing effective therapy

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. Fear of hyperkalemia in patients with CKD and heart failure leads to undertreatment with RAAS inhibitors and MRAs. The article draws on an educational activity built around the case of a 45-year-old woman with CKD stage 3b, eGFR below 40 mL/min, proteinuria, and HFrEF NYHA class II. The CARE-HK registry shows inadequate use of RAASi or MRAs in about one third of patients. According to KDIGO 2024, hyperkalemia alone should not be a reason to discontinue RAAS inhibition; the solution is to add a potassium binder, and patiromer and SZC differ in their safety profiles in heart failure.

Summary

- According to KDIGO 2024, as interpreted by the activity, hyperkalemia alone should not lead to discontinuation of RAAS inhibition; RAASi and MRAs should be maintained and potassium-binding therapy added, possibly with diuretics to promote potassium excretion.
- In the CARE-HK registry (HFrEF, often CKD stage 3b with eGFR below 40), RAASi or MRAs were inadequately used in about one third of patients, mainly because of concerns about hyperkalemia; episodes were recurrent, and potassium binders were underused.
- Patiromer exchanges potassium for calcium and SZC for sodium; the sodium load with SZC may worsen fluid retention and heart failure in some patients, as suggested by smaller RCTs and pooled analyses with regulatory updates.
- In REALIZE-K, SZC with spironolactone was associated with more serious adverse events due to heart failure, whereas in a DIAMOND subanalysis, NT-proBNP decreased over time with patiromer.
- The case of a 45-year-old woman with CKD stage 3b and HFrEF NYHA class II shows that concern about hyperkalemia leads to undertreatment, and GDMT optimization declines further as kidney function decreases. In practice, RAAS blockade should not be discontinued for hyperkalemia alone, fear of potassium should be overcome through treatment and monitoring, and the binder should be chosen in multidisciplinary collaboration among the cardiologist, nephrologist, and internist.

10.6 Optimizing RAASi/MRA therapy in patients with heart failure, CKD, and hyperkalemia (a practical approach)

Published June 2, 2026 · [online version \(Slovak\)](#)

Abstract. Hyperkalemia is a common barrier that prevents patients with heart failure and CKD from reaching target doses of RAASi and MRAs or leads to discontinuation of these drugs. A practical review without original study data proposes a dual-track working framework: titrate RAASi or MRAs while actively managing potassium. It includes risk stratification before titration, early laboratory checks, adjustment of diuretics, diet, and metabolic acidosis, and potassium binders for recurrent K⁺ elevation. Success means stable K⁺, therapy maintained at the highest tolerated dose, and no acute escalations.

Summary

- In practice, RAASi and MRAs are undertitrated or interrupted mainly because of high or fluctuating K⁺, worsening kidney function or dehydration, drug interactions, lack of prevention before titration, and late monitoring after dose adjustment.
- Before titration, the author recommends assessing current K⁺ and its trend, eGFR and creatinine dynamics, volume status and diuretics, potassium-raising drugs, metabolic acidosis, and diet; after a dose change, K⁺ should be checked within the first days to weeks according to the local protocol and risk.
- When K⁺ rises, the recommendations are to optimize diuretics in patients prone to fluid retention, educate about potassium sources instead of imposing blanket prohibitions, correct metabolic acidosis, and, for recurrent or persistent elevation, start potassium binders, which international guidelines cite as a tool for maintaining RAASi and MRAs.
- Success is defined on three levels: consistently acceptable K⁺, RAASi or MRA maintained at the highest tolerated dose, and no repeated hospitalizations or urgent treatment changes due to hyperkalemia.
- Common errors include reacting only to a marked K⁺ rise, complete withdrawal without a potassium control plan, overlooking interactions and kidney function dynamics, poor education, and insufficient monitoring after each dose adjustment; this is a working framework, not the results of a specific study.

10.7 Potassium control in kidney disease: educate, don't scare

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Traditional education on hyperkalemia relies on lists of forbidden foods that cause fear, resignation, and unnecessary restrictions in patients with CKD and on dialysis. A practical review advocates individualized education based on serum potassium, treatment type, residual urine output, medications, and nutritional status. Boiling diced potatoes with a change of water can reduce their potassium content by approximately 50% to 75%, and a cup of melon or green grapes contains about 180 mg of potassium. Portion size and treatment of constipation are decisive, but severe hyperkalemia warrants a stricter approach.

Summary

- A single universal list of forbidden foods is not sensible; recommendations should take into account current potassium, treatment type, residual urine output, dialysis dose, medications, nutritional status, constipation, and clinical context.
- Potatoes should be peeled, diced, and boiled in a large volume of water; after approximately 10 minutes the water is drained and cooking is finished in fresh water. According to the cited source, this reduces potassium by approximately 50% to 75%, whereas soaking alone has only a limited effect. The principle can also be applied to legumes or squash, but the evidence is strongest for tubers.
- A cup of melon or green grapes contains approximately 180 mg of potassium, less than the usual threshold of 200 mg per serving for a high-potassium food; freezing fruit and making fruit salad with half a portion of high-risk fruit can help.
- As kidney excretion declines, the gut contributes compensatorily to potassium excretion, so constipation worsens potassium control; recommended measures include fluid adjustment, lower-potassium fiber, suitable laxatives agreed with the team, and physical activity.

- In stable patients, the author prefers targeted education, portion management, and modified food preparation; severe or recurrent hyperkalemia requires a stricter approach guided by laboratory results, ECG, treatment, residual function, and the dialysis regimen.

10.8 Anemia of CKD according to KDIGO 2026: what the KDOQI US Commentary brings to practice

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. A KDOQI work group prepared a US commentary on the KDIGO 2026 clinical practice guideline for anemia of CKD, with an emphasis on implementation in practice. The commentary endorses the iron thresholds (in hemodialysis, ferritin up to 500 ng/mL and TSAT up to 30%, with a preference for IV iron), the PIVOTAL safety limits (ferritin above 700 ng/mL or TSAT of 40% or higher), and ESAs as first choice ahead of HIF-PHIs. It draws attention to hypophosphatemia after ferric carboxymaltose. In practice, this means a stepwise diagnostic evaluation, treatment of iron deficiency, and shared decision-making about ESAs.

Summary

- The basic panel comprises a complete blood count, reticulocytes, ferritin, and TSAT; when the cause is unclear, the panel is expanded (hemolysis, CRP, B12, folate, paraproteins, TSH, PTH). Ferritin below 45 ng/mL, MCV below 80 fL, or TSAT below 20% with unexplained deficiency should prompt a search for bleeding.
- In hemodialysis, iron is started at ferritin up to 500 ng/mL and TSAT up to 30%, and the IV form is preferred. Outside HD (including PD), it is started at ferritin below 100 ng/mL and TSAT below 40%, or at ferritin from 100 to below 300 ng/mL and TSAT below 25%.
- It is reasonable to withhold iron temporarily at ferritin above 700 ng/mL or TSAT of 40% or higher, which corresponds to the limits of the proactive arm of PIVOTAL; the commentary considers them justified mainly in HD, whereas safety data for PD and nondialysis CKD are limited.
- Hb, ferritin, and TSAT are checked every 3 months outside dialysis and in PD, and every 1 to 3 months in HD; if oral iron fails after 1 to 3 months, IV iron is used instead, and iron is paused during systemic infection.
- Ferric carboxymaltose has the highest incidence and longest duration of FGF23-mediated hypophosphatemia; phosphate should be measured in patients with fatigue, bone pain, or weakness and before repeat doses.
- After correction of causes, an ESA rather than a HIF-PHI is the first choice; in dialysis, an ESA is started at an Hb of 90 to 100 g/L, outside dialysis according to symptoms and risks, and maintenance Hb should be below 115 g/L, within shared decision-making.

10.9 Roxadustat in ESA hyporesponsiveness: more frequent repeated achievement of target Hb, not lower variability

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. ESA hyporesponsiveness in hemodialysis signals correctable causes rather than being a standalone diagnosis. A secondary analysis of a retrospective multicenter cohort (Demir et al., Hemodialysis International 2026) compared roxadustat (78 patients) with continued ESA therapy (30 patients). Roxadustat was associated with a higher adjusted probability of achieving an Hb of at least 10 g/dL in three of six months (OR 4.44; 95% CI, 1.24 to 15.84), but it did not significantly reduce Hb variability. These are nonrandomized data, and KDIGO 2026 continues to prefer ESAs as first choice.

Summary

- The analysis included 108 adults on maintenance hemodialysis with complete monthly Hb values over six months; treatment allocation was not randomized, and the target of an Hb of at least 10 g/dL in three of six months did not require consecutive months.

- The target was achieved by 52 of 108 patients, with an adjusted OR for roxadustat versus ESA of 4.44 (95% CI, 1.24 to 15.84; $p = 0.022$); baseline Hb was a strong predictor (OR 2.95 per 1 g/dL). At month 6, 50.0% of patients on roxadustat and 33.3% on ESA had an Hb of at least 10 g/dL; this is a descriptive finding.
- Measures of month-to-month Hb variability were only numerically lower with roxadustat, without statistical significance; in the original cohort, after adjustment for baseline Hb, treatment group was not independently associated with final Hb, so the OR of 4.44 does not represent a fourfold causal effect.
- The small sample and six-month follow-up do not allow safety to be assessed; according to the EMA, known risks of roxadustat are hypertension, hyperkalemia, edema, and vascular access thrombosis.
- KDIGO 2026 suggests an ESA rather than a HIF-PHI as first choice (2D); a trial of a HIF-PHI in hyporesponsiveness is possible, although the evidence is weak, and ESAs and HIF-PHIs are not combined. Hb is checked 2 to 4 weeks after a change and then every four weeks, and with roxadustat, thyroid function is also monitored during the first three months.
- The author recommends first verifying dosing and adherence and evaluating iron, blood loss, inflammation, dialysis adequacy, and other causes, and only then individually choosing an ESA, a HIF-PHI, or transfusion; the EMA does not recommend switching a stable patient on an ESA without a clinical reason.

10.10 Vitamin D in clinical practice: whom to test, whom to treat, and when treatment can cause harm

Published August 13, 2026 · [online version \(Slovak\)](#)

Abstract. A 25(OH)D concentration is not a diagnosis, and a higher dose of vitamin D is not automatically more effective. The review draws mainly on the Endocrine Society 2024 guideline, KDIGO, and the randomized VITAL and VITAL-DKD trials. Routine screening of healthy adults is not recommended; empirical supplementation is supported in children aged 1 to 18 years, pregnant women, people aged 75 years or older, and those with high-risk prediabetes. VITAL-DKD in 1,312 patients with diabetes did not confirm preservation of eGFR with 2,000 IU daily. In CKD, nutritional vitamin D must be distinguished from calcitriol, and decisions should be based on CKD-MBD trends.

Summary

- The Endocrine Society 2024 suggests not testing 25(OH)D in healthy asymptomatic adults, or in those with obesity or darker skin pigmentation without an indication, and does not endorse a target of at least 30 ng/mL (75 nmol/L); as a rough guide, the risk of deficiency rises below 12 ng/mL, and 20 ng/mL is sufficient for most healthy people (1 ng/mL = 2.5 nmol/L).
- Empirical supplementation is suggested in children aged 1 to 18 years, pregnant women, people aged 75 years or older, and those with high-risk prediabetes (a relative reduction in progression to diabetes of approximately 15% and an absolute reduction of 3.3 percentage points over three years); the optimal dose is uncertain.
- In healthy people younger than 75 years, empirical supplementation above the reference intake is not recommended; in VITAL, 2,000 IU daily did not reduce invasive cancers or major cardiovascular events. Daily dosing is preferred: an annual bolus of 500,000 IU in 2,256 women aged 70 years or older increased both falls and fractures.
- In CKD G3a to G5D, 25(OH)D may be considered as part of the assessment of calcium, phosphate, PTH, and alkaline phosphatase; calcitriol and its analogs should not be used routinely in CKD G3a to G5 not on dialysis and are reserved for G4 to G5 with severe progressive secondary hyperparathyroidism.
- Vitamin D is not proven kidney protection: in VITAL-DKD, 2,000 IU daily for five years in 1,312 adults with type 2 diabetes did not significantly preserve eGFR.
- The tolerable upper intake level for a healthy adult is 4,000 IU daily; toxicity is associated with 25(OH)D concentrations often above 150 ng/mL (375 nmol/L) and manifests as hypercalcemia; all sources of intake should be added up.

10.11 Hyperkalemia in CKD in real-world practice: recurrence is the rule, and mineralocorticoid receptor antagonists pay the highest price

Published August 24, 2026 · [online version \(Slovak\)](#)

Abstract. Hyperkalemia in CKD often leads to weakening of cardiorenal-protective therapy. A Spanish prospective multicenter observational study of 429 patients with serum potassium above 5.5 mmol/L from fifteen hospitals describes real-world practice over two years. Recurrence occurred in 43.8% of patients, and mineralocorticoid receptor antagonists were discontinued in 21.82% of users versus 6.80% for ACE inhibitors. However, the design does not allow causal conclusions or comparison of binders. The goal of management is normal potassium with effective therapy maintained.

Summary

- The cohort comprised 429 patients (mean age 71.5 years, 71.3% men, 88.6% with hypertension, 55.9% with diabetes); 19.1% had moderate and 3.7% severe hyperkalemia; use of dietary modifications increased by 39.9%, sodium zirconium cyclosilicate by 24.6%, patiromer by 10.9%, and resins by only 2.6%.
- ACE inhibitors were reduced in 12.00% and discontinued in 6.80% of users of the respective class, ARBs in 5.49% and 7.32%, and mineralocorticoid receptor antagonists in 12.73% and 21.82%.
- Recurrence occurred in 43.8% of patients, more often at eGFR below 30 mL/min (49.1% vs. 38.8%; $p = 0.03$), and was associated with a decline in filtration of 2.4 mL/min per year; no difference in survival was found.
- Limitations: only unadjusted proportions without hazard ratios or confidence intervals, no definition of recurrence, possible immortal time bias and bias from sampling intensity, a nonrandomized comparison of binders with resins, and unreported cardiovascular events; the authors themselves describe causality as unclear.
- Maintaining RAAS blockade with a binder is supported by the randomized AMBER (patiromer with spironolactone) and DIAMOND trials, not by this study.
- The author recommends ruling out pseudohyperkalemia, eliminating causes (acidosis, constipation, dehydration, NSAIDs, trimethoprim, heparin, salt substitutes), proceeding stepwise from correction of acidosis and volume through diet, a loop diuretic, and a binder, with dose reduction only as a last resort, planning reintroduction of therapy, and documenting the reason for the intervention.

10.12 60-minute iron dextran infusion at a smaller center: benefits, risks, and limits of the evidence

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. Shortening the administration of intravenous iron dextran can save time and staff resources. A single-center retrospective observational study from a rural Veterans Health Administration facility compared 123 administrations before and 123 after introduction of a 60-minute protocol. Mean total time decreased from 189.2 to 80.1 minutes, but under the new regimen a grade 4 anaphylactic shock occurred during the test dose. The study demonstrates feasibility, not safety equivalence, and the US protocol, with doses outside approved dosing, cannot be transferred to European nephrology practice without modification.

Summary

- The study evaluated 160 patients and 246 administrations (2020 to 2025); the extended regimen used a 15-minute test dose, 45 minutes of observation, and a 120- to 180-minute infusion, whereas the new protocol used fixed doses of 500, 750, or 1,000 mg, a 5-minute test dose, 15 minutes of observation, and a rate of 300 mL/h.
- Total time decreased from 189.2 to 80.1 minutes ($P < 0.01$), from 218.5 to 86.7 minutes in patients without prior exposure, and from 169.7 to 67.3 minutes in previously treated patients.
- Under the new regimen, six events occurred in 123 administrations (4.9%): five mild gastrointestinal events and one anaphylactic shock requiring vasopressors and ICU admission; under the extended regi-

men, there was one grade 3 hypertension event (0.8%). The difference was not significant, but with a single serious case in the group the estimate is very imprecise.

- In a JAMA Network Open cohort (35,737 administrations), reactions occurred with 3.9% of all administrations and 3.8% of iron dextran administrations; an older analysis of 20,213 patients reported reactions requiring resuscitation drugs in 0.035%, all with the test or first dose.
- Doses of 500 to 1,000 mg are outside the approved dosing of INFeD in the US; the EMA does not routinely recommend a test dose but requires observation for at least 30 minutes after administration, and KDIGO 2026 does not consider routine premedication necessary.
- The study did not focus on CKD or dialysis; an internal protocol must respect the specific product and provide trained staff, epinephrine and resuscitation equipment, an infusion-stopping algorithm, and standardized documentation, without shortening the regulatory observation period.

10.13 Iron in anemia of CKD according to KDIGO 2026: initiation thresholds, proactive IV repletion, and the European ERBP perspective

Published October 1, 2026 · [online version \(Slovak\)](#)

Abstract. Disordered iron homeostasis is the most common and least expensive correctable cause of anemia of CKD. The review discusses the iron chapter of the KDIGO 2026 guideline, all four recommendations of which are graded 2D, and the 2026 European ERBP commentary. Proactive IV iron in hemodialysis rests on the PIVOTAL RCT (2,141 patients), in which it reduced the primary composite endpoint (29.3% vs. 32.3%; HR 0.85). Initiation thresholds differ by dialysis status, and ERBP points out the limited validity of the safety limits outside hemodialysis.

Summary

- All recommendations in the chapter are graded 2D, and the eleven practice points are ungraded; absolute deficiency is termed systemic iron deficiency, and functional deficiency is termed iron-restricted erythropoiesis.
- In G5HD, iron is started at ferritin up to 500 ng/mL and TSAT up to 30%, with a preference for the IV form; outside HD and in PD, at ferritin below 100 ng/mL and TSAT below 40%, or at ferritin from 100 to 299 ng/mL and TSAT below 25%.
- In PIVOTAL, proactive iron sucrose 400 mg monthly reduced the primary endpoint (29.3% vs. 32.3%; HR 0.85; 95% CI, 0.73 to 1.00) as well as the ESA dose, without an increase in infections; ERBP notes that the trial involved incident patients treated with ESAs and compared strategies.
- In FIND-CKD (626 nondialysis patients), ferric carboxymaltose with a target ferritin of 400 to 600 µg/L reduced the primary endpoint compared with oral iron (23.5% vs. 31.8%; HR 0.65); ERBP considers ferritin of 400 to 600 ng/mL a reasonable target in this setting.
- Iron is withheld at ferritin above 700 ng/mL or TSAT of 40% or higher (the PIVOTAL limits) and during systemic infection; after 1 to 3 months of ineffective or poorly tolerated oral therapy, IV iron is used instead.
- Neither a test dose nor premedication is routinely required; according to ERBP, ferric carboxymaltose is not recommended in recent kidney transplant recipients because of the risk of severe hypophosphatemia. At ferritin below 30 ng/mL and TSAT below 20%, iron should be considered even without anemia.

CHAPTER 11

Differential diagnosis, genetics, onconeurology, and rare diseases

Diagnostic reasoning, hematuria, genetics of kidney disease, onconeurology, monoclonal gammopathies, and rare conditions that should be considered early. Number of articles: 18.

Key messages of the chapter

The common denominator of this chapter is disciplined diagnostic reasoning. A good history remains the key tool, yet according to the cited studies, physicians interrupt the patient after 11 to 18 seconds (11.1). A small qualitative study with nine nephrologists showed that experienced physicians switch between Occam's razor and Hickam's dictum according to case complexity, and time pressure pushes them toward the simplest defensible diagnosis; however, the benefit of explicit teaching for outcomes has not been verified (11.10). The Klug entscheiden initiative complements this framework with a parsimonious laboratory panel: quantification of albuminuria at an eGFR below 60 mL/min/1.73 m² and measurement of Ca, P, iPTH, and 25-OH vitamin D3 at an eGFR below 45, although these are recommendations of professional societies (11.5).

Genetics is becoming a routine part of nephrology diagnostics but is little used in practice. In a self-reported NKF survey, 61.4% of respondents had an indication, yet only 8.6% were tested (11.4). A genetic diagnosis can change treatment, for example in Fabry disease or primary hyperoxaluria type 1, and influence the choice of a related donor (11.4). A negative test, however, does not rule out a genetic cause: in a small retrospective study, exome sequencing after a negative panel yielded a new finding in 13% of patients, which is not a universal probability of success (11.14). In Bartter syndrome, the current classification requires a panel that also covers deletions, particularly in CLCNKB, and a small Korean cohort showed CKD G3 to G5 in 11% of patients (11.12). In ADPKD, active, risk-oriented management with tolvaptan in rapid progressors is gaining ground (11.2); preclinical data and a small phase 2 study with probenecid suggest attenuation of aquaresis, but this is a mechanistic finding without a recommendation for practice (11.6). In Prader-Willi syndrome, diazoxide choline has randomized evidence only for maintenance of effect in the withdrawal phase and is not authorized in the EU (11.9).

Onconeurology forms the second major axis. Immune checkpoint inhibitors mainly cause acute tubulointerstitial nephritis, which must be distinguished from prerenal causes, sepsis, obstruction, and nephrotoxic drugs (11.7, 11.8, 11.15). In an observational cohort of 429 patients with ICI-associated AKI, nephritis was found in 82.7% of biopsies, and recurrent AKI after rechallenge occurred in 16.5% (11.15). The randomized KEYNOTE-564 trial confirmed the overall survival benefit of adjuvant pembrolizumab (HR 0.66) but excluded patients on dialysis and transplant recipients (11.15); in transplant recipients, ICIs may increase the risk of rejection (11.8). In myeloma, rapid reduction of light chains is decisive (11.7), and the concept of monoclonal gammopathies of clinical significance, based predominantly on case reports, highlights that even a small clone can be pathogenic (11.11). In iTTP, the evidence is strong: in the randomized HERCULES trial, caplacizumab reduced the composite endpoint from 49% to 12%, whereas the plasma exchange-free regimen from the single-arm MAYARI study belongs only in specialized centers (11.13).

For hematuria and urolithiasis, this chapter emphasizes stratification instead of automatic workup. Microhematuria requires confirmation of at least 3 RBC/HPF and a risk-appropriate urologic evaluation in parallel with nephrologic assessment (11.17). Radiologic compression of the left renal vein is

common, nutcracker syndrome has no validated criteria, and evidence on treatment comes only from nonrandomized series (11.18). According to AUA 2026, prevention of nephrolithiasis rests on a 24-hour urine collection, a urine volume of at least 2.5 liters per day, and phenotype-directed therapy (11.16). A urinary molecular fibrosis test achieved a sensitivity of 84% and a specificity of 94% in a pilot study, but confirmation in large cohorts is lacking (11.3).

A practical framework for the outpatient clinic: let the patient speak and name in advance the threshold for revising the diagnosis (11.1, 11.10); in CKD of unclear etiology, early onset, a positive family history, or before related-donor transplantation, actively consider genetic testing and read a negative result in the context of the test's scope (11.4, 11.14); before cancer therapy, know the creatinine, eGFR, albuminuria, and sediment, and involve the nephrologist early (11.7, 11.8); with a paraprotein, perform immunofixation and free light chain testing (11.11); in thrombocytopenia with microangiopathic hemolysis, urgently consider TTP (11.13); and do not close the workup of hematuria or stones without risk stratification (11.16, 11.17).

11.1 Five practical tips for better history-taking and a more accurate diagnosis

Published May 13, 2026 · [online version \(Slovak\)](#)

Abstract. A well-taken history is among the most important diagnostic tools, yet according to the cited studies, physicians often interrupt the patient after only 11 to 18 seconds. A short practical article without original data offers five communication tips: let the patient speak for at least a minute, move on to structured questions, specifically look for red flags, do not let the computer get in the way of the conversation, and finally summarize what was heard. The aim is to reduce diagnostic errors arising from misunderstanding and to improve clinical decision-making.

Summary

- According to studies, physicians interrupt the patient after 11 to 18 seconds; a minute of uninterrupted speech often yields key information, after which the clinician should move from open to closed questions.
- Red flags should be specifically sought: unexplained weight loss, night sweats and low-grade fever, a sudden change in the character of pain, neurologic deficits, and fatigue not relieved by rest.
- Eye contact and empathy, even while working with the electronic health record, increase trust, and a final recap allows the patient to correct or add information.

11.2 Optimizing treatment outcomes in polycystic kidney disease

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. Autosomal dominant polycystic kidney disease (ADPKD) is the most common inherited kidney disease, with a variable rate of progression. A review based on a Medscape educational program describes a shift from passive observation to active, risk-oriented management. It emphasizes early identification of rapid progressors, rigorous treatment of hypertension with ACE inhibitors or ARBs, lifestyle modification, and, in selected patients, tolvaptan under the supervision of an experienced nephrologist. The article does not report specific trial figures; its value lies in a practical framework for long-term care.

Summary

- Not every patient with ADPKD progresses at the same rate: some have stable function for years, whereas others reach advanced CKD by middle age. Assessment includes family history, ultrasonography, eGFR, urinalysis, blood pressure, kidney size, and cyst burden, with genetic testing in selected patients; the goal is to identify the risk of rapid progression.
- Hypertension often appears before a marked decline in GFR and accelerates kidney damage; it is treated mainly with ACE inhibitors or ARBs, and adequate fluid intake, less salt, not smoking, a healthy weight, and avoidance of nephrotoxic drugs are recommended.

- Tolvaptan can slow cyst growth and kidney function decline in patients at risk of rapid progression, but it requires liver function monitoring and counseling about polyuria and thirst; it should be prescribed by an experienced nephrologist.
- Extrarenal manifestations include liver cysts, heart valve abnormalities, diverticulosis, and intracranial aneurysms. Screening for intracranial aneurysms is considered individually in patients with a family history of aneurysm or subarachnoid hemorrhage, a high-risk occupation, before major surgery or transplantation, and in the presence of neurologic symptoms.
- Regular monitoring of blood pressure, kidney function, albuminuria, and complications is needed, along with timely preparation for kidney replacement therapy, for which transplantation is a suitable option for many patients.

11.3 Molecular imaging from urine: a new biomarker test that may detect early kidney fibrosis

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. Kidney fibrosis is a key mechanism of CKD progression, but its assessment currently requires an invasive biopsy. A pilot study in Science Translational Medicine introduced a urine test with a fluorescent reporter activated by the biomarkers transglutaminase 2 (TG2) and allysine (LysAld). In a cohort of 35 participants, it distinguished fibrosis from healthy controls with a sensitivity of 84% and a specificity of 94% and discriminated mild from severe fibrosis, which eGFR, creatinine, and urea could not. The group was small, and confirmation in large longitudinal studies is lacking.

Summary

- Biopsy is invasive and subject to sampling error and interobserver variability, and blood biomarkers often cannot distinguish kidney fibrosis from other conditions.
- The reporter binds to LysAld and is activated by TG2 to fluoresce; in a cohort of 35 participants, it achieved a sensitivity of 84% and a specificity of 94% and distinguished histologically confirmed mild from severe fibrosis.
- The results come from a small group; the authors see the test as a complementary method whose diagnostic and prognostic value and usefulness for monitoring treatment must be confirmed in larger studies.

11.4 Genetic testing in kidney disease is still offered too rarely

Published June 5, 2026 · [online version \(Slovak\)](#)

Abstract. A genetic basis may underlie 20% to 30% of cases of chronic kidney disease, yet genetic testing is rarely offered. A survey of 1,168 patients from the National Kidney Foundation, presented at the NKF 2026 Spring Clinical Meetings, showed that 61.4% of respondents had an indication, but a physician had discussed testing with them in only 11% to 29% of cases, depending on the number of indications, and only 8.6% had been tested. Yet a genetic diagnosis can change treatment and influence the selection of a related donor.

Summary

- Of 1,168 respondents from 49 US states, 717 (61.4%) had at least one indication: familial occurrence (65.1%), a diagnosis for which testing is recommended (64.7%), extrarenal manifestations (27.3%), or CKD of unknown cause before age 50.
- A discussion about testing was reported by 11% of patients without an indication, 16% with one, 25% with two, and 29% with three indications; only 8.6% of all respondents had been tested. This is a self-reported survey within a patient organization.
- Genetic confirmation can enable targeted therapy, for example enzyme replacement in Fabry disease, lumasiran in primary hyperoxaluria type 1, or high-dose coenzyme Q10 in primary coenzyme Q10 deficiency, and can protect a related donor with the same predisposition.

- Barriers include physicians' uncertainty in communicating about genetics, cost concerns, unequal access, low patient awareness, and the rapid evolution of the field.
- The author recommends actively considering testing in CKD of unclear etiology, early onset, a positive family history, extrarenal manifestations, and before living-related transplantation, but not routinely in every patient.

11.5 Klug entscheiden in nephrology: which laboratory tests are really necessary?

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. The German initiative Klug entscheiden, led by the DGIM, seeks to limit blanket laboratory screening without a clinical reason while not omitting tests that change management. For nephrology, it formulates five positive recommendations: creatinine and urinalysis in at-risk patients, quantification of albuminuria at eGFR below 60 mL/min/1.73 m², measurement of Ca, P, iPTH, and 25-OH vitamin D3 at eGFR below 45, careful consideration of contrast, and checking vaccination status. These are recommendations of professional societies; their value lies in a lean, targeted laboratory panel.

Summary

- In patients at increased cardiovascular and kidney risk, creatinine with eGFR calculation and urinalysis are recommended, at an interval of once every 1 to 2 years according to the document.
- In CKD with eGFR below 60 mL/min/1.73 m², albuminuria or proteinuria should be quantified alongside eGFR, for example as ACR, because it substantially adds prognostic information; at eGFR below 45, Ca, P, iPTH, and 25-OH vitamin D3 should be measured at least once and monitored if abnormal.
- eGFR alone may not explain the risk; the purpose of mineral and bone metabolism testing is to detect CKD-MBD early, while it can still be specifically addressed. In CKD stages 4 to 5 (eGFR below 30), the benefit of a contrast study should be weighed against its risk so as to avoid systematic underdiagnosis; in CKD or immunosuppression, vaccination status should be checked regularly.
- The negative recommendations reject automatic high oral fluid intake to improve kidney function, diuretics and fluids without an indication in acute kidney injury, and regular NSAID use in hypertension or CKD; other tests should be tied to symptoms, risk, and a specific clinical question.

11.6 A new vasopressin-independent pathway of water regulation in the kidney and its significance for the treatment of ADPKD

Published July 2, 2026 · [online version \(Slovak\)](#)

Abstract. Tolvaptan slows ADPKD progression, but its aquaretic effect, with polyuria and nocturia, impairs tolerability and adherence. Translational research at Mayo Clinic described a vasopressin-independent urate, AMPK, and AQP2 pathway in the collecting ducts that increases water reabsorption. In preclinical models, probenecid reduced cyst growth and attenuated tolvaptan-induced diuresis, and in a small phase 2 study it reduced urine volume by approximately 30%. This is a mechanistic finding, not a recommendation for routine practice.

Summary

- An increase in intracellular urate in the collecting ducts (uptake via GLUT9b, reduced efflux via ABCG2) triggers a cascade involving PDE4, a decrease in cAMP, and AMPK activation, which accumulates AQP2 at the apical membrane independently of the V2 receptor axis.
- In preclinical models, probenecid reduced cyst growth and markedly attenuated the tolvaptan-induced increase in urine volume, thereby separating cyst suppression from the aquaretic effect.
- In a small phase 2 study in patients treated with tolvaptan, probenecid reduced urine volume by approximately 30% and nocturia to approximately once per night, and most patients reported better quality of life; the data come from Mayo Clinic material for the public.
- Clinically, the ability to slow cystic progression must be distinguished from the tolerability of the aquaretic effect; mitigating aquaresis without losing the anticystic benefit could improve adherence and

the sustainability of treatment. Probenecid does not act selectively, affects uric acid, and has drug interactions, so the author does not recommend adding it off label to tolvaptan outside protocols; in the future, urine volume and nocturnal diuresis may become measurable clinical treatment targets.

11.7 Post-congress notes from EHA and ASCO 2026: what oncology updates have in common with nephrology

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. Modern anticancer therapy, namely immunotherapy, targeted drugs, CAR T cells, and bispecific antibodies, improves survival but increases the demands for monitoring kidney complications. A commentary on the EHA and ASCO 2026 post-congress format from a nephrologist's perspective does not cite specific studies or figures, because the content of the source video could not be verified. It summarizes the main onconeurology risks: immune-mediated nephritis, hypertension and proteinuria with anti-VEGF therapy, myeloma kidney, tumor lysis syndrome, and drug dosing. For practice, it recommends early nephrology involvement and systematic monitoring.

Summary

- With immune checkpoint inhibitors, the best-known complication is acute tubulointerstitial nephritis with a nonspecific presentation; however, a rise in creatinine must first be distinguished from prerenal causes, sepsis, obstruction, contrast, nephrotoxic drugs, proton pump inhibitors, and NSAIDs, and biopsy helps in unclear cases.
- VEGF inhibitors cause hypertension, proteinuria, nephrotic syndrome, and thrombotic microangiopathy; with tyrosine kinase inhibitors, creatinine, eGFR, urine, blood pressure, potassium, magnesium, and interactions should be monitored.
- In myeloma, kidney involvement is signaled by an unexplained rise in creatinine, proteinuria with disproportionately low albuminuria, hypercalcemia, anemia, and high free light chains; rapid reduction of light chains improves the chance of recovery, so time is critical.
- AKI with CAR T cells and bispecific antibodies is often multifactorial; prevention of tumor lysis syndrome includes risk stratification, hydration, electrolyte monitoring, and a xanthine oxidase inhibitor or uricolytic therapy according to risk.
- In sarcopenia or cachexia, creatinine-based eGFR overestimates function; the author recommends following the trend and considering cystatin C, therapeutic drug monitoring, and consultation with a pharmacologist.
- A nephrologist should be involved early in CKD G3b to G5, significant albuminuria, previous AKI, myeloma, hypertension during anti-VEGF therapy, or a high risk of tumor lysis; this is a commentary, not a summary of specific congress data.

11.8 Modern treatment of melanoma: what internists and nephrologists should know

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. The incidence of melanoma is rising, and by 2040 it may reach approximately 510,000 new cases and 96,000 deaths per year. The article is based on a Medscape educational clinical quiz, not an original study, and summarizes current treatment from the perspective of internists and nephrologists: individualized sentinel lymph node biopsy, abandonment of routine completion lymph node dissection after MSLT-II, adjuvant anti-PD-1 therapy in stages IIB to IIC, and BRAF plus MEK combinations. For nephrologists, it is essential to recognize immune-mediated nephritis, differentiate common causes of AKI, and manage the risk of rejection in transplant recipients.

Summary

- Sentinel lymph node biopsy is considered at a tumor thickness of approximately 1.0 mm or more, and for thinner lesions of about 0.8 mm with risk features; after MSLT-II, routine immediate completion dissec-

tion is not recommended because it does not improve melanoma-specific survival and increases the risk of lymphedema.

- For stage IIA, surveillance is recommended; for IIB and IIC, pembrolizumab or nivolumab may be considered individually, as they improved recurrence-free survival in KEYNOTE-716 and CheckMate 76K; adjuvant immunotherapy is also an option in completely resected stage IV disease.
- In metastatic melanoma with a BRAF V600 mutation, combinations of dabrafenib with trametinib, vemurafenib with cobimetinib, and encorafenib with binimetinib are used; interferon and chemotherapy have receded, and larotrectinib has a role only in NTRK fusion.
- The typical kidney complication of immunotherapy is acute tubulointerstitial nephritis; the risk is increased by CKD and concomitant proton pump inhibitors, NSAIDs, or antibiotics, so creatinine, eGFR, albuminuria, and urine sediment should be known before treatment.
- When creatinine rises, hypovolemia, sepsis, obstruction, contrast, and nephrotoxic drugs should be ruled out first; during BRAF and MEK therapy, temperature, hydration, electrolytes, and muscle symptoms should be monitored.
- In kidney transplant recipients, checkpoint inhibitors may increase the risk of rejection, so the risk of graft loss and return to dialysis should be discussed in advance with the oncologist and the transplant nephrologist.

11.9 Prader-Willi syndrome across the lifespan: genetics, hyperphagia, and coordinated care

Published August 4, 2026 · [online version \(Slovak\)](#)

Abstract. Prader-Willi syndrome (PWS) is a lifelong neurodevelopmental and endocrine disorder whose priorities shift from neonatal hypotonia to hyperphagia, sleep disorders, and metabolic complications. A commentary on a Medscape educational activity (a CME synthesis, not a systematic review) adds the regulatory status and the strength of evidence. Diazoxide choline extended-release was approved by the FDA in 2025 and, in the randomized withdrawal phase, improved the HQ-CT score by 5.0 points, but it is not authorized in the EU. For nephrologists, the key issues are secondary kidney risk and safe treatment monitoring.

Summary

- DNA methylation analysis together with an oligonucleotide SNP array establishes the diagnosis and, in most cases, the mechanism (deletion of the paternal 15q11.2-q13 region, maternal uniparental disomy, imprinting defect); the mechanism is important for genetic counseling.
- The foundation is a structured environment with controlled access to food, appropriate energy intake, and physical activity; growth hormone improves growth and body composition, and GeneReviews recommends polysomnography before and after its initiation.
- Vykat XR was approved by the FDA on March 26, 2025, for patients aged four years and older; in a 16-week withdrawal phase with 77 previously treated participants, the difference on the HQ-CT scale was -5.0 points (95% CI, -8.1 to -1.8), which demonstrates maintenance of effect, not efficacy of treatment initiation.
- The drug can cause hyperglycemia, including ketoacidosis, and fluid retention, including pulmonary edema; it has not been studied in kidney impairment, and in the EU, the marketing authorization application for Viokat was withdrawn in April 2026.
- Liraglutide in 55 children and adolescents did not meet the coprimary BMI z-score endpoints, data on semaglutide come mostly from case reports, topiramate failed on the primary outcome in the TOPRA-DER trial (62 participants), and the phase 3 pitolisant trial has not yet reported results.
- Depending on risk, the nephrologist should screen for CKD (blood pressure, eGFR, albuminuria) and monitor bicarbonate, acidosis, and nephrolithiasis with topiramate, volume status with diazoxide choline, and hypovolemia with the risk of AKI with GLP-1 receptor agonists.

11.10 Occam or Hickam? How nephrologists switch between diagnostic frameworks

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. Patients in nephrology often have several concurrent diseases, so the choice between Occam's razor and Hickam's dictum is an everyday question. A Canadian qualitative study using semistructured interviews with nine nephrologists showed that diagnostic reasoning is learned implicitly, that experienced physicians switch between frameworks depending on case complexity, and that the publicly funded system pushes toward the simplest defensible diagnosis. The authors propose explicit teaching of heuristic switching and structured reflection, but the benefit of these measures for outcomes has not yet been verified.

Summary

- A team from Western University in Ontario analyzed interviews with nine nephrologists using an inductive narrative approach across three domains: education, individual experience, and system constraints.
- Diagnostic reasoning is acquired through exposure to cases rather than explicit teaching; residents' questions force supervisors to defend their hypotheses, reducing the risk of premature diagnostic closure.
- With experience, nephrologists deliberately alternate between frameworks, applying parsimony to routine presentations and multifactorial reasoning to complex patients.
- According to participants, time pressure and resource stewardship push toward the simplest defensible diagnosis regardless of case complexity, so the choice of framework comes to depend on operational conditions.
- The author of the article adds situations in which coexisting conditions are the rule in nephrology: AKI in a hospitalized patient, proteinuria in diabetes, anemia in CKD, hyponatremia, and monoclonal gammopathy with kidney findings.
- Limitations include nine participants, reported rather than observed practice, the Canadian publicly funded system, the authors' pediatric setting, and the unverified effect of the proposed changes on clinical outcomes.
- In practice, it is recommended to define in advance a threshold for revising the diagnosis, to avoid closure when findings are not fully concordant, to distinguish working from definitive diagnoses in documentation, and to regard time pressure as a risk factor for error.

11.11 Monoclonal gammopathy of clinical significance beyond MGRS: what nephrologists should know

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. A monoclonal protein in an older patient is usually insignificant, but in rare cases it directly causes disease even without meeting criteria for malignancy. A review in Nephrology Dialysis Transplantation introduces a classification of monoclonal gammopathies of clinical significance (MGCS) based on the pathogenicity of the paraprotein rather than clone size and, beyond MGRS, highlights POEMS, TEMPI, systemic capillary leak syndrome, and autoantibody-mediated disorders. The data are mostly case-based, but recognition has direct therapeutic implications, because clone-directed therapy can reverse organ damage.

Summary

- In both MGUS and myeloma, the paraprotein is usually only a marker of tumor burden, with the exception of light chain cast nephropathy; in MGCS, the clone is small yet pathogenic.
- POEMS syndrome is mediated by VEGF overproduction and leads to progressive demyelinating polyneuropathy; the kidneys show a membranoproliferative pattern with mesangiolytic resembling thrombotic microangiopathy, and the monoclonal protein is usually small, almost always with lambda light chain restriction.
- In TEMPI syndrome, perirenal fluid collections together with unexplained erythrocytosis should suggest the diagnosis; the syndrome responds well to therapy directed against the plasma cell clone.

- Systemic capillary leak syndrome presents with episodes of hypovolemic shock with hemoconcentration and hypoalbuminemia, with AKI in the leak phase and a risk of volume overload in the recovery phase.
- Among autoantibody-mediated disorders, the most important for nephrologists is cold agglutinin disease, in which hemolysis leads to pigment nephropathy and AKI.
- This is a review of small, mostly case-based series, and the full text was not available when the article was prepared, so the clinical descriptions are based on established knowledge rather than on the authors' arguments.
- Serum immunofixation and serum free light chains with their ratio should be measured, because electrophoresis alone is insufficient; clinicians should look for a systemic pattern and collaborate early with a hematologist.

11.12 Bartter syndrome: diagnosis, genetic forms, and long-term treatment of renal salt wasting

Published August 13, 2026 · [online version \(Slovak\)](#)

Abstract. Bartter syndrome is a group of rare inherited tubulopathies characterized by renal sodium chloride wasting, hypokalemic hypochloremic metabolic alkalosis, and normal or low blood pressure. A review based mainly on the European consensus describes the current genetic classification (types 1 to 5), the diagnostic approach, and treatment, the evidence for which comes mostly from small observational series and expert consensus. In a Korean cohort of 54 patients, 11% had CKD G3 to G5 after a median of eight years. The disease therefore cannot be considered benign and requires genetic confirmation and long-term follow-up.

Summary

- Type 1 (SLC12A1, NKCC2) and type 2 (KCNJ1, ROMK, with possible transient neonatal hyperkalemia) present with polyhydramnios, hypercalciuria, and nephrocalcinosis; type 3 (CLCNKB) can mimic Gitelman syndrome; types 4A and 4B are associated with sensorineural hearing loss; and type 5 is a transient X-linked form associated with MAGED2.
- Activating CASR variants are no longer classified as type 5 but as autosomal dominant hypocalcemia with a Bartter phenotype.
- Diagnosis requires confirmation of metabolic alkalosis and renal chloride wasting and exclusion of diuretics (urine toxicology screening), vomiting, laxatives, and, in hypertension, mineralocorticoid excess; the genetic panel should also cover deletions and duplications, particularly in CLCNKB.
- The consensus recommends considering NaCl at approximately 5 to 10 mmol/kg/day in divided doses, but not in secondary nephrogenic diabetes insipidus; potassium is replaced as potassium chloride, and hypomagnesemia is corrected.
- NSAIDs can reduce polyuria and support growth but carry a risk of AKI and bleeding and, with PPIs, of worsening hypomagnesemia; potassium-sparing diuretics, RAAS blockade, and thiazides are not routinely recommended.
- Follow-up visits are recommended every 3 to 6 months in infants and every 6 to 12 months in stable patients, with kidney ultrasonography approximately every 12 to 24 months; the Korean cohort was small and genetically incomplete.
- In new-onset hypokalemic alkalosis in adulthood, an acquired cause is much more likely, and a plan for acute illness with vomiting or diarrhea should be prepared in advance.

11.13 Immune-mediated thrombotic thrombocytopenic purpura: an urgent diagnosis in nephrology practice too

Published September 12, 2026 · [online version \(Slovak\)](#)

Abstract. Immune-mediated thrombotic thrombocytopenic purpura (iTTP) is a life-threatening thrombotic microangiopathy with severe ADAMTS13 deficiency that nephrologists must distinguish from HUS and other TMAs. The review draws on the ISTH guidelines, the SmPC, and the HERCULES and MAYARI trials. In the

randomized HERCULES trial, caplacizumab reduced the composite endpoint of TTP-related death, recurrence, or thromboembolism from 49% to 12%. When clinical probability is high, plasma exchange and corticosteroids are started without waiting for the ADAMTS13 result.

Summary

- Unexplained thrombocytopenia with microangiopathic hemolytic anemia requires urgent evaluation for TTP without waiting for the full pentad, and the workup should include ECG and troponin.
- ADAMTS13 activity below 10% with a compatible clinical picture strongly supports TTP, 10% to 20% is a borderline range, and above 20% acute TTP is less likely; the sample is drawn before plasma administration, but sampling must not delay treatment.
- A PLASMIC score of 0 to 4 indicates low, 5 intermediate, and 6 to 7 high risk of severe deficiency; severe kidney injury is less typical in iTTP than in complement-mediated HUS, but it is not an absolute distinguishing feature.
- ISTH 2025 retained a strong recommendation for corticosteroids added to TPE and conditional recommendations for rituximab (off label, with HBsAg and anti-HBc testing before administration) and for caplacizumab, which is given as 10 mg IV before TPE and then 10 mg subcutaneously daily for 30 days after the last exchange, without dose adjustment in kidney impairment.
- In the single-arm MAYARI trial, 93.5% of 46 evaluable patients achieved remission without TPE (95% CI, 82.5% to 97.8%); there was no control group, so a TPE-free regimen belongs only in specialized centers.
- Platelets are not given prophylactically, thromboprophylaxis may be considered at counts above $50 \times 10^9/L$, dialysis is indicated according to standard criteria, and complement inhibition is not standard in confirmed iTTP.
- After remission, ADAMTS13 is checked approximately monthly for three months, then every three months during the first year, and thereafter every 6 to 12 months; the nephrologist monitors blood pressure, kidney function, and proteinuria.

11.14 A negative genetic test does not rule out a genetic cause of kidney disease

Published September 13, 2026 · [online version \(Slovak\)](#)

Abstract. A negative result of genetic testing for kidney disease means only that the method used, within its scope and given current knowledge, did not find a causal variant. A retrospective single-center study from Cleveland Clinic showed that exome sequencing after a negative or inconclusive panel yielded a new or reclassified pathogenic finding in 7 of 54 patients (13%). Given the small, selected sample, this figure is not a universal probability of success, but it confirms that when clinical suspicion persists, a nondiagnostic first test need not end the diagnostic workup.

Summary

- Of 584 patients tested from January 2019 to June 2024, 457 underwent panel testing and 98 exome sequencing; the diagnostic yields of 33% for panels and 31% for exomes cannot be compared, because the groups were different and nonrandomized.
- The cause may be missed because of a narrow panel, technical blind spots (CNVs, structural rearrangements, deep intronic variants, mosaicism), difficult-to-analyze regions such as PKD1 or the MUC1 VNTR region, outdated variant interpretation, or a mismatch between phenotype and test.
- The laboratory report should state the genes and regions analyzed, the technology, limitations, CNV analysis, and the year of testing; a VUS is not an almost-positive result and should not be used for treatment decisions or donor exclusion.
- The study does not support automatic exome sequencing after every negative panel, but rather targeted reassessment in early-onset or unexplained CKD, a family history, cystic anomalies, or extrarenal manifestations; subgroup yields are exploratory.
- Clinicians should return to the phenotype and pedigree, read the entire report, consider reanalysis of the original data, targeted supplementary testing, or exome or genome sequencing, and address com-

plex findings with a clinical geneticist; for a related living donor, a generically negative panel may not suffice.

11.15 New trends in the diagnosis and treatment of kidney cancer: a nephrology-oriented review

Published September 27, 2026 · [online version \(Slovak\)](#)

Abstract. In renal cell carcinoma (RCC), the nephrologist should recognize the kidney toxicity of treatment, help preserve kidney parenchyma, and ensure safe imaging. A review based on EAU guidelines and primary studies summarizes diagnosis, adjuvant pembrolizumab, combinations of ICIs with targeted therapy, and belzutifan. In the five-year analysis of KEYNOTE-564, overall survival was 87.7% with pembrolizumab versus 82.3% (HR 0.66). The benefit applies to a defined population and requires weighing toxicity and rigorous nephrology monitoring during treatment.

Summary

- CKD is not a reason to refuse a necessary contrast-enhanced CT; in AKI or eGFR < 30 mL/min/1.73 m², IV prophylaxis with isotonic solution is considered in patients not on dialysis, and at eGFR 30 to 44 mL/min/1.73 m², it is considered individually.
- The EAU recommends partial nephrectomy for T1 tumors; tumor biopsy is worthwhile if it will change treatment, and genetic testing is indicated particularly with diagnosis at age ≤ 46 years or with bilateral tumors.
- In KEYNOTE-564 (994 participants), five-year disease-free survival was 60.9% versus 52.2%, treatment-related grade 3 or 4 adverse events occurred in 18.6% versus 1.2%, and the trial excluded patients on dialysis and transplant recipients.
- In LITESPARK-005 (746 patients), belzutifan increased the objective response rate to 21.9% versus 3.5% but did not significantly improve overall survival (HR 0.88; 95% CI, 0.73 to 1.07); typical risks are anemia and hypoxia.
- ICIs can cause tubulointerstitial nephritis, and anti-VEGF drugs can cause hypertension and proteinuria; the pembrolizumab dose is not reduced, and cabozantinib is not recommended in severe kidney impairment.
- In an observational cohort of 429 patients with ICI-associated AKI, the median time to onset was 16 weeks, tubulointerstitial nephritis was found in 82.7% of biopsies, and recurrent AKI after rechallenge occurred in 16.5%.
- According to SITC, ICIs are temporarily withheld in AKI; at creatinine ≥ 2 times baseline, expedited nephrology evaluation is needed and biopsy should be considered; glucocorticoids are the mainstay of nephritis treatment.

11.16 Kidney stones according to AUA 2026: metabolic evaluation, prevention, and treatment

Published September 27, 2026 · [online version \(Slovak\)](#)

Abstract. Prevention of recurrent nephrolithiasis should not mean a universal ban on oxalate-containing foods or an automatic drug prescription. The updated 2026 American Urological Association (AUA) guideline combines screening, stone analysis, metabolic evaluation with 24-hour urine collection, phenotype-directed diet, and pharmacotherapy with monitoring of response. The key goals are a urine volume of at least 2.5 liters per day and a normal calcium intake of 1,000 to 1,200 mg. For nephrologists, it is essential

to distinguish general prevention from treatment of a specific phenotype and to use thiazides and alkali safely.

Summary

- Obstruction with fever or sepsis, anuria, a solitary kidney with obstruction, or significant AKI requires urgent urologic assessment; extended metabolic evaluation is reserved for recurrent and high-risk patients.
- The AUA recommends one or two 24-hour urine collections on a usual diet, with a minimum panel of volume, pH, calcium, oxalate, urate, citrate, sodium, potassium, and creatinine.
- In hypercalciuria, sodium is restricted and calcium intake maintained, and a thiazide may be added for recurrence; in hypocitraturia, potassium citrate with monitoring of serum potassium is standard; and in hyperuricosuria with normal calcium, allopurinol is used.
- For uric acid stones, alkalization targeting a pH of approximately 6.0 to 6.5 is first line; in cystinuria, the targets are urine output of at least 3 liters and a pH of about 7.0 to 7.5, with tiopronin for recurrence; and for struvite stones, stone removal and infection control are the mainstays.
- Pyridoxine helps only in responsive genotypes of primary hyperoxaluria type 1, and lumasiran and nedosiran have specific indications; in heart failure, advanced CKD, or hyponatremia, the urine output target should be individualized.
- A follow-up 24-hour urine collection is obtained within six months of starting interventions and approximately once a year when stable, and blood tests target the adverse effects of the specific treatment.

11.17 Evaluation of hematuria in nephrology practice: confirmation, emergencies, and identification of the source

Published September 27, 2026 · [online version \(Slovak\)](#)

Abstract. A positive dipstick, red urine, and microscopically confirmed hematuria are not synonymous. A review based on the 2026 Core Curriculum in Am J Kidney Dis and the AUA/SUFU guideline (2020, amended 2025) describes the approach from confirmation of the finding through emergencies to identification of the source. Microhematuria is defined as ≥ 3 RBC/HPF in a properly collected specimen, and further urologic evaluation is guided by risk stratification. Nephrologic assessment and risk-appropriate urologic evaluation proceed in parallel, because both causes may coexist.

Summary

- A positive dipstick without red cells suggests hemoglobinuria or myoglobinuria, and a negative dipstick with red urine suggests pigmenturia; high vitamin C intake can cause a false-negative result.
- Urgent evaluation is required for gross hematuria with clots or retention, hemodynamic instability, oliguria or suspected rapidly progressive glomerulonephritis, and infected obstruction.
- Acanthocytes, red cell casts, albuminuria, hypertension, and declining eGFR suggest a glomerular source, but the absence of dysmorphic red cells does not exclude it.
- Low risk (3 to 10 RBC/HPF, women < 60 or men < 40 years, < 10 pack-years) requires repeat urinalysis within 6 months; intermediate risk requires cystoscopy and ultrasonography; and high risk (> 25 RBC/HPF, gross hematuria, men ≥ 60 years, > 30 pack-years) requires cystoscopy and, preferably, CT urography.
- Anticoagulants, physical exertion, and sickle cell trait should not be accepted as a definitive explanation, and negative ultrasonography in high-risk patients does not replace cystoscopy.
- After a negative evaluation and a negative repeat urinalysis, follow-up can be discontinued; new gross hematuria or an increase in red cells warrants repeat evaluation.

11.18 Nutcracker syndrome: when compression of the left renal vein really explains hematuria

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. Compression of the left renal vein is a common CT finding, but symptomatic nutcracker syndrome is rare and may explain otherwise unexplained hematuria, orthostatic proteinuria, or flank pain. A review in Clinical Kidney Journal 2026, a 2025 Delphi consensus, and a systematic review of 24 observational studies with 578 patients show that no validated diagnostic criteria exist. Left renal vein transposition achieved symptom resolution in 92%, with the highest reintervention rate. The diagnosis rests on concordance of the clinical picture with anatomy and hemodynamics after other causes have been excluded.

Summary

- Among 324 asymptomatic donor candidates, 30.5% had an aortomesenteric angle below 41°, but only 0.7% met a diameter ratio of ≥ 4.9 , so radiologic evidence of compression is never sufficient for the diagnosis.
- The hematuria is nonglomerular and isomorphic, orthostatic symptoms resolve in the supine position, left-sided varicocele occurs in 20% to 40% of men, and secondary hypertension in 5% to 15%.
- Biopsy-proven IgA nephropathy does not exclude the syndrome if the finding does not respond to treatment and is orthostatic or unilateral in character; however, dysmorphic red cells, casts, or a decline in eGFR suggest a glomerular cause.
- Doppler ultrasonography (peak velocity ratio ≥ 4.2 to 5.0) is the first choice but not the only test; the Delphi consensus recommends a cross-sectional and a functional modality and, if these are positive, venography with pressure gradient measurement and intravascular ultrasound.
- In children and adolescents, a conservative approach for at least one to two years is justified; RAAS inhibitors and acetylsalicylic acid are the drugs mentioned, and the conservative approach led to symptom resolution in 52%.
- Endovascular stenting achieved resolution in 76% with reintervention in 11.3%, and autotransplantation in 69%; these are nonrandomized series, and experts prefer surgery because of the risk of stent migration.

CHAPTER 12

Clinical pharmacology and drug safety

Drug safety and dosing in reduced kidney function, polypharmacy, analgesics, psychotropic drugs, and dietary supplements with a nephrological dimension. Number of articles: 27.

Key messages of the chapter

According to expert interpretation, chronic kidney disease (CKD) is an independent driver of polypharmacy: as the glomerular filtration rate declines, complications requiring treatment accumulate, while pharmacokinetics and susceptibility to adverse effects change (12.13). Not all polypharmacy is harmful, since the combination of a RAAS inhibitor, SGLT2 inhibitors, and a statin has a clear benefit; the risks lie in drugs without an indication, without eGFR-based dose adjustment, and with interactions (12.13). Drug approval does not guarantee long-term safety; safety issues emerged in almost one third of drugs approved by the FDA from 2001 to 2010, so adverse reactions should be actively reported (12.7).

The strongest evidence in this chapter is randomized and often negative. In both CKD-FIX and PERL, allopurinol did not slow the decline in kidney function despite effective urate lowering, whereas a positive retrospective cohort is burdened by confounding by indication; KDIGO 2024 therefore suggests not using urate-lowering drugs for kidney protection in asymptomatic hyperuricemia (12.16). For PPIs, the randomized COMPASS trial illustrates the difference between types of evidence: pantoprazole did not increase all-cause mortality (HR 1.03), unlike observational signals burdened by protopathic bias (12.24). A South Korean cohort linked PPIs during NSAID use with CKD progression (HR 1.38), but a signal after only 1 to 15 days of treatment suggests reverse causality (12.23, 12.24); similarly observational is the association of combining PPIs with GLP-1 receptor agonists with gastrointestinal adverse effects (12.1). An observational Korean cohort linked beta-blockers in CKD without cardiovascular disease with higher mortality (HR 2.09), but this does not change their role when there is a clear indication (12.12). For itch in hemodialysis, difelikefalin achieved a response in 51% versus 35% with placebo in the phase 3 KALM-1 and KALM-2 trials, although the data are presented by an educational activity (12.4).

Pain management in CKD is a series of trade-offs. In a retrospective cohort, a single perioperative dose of ketorolac in patients with CKD was associated with AKI in 12.7% versus 2.0%, and NSAID thresholds are not supported by randomized data (12.26). Morphine and codeine accumulate active metabolites, and duloxetine is not recommended at a clearance below 30 mL/min (12.15, 12.26). Gabapentin above 300 mg per day was associated in hemodialysis cohorts with a 55% higher risk of falls and in CKD with a higher 30-day risk of a composite outcome (12.25); gabapentinoid doses are guided by creatinine clearance (12.22), but a preprint on dementia should not change practice (12.25).

Antimicrobial therapy requires precision, not automatically a stronger drug: cultures before treatment, eGFR-based dosing, and de-escalation (12.9). In frail older men, trimethoprim-sulfamethoxazole was associated in a retrospective study with more frequent adverse events, particularly AKI (12.3). Gentamicin can cause permanent vestibulotoxicity even with a normal audiogram (12.17), and in OPAT, veins should be protected from PICCs in CKD G3b to G5 (12.14). In HIT, the 4Ts score is most powerful for its negative predictive value (12.11). Lithium carries real kidney toxicity, with a concentrating defect in at least 54% of users (12.19).

Supplements and nonprescription substances form a separate group. Wellness peptides have no controlled studies in humans (12.2), Pielotax has no evidence of increasing the glomerular filtration rate (12.20), and ursolic acid should not replace ursodeoxycholic acid (12.21). St. John's wort lowers

tacrolimus and cyclosporine levels through CYP3A4 induction (12.8). In a randomized crossover study, cannabis modestly reduced atrial ectopy, which is not evidence for treatment (12.18), and after inhalation, driving should be delayed for at least 5 hours (12.27). Mouse data on mRNA vaccines and a proof-of-concept study with prucalopride are only hypotheses (12.5, 12.10), as are inflammation-affected eGFR estimates in rheumatoid arthritis without directly measured GFR (12.6).

A practical framework for the outpatient clinic: with every change in eGFR, hospitalization, or AKI, review the indication and dose of every drug and deprescribe in a planned manner (12.13); actively ask about supplements, herbs, and cannabis (12.2, 12.8, 12.27); tie PPIs to a clear indication and reassess NSAIDs first (12.23, 12.24); start gabapentinoids low and taper them gradually (12.22, 12.25); and give allopurinol only for gout or urate nephrolithiasis (12.16).

12.1 Concurrent use of proton pump inhibitors (PPIs) and GLP-1 receptor agonists may increase gastrointestinal adverse effects

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. GLP-1 receptor agonists are frequently combined with proton pump inhibitors (PPIs). A real-world analysis of TriNetX network data presented at the SGIM 2026 meeting showed that patients taking both drug classes had a more than twofold higher incidence of upper gastrointestinal adverse effects than patients on a GLP-1 receptor agonist alone (RR 2.37). This is an observational association without proof of causality. In practice, it supports systematic reassessment of the indication and duration of PPI therapy, not blanket discontinuation.

Summary

- With the combination, the risk of upper gastrointestinal adverse effects was RR 2.37 (95% CI, 2.30 to 2.44), of nausea and vomiting RR 1.81, and of dyspepsia RR 3.50 (95% CI, 3.28 to 3.75); acute pancreatitis, gallstones, and upper endoscopy were also reported more often.
- The data lack the reason for PPI initiation, the groups may differ in drug type and prior symptoms, and GLP-1 receptor agonists themselves cause reflux-like symptoms, so causality cannot be claimed.
- Differences were also observed in the lower gastrointestinal tract, in the form of diarrhea, constipation, and slowed intestinal motility.
- In patients on GLP-1 receptor agonists with new or worsening symptoms, the article recommends reassessing the indication and duration of PPI therapy, not assuming PPIs are automatically safe without reviewing treatment duration, and individualizing justified reflux treatment rather than discontinuing it across the board.
- A patient with new or worsening gastrointestinal symptoms should not change treatment on their own but should discuss with a physician the need for a PPI and any adjustment of treatment or further investigation.
- As general clinical caution rather than a study finding, the author adds the prevention of dehydration during vomiting or diarrhea, particularly in patients with CKD.

12.2 “Wellness” peptides: reliable verification of safety and efficacy is lacking

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Abstract. ISMP and ECRI warn of a growing market for “wellness” peptides sold without clear regulation, prescription, or quality control. For substances such as BPC-157, TB-500, and epitalon, data come mostly from animal studies, controlled human studies are lacking, and product testing has found variable purity as

well as arsenic and lead contamination. Cardiac events, kidney dysfunction, melanoma, and infections have been reported. Physicians should actively ask about peptide use.

Summary

- Many peptides are not approved for human use, have not even undergone phase 1 safety testing, and are promoted for muscle growth, recovery, or immune support.
- Gray-market products showed wide variability in purity on testing, and some contained arsenic and lead above levels acceptable for injectable drugs.
- Clinicians should ask about the source and use of peptides as they do for dietary supplements and monitor for infections, immunologic reactions, and kidney dysfunction.

12.3 Co-trimoxazole for UTI in older patients causes adverse events more often than fluoroquinolones

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Abstract. A retrospective study from eight French hospitals compared co-trimoxazole with fluoroquinolones in 228 older men with urinary tract infection (median age, 85 years). Adverse events occurred in 48.5% of patients on co-trimoxazole versus 29.7% (adjusted OR 2.10), most often AKI and metabolic disturbances. In frail older patients, tolerability should also guide antibiotic choice, although the results are limited by the observational design.

Summary

- Treatment was discontinued because of adverse events in 11.3% of patients on co-trimoxazole versus 2.3% on fluoroquinolones ($P = 0.009$).
- The most frequent adverse events were AKI and metabolic disturbances, both significantly more common with co-trimoxazole.
- This is a retrospective study of a small number of frail, multimorbid patients, which limits generalizability.

12.4 Stuck on antihistamines for CKD-aP? Time to reconsider

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. CKD-associated pruritus (CKD-aP) is often overlooked in patients on dialysis and treated with antihistamines, although it is not primarily histamine mediated. Based on an educational activity, the article describes mechanisms involving nonhistaminergic pathways, IL-31, and an imbalance of mu and kappa opioid receptors. In the phase 3 KALM-1 and KALM-2 trials, difelikefalin, a peripheral kappa opioid receptor agonist, reduced WI-NRS by at least 3 points in 51% of patients versus 35% with placebo. Pruritus should be actively and regularly sought and treated in a targeted way.

Summary

- The case of a 62-year-old man on hemodialysis with polycystic kidney disease shows that severe pruritus (WI-NRS 7/10) can go unnoticed for months and that antihistamines may be ineffective.
- CKD-aP is mostly generalized and transmitted via nonhistaminergic neural pathways, so antihistamines can at most induce sedation without addressing the core pathophysiology.
- Daily emollients and adequate dialysis targeting a single-pool Kt/V of 1.4 remain the foundation.
- Difelikefalin is a selective, peripherally acting kappa opioid receptor agonist on sensory neurons and immune cells that restores the balance against mu receptors; the activity describes it as the standard of care for CKD-aP in patients on hemodialysis.
- In KALM-1 and KALM-2, with 851 participants, sleep and quality of life on the 5-D Itch and Skindex-10 scales also improved, with separation from placebo from week 1; in responders, inflammatory markers including IL-31 decreased after 12 weeks, supporting the hypothesis of a neuroimmune effect.

- According to the activity, kappa opioids are not addictive and lack the typical central opioid adverse effects; these are claims of the educational activity, not an independent assessment.
- In practice, physicians should ask patients on dialysis about pruritus at regular intervals, typically every 3 months, measure its severity, and, once CKD-aP is confirmed, move from antihistamines to targeted therapy.

12.5 mRNA vaccines apparently work more broadly than we thought

Published June 5, 2026 · [online version \(Slovak\)](#)

Abstract. The effect of mRNA vaccines has been explained mainly by their entry into dendritic cells. Experimental work in mice, above all a study in Nature Biotechnology, suggests that expression in nonimmune cells, especially muscle cells, can also elicit an immune response. Silencing expression in muscle cells reduced the T-cell response, whereas silencing it in liver cells increased it. Direct presentation, cross-presentation, and cross-dressing probably act together, which may aid the development of cancer vaccines and mRNA therapies, but clinical relevance in humans remains to be verified.

Summary

- In mouse experiments, mRNA expression was selectively silenced in different cell types: silencing in muscle cells reduced the T-cell response, whereas silencing in liver cells increased it, suggesting a suppressive role for the latter.
- Silencing expression in dendritic cells did not abolish T-cell activation, although for some antigens it reduced the number of cytotoxic T lymphocytes; dendritic cells therefore remain important.
- In cross-presentation, a dendritic cell captures an antigen produced by another cell, processes it, and presents it; in cross-dressing, a nonimmune cell loads the antigen onto its own MHC molecules and a dendritic cell takes over these complexes and presents them to T lymphocytes, and the researchers assume that several pathways, including direct presentation, operate simultaneously.
- Cancer vaccines aim for strong activation of CD8⁺ T lymphocytes, whereas therapies for genetic diseases aim to dampen the immune response so that the body does not destroy therapeutically modified cells; a more precise understanding of presentation pathways may lead to more targeted and safer mRNA drugs.
- These are mostly experimental data from mouse models, published in Nature Biotechnology, Nature, and Nature Communications, and their clinical relevance in humans will need further verification.

12.6 Cystatin C and creatinine-based eGFR estimates vs. biomarkers of rheumatoid arthritis activity: insights from a secondary analysis of a randomized trial

Published June 20, 2026 · [online version \(Slovak\)](#)

Abstract. In rheumatoid arthritis, inflammation, sarcopenia, and treatment can all affect creatinine and cystatin C and thus eGFR estimates. A secondary observational analysis of a randomized trial in 157 patients with active RA showed that eGFR_{cys} was 20.9 mL/min/1.73 m² lower than eGFR_{cr} at baseline and that neither estimate changed significantly over 24 weeks. Several inflammatory biomarkers were inversely associated with eGFR. Without directly measured GFR, a true change in filtration cannot be distinguished from the effect of inflammation on the markers.

Summary

- The trial compared a TNF inhibitor plus methotrexate with triple therapy of methotrexate, sulfasalazine, and hydroxychloroquine in patients with a median age of 58 years, 75% of whom were women.
- Baseline eGFR_{cys} was 63.3 and eGFR_{cr} 84.2 (difference, -20.9 mL/min/1.73 m²; 95% CI, -24.7 to -17.0); the change over 24 weeks was +1.74 for eGFR_{cys} and -0.28 mL/min/1.73 m² for eGFR_{cr}, both nonsignificant.

- At baseline, after multivariable adjustment, VCAM-1, IL-6, TNFR1, leptin, and resistin were inversely associated with eGFR_{cys} or eGFR_{cr}, supporting the notion that the inflammatory milieu may affect the parameters from which eGFR is calculated.
- Over time, only the change in TNFR1 was related to the change in eGFR_{cys} (−2.91 mL/min/1.73 m²; 95% CI, −4.48 to −1.33), and no biomarker was related to the change in eGFR_{cr}, suggesting a different sensitivity of cystatin- and creatinine-based estimates to inflammatory dynamics.
- Limitations include the lack of directly measured GFR, the short 24-week follow-up, and the risk of false-positive associations due to multiple testing.
- In active RA, the difference between eGFR_{cys} and eGFR_{cr} should be interpreted cautiously, particularly when assessing the renal safety of treatment; the authors emphasize the need for studies with directly measured GFR that can distinguish a true change in function from an inflammatory artifact.

12.7 Postmarketing drug safety failures: an in-depth analysis of 10 drugs with fatal consequences in the FDA and EMA regulatory environment

Published June 22, 2026 · [online version \(Slovak\)](#)

Abstract. FDA or EMA approval does not guarantee a drug's long-term safety. This review of ten drugs with fatal postmarketing failures (for example, rofecoxib, cerivastatin, sibutramine, natalizumab, and ranitidine) draws on a systematic review of 462 withdrawn drugs and a cohort study from the Yale School of Medicine. Safety issues emerged for almost one third of drugs approved by the FDA from 2001 to 2010, at a median of 4.2 years after approval, and case reports were the basis for withdrawal in 71% of cases. In practice, adverse reactions should be actively reported and caution maintained with new drugs.

Summary

- In the placebo-controlled APPROVe trial, rofecoxib doubled the risk of cardiovascular events; worldwide it is associated with an estimated 88,000 to 140,000 serious cardiovascular events, 30% to 40% of which were fatal.
- Cerivastatin had a 10- to 50-fold higher incidence of rhabdomyolysis than other statins; the FDA recorded 31 deaths, 12 of them with concomitant gemfibrozil, which increased plasma cerivastatin levels 5- to 8-fold.
- The SCOUT trial in 10,744 patients showed a 16% increase in the risk of major cardiovascular events with sibutramine (HR 1.16; 95% CI, 1.03 to 1.31); because of a PML risk of 1:1,000 to 1:100, natalizumab was reintroduced only within the TOUCH program, and ranitidine was withdrawn in 2020 because of formation of carcinogenic NDMA.
- In the review of 462 withdrawn drugs, the most common reasons were hepatotoxicity (18%) and immune-mediated reactions (17%); deaths were associated with withdrawal in 25% of cases, and only 9.34% of drugs were withdrawn worldwide.
- The median time from marketing to withdrawal was 18 years and from the first reported reaction to withdrawal 6 years; the interval between the first report and regulatory action is not shortening substantially.
- The evidence base is weakened by the predominance of case reports, voluntary and underreported adverse reaction reporting, and premarketing trials that mostly enrolled fewer than 1,000 patients and lasted up to six months.
- The author recommends actively reporting suspected adverse reactions, strengthening pharmacovigilance and the transparency of clinical data, and not regarding drug approval as a guarantee of long-term safety.

12.8 St. John's wort for depression: when to consider it and when nephrology is better off without it

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Patients often do not report St. John's wort as a medication, although it is a biologically active substance. This review, based on a Medscape article, assesses when a standardized extract may be a rational option for mild to moderate depression and why it is problematic in nephrology. The evidence of efficacy is inconsistent: some meta-analyses and trials support an effect, whereas other studies have not shown a convincing difference from placebo. The main risk is induction of CYP3A4 and P-glycoprotein, with decreased tacrolimus and cyclosporine levels and even acute rejection, so any recommendation must be preceded by a review of the entire medication list.

Summary

- The antidepressant effect is attributed mainly to hyperforin; hyperforin and hypericin content varies widely between products, so tea, oil, or an over-the-counter supplement cannot be considered equivalent to a standardized extract, and their interaction potential is unpredictable.
- Induction of CYP3A4 and P-glycoprotein lowers cyclosporine and tacrolimus concentrations (cases of acute rejection have been described) as well as exposure to steroid hormones, with a risk of contraceptive failure; after discontinuation, induction wanes and drug levels may rise, which requires coordinated level monitoring.
- Combination with SSRIs, SNRIs, MAOIs, triptans, tramadol, linezolid, or dextromethorphan carries a risk of serotonin syndrome; warfarin, direct oral anticoagulants, digoxin, some statins, antiepileptic drugs, antiretroviral drugs, and cancer therapy also require particular caution.
- St. John's wort is not appropriate for severe depression, suicidal ideation, psychosis, suspected bipolar disorder, an unclear diagnosis, pregnancy and breastfeeding, after transplantation, or with immunosuppressants that have a narrow therapeutic window.
- Reported adverse effects include gastrointestinal complaints, dry mouth, fatigue, restlessness, headache, dizziness, and skin reactions; photosensitivity is less common at usual antidepressant doses but cannot be ignored.
- The author recommends actively asking about herbs, teas, and supplements, recording the exact product and dose, coordinating with the transplant center in transplant recipients, and considering St. John's wort only as a standardized product after an interaction check and with a clear monitoring plan.

12.9 Antimicrobial resistance in urinary tract infections: global burden and a nephrology perspective

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Antimicrobial resistance in urinary tract infections (UTIs) no longer concerns only hospitals but also outpatients, multimorbid patients, and transplant recipients. A review in Nephrology Dialysis Transplantation views it as the result of the interplay of the bacterium, the patient, urine flow, the microbiome, and drug pharmacokinetics. It describes resistance mechanisms of gram-negative uropathogens, high-risk patient groups, and future approaches including prediction models, antimicrobial peptides, and bacteriophages. For nephrology practice, it calls for more precise rather than automatically stronger treatment: culture before antibiotics, dosing according to eGFR, and timely de-escalation.

Summary

- Most UTIs are caused by gram-negative bacteria, particularly uropathogenic strains of *Escherichia coli*, which acquire resistance through β -lactamase production, target modification, porin changes, and efflux pump activation, and resistance may be linked to greater virulence.
- Patients at higher risk of resistant UTI include those with recent or repeated antibiotic treatment, recurrent infections, a catheter, obstruction, or residual urine, those with CKD, diabetes, or frailty, kidney transplant recipients, and those with frequent hospitalizations.

- A catheter allows biofilm formation, which reduces antibiotic efficacy and often prevents eradication without removal or replacement of the device; in transplant recipients, infection may present atypically, and inadequately treated graft pyelonephritis threatens graft function.
- Urinary stasis, residual urine, reflux, or a foreign body promote persistence of infection and selection of resistant strains; in pyelonephritis, a drug achieving adequate concentrations in the renal parenchyma should be chosen, and in CKD the dose should be adjusted to eGFR without subtherapeutic dosing.
- The authors recommend not treating asymptomatic bacteriuria without a clear indication, obtaining a culture before treatment in complicated UTI, pyelonephritis, catheter-associated infection, transplantation, or failure of empirical therapy, taking previous cultures and local resistance data into account, de-escalating according to susceptibility, and addressing urologic factors.
- Prediction models including artificial intelligence, antimicrobial peptides, and bacteriophage therapy are promising, but their wider clinical use still requires standardization, high-quality clinical data, accessible diagnostics, and a regulatory framework.

12.10 Prucalopride and “brain fog” after depression: can 5-HT₄ agonism improve cognition?

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Abstract. About 80% of patients have cognitive difficulties during a depressive episode, and these may persist after remission. A small double-blind, placebo-controlled proof-of-concept study in Psychological Medicine in 49 adults in remission from recurrent depression showed that prucalopride, a selective 5-HT₄ receptor agonist titrated from 1 mg to 2 mg daily over 7 to 10 days, improved performance on several objective cognitive tasks. This is a preliminary signal on cognitive tests without proof of functional benefit; in CKD, prucalopride remains a drug for chronic constipation, dosed according to kidney function.

Summary

- The study enrolled 50 adults aged 18 to 40 years with a history of at least two depressive episodes, at least 6 months in remission, and no psychotropic treatment; 49 participants randomized to prucalopride or placebo were analyzed.
- Prucalopride improved immediate word recall on an auditory verbal learning test, shortened reaction times on a complex working memory task without loss of accuracy, and improved recognition of briefly presented facial expressions and a composite score of nonemotional tasks, independent of baseline mood.
- The proposed mechanism includes enhanced synaptic plasticity in the hippocampus, increased acetylcholine release, and modulation of glutamatergic neurotransmission, but it rests mainly on preclinical and earlier human data.
- Limitations include the small sample, treatment for only 7 to 10 days, a young and predominantly highly educated population, enrollment without a requirement for cognitive deficit, and uncertainty whether test improvement means real functional improvement; prucalopride therefore cannot yet be recommended for cognitive symptoms outside research.
- Brain fog in CKD has many causes, including uremic toxins, anemia, sleep disorders, depression, drug adverse effects, inflammation, and dialysis-related fluctuations in volume and osmolality, so the results cannot be transferred mechanically to patients with CKD or on dialysis.
- For chronic constipation, the prucalopride dose is usually not adjusted in mild to moderate renal impairment, 1 mg daily is recommended in severe impairment, and in end-stage kidney failure on dialysis its use is not recommended or is contraindicated, depending on the specific marketing authorization.

12.11 The 4Ts score in suspected heparin-induced thrombocytopenia: a simple tool of great clinical importance

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. Heparin-induced thrombocytopenia (HIT) is an immune-mediated, strongly prothrombotic reaction to heparin that must be neither missed nor overdiagnosed. This practical review describes the 4Ts score, which estimates the pretest probability of HIT from four clinical domains scored 0 to 2 points each. A score of 0 to 3 indicates low, 4 to 5 intermediate, and 6 to 8 high probability, and the tool's greatest strength is the negative predictive value of a low score. In nephrology and dialysis, where heparin exposure is common and thrombocytopenia has many causes, it helps avoid both unnecessary testing and missed HIT.

Summary

- The platelet fall scores 2 points for a decrease of more than 50% with a nadir of approximately $\geq 20 \times 10^9/L$, 1 point for a decrease of 30% to 50% or a nadir of 10 to $19 \times 10^9/L$, and 0 points for a decrease of less than 30% or a nadir below $10 \times 10^9/L$; the relative change from baseline is decisive.
- Typical onset is on days 5 to 10 after starting heparin, but with exposure in the previous 30 days the fall may occur within 24 hours; the history must also include dialysis and catheter flushes.
- New thrombosis, skin necrosis, or an acute systemic reaction after intravenous heparin scores two points; dialysis access thrombosis or repeated circuit clotting are suspicious but do not by themselves prove HIT, and a clear alternative cause of thrombocytopenia (sepsis, DIC, drugs, or pseudothrombocytopenia) lowers the probability of HIT.
- Immunoassays for antibodies against the PF4/heparin complex are sensitive, but at low pretest probability they yield false-positive or clinically irrelevant results; functional assays are more specific but less available, so the result is always interpreted together with the clinical score.
- With an intermediate or high score, all sources of heparin, including catheter flushes, should be stopped, low-molecular-weight heparin should not be used as a substitute, nonheparin anticoagulation should be started with attention to renal elimination, tests should be sent, thromboses should be sought, and a hematologist consulted when in doubt.
- Options for dialysis include regional citrate anticoagulation, heparin-free dialysis with more frequent saline flushing, or systemic nonheparin anticoagulation; heparin should not be continued merely because dialysis requires it.

12.12 Beta-blockers in chronic kidney disease without known cardiovascular disease: caution is warranted

Published July 9, 2026 · [online version \(Slovak\)](#)

Abstract. The role of beta-blockers in primary prevention in patients with CKD without known cardiovascular disease is unclear. A nationwide Korean cohort study in Clinical Kidney Journal compared, after propensity score matching, 9,067 beta-blocker users with 17,094 nonusers with $eGFR < 60 \text{ mL/min/1.73 m}^2$. Beta-blocker use was associated with higher all-cause mortality (HR 2.09) and a higher risk of major adverse cardiovascular events (HR 1.33). These are observational data subject to possible confounding by indication, but they support prescribing beta-blockers in CKD only when there is a clear indication.

Summary

- The analysis included patients identified from 2012 to 2015 with $eGFR < 60 \text{ mL/min/1.73 m}^2$ on at least two measurements and no established cardiovascular disease; users were defined as patients with a beta-blocker prescription for at least 6 months, and matching was performed in a 1:2 ratio.
- Beta-blocker users had a higher risk of all-cause mortality (HR 2.09; 95% CI, 1.96 to 2.24) and 4-point MACE (HR 1.33; 95% CI, 1.26 to 1.41), as well as cardiovascular death (HR 2.07), myocardial infarction (HR 1.29), stroke (HR 1.21), and hospitalization for heart failure (HR 1.58).
- The mortality risk was more pronounced at lower $eGFR$, and less favorable outcomes were observed with carvedilol, although this may reflect patient selection.

- As an observational study based on administrative data, it cannot exclude residual confounding by indication or undiagnosed subclinical disease; the study does not directly demonstrate the proposed mechanisms, such as bradycardia, orthostatic hypotension, or masking of hypoglycemia, and the results may not apply outside the Korean health system.
- The study does not change the role of beta-blockers in heart failure with reduced ejection fraction, after myocardial infarction, in tachyarrhythmias, in symptomatic angina pectoris, or when rate control is needed; in uncomplicated hypertension with CKD, RAAS inhibitors, calcium channel blockers, and diuretics according to volume status are usually preferred.
- In patients taking a beta-blocker, the author recommends verifying the original indication, ejection fraction, arrhythmias, heart rate, and orthostatic blood pressure changes, reassessing treatment with a cardiologist or internist when the indication is unclear, and tapering gradually to avoid rebound tachycardia and worsening angina.

12.13 Chronic kidney disease as an independent driver of polypharmacy

Published July 10, 2026 · [online version \(Slovak\)](#)

Abstract. Polypharmacy, defined as the use of five or more drugs, is common in CKD as early as the moderately advanced stages. A brief letter in *Clinical Kidney Journal* argues that CKD is not merely a coexisting comorbidity but an independent driver of polypharmacy: declining eGFR brings complications that require treatment and at the same time alters pharmacokinetics and susceptibility to adverse effects. This is an expert interpretation without new primary data. In practice, the medication list should be regarded as a dynamic document and reviewed at every change in eGFR, hospitalization, or episode of AKI.

Summary

- As eGFR declines, complications requiring treatment accumulate, such as anemia, mineral and bone disorder, metabolic acidosis, and hyperkalemia, while the volume of distribution, protein binding, and renal elimination of drugs also change.
- Not all polypharmacy is harmful; a combination of a RAAS inhibitor, an SGLT2 inhibitor, a statin, and appropriate antihypertensive therapy can have clear benefit, whereas the risky drugs are those without a current indication, without dose adjustment to eGFR, duplicated, or with a significant interaction.
- The main risks include AKI, hyperkalemia, hyponatremia, hypotension with falls, bleeding, hypoglycemia, and delirium; nonsteroidal anti-inflammatory drugs, inappropriately dosed antibiotics, sedatives, opioids, anticholinergics, digoxin, and lithium are particularly risky.
- The source is a brief letter without new primary data, so its conclusion represents an expert interpretation, not quantitative evidence of causality between CKD and the number of drugs.
- At every change in eGFR, hospitalization, AKI, or transition to dialysis, the authors recommend verifying the indication and dose of each drug, duplications, effects on potassium, acidosis, and blood pressure, the patient's understanding, and the sick-day rules for temporary discontinuation during dehydration, infection, or vomiting.
- Deprescribing is the planned and documented discontinuation of a drug with an unfavorable benefit-risk ratio and must not be abrupt for beta-blockers, benzodiazepines, corticosteroids, or antiepileptic drugs; a clinical pharmacist helps with complex regimens, and medication reconciliation helps at discharge.

12.14 Outpatient parenteral antimicrobial therapy: a safe alternative to hospitalization only with proper patient selection

Published August 13, 2026 · [online version \(Slovak\)](#)

Abstract. Outpatient parenteral antimicrobial therapy (OPAT) allows selected patients to complete treatment outside the hospital. This review draws on the 2026 UK good practice recommendations and IDSA guidelines and emphasizes that OPAT must be a formally governed process with patient selection, monitoring, and available escalation of care. Several specific recommendations rest on low- or very low-quality

evidence. Key points for nephrology are vein preservation before PICC placement in CKD G3b to G5, ongoing dose adjustment to changing kidney function, and laboratory checks at least once weekly.

Summary

- Before enrollment, the continuing indication, source control, and an effective oral alternative, which the 2026 recommendations prefer, should be verified; the first dose of a new drug is given under the supervision of a person able to manage anaphylaxis.
- Age, substance use, and insecure housing are not automatic contraindications; clinical stability, the abilities of the patient or caregiver, and access to emergency help are decisive.
- In advanced CKD, IDSA recommends a tunneled central venous catheter rather than a PICC (strong recommendation, low-quality evidence), and in CKD G3b to G5, PICC placement should be preceded by nephrology consultation to preserve veins for future vascular access.
- In patients on hemodialysis, selected drugs can be given during or after dialysis; the dose should be adjusted continuously according to the trend in kidney function, because a static estimate may not reflect reality when creatinine is unstable.
- For vancomycin in serious MRSA infections, AUC monitoring targeting 400 to 600 mg·h/L at an MIC of 1 mg/L is preferred; cefepime can cause neurotoxicity in renal dysfunction even at an adjusted dose, and aminoglycosides require a pharmacokinetic plan and hearing monitoring.
- The UK standard requires a blood count, kidney and liver function, CRP, and drug levels as needed at least once weekly, but in advanced CKD or dialysis checks should be more frequent, and it should be decided in advance who will review the results and adjust the dose.
- The patient needs a written plan with a contact available around the clock for warning symptoms; OPAT should not prolong treatment merely because it is logistically feasible.

12.15 Low back pain and sciatica: diagnosis, treatment, and analgesic safety in CKD

Published August 13, 2026 · [online version \(Slovak\)](#)

Abstract. Low back pain is mostly nonspecific and has a favorable course, whereas a smaller proportion of patients have radicular syndrome or a serious cause. This practical review, based on NICE, IDSA, ACR, WHO, and KDIGO guidelines, describes classification, red flags, indications for imaging, and pharmacotherapy with attention to CKD. An active approach is the foundation; routine MRI is not performed and is considered after about six weeks if symptoms persist. In CKD, NSAIDs should be assessed individually, gabapentinoids, corticosteroids, and benzodiazepines should not be used routinely for sciatica, and opioids should be chosen with regard to metabolite accumulation.

Summary

- Suspected cauda equina syndrome, spinal infection, malignancy, fracture, or a vascular cause requires urgent assessment; hemodialysis, an intravascular catheter, and recent *Staphylococcus aureus* bacteremia increase the risk of spinal infection, and fever may be absent.
- Pain from the kidneys and urinary tract tends to be located in the flank; bilateral obstruction or obstruction of a solitary kidney can cause acute kidney injury even without marked pain, and the absence of hematuria does not exclude a stone.
- NICE does not recommend routine imaging or bed rest; ACR recommends MRI when symptoms persist after about six weeks of appropriate treatment, particularly in candidates for intervention, but imaging is not delayed in the presence of red flags.
- NSAIDs may be considered short term at the lowest effective dose but should generally be avoided in active AKI, volume depletion, hyperkalemia, or decompensated heart failure; combination with a diuretic and an ACE inhibitor or angiotensin receptor blocker is particularly risky.
- Acetaminophen has a more favorable renal profile in CKD but a modest effect; morphine and codeine form renally excreted active metabolites, and gabapentin and pregabalin are not used routinely for sciatica.

ica and, for other indications, require dose adjustment because of the risk of sedation, myoclonus, and encephalopathy.

- The author proposes a seven-step algorithm: exclude a time-critical condition, classify the syndrome, switch the diagnostic pathway when kidney symptoms are present, image with a specific question, start an active plan, check eGFR, potassium, hydration, and concomitant medications before prescribing, and set a date for reassessment.

12.16 Allopurinol in chronic kidney disease: why randomized trials and retrospective data point in opposite directions

Published August 23, 2026 · [online version \(Slovak\)](#)

Abstract. Lowering uric acid is a biologically plausible target in CKD, but randomized evidence does not confirm kidney protection. This analysis of a 2026 editorial in the Brazilian Journal of Nephrology compares the CKD-FIX and PERL trials with a retrospective cohort of 80 patients. In CKD-FIX, the difference in annual eGFR decline was $-0.10 \text{ mL/min/1.73 m}^2$ ($p = 0.85$), and in PERL the difference in measured GFR was $0.001 \text{ mL/min/1.73 m}^2$ ($p = 0.99$), despite effective urate lowering. The author attributes the positive result of the cohort to confounding by indication and lack of adjustment. Allopurinol should therefore not be used for kidney protection in asymptomatic hyperuricemia.

Summary

- CKD-FIX enrolled 369 patients with CKD G3 to G4 without gout; the annual eGFR decline was -3.33 with allopurinol versus $-3.23 \text{ mL/min/1.73 m}^2$ with placebo, and the confidence interval of the difference (-1.18 to 0.97) excludes a clinically meaningful benefit, although the trial ended early because of slow recruitment.
- PERL, in 530 patients with type 1 diabetes, showed no difference in iohexol-measured GFR after 3 years despite a fall in urate from 6.1 to 3.9 mg/dL , and albumin excretion was 40% higher with allopurinol; thus the hypothesis failed, not the treatment.
- In recommendation 3.14.2 (grade 2D), KDIGO 2024 suggests not using urate-lowering drugs to slow CKD progression in asymptomatic hyperuricemia, but this does not apply to gout, uric acid nephrolithiasis, or prevention of tumor lysis syndrome.
- A retrospective cohort of 40 treated and 40 untreated patients with CKD G3 to G4 describes a rise in mean eGFR from about 35 to $47 \text{ mL/min/1.73 m}^2$ and 0 versus 4 progressions to kidney failure over 24 months; the author attributes this to confounding by indication, regression to the mean, and unadjusted analysis.
- When treatment is indicated in advanced CKD, it is started at 50 mg daily or every other day and titrated according to serum uric acid, with a target below $360 \text{ }\mu\text{mol/L}$ in gout; HLA-B*58:01 testing should be considered, and combination with azathioprine or 6-mercaptopurine should be avoided because of the risk of severe myelosuppression.
- The author recommends regarding hyperuricemia in CKD as a marker of metabolic and hemodynamic burden and addressing its causes, such as diuretic therapy, obesity, and diabetes control, rather than treating the number itself.

12.17 Gentamicin vestibulotoxicity: recognition, diagnosis, and management

Published August 24, 2026 · [online version \(Slovak\)](#)

Abstract. Gentamicin can cause severe, sometimes permanent bilateral vestibular hypofunction that may be accompanied by neither hearing loss nor vertigo. This review bases the clinical picture, diagnosis, and management on the SmPC, Bárány Society criteria, small retrospective and randomized studies, and clinical guidelines. In a series of 35 patients with gentamicin ototoxicity, all 35 had imbalance and 33 had oscillopsia, but only three had hearing impairment. In CKD and dialysis, the regimen, timing of sampling, and cu-

mulative dose should be evaluated; when toxicity is suspected, treatment should be reassessed, the deficit objectified with vHIT, and vestibular rehabilitation started.

Summary

- Warning signs are new unsteadiness when walking, worse stability in the dark, and oscillopsia; diagnosis requires objective evidence, such as a bilateral horizontal VOR gain < 0.6 on vHIT, and a normal audiogram does not exclude toxicity.
- According to the SmPC, the trough concentration should be ≤ 2 mg/L with twice-daily dosing and ≤ 1 mg/L with once-daily dosing, which is not recommended in severe kidney failure; treatment should last 7 to 10 days whenever possible.
- Toxicity has also been described at recommended doses and concentrations; in a small 1978 cohort of 23 patients on hemodialysis, age, total dose, and treatment duration, not concentrations, were decisive, with risk increasing at a cumulative dose of about 17.5 mg/kg, which is, however, not a universal threshold.
- When toxicity is suspected, gentamicin should be stopped or replaced if the infection allows, and doses, sampling times, and the cumulative dose in mg/kg should be verified; hemodialysis can lower the concentration but does not restore damaged vestibular function, and no antidote exists.
- NAC 600 mg twice daily reduced audiometrically defined ototoxicity in patients on hemodialysis, and ISPD 2022 suggests it (grade 2B) in peritoneal dialysis, but the data concern hearing, not vestibular function.
- The mainstay of treatment is vestibular rehabilitation, supported by strong evidence, with gaze stabilization exercises roughly 3 to 5 times daily for 5 to 7 weeks and balance training for at least 6 to 9 weeks, together with fall prevention; long-term vestibular suppressants hinder compensation.

12.18 Inhaled cannabis and cardiac ectopy: what a randomized crossover trial shows (and why it does not yet support a recommendation to “treat” arrhythmias)

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. Observational data on cannabis and cardiac arrhythmias are conflicting. The randomized crossover MARY-JANE trial, published in JACC in 2026, alternated days of inhalation and abstinence over 14 days in 108 regular users without a history of arrhythmia and measured ectopy with a patch ECG. On inhalation days, daily ectopy was about 9% lower (RR 0.91), mainly because of premature atrial contractions, with no change in ventricular ones. This is an acute surrogate endpoint in a young cohort; the trial demonstrates neither a therapeutic effect nor the cardiovascular safety of cannabis and cannot be extrapolated to CKD or dialysis.

Summary

- The cohort comprised 108 adults with a mean age of 32 ± 10 years, 59% of them women, with 713 days randomized to inhalation and 616 to abstinence; participants obtained cannabis themselves, so THC and CBD content was not standardized.
- Daily ectopy (PAC + PVC) had an RR of 0.91 (95% CI, 0.85 to 0.98; $P = 0.02$), PACs an RR of 0.90 (95% CI, 0.82 to 0.97), and each inhalation was associated with about 2% less ectopy (RR 0.98); PVCs, steps, sleep, and glucose did not change significantly.
- The absolute effect may be small, because the abstract does not report the baseline number of extrasystoles and the burden tends to be low in a young cohort without arrhythmia.
- According to the authors, causal interpretation is limited by withdrawal effects in habitual users, moderate adherence, and cointervention; the trial did not assess atrial fibrillation, hospitalizations, heart failure, deaths, or long-term safety.
- People with atrial fibrillation, heart failure, or a pacemaker and those taking antiarrhythmic drugs or insulin were excluded, so the results cannot be transferred to patients with CKD or on dialysis.

- The author recommends asking about cannabis nonjudgmentally because of cardiovascular risk, drug interactions including immunosuppression, and adherence, while cannabis should not be prescribed as an antiarrhythmic or kidney-protective agent.

12.19 Lithium: seven myths that may stand in the way of its use, from a nephrology perspective

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. The author checks Nassir Ghaemi's commentary on seven myths that hinder lithium use against primary sources from a nephrology perspective. This is a critical review of randomized trials (Gelenberg, BALANCE), cohorts, meta-analyses, and ISBD/IGSLI and NICE guidelines. Most of the psychiatric corrections hold up, but not the claim of low renal toxicity: at least 54% of users have a concentrating defect, the risk of stage 3 CKD is increased (HR 1.93), and ESRD affects about 1.5%. In practice, the lowest effective level and regular monitoring of eGFR, lithium, TSH, and calcium apply.

Summary

- ISBD/IGSLI recommends 0.60 to 0.80 mmol/L for maintenance treatment, individualized to 0.40 to 0.60 or 0.80 to 1.00 mmol/L; in older patients usually 0.40 to 0.60 mmol/L, and above 80 years at most 0.70 mmol/L. Samples are drawn in the morning, 12 ± 1 hours after the evening dose.
- The renal entities must be distinguished: nephrogenic diabetes insipidus is common (concentrating defect in at least 54% of 1,105 patients), stage 3 CKD has an HR of 1.93 (95% CI, 1.76 to 2.12), and ESRD affects about 1.5% (relative risk 7.8). Ghaemi's figure of 1% to 5% corresponds rather to ESRD or severe CKD.
- Toxicity depends on exposure: Presne et al. described a mean annual decline in creatinine clearance of 2.29 mL/min and hypercalcemia with hyperparathyroidism in about 35% of patients, and Shine et al. confirmed an association with hypothyroidism (HR 2.31).
- Once-daily evening dosing is rational and does not impair prophylactic efficacy, but a lower tubular burden has not been demonstrated on hard kidney outcomes. In the BALANCE trial, lithium prevented relapse better than valproate (HR 0.71), but superiority at the depressive pole cannot be confirmed from the abstract.
- A meta-analysis by Gomes-da-Costa et al. showed a nonsignificant weight gain of +0.462 kg, and cognition is not significantly impaired with therapeutic use; "blunted creativity" is merely a clinical opinion.
- The author recommends monitoring eGFR, lithium, TSH, and calcium, avoiding over-the-counter NSAIDs, using caution with ACEIs/ARBs and thiazides, and temporarily withholding lithium during vomiting, diarrhea, or fever. Decisions about continuation when eGFR declines should be made jointly by a psychiatrist and a nephrologist.

12.20 Pielotax and "kidney peptides": an analysis of composition, efficacy, and safety from a nephrologist's perspective

Published September 5, 2026 · [online version \(Slovak\)](#)

Abstract. Pielotax is a dietary supplement containing the A-9 peptide complex of animal origin, promoted as a kidney bioregulator. A critical analysis of its composition and the only available clinical report (42 patients with gouty nephropathy, 30 days, no placebo, and undescribed randomization) shows that the control group also improved markedly, for example with urea falling by about 20%, and that neither group reached the target serum uric acid. Evidence of an increase in glomerular filtration rate or of slowed CKD progression is lacking, so the product cannot be recommended as treatment.

Summary

- One capsule contains approximately 10 mg of the A-9 complex, which the label declares only by three amino acids (glutamic acid, aspartic acid, and alanine); peptide sequences, their concentrations, and

batch purity are not publicly characterized. English promotional material describes it differently, as peptides up to 5,000 Da from calf kidneys.

- Pharmacokinetics, oral bioavailability, and selective action on the kidney are not documented; for comparison, even oral semaglutide with SNAC has a bioavailability on the order of about one percent.
- In the 2011 report (15 controls, 27 treated), urea fell from 13.8 to 9.2 mmol/L with Pielotax but also from 14.7 to 11.8 mmol/L in the control group. Creatinine, eGFR, albuminuria, blinding, placebo, registration, and a description of the statistics were all lacking, and the report was published by the vendor.
- Men in the treated group reached a serum uric acid of 0.44 mmol/L and women 0.37 mmol/L, which is above the ACR and EULAR target of below 0.36 mmol/L (360 μ mol/L). The tested dose of up to 60 mg daily was as much as three times the reference daily dose of 20 mg.
- Safety cannot be inferred from 42 patients followed for one month; the potassium acetate content is not quantified, which is a separate shortcoming in advanced CKD and hyperkalemia. Notification of the supplement to the Public Health Authority of the Slovak Republic is a registration step, not an assessment of efficacy.
- The author does not recommend Pielotax for CKD, gouty kidney damage, or hyperuricemia; patients should not use it in place of treatment with proven efficacy, such as RAAS blockade, SGLT2 inhibitors, or treat-to-target urate-lowering therapy.

12.21 Ursodeoxycholic acid and ursolic acid: similar names, fundamentally different clinical significance

Published September 8, 2026 · [online version \(Slovak\)](#)

Abstract. The similar names of ursodeoxycholic acid (UDCA) and ursolic acid (UA) can lead to a drug being confused with a dietary supplement. This review, based on EASL guidelines, the Slovak SmPC, a phase I study, and a meta-analysis, shows that UDCA is a bile acid and the first-line treatment for primary biliary cholangitis at a dose of 13 to 15 mg/kg/day. UA is a plant triterpenoid with low bioavailability, and a meta-analysis of six studies showed no change in metabolic parameters. UA should therefore not replace prescribed UDCA.

Summary

- UDCA (C₂₄H₄₀O₄) is the 7 β -epimer of chenodeoxycholic acid, is less hydrophobic and less cytotoxic, and enters the enterohepatic circulation; UA (C₃₀H₄₈O₃) is a pentacyclic triterpenoid from apple peel, rosemary, or thyme with low solubility and variable oral bioavailability.
- For PBC, EASL recommends 13 to 15 mg/kg/day and the Ursofalk SmPC 14 $\pm$ 2 mg/kg/day; response is assessed after 12 months by alkaline phosphatase and bilirubin, and an inadequate response leads to specialist reassessment, not substitution with ursolic acid.
- Gallstone dissolution is possible only for radiolucent stones up to 15 mm with a functioning gallbladder, usually over 6 to 24 months; if there is no reduction after 12 months, treatment is stopped. The SmPC requires monitoring of AST, ALT, and GGT every four weeks for the first three months and every three months thereafter.
- The most common adverse effect is soft stools or diarrhea. Cholestyramine, colestipol, and aluminum-containing antacids reduce UDCA absorption and should therefore be taken at least two hours apart; an interaction with cyclosporine may also be significant.
- Preclinical data predominate for UA; a phase I study of an intravenous nanoliposomal formulation did not assess efficacy, and a meta-analysis of six studies with doses of 51 to 450 mg/day showed no change in anthropometric, glycemic, lipid, or blood pressure parameters.
- In practice, the full name of the active substance should be checked, the dietary supplement recorded in the medication history, and prescribed UDCA therapy not changed on the patient's own initiative.

12.22 Polyneuropathy in chronic kidney disease: diagnosis, treatment, HTEMs, and safe dosing

Published September 8, 2026 · [online version \(Slovak\)](#)

Abstract. Polyneuropathy in advanced CKD is typically a distal symmetric sensorimotor axonal neuropathy, but diagnosis requires exclusion of diabetes, vitamin B12 deficiency, paraproteinemia, and drug-related causes. This review combines AAN and KDIGO guidelines, gabapentinoid SmPCs, and small uncontrolled studies of high-tone external muscle stimulation (HTEMs), in which 73% of patients improved in at least one symptom. HTEMs remains an experimental adjunct; the foundation is treatment of the cause, optimization of dialysis, and dosing of gabapentin and pregabalin according to creatinine clearance.

Summary

- Rapid progression, asymmetry, proximal or marked motor weakness, demyelination, or acute symptoms after creation of an arteriovenous access are not typical of uremic neuropathy and require urgent investigation.
- According to AAN, the highest diagnostic yield comes from testing glucose metabolism, vitamin B12 (with methylmalonic acid if borderline), and electrophoresis with immunofixation. In progression despite intensive dialysis, KDIGO suggests considering urgent transplantation, with very low certainty of evidence.
- Gabapentin according to the Slovak SmPC: 300 to 900 mg/day at a creatinine clearance of 30 to 49 mL/min, 150 to 600 mg/day at 15 to 29 mL/min, and 150 to 300 mg/day below 15 mL/min; in an anuric patient on HD, a loading dose of 300 to 400 mg followed by 200 to 300 mg after each four-hour hemodialysis session.
- Pregabalin according to EMA: 25 to 50 mg/day with a maximum of 150 mg/day at a clearance of 15 to 30 mL/min, 25 mg/day with a maximum of 75 mg/day below 15 mL/min, and a supplementary dose of 25 to 100 mg after hemodialysis; the maximum dose is not a therapeutic target.
- In three small nonrandomized studies with 25 to 40 patients, HTEMs (60 minutes during dialysis three times weekly for 12 weeks) improved symptoms and increased ulnar nerve conduction velocity from 48.53 to 51.50 m/s, but it does not prove nerve regeneration, and routine availability in Slovakia could not be confirmed.
- With gabapentinoids, sedation, falls, respiratory depression (particularly in combination with opioids), edema, and dependence should be monitored; they should be tapered over at least one week, and ineffective treatment should be deprescribed.

12.23 Proton pump inhibitors vs. H2-receptor antagonists in patients with CKD taking nonsteroidal anti-inflammatory drugs

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. In patients with CKD taking traditional NSAIDs, PPIs are often added for gastroprotection, although observational studies associate them with CKD progression. A South Korean retrospective cohort (5,315 patients on a PPI and 7,112 on an H2-receptor antagonist, 2018 to 2023) found a higher risk of CKD progression with PPIs, adjusted HR 1.38 (95% CI, 1.10 to 1.74). Because of the observational design and confounding by indication, this is not proof of causality; the result supports reviewing the indication for PPIs, but the NSAID itself remains the main modifiable risk.

Summary

- Only 12,427 of 141,093 patients with CKD starting an NSAID received gastroprotection; the groups were balanced with stabilized inverse probability weighting, but neither the absolute risk nor the exact definition of CKD progression is given in the summary.
- In the model, hypertension (HR 2.03), prior diuretic use (HR 2.00), diabetes (HR 1.69), and a history of kidney disease (HR 1.43) increased the risk, whereas dapagliflozin or empagliflozin reduced it (HR 0.49); these are associations, not independent causal effects.

- The association with PPIs was strongest in women, in CKD G3, at age 71 years and older, and already with 1 to 15 days of treatment, which is biologically implausible and points instead to AKI, acute illness, or reverse causality.
- The best documented renal effect of PPIs is acute tubulointerstitial nephritis, often without the classic triad of fever, rash, and eosinophilia; another is hypomagnesemia, particularly with a concomitant diuretic.
- An H2-receptor antagonist may be considered where a PPI is not essential, with dose adjustment to kidney function and monitoring for confusion in older patients; in practice this mainly means famotidine, since ranitidine was withdrawn in 2020. PPIs remain appropriate, for example, for a bleeding peptic ulcer or severe erosive esophagitis.
- The author recommends first reassessing the need for the NSAID, avoiding combination with a diuretic and a renin-angiotensin system inhibitor, monitoring creatinine, eGFR, potassium, and volume status, and discontinuing PPIs in a controlled manner.

12.24 Long-term use of proton pump inhibitors: what we actually know and how to translate it into practice

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. Long-term PPI use has been associated in observational studies with CKD, AKI, fractures, dementia, and mortality, but these signals are burdened by confounding by indication and protopathic bias. A review of the ACG guideline, the randomized COMPASS trial, and a systematic review of mortality shows that pantoprazole 40 mg did not increase all-cause mortality after about 3 years (HR 1.03) and increased only enteric infections (1.4% vs. 1.0%). With a clear indication, benefit prevails; PPIs without an indication should be deprescribed regularly and in a controlled manner.

Summary

- In COMPASS, in patients without an indication for a PPI, the HR for all-cause mortality was 1.03 (95% CI, 0.92 to 1.15), and enteric infections occurred in 1.4% versus 1.0% (odds ratio 1.33); *Clostridioides difficile* infection was roughly doubled, but with 13 cases it was not statistically significant. The trial was not designed to detect rare or late events.
- Ben-Eltriki et al. (2020) compared observational studies (OR 1.68), a veterans cohort versus H2-receptor antagonists (HR 1.17), and COMPASS (HR 1.03); the convergence is a pharmacoepidemiologic hypothesis, not proof that PPIs increase mortality.
- Acute tubulointerstitial nephritis is a recognized idiosyncratic reaction. ACG does not recommend routine creatinine monitoring in the absence of risk factors but does recommend close monitoring of kidney function or nephrology consultation in renal insufficiency.
- In a South Korean cohort of patients with CKD taking NSAIDs, PPI users had an aHR of 1.38 for CKD progression compared with H2-receptor antagonist users, but a signal already at 1 to 15 days of treatment suggests time-related bias.
- An interaction with clopidogrel via CYP2C19 is plausible, but clinical data are inconsistent, and regulatory warnings exist for omeprazole and esomeprazole. ACG does not support routine screening for magnesium, vitamin B12, or fractures in the absence of other risk factors.
- In practice, the indication should be verified; for typical GERD, an 8-week course should be given with an attempt at discontinuation, the lowest effective dose taken 30 to 60 minutes before the first meal, eGFR, potassium, and magnesium monitored in CKD, and deprescribing done gradually.

12.25 Gabapentin and its “dark side”: safety in the nephrology setting (CKD, hemodialysis)

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. In 2011, 19% of patients on hemodialysis in the USRDS registry were taking gabapentin, which is excreted unchanged by the kidneys and has a half-life of up to 132 hours on dialysis. This review of FDA warnings, cohort studies (Ishida 2018, Muanda 2022), and a systematic review of misuse distinguishes four lines of safety concern. On hemodialysis, doses above 300 mg daily increased the risk of falls by 55%, and a higher starting dose in CKD increased the 30-day risk by 0.40 percentage points. A clear indication and a low starting dose are key.

Summary

- In 2019, the FDA warned of serious respiratory depression with gabapentinoids; from 2012 to 2017, 49 cases with 12 deaths were reported, and risk factors include opioids and other CNS depressants, COPD, and older age.
- Ishida et al. (140,899 patients on HD): gabapentin above 300 mg/day increased the risk of altered mental status by 50%, falls by 55%, and fractures by 38%, risk was increased even at lower doses, and pregabalin increased the risk of falls by as much as 68%. Almost one third of users had no documented indication.
- Muanda et al. (74,084 older adults with eGFR below 60 mL/min/1.73 m²): starting gabapentin above 300 mg/day or pregabalin above 75 mg/day increased the 30-day incidence of the composite endpoint to 1.9% versus 1.5% (RR 1.27; difference, 0.40%), corresponding to an NNH of about 250.
- Misuse is reported in 40% to 65% of people with a prescription and in 15% to 22% of people who misuse opioids, but this does not represent dependence, and the data are heterogeneous; the United Kingdom has controlled gabapentinoids since 2019, Slovakia has not.
- A 2026 preprint reports an HR of 7.39 for dementia in CKD, but this is a non-peer-reviewed and probably biased signal that should not change practice; the main data are observational, and the optimal dose on dialysis has not been established.
- The author recommends documenting the indication, starting low, reviewing CNS-depressant drugs, monitoring for confusion and falls, considering gabapentin toxicity in a patient on dialysis with unexplained confusion, and tapering over at least one week.

12.26 Chronic pain in chronic kidney disease: which analgesic, and at what cost

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. Chronic pain in CKD is underdiagnosed and undertreated because every common analgesic carries a different risk in impaired kidney function. This review draws on the 2022 CDC guideline, a debate in Kidney360, a retrospective cohort, and prescribing information. A single perioperative dose of ketorolac was associated with AKI in 12.7% versus 2.0% of patients with CKD. The author recommends preferring non-pharmacologic and nonopioid treatment, using NSAIDs only short term, avoiding duloxetine at a creatinine clearance below 30 mL/min, and exercising caution with renally eliminated opioid metabolites.

Summary

- The 2022 CDC guideline is based on GRADE methodology, prefers nonpharmacologic and nonopioid treatment, and is explicitly not meant to be applied as inflexible standards; an opioid may be considered if the expected benefit outweighs the harm.
- In CKD, NSAIDs increase the risk of AKI, hyperkalemia, and worsening hypertension, particularly in combination with a diuretic and RAAS blockade; after total knee arthroplasty, ketorolac was associated with AKI in 12.7% versus 2.0% of patients with CKD (n = 114), with no difference in patients without CKD. If NSAIDs are unavoidable, they should be used for the shortest possible time, with monitoring of creatinine, potassium, and blood pressure and with sick-day rules.

- Duloxetine can be used without dose adjustment at a creatinine clearance of 30 to 80 mL/min but is not recommended in severe impairment and end-stage kidney failure: on hemodialysis, C_{max} and AUC are approximately doubled and metabolite exposure is 7- to 9-fold higher.
- Morphine and codeine accumulate active glucuronides with a risk of myoclonus, hyperalgesia, and delirium, and meperidine accumulates neurotoxic normeperidine; combining an opioid with a gabapentinoid increases the risk of serious respiratory depression.
- In practice, the type and duration of pain should be determined, nonpharmacologic treatment (exercise, psychological intervention, and topical capsaicin) started immediately, eGFR and the full medication list checked before prescribing, and function and tolerability assessed as well. NSAID thresholds in CKD are not supported by randomized data.

12.27 Cannabis, polypharmacy, and driving in older patients: what the data show and what to ask

Published October 2, 2026 · [online version \(Slovak\)](#)

Abstract. During medication review in older multimorbid patients, questions about cannabis and driving are rarely asked. A meta-analysis of 80 publications showed that most driving-related abilities recover within about 5 hours after inhalation of 20 mg of Δ9-THC, and later after oral use. A simulator study in 31 older regular users found impairment 30 minutes after smoking but not after 180 minutes, and blood THC levels did not correlate with impairment. Patients should be advised to wait at least 5 hours.

Summary

- A meta-analysis in *Neuroscience and Biobehavioral Reviews* (2021) showed impairment of lane keeping, tracking, and divided attention; regular users were less impaired ($p = 0.003$), so their data tend to underestimate the risk in a typical patient.
- In a 2024 JAMA Network Open study (mean age, 68.7 years; cannabis with a mean THC content of 18.74%), lane weaving increased 30 minutes after smoking (effect size, 0.30) and mean speed decreased, whereas the subjective effect persisted for 5 hours.
- Limitations: a small, predominantly male sample and simulator driving, with no data on actual crashes, drug interactions, or patients with CKD; the kidney section of the article is a pharmacologic extrapolation.
- Polypharmacy with benzodiazepines, opioids, sedating antidepressants, first-generation antihistamines, or gabapentinoids amplifies the risk, and in CKD exposure to renally eliminated drugs may be higher.
- The author recommends asking older patients who drive and have polypharmacy, CKD, or a balance disorder, advising them not to drive for at least 5 hours after inhalation and longer after oral use, not to rely on blood levels or on how they feel, and recording the answer in the medical record.

CHAPTER 13

Nutrition, lifestyle, and the environment

Protein, fiber, plant-based diets, physical activity, heat waves, and popular nutrition trends. What holds up against the evidence and what it means for patients with kidney disease. Number of articles: 30.

Key messages of the chapter

The most consistent message of this chapter is a shift toward a predominantly plant-based, minimally processed diet, which, however, rests almost exclusively on observational data. In the UK Biobank cohort, a vegetarian diet was associated with a lower risk of incident chronic kidney disease (CKD; HR 0.81), and an ERA conference abstract linked higher adherence to the Planetary Health Diet with lower mortality in patients with CKD (HR 0.83), whereas ultraprocessed foods and red meat were associated with a higher risk of death (13.24, 13.5). Associations of colorants and preservatives with type 2 diabetes and hypertension in the NutriNet-Santé cohort, cross-sectional NHANES data on hepatic steatosis, and expert opinion favoring predominantly plant-based eating point in the same direction (13.4, 13.20, 13.3). In practice, the Mediterranean and DASH diets are more useful than counting fatty acids, and according to a review, no higher risk of hyperkalemia was observed with them (13.2). Causality has not been proven; vegetarians were younger, leaner, and smoked less (13.24).

The greatest tension between population and nephrology recommendations concerns protein. In people without CKD, the 10-year RENIS cohort with repeatedly measured GFR found no faster decline in filtration with higher intake, but the conclusion does not apply to extreme intake or to CKD (13.16). In CKD G3 to G5 not on dialysis, KDIGO 2024 recommends approximately 0.8 g/kg/day and avoiding intake above 1.3 g/kg/day, which the upper limit of the US DGA recommendations and the fitness goals of trends such as gutmaxxing clearly exceed (13.20, 13.21, 13.19). The evidence for restriction is mixed: an umbrella review showed moderate certainty of benefit of a low-protein diet for serum phosphate and the risk of kidney failure, but the pragmatic ERIKA RCT did not demonstrate a benefit of a very low-protein diet with ketoanalogues and revealed low adherence (13.30). On dialysis, requirements are higher, approximately 1.0 to 1.2 g/kg/day, and the main risk is malnutrition and loss of muscle mass (13.30, 13.21).

A cross-cutting theme is the diagnostic trap of creatinine. A meta-analysis of randomized trials showed that creatine raises serum creatinine by an average of 0.13 mg/dL with no difference in eGFR, making it a determinant independent of the glomerular filtration rate, although safety in advanced CKD has not been demonstrated (13.15). High protein intake and greater muscle mass act similarly, whereas in an RCT of a ketogenic diet, eGFR rose only because of a decrease in cystatin C, which probably reflects a change in the marker (13.21, 13.28). The cognitive use of creatine is supported only by small and inconsistent studies (13.9, 13.8). When creatinine is discordant with the clinical picture, KDIGO 2024 recommends the combined eGFR with cystatin C (13.15, 13.16).

The second prominent theme is heat stress. Modeled estimates report 62,775 heat-attributable deaths in Europe for 2024, and in patients on dialysis, extreme humid heat was associated with an HR of 1.18; a post hoc analysis of DAPA-CKD linked hot days with a faster eGFR decline, all at the level of observational associations (13.13, 13.7). In chronic kidney disease of nontraditional etiology, piece-rate pay was a stronger predictor of AKI than heat stress itself (13.27), and AKI with rhabdomyolysis also threatens healthy athletes (13.10). A universal recommendation to drink more is wrong,

and temporary drug withholding according to KDIGO 2024 applies only to acute dehydrating illness (13.13).

For beverages and supplements, individualization prevails over prohibitions. In the UK Biobank, coffee was associated with a lower risk of cirrhosis (HR 0.68), and in CKD, additives and volume are what matter most (13.23, 13.12). Alcohol has no net protective effect and is carcinogenic (13.11). The kidney associations of sweeteners are observational, and the signal for xylitol comes from a non-peer-reviewed conference abstract (13.26, 13.29). Meta-analyses support dietary fiber but excluded the chronically ill (13.22, 13.6), bowel cleansing with sodium phosphate can cause acute phosphate nephropathy (13.19), and the relationship of the microbiome to the kidneys remains at the level of cross-sectional associations (13.1). Physical activity has favorable observational associations from as few as 7,000 steps per day, and in atrial fibrillation it does not replace anticoagulation (13.25, 13.18).

This yields a simple framework for the outpatient clinic: promote a plant-dominant, minimally processed diet; set the protein target according to CKD stage and dialysis status; actively ask about supplements and trends; verify a rise in creatinine with cystatin C and albuminuria; prepare an individual fluid and medication plan before a heat wave; and increase physical activity gradually and measurably (13.30, 13.21, 13.15, 13.13, 13.25). When restrictions accumulate, malnutrition and sarcopenia should be considered and a dietitian involved (13.17, 13.2).

13.1 The gut microbiome and kidney health: what the largest study to date brings

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. The relationship between the gut microbiome and kidney health has so far been supported mainly by associations. A 2026 commentary evaluates the largest cross-sectional study to date, which combined a large sample, a discovery and validation approach, shotgun metagenomics, and serum metabolomics. Methodologically, this is a significant advance, but the cross-sectional design does not allow determination of whether microbiome changes precede kidney damage or result from it. In practice, the microbiome remains a field for research and risk models, not a basis for treatment rules.

Summary

- The study combines shotgun metagenomics, which provides a detailed view of the genetic content of the microbiome, with serum metabolomics capturing the metabolic profile, and validates the findings in an independent validation cohort.
- In the cross-sectional design, the microbiome and kidney status were measured at the same time, so cause cannot be distinguished from effect, nor can a bidirectional relationship mediated by other factors be excluded.
- The commentary regards the association as more credible both biologically and analytically and the microbiome as potentially relevant to kidney health rather than a fringe hypothesis, but it does not confirm direct causality, and a “one intervention, one effect” rule is not yet supported.
- The authors call for prospective cohorts with longitudinal measurements, or designs that better capture the direction of the relationship, to reveal the actual temporal relationships, that is, whether microbiome changes precede deterioration of kidney function.
- The discovery and validation approach reduces the risk of a chance finding in a single cohort, and combining metagenomics with metabolomics shows how differences in the microbiome may translate into a person's metabolic profile. Microbiome composition alone may not suffice to understand what happens in the body.

13.2 Diet, fats, and cardiovascular risk in chronic kidney disease (CKD): what makes sense in practice

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. Cardiovascular disease is the most common cause of death in CKD, yet trials have often excluded patients with CKD, and especially those on dialysis. This review focuses on dietary fats: in CKD, levels of polyunsaturated fatty acids (PUFAs), which may slow progression and reduce cardiovascular risk, are lower. In practice, recommending a Mediterranean or DASH diet, neither of which has been associated with a higher risk of hyperkalemia, is more useful than counting fatty acid percentages. Individualized care by a dietitian offers the greatest benefit.

Summary

- CKD alters the background of cardiovascular disease through mineral and bone disorders, uremic toxins, and hemodynamic changes; improvement in surrogate markers such as LDL, blood pressure, phosphorus, or CRP may not translate into lower cardiovascular mortality.
- Current recommendations favor increasing PUFA intake, and some studies describe lower mortality with regular fish consumption; however, the picture is not clear-cut, because improved lipids are not the same as reduced mortality.
- The Mediterranean and DASH diets increase PUFAs and provide fiber, antioxidants, and polyphenols; according to the article, they have better tolerability and adherence than strict renal restrictions, and the correlation between potassium intake and serum potassium is weak.
- The Mediterranean diet derives at least about 35% of energy from fat, so it can help maintain energy intake and prevent malnutrition and muscle loss.
- The patient's priorities may be the affordability of food, abstinence from smoking and alcohol, or glycemic control; the plan should be developed jointly by the nephrologist and a registered dietitian with regard to cultural and economic factors. The dietitian performs a nutritional assessment, sets realistic goals, and harmonizes recommendations across the team.

13.3 How “plant-based” is the current direction of dietary recommendations, and why it makes sense

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. Dietary recommendations have changed over time and have often contradicted each other. A Medscape interview with cardiologist Kim Williams, former president of the ACC, explains why AHA recommendations are converging on a diet dominated by plant foods. This is expert opinion, not a new study: it argues from the substitution effect, the effect of fiber on the microbiome and inflammation, and improvement in risk factors. There is consensus on limiting ultraprocessed foods and preferring whole-grain sources; without access to healthy food, however, the recommendations remain on paper.

Summary

- According to Williams, a plant-based diet lowers cardiovascular risk by replacing unfavorable foods, and fiber changes the composition of gut bacteria and may dampen inflammatory mechanisms linked to atherosclerosis.
- According to the interview, the current recommendation rests on a growing body of long-term data, and a plant-based diet of whole, unprocessed foods improves both lipids and inflammation; Williams also mentions an association with plaque regression during LDL cholesterol-lowering therapy.
- Williams links an emphasis on animal protein to chronic kidney disease and gout; this is his interpretation, not an analysis of causal evidence.
- Ultraprocessed foods should be limited, ideally eliminated; saturated fat, sodium, sugars, and fiber should be checked on products, and refined sources such as white flour, white rice, and white pasta replaced with whole grains. According to the interview, whole foods and whole grains are consistently associated with better health outcomes.

- Williams describes a conflict between the US federal dietary guidelines and the cardiology community; in his view, the final guidelines in some periods were less consistent with the conclusions of expert advisory committees, for example on saturated fat, sugars, and refined grains.
- O'Donoghue and Williams emphasize the systemic problem of access to healthy food, so alongside physician counseling, the food environment in schools and communities must also change.

13.4 Food additives under the microscope: new data link colorants and preservatives with a higher risk of chronic diseases

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Colorants and preservatives are common in ultraprocessed foods; in 2024, more than 139,000 products contained a colorant from E100 to E199. Three analyses of the prospective French NutriNet-Santé cohort (105,260 to 112,395 participants, followed since 2009) showed that the highest colorant intake was associated with a 38% higher risk of type 2 diabetes and the highest preservative intake with a higher risk of hypertension (HR 1.24). These are observational data without proof of causality; in practice, limiting ultraprocessed foods is recommended.

Summary

- The main analyses included 10 colorants and 17 preservatives consumed by at least 10% of participants; exposure was determined from repeated 24-hour dietary records including product brands.
- Type 2 diabetes (108,723 participants): total colorants HR 1.38 (95% CI, 1.17 to 1.63), plain caramel E150a HR 1.46, beta-carotene E160a HR 1.44, and curcumin E100 HR 1.49.
- Cancer (105,260 participants, 4,226 cases): high colorant intake was associated with a 14% higher risk of overall cancer, a 21% higher risk of breast cancer, and a 32% higher risk of postmenopausal breast cancer; this concerns additives, not natural beta-carotene from foods.
- Preservatives (112,395 participants): hypertension HR 1.24, and HR 1.29 for nonantioxidant preservatives; nonantioxidant preservatives were also associated with a 16% higher cardiovascular risk, and potassium sorbate E202 with a 39% and citric acid E330 with a 25% higher risk of hypertension.
- Despite adjustment for confounders, additives may be a marker of an ultraprocessed diet, and exposure is difficult to measure; the results do not show that lemon or vitamin C is harmful.
- Patients do not need to track individual E numbers; it is more sensible to cook from minimally processed ingredients and to regard a long list of additives as a sign of more extensive industrial processing; the authors call for a reassessment of the safety of selected substances.

13.5 Plant-based diet and lower mortality in patients with CKD: what to use in practice

Published June 16, 2026 · [online version \(Slovak\)](#)

Abstract. A conference abstract from the 63rd ERA Congress evaluated, in a multicohort observational analysis (NHANES and HCNS), 8,165 adults with CKD followed for about 7 to 8 years and four dietary patterns. The Planetary Health Diet had the strongest association: a 1-standard-deviation increase in adherence corresponded to 17% lower all-cause mortality (HR 0.83). The Inverted Pyramid Diet was associated with higher mortality. The results do not prove causality but support a predominantly plant-based, minimally processed diet.

Summary

- With the PHD, kidney-related mortality was 24% lower (HR 0.76; 95% CI, 0.64 to 0.90), the DASH and Mediterranean diets had a smaller effect, whereas the Inverted Pyramid Diet increased all-cause mortality by 18% and kidney-related mortality by 32% (HR 1.32).
- Ultraprocessed foods accounting for at least 30% of energy intake increased the risk of death by 24%, and red meat at 4 or more servings per week by 19%.

- This is an observational analysis from a conference abstract; with an excessively plant-based regimen, protein intake and the risk of sarcopenia must be considered, so nutritional status screening and collaboration with a dietitian are recommended.

13.6 The “chia seed internal shower”: what is true and what is not from a medical perspective

Published June 17, 2026 · [online version \(Slovak\)](#)

Abstract. The “chia seed internal shower” trend promises to flush the bowel with chia seeds soaked in water. An analysis of the source article shows that chia has real nutritional value (2 tablespoons contain about 10 g of fiber), but there is no peer-reviewed evidence of bowel cleansing beyond the effect of ordinary fiber. A sudden increase in intake can cause bloating, cramps, and paradoxical worsening of constipation within about 72 hours. Gradual titration, adequate fluids within the individual limit, and, for chronic constipation, psyllium rather than chia are recommended.

Summary

- The soluble fiber in chia forms a viscous gel with water that increases stool volume, softens its consistency, and promotes transit; this is ordinary fiber physiology, not detoxification.
- Psyllium has the strongest evidence base for chronic constipation; insoluble fiber, such as wheat bran, can worsen bloating in IBS, and chia, which is predominantly soluble, may not suit everyone.
- Starting with 0.5 to 1 tablespoon daily is recommended, increasing the amount over several weeks with adequate fluids and aiming overall for 25 to 35 g of fiber daily from a variety of foods, such as prunes, kiwifruit, or oats.
- Constipation lasting months requires a search for a drug-related, endocrine, or functional cause rather than simply more fiber, because the wellness trend may delay necessary investigation.
- An abrupt increase in intake is risky because the gut microbiome needs time to adapt to a larger supply of fermentable substrate; fiber without fluids can actually worsen transit.
- In CKD and dialysis, fluid advice must be adapted to the individual limit, chia should be tried only after agreement with the treating team, and it should be stopped if bloating, cramps, or transit worsen.

13.7 Extreme heat and underestimated health risks: what it also means for nephrology

Published June 26, 2026 · [online version \(Slovak\)](#)

Abstract. Heat waves are a growing environmental health risk whose impact is underestimated because heat is rarely recorded as a direct cause of illness or death. A Medscape review (a journalistic overview, not an original study) cites a modeled estimate of about 2,500 heat-attributable deaths in Germany in 2025 and an increase in sick leave of 3.5% after one day above 30 °C and up to 10.8% after seven days. Patients with kidney disease are particularly vulnerable because of dehydration, electrolyte disturbances, and the effects of diuretics, which calls for prevention and individualized adjustment of treatment.

Summary

- The impact of heat is estimated indirectly with statistical models: the Robert Koch Institute estimated about 2,500 deaths in Germany in 2025, and sick-leave reports rise by 3.5% after one day above 30 °C and by 10.8% after seven consecutive days.
- Heat strains the cardiovascular system (vasodilation, falling blood pressure, and risk of angina pectoris, myocardial infarction, heart failure, and arrhythmias), the respiratory system when combined with ozone and particulate matter, and the nervous system, including a higher risk of stroke and of hospitalization in dementia.
- Dehydration and electrolyte abnormalities increase the risk of circulatory collapse and thrombosis, and diuretics and beta-blockers can further impair circulatory stability and hydration.

- The most vulnerable are older people, people with low socioeconomic status, limited mobility, or cardiopulmonary disease, people taking sedatives, and patients with kidney disease, who have less physiologic reserve.
- The source does not directly address heat-related AKI; the link to reduced renal perfusion and acute worsening of kidney function in CKD is the author's interpretation based on known mechanisms.
- By the end of 2024, only 20 of several thousand German municipalities had a heat-health action plan, indicating insufficient system preparedness.
- The author recommends identifying at-risk patients with CKD in advance, giving them a care plan (fluids, symptoms, and when to contact a physician), responding promptly to weakness, dizziness, or changes in urination, and adjusting treatment, including diuretics, individually rather than across the board.

13.8 Protein and creatine are no longer exclusively a “fitness topic”: what follows for nephrology

Published June 27, 2026 · [online version \(Slovak\)](#)

Abstract. Protein and creatine supplements have moved from sports into everyday outpatient conversations, and patients often fear kidney damage. An educational Medscape review states that concerns about high-protein diets in healthy individuals have repeatedly been challenged and that creatine has a very favorable safety profile in sports nutrition. For adults, about 1.2 to 1.6 g of protein/kg/day is recommended, and during GLP-1 receptor agonist therapy often 1 to 1.2 g/kg/day together with resistance training. In CKD, the dose should be individualized and creatinine interpreted cautiously.

Summary

- The recommended protein intake for adults is about 1.2 to 1.6 g/kg/day (about 100 to 130 g daily at 82 kg); during GLP-1 receptor agonist therapy with reduced appetite, a target of 1 to 1.2 g/kg/day is often used in practice.
- Muscle mass is protected primarily by resistance training, with protein only second; supplements without training make sense if the patient cannot meet requirements through diet (reduced appetite, older age, or restricted intake).
- Stimulating muscle protein synthesis usually requires about 20 g per meal in younger and about 30 g in older adults, and 40 g may maximize it in most people; a larger single dose is not automatically wasted, since total daily intake and quality are decisive.
- For creatine, creatine monohydrate from established manufacturers with independent third-party testing is preferred; the best protein is the one the patient tolerates and takes regularly, with whey isolate being practical and plant blends working similarly if they contain leucine.
- Possible cognitive and neuroprotective uses of creatine are not confirmed, and the evidence is still evolving.
- For nephrology practice, the author recommends distinguishing advice for people without kidney disease from that for people with CKD, taking into account stage, eGFR trend, albuminuria, and metabolic profile in CKD, and monitoring laboratory parameters with creatine and interpreting creatinine changes cautiously.

13.9 Creatine and brain health: is it worth a try, or is it just another supplement?

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. Creatine is moving from sports nutrition into neurologic debate because of its possible effects on cognition, mood, and sleep. A Medscape review summarizes mainly small and inconsistent studies: in 29 sleep-deprived adults, creatine monohydrate (0.2 g/kg) attenuated cognitive decline by about 12%, and in a pilot trial in Alzheimer's disease (n = 20) it increased total brain creatine. The evidence for dementia is not

robust, and safety is not fully settled. In CKD, the effect on creatinine must be kept in mind and a conservative approach taken.

Summary

- Creatine is synthesized in the liver and kidneys but also in the brain and retina via guanidinoacetate methyltransferase expressed in glial cells; creatine transporter deficiency (SLC6A8 gene) causes marked neurologic manifestations, and no specific treatment for it exists in practice.
- Small studies in healthy adults suggest variable improvements in mood, cognition, and sleep; in a 2026 study of 29 sleep-deprived individuals, creatine monohydrate 0.2 g/kg attenuated cognitive decline by about 12%, but this is a short-term effect, not dementia prevention.
- In Alzheimer's disease, a single-arm pilot trial (n = 20, 20 g/day for 8 weeks) showed an increase in total brain creatine and slight improvement on cognitive tests, and a 2022 review in *Nutrients* suggests possible benefit in traumatic brain injury and concussion; these are signals rather than definitive answers.
- Standard doses and dietary recommendations for brain health have not been established, and safety questions are not fully resolved; a trial may make sense with realistic expectations and personal evaluation of the effect.
- In CKD, the author points out that creatine is converted to creatinine and may complicate interpretation of results, and in advanced CKD recommends consultation before starting, monitoring, and avoiding empirical dose escalation.

13.10 Hydration breaks in soccer: a marketing move or a medical necessity?

Published July 10, 2026 · [online version \(Slovak\)](#)

Abstract. At the 2026 FIFA World Cup, FIFA introduced three-minute hydration breaks in the middle of each half regardless of the weather. A journalistic and educational *Medscape* article, supported by a review in *Sports Medicine*, reports historical WBGT values of 18.8 to 29.4 °C in host cities, with 14 of 16 cities having days above 28 °C WBGT. Exercise in the heat carries risks of dehydration, hyponatremia, rhabdomyolysis, and acute kidney injury, so the breaks, together with acclimatization and managed hydration, are medically justified.

Summary

- The break, at about the 22nd and 67th minutes, lasts three minutes from whistle to whistle, the time is added to stoppage time, and it applies in every match; it is of greatest medical importance in hot and humid conditions.
- The WBGT index accounts for temperature, humidity, radiant heat, and air movement; maximum values in host cities reach about 21 to 35 °C, and matches in Guadalajara and Mexico City are played at altitudes of about 1,566 m and 2,240 m.
- Fluid loss equivalent to about 2% of body weight already impairs performance and reactions; further risks include hyperthermia, exertional heat stroke, hyponatremia from excessive free water intake, rhabdomyolysis, and acute kidney injury.
- AKI can develop even in a healthy athlete through a combination of reduced renal perfusion with dehydration, sympathetic and RAAS activation, and rhabdomyolysis with myoglobinuria, which damages the tubules particularly in the setting of hypovolemia and acidosis.
- The break allows replacement of fluids and electrolytes, cooling, and rapid clinical assessment of the player; some experts consider three minutes too short under high heat stress, and acclimatization remains the key layer of prevention.
- The source is journalistic and based mainly on an interview with an expert; it is not an original clinical study.
- Outside professional sport, those at higher risk include unsupervised people, children, older adults, patients with hypertension, diabetes, CKD, or heart failure, and users of diuretics, RAAS inhibitors, or non-

steroidal anti-inflammatory drugs; drinking adequately but not excessively, replacing sodium, and, if heat stroke is suspected, stopping exercise immediately and cooling are recommended.

13.11 Alcohol as a “healthy drink”: why medical recommendations are changing

Published July 10, 2026 · [online version \(Slovak\)](#)

Abstract. The decades-old notion of a cardioprotective glass of wine was based on observational studies with a J-shaped curve, which were biased by classifying former drinkers as abstainers and by inadequate adjustment for confounders. An educational Medscape article, drawing on newer epidemiologic work, states that the Alcohol Intake and Health Study synthesis (2026) found no net protective effect of low consumption on mortality and that a causal link with cancer has been established for at least seven sites. Alcohol cannot be recommended for prevention; in CKD, advice should be individualized.

Summary

- After separating lifelong abstainers from former drinkers and adjusting for socioeconomic and lifestyle factors, the protective effect of low consumption weakens or disappears; however, this is a population estimate, not proof of equal risk for every individual.
- Ethanol and acetaldehyde are carcinogenic, no safe threshold has been defined for cancer, and a causal link has been established for cancers of the oral cavity, pharynx, larynx, esophagus, liver, colorectum, and female breast, regardless of beverage type.
- A 2025 American Heart Association statement rates the data on low to moderate consumption as mixed, and Mendelian randomization studies have not confirmed a protective effect; heavy and binge drinking are consistently associated with hypertension, atrial fibrillation, stroke, and heart failure.
- In CKD, indirect effects matter more than direct nephrotoxicity: poorer blood pressure control, volume depletion, a higher risk of hypotension and AKI with diuretics, renin-angiotensin system blockers, SGLT2 inhibitors, or NSAIDs, and hypoglycemia with insulin; in patients on dialysis, drinks count toward fluid intake.
- A standard drink contains about 10 to 14 g of ethanol depending on the country, so type, volume, frequency, and maximum amount per occasion should be ascertained, or the AUDIT-C questionnaire used.
- The evidence is limited by self-reported consumption and residual confounding, long-term randomized trials are not ethically feasible, and the source is a journalistic text.
- The author recommends that nondrinkers not start, that drinkers reduce consumption and avoid binge drinking, and that, when physical dependence is suspected, the risk of withdrawal syndrome be assessed first, because abrupt cessation can cause seizures or delirium.

13.12 Coffee and the gut microbiome: the effect goes beyond caffeine

Published July 10, 2026 · [online version \(Slovak\)](#)

Abstract. The intestinal effects of coffee have traditionally been attributed to caffeine, but more recent work also points to polyphenols, chlorogenic acids, and melanoidins. A Medscape review summarizes mainly observational and small interventional studies: a multicohort analysis of 22,867 participants showed a reproducible association of coffee, including decaffeinated coffee, with the bacterium *Lawsonibacter asaccharolyticus*, and a meta-analysis of 1,246 patients showed earlier return of bowel function after surgery. The clinical significance of microbiome changes is unknown. In CKD, coffee does not need to be prohibited across the board; volume and additives are what matter.

Summary

- A meta-analysis of 13 studies with 1,246 patients showed earlier return of bowel function and less post-operative ileus with coffee after abdominal surgery, with no significant difference between caffeinated and decaffeinated coffee, at low to moderate certainty of evidence.

- A study in *Nature Microbiology* (2024) with 22,867 participants found a higher prevalence and abundance of *L. asaccharolyticus* in coffee drinkers, including drinkers of decaffeinated coffee, and coffee promoted its growth in vitro; this is evidence of a food–microorganism interaction, not of a health benefit.
- In a study in *Nature Communications* (2026) of 62 healthy adults, coffee drinkers and nondrinkers differed in microbiota composition and fecal metabolites, including GABA, and some of the changes were independent of caffeine, but behavioral outcomes showed no clear benefit; the study was funded by the Institute for Scientific Information on Coffee.
- A meta-analysis of 40 studies with 122,074 participants found a small association of coffee with reflux (OR 1.18) with high heterogeneity and no association with Barrett esophagus; data on constipation are conflicting, and an association with IBD was not confirmed.
- Usual amounts of coffee did not impair hydration compared with water, but in patients on dialysis or with heart failure it counts toward fluid intake, and caffeine can transiently raise blood pressure.
- In advanced CKD, additives (milk, cream, creamers, syrups) carry more risk because of potassium, phosphorus, and sugar; the microbiome studies did not include patients with advanced CKD, on dialysis, or after transplantation.
- The author recommends not prohibiting coffee when it is well tolerated, asking about the type, servings, and additives, and not recommending increased consumption to treat the microbiome, constipation, or cognition.

13.13 Extreme heat: a growing risk for patients with kidney disease

Published July 17, 2026 · [online version \(Slovak\)](#)

Abstract. In vulnerable patients, heat waves can lead to dehydration, hypotension, AKI, electrolyte and acid-base disturbances, or decompensated heart failure. A Medscape commentary, together with epidemiologic data, reports that the Lancet Countdown estimated a total of 62,775 heat-attributable deaths in Europe in 2024, and in a US cohort of 945,251 patients on dialysis, extreme humid heat was associated with a higher risk of death (HR 1.18). These are observational associations. Prevention must be neither a universal “drink more” nor blanket discontinuation of medications, but an individual plan prepared before the heat arrives.

Summary

- Heat-attributable mortality increased between the periods 1991 to 2000 and 2015 to 2024 in 820 of 823 European regions, by an average of 52 deaths per million population per year; these are modeled estimates.
- In a post hoc analysis of DAPA-CKD in 4,017 participants, more days with a heat index above 30 °C were associated with faster eGFR decline, but this was an observational analysis that does not prove causality.
- In patients on dialysis, extreme humid heat was associated with an HR of 1.18 (95% CI, 1.15 to 1.20), and in the REIN registry, temperatures above 32.5 °C were associated with an IR of 1.09, but not in transplant recipients; these data do not imply an automatic increase in interdialytic weight gain or intradialytic hypotension.
- The fluid regimen must be tailored to the situation: with preserved urine output and no congestion, prevent hypovolemia and AKI; in advanced CKD, heart failure, anuria, and hemodialysis, do not liberalize intake arbitrarily; and in peritoneal dialysis, monitor the ultrafiltration balance.
- KDIGO 2024 allows temporary interruption of RAAS inhibitors, diuretics, SGLT2 inhibitors, metformin, and NSAIDs only during acute dehydrating illness, not on every hot day; evidence for sick-day rules is limited, and a common error is failure to restart the medications.
- The author recommends identifying high-risk patients (CKD G4 to G5, previous AKI, dialysis, frailty), recording a personal fluid and medication plan, monitoring weight, blood pressure, and urine output, and designating an escalation contact.
- Confusion, syncope, new dyspnea, a marked decrease in urine output, or rapid weight change require urgent evaluation, and dark urine with muscle pain may indicate myoglobinuria.

13.14 Can the smell of chocolate improve performance in strength training?

Published July 24, 2026 · [online version \(Slovak\)](#)

Abstract. Food odors influence appetite, emotions, and reward anticipation, which raises the question of their effect on performance. A small randomized, double-blind crossover study in 23 healthy, recreationally trained men exercising in the fasted state showed that the smell of dark chocolate (90% cocoa) increased the number of knee extensions by an average of about 18 repetitions and that of milk chocolate (60% cocoa) by about 9 compared with control, with no difference in perceived exertion. The results are preliminary, the mechanism is hypothetical, and they have no direct clinical relevance for patients with CKD.

Summary

- Participants, with a mean age of about 23 years, were exposed to the odor after fasting for at least 10 hours, before exercise and between sets, and performed knee extensions at 80% of 10RM to exhaustion; the primary outcome was the total number of repetitions.
- Dark chocolate suppressed subjective hunger more strongly, whereas milk chocolate was rated as more pleasant; the study assessed submaximal load, that is, local muscular endurance rather than maximal strength.
- A mechanism via the limbic system, hunger suppression, or endorphins is hypothetical, because endorphins, catecholamines, cortisol, dopaminergic activity, and neuromuscular parameters were not measured.
- Limitations include a small, homogeneous sample of young men, a single type of exercise, fasted testing, probably inadequate blinding with a possible placebo effect, and a lack of data on long-term effects on strength and muscle mass.
- Without the context of total volume and a standardized effect size, the figure of 18 repetitions does not allow assessment of athletic relevance; larger studies including women, different age groups, and a credible control odor are needed before any practical recommendation.
- The study did not include patients with CKD or on dialysis; smelling chocolate does not mean ingesting potassium, phosphorus, or sugar, but it cannot be claimed to improve performance or rehabilitation, and in patients with nausea or migraine an intense aroma may cause symptoms.

13.15 Creatine in kidney disease: harm, benefit, or diagnostic trap?

Published August 4, 2026 · [online version \(Slovak\)](#)

Abstract. Creatine has spread from sports to aging, sarcopenia, and chronic diseases, in which CKD is common and data are scarce. A narrative review by Pereira and coauthors in *Nephrology Dialysis Transplantation* (2026), supplemented by a meta-analysis of randomized trials, shows that creatine raises serum creatinine by an average of 0.13 mg/dL without a significant difference in eGFR or urea. This mainly reflects bias of the creatinine-based GFR estimate, but safety in advanced CKD has not been demonstrated. A rise in creatinine should be evaluated with multiple markers, and creatine is not a standard treatment for CKD.

Summary

- Supplementation increases the body creatine pool and creatinine generation, so serum creatinine may rise and creatinine-based eGFR may appear to fall without any change in filtration; this is a non-GFR determinant of creatinine, not analytical interference.
- A meta-analysis of 19 RCTs and one crossover study found a rise in creatinine of 0.13 mg/dL (95% CI, 0.07 to 0.18) with no difference in eGFR, but eGFR was assessed in only eight studies, often from creatinine, and studies longer than one year are lacking.
- The most robust safety data concern creatine monohydrate at doses of approximately 3 to 5 g daily in persons without kidney disease; in CKD, long-term RCTs by G category and data with measured GFR, cystatin C, and after transplantation are lacking.

- In an exploratory one-year study in patients on hemodialysis (5 g daily), fat-free mass by bioimpedance increased, but so did total and intracellular water, and the malnutrition-inflammation score did not improve, so clinical benefit has not been demonstrated.
- When creatinine rises, look for AKI: oliguria, albuminuria, active sediment, hyperkalemia, acidosis, rhabdomyolysis, NSAIDs or anabolic agents, and a decline in cystatin C-based eGFR; KDIGO 2024 recommends combined creatinine-cystatin C eGFR and, for critical decisions, measured GFR.
- Before supplementation in CKD, the author recommends clarifying the goal, assessing the G category, albuminuria, and medications, recording baseline creatinine, eGFR, and UACR, documenting the exact product, and agreeing on monitoring; do not start during AKI, unstable kidney function, or decompensation, and in advanced CKD or after transplantation decide together with a nephrologist.

13.16 Higher protein intake and kidney function: a ten-year cohort found no faster decline in measured GFR

Published August 14, 2026 · [online version \(Slovak\)](#)

Abstract. The concern that higher protein intake damages healthy kidneys is common, but population studies usually rely on creatinine-based eGFR. An observational analysis of the Norwegian RENIS cohort (1,324 participants, mean age 63.6 years) with repeated iohexol GFR measurements over a median of 10 years found no association of higher reported protein intake with faster mGFR decline (-0.01 mL/min/1.73 m² per year per 0.1 g/kg/day). The result weakens the case for preventive protein restriction in people without CKD, but it does not apply to extreme intake or to patients with CKD.

Summary

- Participants had a mean baseline mGFR of 89.1 mL/min/1.73 m², 52.1% had hypertension, and 47.8% had prediabetes; mean reported protein intake was 1.2 ± 0.5 g/kg/day, and the highest quartile had a mean of 1.8 g/kg/day.
- Per 0.1 g/kg/day, there was no association with annual change in mGFR (95% CI, -0.04 to 0.02), with accelerated decline (OR 0.97), or with incident mGFR below 60 mL/min/1.73 m² (HR 1.09; 95% CI, 0.96 to 1.21), which occurred in 118 of 1,064 persons.
- Limitations include the observational design with possible residual confounding, a single questionnaire completed a median of 20.5 months from the measurement, the lack of objective validation of intake and protein sources, and a predominantly homogeneous population from Tromsø.
- The study did not assess supplements or extreme intake, so no safe upper limit can be set, nor can it be claimed that an intake of 2 to 3 g/kg/day carries no risk.
- In CKD, KDIGO 2024 applies: approximately 0.8 g/kg/day in G3 to G5 and avoidance of intake above 1.3 g/kg/day in patients at risk of progression; requirements are higher on dialysis, and the RENIS conclusion does not apply to it.
- The author recommends not restricting protein preventively in people without CKD; in diabetes, hypertension, albuminuria, or a solitary kidney, eGFR and UACR should be known before a substantial increase in intake, and when creatinine is inconsistent with the clinical picture, combined eGFR with cystatin C should be used.

13.17 Non-celiac wheat sensitivity: gluten may not be the only trigger

Published August 14, 2026 · [online version \(Slovak\)](#)

Abstract. Improvement after wheat elimination does not confirm gluten intolerance, because non-celiac wheat sensitivity (NCWS) is a heterogeneous syndrome without a validated biomarker. A review based on meta-analyses, RCTs, and the ESSCD 2025 guideline reports that 10.3% of the population report subjective sensitivity, but on blinded challenge only 16% had gluten-specific symptoms and 40% reacted equally or

more to placebo. Fructans or a nocebo effect may also be triggers. Celiac disease must be ruled out while the patient is still consuming gluten.

Summary

- In a crossover RCT, fructans induced more gastrointestinal symptoms than gluten, with no difference between gluten and placebo; the role of amylase-trypsin inhibitors is hypothetical, and in a study with 84 participants the combination of expectation and gluten consumption had the greatest effect.
- The first test for celiac disease is IgA anti-TG2 with total IgA during gluten consumption; ESSCD 2025 conditionally allows a nonbiopsy diagnosis in selected adults up to 45 years of age with a titer of at least ten times the upper limit of normal confirmed in a second sample.
- If the patient has already eliminated gluten, negative HLA-DQ2/DQ8 practically rules out celiac disease, and for confirmation a challenge with at least 3 g of gluten daily for 6 weeks under the supervision of a gastroenterologist is considered.
- Wheat allergy must be ruled out in patients with rapid-onset urticaria, angioedema, or anaphylaxis; food IgG panels, zonulin, microbiome analyses, and intestinal permeability tests should not be used to diagnose NCWS.
- Only after celiac disease, allergy, and organic disease have been ruled out does an approximately six-week structured elimination with planned reintroduction follow; when fructans are suspected, a low-FODMAP approach with a restriction phase of 2 to 6 weeks is chosen, and the goal is the least restriction that is effective.
- KDIGO 2025 does not include a gluten-free diet among the standard treatments for IgA nephropathy; it is indicated only in confirmed celiac disease; in anemia of CKD, celiac disease should also be considered, and the accumulation of kidney and gluten-free restrictions increases the risk of malnutrition and sarcopenia.

13.18 Physical activity in atrial fibrillation: lower risk of stroke and death, not a substitute for anticoagulation

Published August 14, 2026 · [online version \(Slovak\)](#)

Abstract. It was unclear whether the favorable association of physical activity with stroke and mortality persists after atrial fibrillation develops. An observational analysis of the Norwegian HUNT and Tromsø cohorts (87,340 adults, 6,539 with atrial fibrillation, median follow-up 13.5 years) showed that even low activity was associated with a 9% lower risk of stroke and an 11% lower risk of death compared with inactivity, similarly in persons with and without the arrhythmia. Causality has not been proven, and physical activity does not replace indicated anticoagulation.

Summary

- Compared with inactive persons, the risk of stroke was lower by 9%, 19%, and 18% with low, moderate, and high activity, and the risk of death by 11%, 18%, and 22%; activity was assessed by questionnaire in MET-min/week.
- In atrial fibrillation, the 15-year stroke risk ratios were 0.80, 0.88, and 0.80, without a monotonic dose-response relationship, and the modeled gain in life expectancy was 0.50 to 1.15 years, which is a population estimate, not an individual promise.
- For intracerebral hemorrhage, a significant association was not confirmed, probably because of insufficient statistical power.
- Limitations include the observational design with residual confounding, possible reverse causality, self-reported activity, incomplete data on anticoagulation, and limited generalizability of the Norwegian cohorts.
- According to ESC 2024, anticoagulation is considered with a CHA₂DS₂-VA score of 1 and recommended with a score of at least 2; patients should not discontinue it because of exercise, weight loss, or ablation.

- In general, 150 to 300 minutes of moderate-intensity activity per week and resistance training at least twice a week are recommended, and KDIGO recommends at least 150 minutes in CKD as tolerated; in frail patients, the first goal may be several short bouts of walking a day.
- In CKD, direct oral anticoagulants are dosed according to creatinine clearance (Cockcroft-Gault equation), which is not interchangeable with eGFR, and in patients on dialysis, blood pressure, volume status, anemia, fall risk, and vascular access must be taken into account.

13.19 Diet and gut health according to influencers: where the most common errors arise and what they mean for nephrology practice

Published August 20, 2026 · [online version \(Slovak\)](#)

Abstract. Influencers' nutritional claims often have a true core that is overgeneralized, and in CKD they run into different limits. A review based on meta-analyses, case series, and position statements of professional societies examines four myths: gains in fat-free mass do not increase above approximately 1.62 g protein/kg/day, whereas KDIGO 2024 recommends 0.8 g/kg/day in CKD G3 to G5; colon cleansing has no evidence of benefit, and sodium phosphate can cause acute phosphate nephropathy; IgG tests reflect exposure; and BMI does not capture functional status.

Summary

- A meta-analysis by Morton et al. (49 RCTs, 1,863 participants) showed that fat-free mass no longer increases above approximately 1.62 g/kg/day, and a meta-analysis by Nunes et al. (74 RCTs) found only a small effect (SMD 0.22; 95% CI, 0.14 to 0.30).
- KDIGO 2024 recommends 0.8 g/kg/day in CKD G3 to G5 not on dialysis and not exceeding 1.3 g/kg/day in patients at risk of progression; a very low intake of 0.3 to 0.4 g/kg/day with ketoanalogs is reserved for selected patients under supervision, protein is not restricted in children, and requirements are higher on dialysis.
- No reliable study shows a benefit of colon cleansing, but electrolyte disturbances, sepsis, colitis, perforation, and deaths have been documented; Markowitz et al. described 21 patients with acute phosphate nephropathy after oral sodium phosphate, with possible permanent damage.
- Neither CSACI nor AAAAI recommends food-specific IgG testing, because it reflects exposure, not a pathologic reaction; in CKD, where phosphorus, potassium, and sodium are already restricted, further restrictions increase the risk of malnutrition and sarcopenia.
- BMI does not distinguish fat from muscle; in CKD, handgrip strength, gait speed, and nutritional status are more informative, and unplanned weight loss in a patient on dialysis is a warning sign.
- The author recommends not starting with rejection, naming a specific limit instead of a prohibition, explaining the mechanism of risk, and actively asking about supplements and regimens.

13.20 US Dietary Guidelines 2025 to 2030: the risk of misinterpretation in MASLD and CKD

Published August 22, 2026 · [online version \(Slovak\)](#)

Abstract. The US Dietary Guidelines for Americans 2025 to 2030 (DGA) emphasize minimally processed foods, but the target of 1.2 to 1.6 g protein/kg/day and the general advice to “consume less alcohol” without a numerical limit call for caution in MASLD and CKD. A cross-sectional NHANES analysis (10,944 adults) associated fruit, vegetables, legumes, and fish with a lower likelihood of steatosis and fibrosis, and red and processed meat and total protein with a higher likelihood. However, it did not test the effect of the new DGA. In CKD G3 to G5 not on dialysis, KDIGO 2024 applies, with an intake of approximately 0.8 g/kg/day.

Summary

- The DGA also recommend fish, poultry, eggs, legumes, nuts, soy, vegetables, and whole grains, saturated fat below 10% of energy, and sodium below 2,300 mg daily from age 14; the claim that plant sources are omitted is incorrect, and the document itself calls for adapting the recommendations in chronic disease.

- The NHANES 2017 to 2023 analysis with transient elastography is cross-sectional, with 24-hour dietary recall, residual confounding, and possible reverse causality; food components were mapped retrospectively onto the new DGA, and the association of the alcohol limit score is not evidence of a protective effect of alcohol.
- The lower DGA limit of 1.2 g/kg/day approaches, and the upper limit of 1.6 g/kg/day clearly exceeds, the threshold of 1.3 g/kg/day that a patient with CKD at risk of progression should avoid according to KDIGO 2024.
- For maintenance hemodialysis and peritoneal dialysis, KDOQI 2020 recommends approximately 1.0 to 1.2 g/kg/day, in cirrhosis 1.2 to 1.5 g/kg/day is generally recommended, and in frailty or sarcopenia the preservation of muscle mass may take priority.
- Coexisting advanced CKD and cirrhosis require a joint plan by the nephrologist, hepatologist, and dietitian; in obesity, ascites, or edema, standard, ideal, or adjusted body weight should be used, and adequate energy intake must be ensured.

13.21 “Gutmaxxing”: when the pursuit of a healthy gut turns into a rigid protocol, and what it means in the nephrology outpatient clinic

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. Gutmaxxing refers to an online trend of maximally optimizing digestion through numerical targets for protein and fiber, combinations of supplements, cleansing cycles, and elimination of food groups. An educational article without original data points out that patients with CKD adopt targets that do not apply to them: the fitness target of 1.6 g protein/kg/day exceeds the ceiling of 1.3 g/kg/day set by KDIGO 2024, while 0.8 g/kg/day is recommended in CKD G3 to G5. A rigid protocol may also conceal an eating disorder. The effect of the trend on CKD progression has not been studied.

Summary

- In an 80-kg patient, the difference between the CKD recommendation and the fitness target amounts to approximately 64 g versus 128 g of protein daily; on dialysis the risk is the opposite, namely undernutrition and loss of muscle mass.
- Supplements may contain hidden potassium, phosphate additives are absorbed better than natural phosphorus, fiber products require fluids, cleansing regimens cause fluid losses with prerenal injury, and some herbal products are nephrotoxic.
- High protein intake, creatine, and greater muscle mass raise creatinine without a decline in filtration; the author therefore recommends adding cystatin C and albuminuria and repeating the test after supplements are stopped.
- Warning signs of an eating disorder include anxiety or guilt after deviations, avoidance of shared meals, an expanding list of forbidden foods, checking behavior, and unintentional weight loss; orthorexia is not a separate diagnosis in DSM-5, and when it is suspected, the patient should be referred to a psychiatrist or clinical psychologist.
- In the outpatient clinic, it is recommended to ask specifically about dietary rules, list all supplements, calculate protein intake per kilogram, test potassium, phosphorus, bicarbonate, and cystatin C as appropriate, and offer a numerical target appropriate to the CKD stage instead of a blanket prohibition.

13.22 “Fibermaxxing”: higher fiber intake brings benefit; the question is where, and for whom, the curve levels off

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. The fibermaxxing trend rests on the premise that the more fiber, the better. A series of meta-analyses in the Lancet (2019; 185 prospective studies and 58 clinical trials) showed a 15% to 30% lower risk of mortality, cardiovascular disease, type 2 diabetes, and colorectal cancer with the highest compared with the lowest intake, most pronounced at 25 to 29 g daily. This range is not a ceiling, but the evidence excluded

persons with chronic disease. In CKD, fiber from food is recommended, with the source chosen according to serum potassium, serum phosphate, and fluid status.

Summary

- The quality of evidence was moderate by GRADE, the trials showed lower body weight, systolic blood pressure, and cholesterol, and a meta-analysis in the BMJ (2013) associated each additional 7 g daily with an HR of 0.91 for cardiovascular disease (95% CI, 0.88 to 0.94).
- The dose-response curves suggested possible further benefit above 25 to 29 g daily, so the argument against extreme doses rests not on their futility but on tolerability, lack of evidence, and the fact that the review excluded persons with chronic disease, so transferring the target to CKD is an extrapolation.
- In CKD, fiber is of interest because it shifts fermentation from proteolytic to saccharolytic, with less generation of indoxyl sulfate and p-cresyl sulfate precursors, but this is a biologically plausible mechanism without studies with hard kidney outcomes.
- Legumes, nuts, and dried fruit are sources of potassium that should be adjusted to serum potassium rather than prohibited across the board; phytate-bound phosphorus is absorbed less than additives, and bulk-forming fiber can worsen constipation in patients on dialysis with fluid restriction.
- There is no convincing evidence of greater benefit from very high doses of supplements, so the author recommends food sources and a gradual increase in intake.

13.23 Coffee and the liver in the UK Biobank: lower risk of cirrhosis and hepatocellular carcinoma, and what applies to the patient with kidney disease

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. The association of coffee with liver health is usually described from a single type of data. A prospective observational analysis of 354,957 UK Biobank participants with a median follow-up of 13 years combined clinical, MRI, and proteomic data: intake of at least 5 cups daily was associated with a lower risk of cirrhosis (HR 0.68), hepatocellular carcinoma (HR 0.53), and liver-related death (HR 0.58), similarly for caffeinated and decaffeinated coffee. Causality has not been proven. In CKD, coffee is generally not a problem; additives and volume are what matter.

Summary

- The confidence intervals were 0.58 to 0.79 for cirrhosis, 0.34 to 0.83 for hepatocellular carcinoma, and 0.45 to 0.74 for liver-related death; higher coffee intake was also associated with lower liver fat and iron content and less fibroinflammation on MRI (28,961 persons).
- Proteomics (Olink, 44,633 persons) showed higher levels of hepatocellular synthesis proteins and complement and lower markers of fibrogenesis; the similar effect of decaffeinated coffee suggests a role for polyphenols, chlorogenic acid, and diterpenes.
- The protective associations persisted even when coffee was sweetened with sugar or sweeteners, but additives were associated with a slightly longer iron-corrected T1 time, that is, a less favorable fibroinflammatory marker.
- Limitations include questionnaire-based exposure, residual confounding, reverse causality, and a healthier population of predominantly European ancestry; the authors' recommendation to drink unsweetened coffee for prevention is worded more strongly than the observational design can support.
- Kidney outcomes were not assessed; the relevance is indirect, through MASLD, which shares risk factors with CKD.
- In CKD, the author recommends not prohibiting coffee across the board, taking the number of cups into account in hyperkalemia, monitoring phosphorus and potassium in milk and creamers, counting the fluid volume in patients on dialysis, allowing an interval before blood pressure measurement, and preferring unsweetened coffee.

13.24 Vegetarian diet and lower risk of incident chronic kidney disease in the UK Biobank: what applies to practice

Published August 28, 2026 · [online version \(Slovak\)](#)

Abstract. The effect of plant-based diets on the kidneys has so far been assessed mainly with post hoc diet quality scores. A prospective observational analysis of the UK Biobank cohort used habitual dietary groups in 416,584 participants without kidney disease followed for a mean of 12.9 years. A vegetarian diet was associated with an approximately one-fifth lower risk of incident chronic kidney disease (HR 0.81), with a graded relationship according to decreasing intake of animal protein. This is an association, not proof of protection; in practice, it supports a shift toward plant sources in primary prevention.

Summary

- Of 23,084 incident cases over 12.9 years, only the low meat consumption group (HR 0.97; 95% CI, 0.943 to 0.996) and vegetarians, including vegans (HR 0.81; 95% CI, 0.711 to 0.927), had a statistically significantly lower risk; in poultry eaters the risk was lower by about 2% and in pescatarians by about 9%, without significance.
- The association in vegetarians persisted after adjustment for baseline eGFR and UACR (HR 0.80) and for polygenic risk (HR 0.79), with which pescatarians also reached significance (HR 0.89); no heterogeneity by sex, kidney function, genetic risk, or ethnicity was demonstrated.
- These are observational data: vegetarians were younger, leaner, better educated, and smoked less, so the result may partly reflect a healthier lifestyle; the cohort is healthier than the general population and of predominantly European ancestry.
- Further limitations include a single dietary assessment by questionnaire, hospital-based case ascertainment by ICD-10 codes, a single measurement of eGFR and UACR, and the pooling of vegans with vegetarians.
- The proposed mechanisms, namely a lower acid load, the gut-kidney axis with lower TMAO, and poorer absorption of phytate phosphate, were not tested by the authors; they are hypotheses.
- In the descriptive data, vegetarians had lower potassium and phosphorus intake than the high meat consumption group, so the label “vegetarian diet” does not automatically mean a potassium-rich diet.
- The finding applies to primary prevention; in existing kidney disease, the diet must be individualized according to stage, serum potassium, serum phosphate, and diabetes control, because evidence from RCTs is still lacking.

13.25 How many steps a day are really enough? What the latest dose-response analysis shows and how to apply it in nephrology

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. The target of 10,000 steps a day lacks a clear dose-response evidence base. A systematic review and meta-analysis of prospective studies with device-measured steps (Ding et al., The Lancet Public Health; 57 studies, 35 cohorts) showed that 7,000 steps compared with 2,000 are associated with a 47% lower risk of all-cause death (HR 0.53), with only modest additional benefit beyond that. These are observational associations. For nephrology, this translates into a realistic, gradually increased, and measurable physical activity target.

Summary

- Compared with 2,000 steps a day, 7,000 steps were associated with a lower risk of cardiovascular disease (HR 0.75), cardiovascular mortality (HR 0.53), cancer mortality (HR 0.63), type 2 diabetes (HR 0.86), dementia (HR 0.62), and depressive symptoms (HR 0.78); the association with falls (HR 0.72) has very low certainty by GRADE.
- For several outcomes, the relationship is nonlinear, with an inflection in the range of 5,000 to 7,000 steps; 10,000 compared with 7,000 steps yielded only about a 10% lower risk of death, whereas even 4,000 compared with 2,000 steps yielded an approximately 36% lower risk.

- Limitations include the observational design with possible residual confounding and reverse causality, few studies for most outcomes, data predominantly from high-income countries, and no separate analysis of CKD, so extrapolation to patients on dialysis is only by analogy.
- According to a meta-analysis by Zhang et al., patients with CKD take an average of about 4,640 steps a day: 5,640 before dialysis, 4,260 on PD, 4,110 on HD (3,410 on dialysis days), and 8,690 after transplantation; these are descriptive means.
- KDIGO 2024 recommends at least 150 minutes of moderate-intensity activity per week in CKD, or a level corresponding to tolerance (1D), and avoidance of prolonged sitting; it does not specify a step count.
- The author recommends measuring the baseline over several days, adding 500 to 1,000 steps a day at a time, combining walking with resistance and balance exercise, and stopping exertion in case of chest pain, syncope, or dyspnea.

13.26 Sweeteners in MASLD, CKD, and incretin therapy: evidence and uncertainties

Published September 5, 2026 · [online version \(Slovak\)](#)

Abstract. Agave syrup, high-intensity non-sugar sweeteners, and polyols are often imprecisely lumped into one group, although they differ fundamentally in their metabolism. A review of randomized trials, meta-analyses, observational cohorts, and regulatory assessments shows that the hepatic risk of fructose relates mainly to energy excess and that replacing sweetened beverages with noncaloric ones brings a small weight loss (−1.06 kg). Associations with CKD are observational, and direct evidence that sweeteners reduce the efficacy of semaglutide or tirzepatide is lacking. In practice, the priority is to limit free sugars and prefer water.

Summary

- Agave syrup is a caloric source of free sugars; increases in liver fat and ALT were observed mainly when high doses of fructose were added on top of energy requirements, not with isocaloric replacement of other carbohydrates.
- According to a network meta-analysis, replacing sugar-sweetened beverages with low-energy ones (12 studies, 601 participants) led to a weight difference of −1.06 kg; no advantage was shown when they replaced water, and WHO 2023 conditionally recommends against non-sugar sweeteners for long-term weight control.
- In February 2026, EFSA confirmed the safety of sucralose with an ADI of 15 mg/kg/day; however, regulatory safety is not evidence of metabolic benefit.
- Kidney data are observational: in the UK Biobank, the HR was 1.26 (95% CI, 1.12 to 1.43) for more than one serving of artificially sweetened beverage per day, and in a meta-analysis the RR was 1.40 with an interval of 0.65 to 3.02; neither causality nor a mechanism has been demonstrated.
- For erythritol, EFSA set an ADI of 0.5 g/kg/day; a small study in 20 volunteers showed an acute increase in platelet reactivity after 30 g, but clinical causality remains unproven, and a validated threshold for eGFR below 30 mL/min/1.73 m² is lacking.
- During incretin therapy, the author recommends addressing tolerability, hydration, and protein intake (in CKD G3 to G5, approximately 0.8 g/kg/day and not above 1.3 g/kg/day); dehydration due to nausea, vomiting, and diarrhea is a better-documented kidney risk than hypothetical antagonism by sweeteners.

13.27 Chronic kidney disease of unclear etiology in heat-exposed workers: is CKDnt arriving in Texas?

Published September 6, 2026 · [online version \(Slovak\)](#)

Abstract. A Texas Monthly report draws attention to kidney damage in immigrant workers in the Texas heat. The review assesses the evidence on chronic kidney disease of nontraditional etiology (CKDnt), a diagnosis of exclusion with a tubulointerstitial phenotype. A California study of 283 agricultural workers found AKI during a single shift in 12.3%, and piece-rate pay was a stronger predictor than heat stress (OR 4.24). Popu-

lation-level evidence of a CKDnt epidemic in Texas is still lacking. Prevention requires labor protection, not just advice to drink more.

Summary

- CKDnt mainly affects younger men performing heavy work in the heat and typically presents with declining eGFR, minimal proteinuria, a bland sediment, hyperuricemia, hypokalemia, and, histologically, tubulointerstitial fibrosis.
- The main suspected mechanism is recurrent heat stress with hypovolemia and episodes of AKI with incomplete recovery; pesticides, heavy metals, NSAIDs, the fructose pathway, and genetics are also being investigated, so the disease is probably multifactorial.
- In the California study, the risk of AKI was associated with heat stress (OR 1.34) and piece-rate pay (OR 4.24); the OR of 102.81 in women has an interval of 7.32 to 1,443.20, so the effect size is practically undetermined.
- A single rise in creatinine after a shift may reflect hemoconcentration or muscular exertion; CKD requires the abnormality to persist for more than three months.
- Screening should include an occupational history, blood pressure, eGFR, UACR, urinary sediment, electrolytes, bicarbonate, and uric acid, repeated after recovery.
- There is no specific treatment, and the benefit of SGLT2 inhibitors in the nonproteinuric phenotype is hypothetical; the cornerstones are ending the exposure, preventing AKI, and avoiding nephrotoxins.
- A federal OSHA heat standard is not yet in force in the United States; for Slovak practice, the author recommends a targeted occupational history in patients working in hot workplaces and follow-up of patients after an episode of AKI.

13.28 Ketogenic diet and metabolic health: promising results for the liver, not proof of universal superiority

Published September 11, 2026 · [online version \(Slovak\)](#)

Abstract. It is unclear whether the metabolic benefit of weight loss also depends on diet composition. An RCT in Cell Metabolism with provided meals compared a ketogenic, a Mediterranean, and a very low-fat diet at an identical weight loss of about 10% in 42 analyzed adults with obesity, prediabetes, and hepatic steatosis. Muscle insulin sensitivity increased equally in all groups ($p = 0.617$), hepatic sensitivity increased two- to threefold more with the ketogenic diet, and liver fat decreased by 67% versus 45%. The results are mechanistic and short-term and do not apply to patients with CKD.

Summary

- Of the two primary clamp-measured endpoints, one was negative: skeletal muscle insulin sensitivity increased by about 50% with no difference between diets, so weight loss was the determining factor; the hepatic insulin sensitivity index increased most with the ketogenic diet ($p < 0.001$).
- The other findings were exploratory: liver fat decreased by a relative 67% versus 45%, de novo lipogenesis decreased most with the ketogenic diet, and prediabetes remission was achieved by 50%, 29%, and 7% of participants, which with 14 persons per group represents a difference of three persons.
- Mechanistically, the ketogenic diet reduced the insulin area under the curve by 74% and increased glucagon by 52%; beta-cell insulin secretion did not improve in any group.
- LDL cholesterol and apolipoprotein B did not differ between diets despite 23% of energy from saturated fat, nor did 24-hour triacylglycerols, owing to higher postprandial values; the sample cannot capture individual LDL increases.
- eGFR in the ketogenic group increased only because of a decrease in cystatin C, not creatinine, which, given the higher protein intake (23% of energy) and reduced inflammation, probably reflects a change in the marker rather than a kidney benefit.

- The study excluded persons with severe kidney disease or treated diabetes; in CKD, the author therefore recommends assessing actual protein intake, volume status, and electrolytes, and warns of the risk of euglycemic ketoacidosis with SGLT2 inhibitors and of hypoglycemia with insulin or sulfonylureas.

13.29 Xylitol and cardiovascular risk: what the new cohort data really show and what applies in nephrology

Published September 17, 2026 · [online version \(Slovak\)](#)

Abstract. Xylitol is a polyol used as a sucrose substitute. A conference abstract from ESC Congress 2026 (CLSA and EPIC-Norfolk cohorts, 17,710 persons in total) associated higher circulating xylitol levels with a higher incidence of MACE (adjusted HR 1.47 and 1.18), building on a 2024 study in the European Heart Journal with an HR of 1.57 in high-risk patients. These are observational, not yet peer-reviewed data with mechanistic support via platelet reactivity. For nephrology, this calls for a cautious position; direct kidney toxicity concerns only parenteral administration.

Summary

- The HR of 1.57 (95% CI, 1.12 to 2.21) comes from the 2024 study (1,157 and 2,149 patients, 3 years, tertiles); for CLSA, sources report both an HR of 1.47 and a 57% increase, so the figure is preliminary and is appropriately expressed as a range of 45% to 60%.
- MACE included all-cause mortality (2,950 events in CLSA, 5,670 in EPIC-Norfolk), so part of the signal may come from noncardiovascular deaths.
- Mechanistically, xylitol enhanced platelet reactivity and increased functional platelet measures in 10 volunteers after a drink; a similar signal has been described for erythritol (HR 1.80 and 2.21), but these are surrogate endpoints.
- Blood levels were measured, not dietary intake; levels may also reflect endogenous production via the polyol pathway in hyperglycemia, so residual confounding and reverse causality remain open questions.
- The 2023 WHO recommendation does not apply to polyols, including xylitol.
- Direct nephrotoxicity, that is, oxalate nephropathy with AKI requiring hemodialysis (for example, after a cumulative dose of 1,560 g), is linked to parenteral administration of hundreds of grams, not to chewing gum or toothpaste.
- In CKD, the author recommends not promoting xylitol as safe, not requiring blanket discontinuation, asking about the daily dose, and preferring a reduction in sweet taste; higher doses carry a risk of osmotic diarrhea with volume depletion.

13.30 Protein in CKD: the question is not only “how much” but also “for whom”

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. Protein intake in chronic kidney disease cannot be reduced to a single number, because the appropriate target depends on the disease phase and nutritional risk. The review compares the KDIGO 2024 and KDOQI 2020 guidelines and the evidence from RCTs and from an umbrella review of 25 meta-analyses, which showed moderate certainty of benefit of a low-protein diet for serum phosphate and the risk of kidney failure, but low certainty for change in GFR. In stable CKD not on dialysis, supervised restriction may help; on dialysis and in undernutrition or catabolism, it is inappropriate.

Summary

- KDIGO 2024 recommends approximately 0.8 g/kg/day in CKD G3 to G5 not on dialysis and avoidance of intake above 1.3 g/kg/day in patients at risk of progression; a very low-protein diet with essential amino acids or ketoanalogues (about 0.6 g/kg/day in total) is reserved for selected, motivated, and stable patients with CKD G4 to G5.
- KDOQI 2020 sets stricter targets of 0.55 to 0.60 g/kg ideal body weight/day without diabetes and 0.6 to 0.8 g/kg/day with diabetes, and 1.0 to 1.2 g/kg/day on maintenance dialysis; it must always be specified which body weight is the denominator.

- In the observational EQUAL study (age at least 65 years, eGFR below 20 mL/min/1.73 m²), a prescribed low-protein diet was not associated with higher mortality; however, this does not confirm safety for every individual.
- The pragmatic ERIKA RCT (223 patients with CKD G4 to G5) did not demonstrate a benefit of a very low-protein diet with ketoanalogs for patient or kidney survival and showed low adherence; a meta-analysis of 16 studies (1,344 participants) suggested a more favorable GFR trajectory, with a median follow-up of only 13 months.
- A plant-dominant diet (PLADO, 0.6 to 0.8 g/kg/day) is a useful planning model; potassium and phosphorus should be assessed individually according to serum potassium, medications, and processed foods.
- The author recommends monitoring actual energy intake, weight trend, handgrip strength, bicarbonate, potassium, and phosphorus, and relaxing rather than tightening the plan when the patient loses weight or becomes weaker.

CHAPTER 14

Internal medicine and related specialties

Hepatology, rheumatology, psychiatry, dermatology, and other specialties a nephrologist meets in everyday internal medicine practice. Number of articles: 19.

Key messages of the chapter

The strongest link in this chapter is metabolic dysfunction-associated steatotic liver disease (MASLD), which shares risk factors with chronic kidney disease (CKD) and, according to the observational NHANES cohort, has the worst prognosis when CKD and advanced fibrosis coexist (14.18). The concept of a cardiovascular-renal-hepatic-metabolic syndrome is so far only a conceptual proposal, not an accepted classification (14.18). There is agreement on stepwise fibrosis triage: FIB-4, then elastography or the ELF test for values of 1.3 to 2.67, and referral to a hepatologist above 2.67 (14.9, 14.18). For the nephrologist, the limitations of these tools are key, because in hemodialysis amino-transferases tend to be lower, FIB-4 thresholds are not validated, and volume overload distorts liver stiffness measurement; in a pilot study of 41 patients on hemodialysis, approximately 10% exceeded the 8 kPa threshold after dialysis (14.9, 14.18). An educational program on hepatology also emphasizes the link between liver disease, kidney dysfunction, and malnutrition (14.2).

The evidence for MASH therapy varies in strength. The phase 3 ESSENCE RCT showed resolution of steatohepatitis with semaglutide 2.4 mg in 62.9% versus 34.3%, tirzepatide has histologic evidence only from phase 2, and retatrutide only an imaging-based decrease in liver fat (14.17). Resmetirom and semaglutide (Kayshild) have conditional authorization in the EU for noncirrhotic MASH with fibrosis F2 to F3 based on surrogate histologic endpoints; they have not demonstrated a reduction in decompensation or mortality, and Kayshild is not recommended at an eGFR below 30 mL/min/1.73 m² (14.9, 14.17). The kidney benefit of GLP-1 receptor agonists is documented separately by the FLOW RCT with semaglutide 1 mg (HR 0.76), which cannot be extrapolated to other incretins (14.17). Pioglitazone has a histologic signal, but PIVENS did not meet its prespecified significance threshold, and in CKD the risk of fluid retention and fractures predominates (14.19). SGLT2 inhibitors, RAS blockade, and finerenone are not treatments for MASH; they are used according to cardiorenal indications (14.18, 14.9).

The second theme is psychiatric diagnoses in patients with somatic illness. Psychosis is a syndrome, not an etiologic diagnosis, and in patients on dialysis, uremic encephalopathy, disorders of sodium, calcium, and magnesium, hypoglycemia, sepsis, and drug toxicity must first be excluded (14.8). When psychosis and substance use coexist, hyperthermia, rhabdomyolysis, and acute kidney injury are a threat, and dehydration or ACE inhibitors and ARBs increase lithium toxicity; according to a Cochrane review, the evidence on psychosocial interventions is weak (14.10). In borderline personality disorder, a population-based cohort showed higher mortality, but an independent kidney risk factor has not been proven, and the main danger is diagnostic overshadowing (14.13). Attention deficits in CKD may be explained by uremia, anemia, or sleep apnea rather than ADHD, and methylphenidate raises BP and pulse rate (14.14).

The chapter also shows how new drugs from neighboring specialties run into kidney-related limitations. Phase 3 RCTs of upadacitinib in severe alopecia areata met their primary endpoint, but the drug is not recommended at an eGFR below 15 mL/min/1.73 m², and figures in the media did not match the primary summary (14.15). The SPARTACUS RCT presented at EULAR 2026 supported early golimumab in peripheral spondyloarthritis, although long-term outcomes are lacking (14.5), and in the axial form, the kidney risk of NSAIDs is an argument for earlier targeted therapy (14.12). Topical

finasteride is supported by a retrospective cohort without a control group, and data in kidney impairment are lacking (14.11). In uncomplicated cystitis, a randomized trial favored 5-day nitrofurantoin over single-dose fosfomycin (14.4), and for recurrences in older women, expert opinion emphasizes accurate confirmation of infection and local vaginal estrogen (14.6).

The common denominator is diagnostic restraint. Inflammatory markers are sensitive but poorly specific (14.7), irritable bowel syndrome is a positive diagnosis according to the Rome criteria, although medical students overused tests anyway (14.16), and the epidemiologic signal linking tattoos with lymphoma and a preclinical erythrocyte glycocalyx test do not yet change practice (14.1, 14.3).

This yields a framework for the outpatient clinic: in patients with CKD and metabolic risk, triage fibrosis in a targeted way, taking volume status into account; choose targeted MASH therapy according to the strength of evidence and the product's kidney-related limitations; exclude metabolic and drug-induced causes before a psychiatric diagnosis; with NSAIDs, lithium, and new immunomodulatory drugs, monitor eGFR, potassium, and interactions; and prescribe antibiotics only after accurate confirmation of infection (14.18, 14.17, 14.8, 14.10, 14.12, 14.15, 14.6).

14.1 Tattoos and the risk of lymphoma or skin cancer: a signal that deserves attention

Published May 17, 2026 · [online version \(Slovak\)](#)

Abstract. Tattoo pigment is transported to lymph nodes and persists there for years. A review of population studies reported by Medscape describes an epidemiologic signal linking tattoos with lymphomas and skin cancers. A Swedish case-control study found a 21% higher, statistically nonsignificant risk of lymphoma, and a Danish twin study found a 62% higher risk of skin cancer, with greater risk for large tattoos. These are observational associations, not proof of causality. In practice, this calls for informed decision-making and monitoring of the skin and lymph nodes.

Summary

- A 2024 Swedish study (1,398 patients with lymphoma, 4,193 controls) found tattoos in 21% versus 18% and, after adjustment, a 21% higher relative risk of lymphoma without statistical significance; the signal was stronger for diffuse large B-cell lymphoma and follicular lymphoma.
- A 2025 Danish twin study (2,367 twins, 316 pairs) showed a 62% higher risk of skin cancer; with larger tattoos, the risk of lymphoma was up to 2.7-fold and the risk of skin cancer more than twofold, and the association persisted within pairs.
- The proposed mechanisms, namely chronic immune activation, carcinogenic ink components (cadmium, lead, azo dyes, polycyclic aromatic hydrocarbons), and nanoparticles, have not been proven in humans; the results may be confounded by smoking, sun exposure, and lifestyle.
- The pigment is deposited in lymph nodes for years to decades, which makes the hypothesis of chronic immune stimulation biologically plausible, although its relevance to cancer development in humans is unknown.
- The author recommends a professional studio and high-quality inks, not tattooing over moles, and seeing a physician in case of persistent lymphadenopathy or changing skin lesions; caution is advisable after lymphoma and in immunosuppression.

14.2 News in hepatology: what the STREAMED UP livestream will offer

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. The STREAMED UP platform offers the hepatology educational livestream *Neues in der Hepatologie*, which, according to the program, links current knowledge with practical decisions at the bedside. This is information about an educational event, not original data. The program covers the cholestatic diseases PBC and PSC, metabolic steatotic liver disease with an inflammatory component (SLD/MASH), and complications of cirrhosis, including hepatic encephalopathy. For nephrology and internal medicine practice,

the topic is relevant because of the overlap of liver disease with kidney dysfunction, undernutrition, and frailty.

Summary

- The program covers PBC and PSC, with emphasis on early diagnosis and long-term follow-up given their often subtle course.
- The author describes SLD/MASH as a potentially progressive disease linked to obesity, type 2 diabetes, and cardiometabolic risk, not as an incidental ultrasound finding.
- In complications of cirrhosis, rapid recognition of deterioration and of the triggers of hepatic encephalopathy is key.

14.3 A new simple blood test could detect vascular damage, and thus heart and kidney risk, earlier

Published May 24, 2026 · [online version \(Slovak\)](#)

Abstract. Direct assessment of the endothelial glycocalyx requires invasive or demanding imaging methods. A review in EMJ Reviews describes the concept of a liquid biopsy according to which the red blood cell glycocalyx reflects the state of the vascular wall glycocalyx because of the continuous exchange of its components. In animal experiments, erythrocyte measurements correlated with glycocalyx damage in the heart and kidneys and with vascular barrier dysfunction. These are preclinical results; human studies and standardization are needed before clinical use in cardiorenal risk assessment.

Summary

- The endothelial glycocalyx forms a permeability barrier and regulates inflammation; its damage is associated with cardiovascular disease, kidney injury, diabetes, and inflammatory diseases.
- The continuous exchange of components between the endothelium and erythrocytes creates a biochemical imprint on the red cells that could allow vascular damage to be detected from a routine blood sample.
- The evidence comes mainly from animal models; human data, reference ranges, and the influence of other diseases and treatments are not yet known.
- If the test is validated, it could help with early cardiorenal risk stratification and treatment monitoring; for now, the author considers it only a complementary marker.

14.4 Nitrofurantoin vs. fosfomycin: what a new Spanish study shows in uncomplicated cystitis in women

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. Uncomplicated cystitis is among the most common reasons for antibiotic prescriptions in women. A Spanish randomized trial in primary care compared single-dose fosfomycin, two doses of fosfomycin, five-day nitrofurantoin, and three-day pivmecillinam in 768 women. Clinical resolution of symptoms by day 7 was achieved by 74% of women with nitrofurantoin versus 59% with single-dose fosfomycin. All regimens were safe, and mild gastrointestinal effects predominated. The result supports preferring a multiday course over single-dose fosfomycin.

Summary

- Women aged 34 to 63 years with typical symptoms and a positive nitrite or leukocyte esterase test received fosfomycin 1 × 3 g or 2 × 3 g, nitrofurantoin 100 mg three times daily for 5 days, or pivmecillinam 400 mg three times daily for 3 days; the culture was positive in 57%.
- On day 7, 74% of women were symptom-free with nitrofurantoin, 70% with pivmecillinam, 67% with two doses, and 59% with single-dose fosfomycin; similar trends were observed on days 14 and 28.
- Adverse events occurred in 19.9% to 26.8% of women; the only serious treatment-related event was pyelonephritis with pivmecillinam, and the sample may have been too small to detect smaller differences.

14.5 Early TNF inhibitor treatment in peripheral spondyloarthritis led to higher remission rates than step-up csDMARD therapy

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. The treatment of peripheral spondyloarthritis has a weaker evidence base and conventionally proceeds from NSAIDs through csDMARDs to biologic therapy. The phase 3 SPARTACUS RCT, presented at EULAR 2026, compared immediate golimumab with a step-up strategy with methotrexate in 97 patients with symptoms for less than one year. Clinical remission at 24 weeks was achieved by 60% versus 33% of patients (OR 3.1; $P = 0.006$), with no new safety signal. The result supports early intensive treatment, but long-term results are not yet available.

Summary

- Patients received golimumab 50 mg subcutaneously every 4 weeks or methotrexate 15 mg weekly, increased to 20 mg; in case of inadequate response after 12 weeks, sulfasalazine 2 g daily could be added.
- Remission was strictly defined as the complete absence of arthritis, dactylitis, and enthesitis; sustained remission was more frequent with golimumab (42% vs. 18%; $P = 0.006$).
- Differences in the resolution of arthritis, enthesitis, and dactylitis were apparent from week 8 and were confirmed by DAPSA and ASDAS-CRP scores; CRP decreased in both groups from approximately 25 to 5 mg/L.
- Rescue sulfasalazine was needed by 8.3% of patients with golimumab and 28.6% with methotrexate ($P = 0.02$); according to the investigator, this is the first evidence of an effect of methotrexate in this diagnosis.
- Over 24 weeks, there was a single serious adverse event, a hospitalization for influenza-like symptoms with golimumab; however, 97 patients and short follow-up are insufficient to assess rare and long-term risks.
- The trial does not answer questions about long-term drug-free remission, radiographic progression, or cost, and TNF inhibitors are not approved for this diagnosis by either the FDA or the EMA.

14.6 Recurrent urinary tract infections in older women and the link to genitourinary syndrome of menopause (GSM): diagnosis, antibiotics, and prevention without them

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. Recurrent urinary tract infections are common in older women, and some symptoms merely mimic bacterial cystitis, which leads to unnecessary antibiotics. A Medscape podcast interview with urologist Dr. Melissa R. Kaufman presents expert opinion on diagnosis and prevention. The keys are a targeted history, pelvic examination, and accurate urine interpretation, with a negative result having a high negative predictive value. The main pillar of prevention is local low-dose vaginal estrogen for genitourinary syndrome of menopause (GSM).

Summary

- The history should include previous urologic and gynecologic procedures, baseline symptoms, bowel symptoms, and the relationship to triggers; pelvic examination may reveal vaginal atrophy, urethral diverticulum, or prolapse.
- A negative urine test without nitrites and leukocytes helps rule out infection; contamination is common in midstream samples, so when the result is unclear, the expert recommends catheterized collection and assessment of pyuria above 5 leukocytes/HPF.
- A positive result does not necessarily mean a clinically relevant infection, and interstitial cystitis, bladder pain syndrome, and overactive bladder must be distinguished; prolonging treatment or broadening the spectrum without evidence increases resistance.
- Antibiotic-free prevention includes a water intake of about 1.5 liters daily, D-mannose and cranberry proanthocyanidins with a theoretical mechanism, and methenamine hippurate, which is converted in acidic urine into a bacteriostatic compound and is often combined with vitamin C.

- Most important is local vaginal estrogen, which in GSM restores protective lactobacilli and the mucosal microenvironment.
- For nephrology, the author emphasizes that older patients with CKD are more vulnerable to the adverse effects of antibiotics and to polypharmacy, so infection must be accurately confirmed and GSM actively treated in recurrent cases.

14.7 How to interpret inflammatory markers in primary care: CRP, ESR, and plasma viscosity (PV) in practice

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. In the outpatient setting, inflammatory markers are often interpreted in isolation or as a diagnosis in themselves. The practical Medscape Primary Care Hack review summarizes the properties of CRP, the erythrocyte sedimentation rate (ESR), and plasma viscosity (PV). CRP rises within 6 to 10 hours, and a value above 10 mg/L almost always indicates acute disease; an ESR above 100 mm/h has a 97% predictive value for underlying disease. All three markers are sensitive but poorly specific, so they must always be interpreted together with the clinical picture.

Summary

- The reference value for CRP is below 5 mg/L; with a half-life of about 19 hours, it normalizes within days after the trigger is removed and is therefore also suitable for monitoring treatment response.
- Mild CRP elevation also occurs in obesity, pregnancy, depression, diabetes, and cardiovascular disease; when ferritin is elevated at the same time, inflammation is more likely than iron overload.
- ESR rises within 24 to 48 hours and normalizes over weeks; the approximate upper limit is $(\text{age} + 10) \div 2$ in women and $\text{age} \div 2$ in men, and it is influenced by hematocrit, pregnancy, medications, and delayed analysis.
- A high ESR is associated mainly with malignancy, for example multiple myeloma and lymphoma, and with giant cell arteritis.
- PV is normal in the range of 1.50 to 1.72 mPa·s, borderline at 1.72 to 1.80 mPa·s with repeat testing recommended, and elevated above 1.80 mPa·s; it is more reliable than ESR but technically more demanding and less available. The common conclusion is that none of the markers should be interpreted in isolation from symptoms, examination, and other tests.

14.8 Primary or substance-induced psychosis? Differential diagnosis and principles of acute management

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. Psychosis is a syndrome, not an etiologic diagnosis, and whether it is substance-induced or primary cannot be determined by a single symptom, a toxicology test, or a fixed time limit. The clinical review bases the differential diagnosis on the timeline, collateral history, exclusion of delirium, and observation during abstinence. According to a meta-analysis of 50 studies with 40,783 persons, substance-induced psychosis transitioned to schizophrenia in 25%, and in 34% after cannabis. In patients on dialysis, a psychotic presentation should first be considered a possible manifestation of a metabolic or toxic cause.

Summary

- Sudden onset after exposure and rapid resolution after it ends support a substance-related etiology, whereas gradual development of negative symptoms suggests a primary disorder; the one-month limit is only a guide for reassessing the diagnosis.
- Visual and tactile hallucinations raise suspicion of intoxication, withdrawal, or delirium but are not diagnostic; a negative toxicology screen does not rule out substance-induced psychosis, and a positive one does not prove the cause.

- Transition to schizophrenia occurred in 34% after cannabis, 26% after hallucinogens, 22% after amphetamines, 12% after opioids, 10% after alcohol, and 9% after sedatives; other brief psychoses transitioned in 36%; these are heterogeneous observational data.
- The first step is to rule out delirium with inattention and a fluctuating course; in alcohol or benzodiazepine withdrawal syndrome, benzodiazepines are the cornerstone, and antipsychotics do not replace them.
- The choice of antipsychotic is guided by the QT interval, extrapyramidal and metabolic risk, and kidney and liver function; there is no uniform renal dose for the entire class, and discontinuation after presumed substance-induced psychosis should be individualized.
- In CKD and in patients on dialysis, uremic encephalopathy, sodium, calcium, and magnesium disorders, hypoglycemia, sepsis, drug toxicity, dialysis disequilibrium syndrome, and thiamine deficiency must be ruled out before a psychiatric diagnosis is made.

14.9 Metabolic dysfunction-associated steatotic liver disease: diagnosis, fibrosis assessment, and relevance for nephrology practice

Published August 13, 2026 · [online version \(Slovak\)](#)

Abstract. Metabolic dysfunction-associated steatotic liver disease (MASLD) is closely linked to diabetes, obesity, cardiovascular disease, and CKD, and the prognosis of liver complications is determined mainly by the stage of fibrosis. The review is based on the 2024 EASL, EASD, and EASO guidelines and describes a stepwise assessment: FIB-4, followed by elastography or the ELF test. Since 2025 and 2026, resmetirom and the semaglutide product Kayshild have held conditional marketing authorization in the EU for noncirrhotic MASH with F2 to F3 fibrosis, based on histologic surrogate endpoints. For the nephrologist, the limitations of the tests in CKD are important.

Summary

- Targeted assessment applies mainly to patients with type 2 diabetes, abdominal obesity with an additional risk factor, persistently elevated liver enzymes, or incidental steatosis; neither normal aminotransferases nor a negative ultrasound rules out advanced fibrosis.
- A FIB-4 below 1.3 indicates low risk (threshold 2.0 above age 65), a value of 1.3 to 2.67 requires elastography or ELF, and a value above 2.67 requires hepatology assessment; a low FIB-4 may miss about one tenth of cases of advanced fibrosis.
- Liver stiffness below 8 kPa supports low risk; measurements are increased by inflammation, cholestasis, venous congestion, and food.
- The recommended weight loss is at least 5% to reduce liver fat, 7% to 10% to improve inflammation, and at least 10% to improve fibrosis, with preservation of muscle mass in CKD and sarcopenia.
- Resmetirom and Kayshild have not yet been shown to reduce decompensation, transplantation, or mortality; resmetirom requires no dose adjustment at an eGFR of 15 to 89 mL/min but interacts with statins and CYP2C8 inhibitors, and Kayshild is not recommended at an eGFR below 30 mL/min/1.73 m².
- Statins are not discontinued because of MASLD; SGLT2 inhibitors are not a targeted treatment for MASH and are used according to cardiorenal indications.
- On dialysis, FIB-4 thresholds are not validated, because AST and ALT tend to be lower and liver stiffness is affected by volume overload; the author recommends standardized elastography and hepatology assessment.

14.10 Psychosis and coexisting substance use: integrated assessment, treatment, and protection of physical health

Published August 13, 2026 · [online version \(Slovak\)](#)

Abstract. Coexisting psychosis and problematic substance use are associated with poorer adherence, relapses, overdose, and higher physical morbidity. The review is based on the NICE CG120 guideline (pub-

lished in 2011, last reviewed in May 2024) and a Cochrane review of 41 randomized trials that found no high-quality evidence of the general superiority of any psychosocial intervention. The key principles are concurrent treatment of both disorders with a single coordinator, reassessment of the diagnosis after the crisis, overdose prevention, and active protection of physical health, including the kidneys, serum sodium, and lithium toxicity.

Summary

- NICE CG120 applies to persons aged 14 years and older with psychosis and problematic substance use; both disorders should be recognized and treated concurrently, and the patient must not be excluded from care for one because of the other. Screening should be bidirectional and specific: substance, amount, route of administration, last dose, combinations, and temporal relationship to symptoms.
- Distinguishing substance-induced psychosis from a primary disorder is uncertain during intoxication or withdrawal, so the diagnosis is reassessed after stabilization. Toxicology testing is performed only for a specific clinical question, because a negative result does not rule out synthetic substances and a positive one does not prove the cause of the psychosis.
- The evidence is weak: a 2023 meta-analysis associates clozapine with higher abstinence and fewer hospitalizations, but these are predominantly low-quality observational data, so clozapine should not be started solely for a substance use disorder.
- Smoking induces CYP1A2 and lowers clozapine and olanzapine concentrations, so toxicity may occur after smoking cessation. According to NICE, planned detoxification should take place in an inpatient setting with an addiction team; abrupt withdrawal of alcohol or benzodiazepines can be life-threatening, and opioid tolerance decreases after abstinence, so a treatment plan and, if possible, naloxone should be in place before discharge.
- Stimulants and synthetic substances can cause hyperthermia, rhabdomyolysis, and acute kidney injury, and cocaine can also cause renal infarction; in agitation, seizures, or prolonged unconsciousness, urine output should be monitored and creatinine, potassium, sodium, and creatine kinase measured. Chronic hyponatremia must not be corrected rapidly in an uncontrolled manner because of the risk of osmotic demyelination syndrome, and dehydration, NSAIDs, ACE inhibitors, and ARBs increase the risk of lithium toxicity.

14.11 Topical finasteride in male androgenetic alopecia: efficacy at 52 weeks and the limits of the evidence

Published August 14, 2026 · [online version \(Slovak\)](#)

Abstract. Topical finasteride for male androgenetic alopecia is intended to achieve efficacy with lower systemic exposure, but long-term data on monotherapy were lacking. A retrospective 52-week cohort of 123 men from the University of Turin showed an increase in hair count of 29.4 hairs/cm² in the temporal area and 21.8 hairs/cm² at the vertex ($p < 0.001$), with local irritation in 7.3% of patients. The uncontrolled design does not prove equivalence with oral finasteride or interchangeability of formulations, because the licensed spray and 1 mL of a compounded 0.25% solution differ in applied dose approximately 5.5-fold.

Summary

- In the cohort of Gallo and coworkers (mean age 32.4 ± 8.1 years), 78 men used a commercial 2.275 mg/mL spray and 45 men used a 0.25% hydroalcoholic solution, 1 mL daily. No systemic adverse events were recorded in the medical records, but the sample lacks the power to rule out rare complications.
- The study lacked a control group, placebo, an oral comparator, and blinding, and the absence of a detected difference between formulations does not prove bioequivalence.
- The licensed product Finjue contains 114 micrograms of finasteride per 50-microliter spray, that is, at most approximately 0.46 mg daily, whereas 1 mL of the 0.25% solution applies about 2.5 mg. Systemic exposure may not be 5.5-fold higher, but the safety data for the spray cannot automatically be transferred to compounded solutions.

- In a phase III RCT (458 men, 24 weeks), hair count increased by 20.2 with topical finasteride and by 6.7 with placebo ($p < 0.001$); plasma concentrations were more than one hundred times lower than with oral 1 mg, but serum DHT decreased by 34.5% versus 55.6%. Sexual adverse events occurred in 2.8% of patients with topical finasteride, 3.3% with placebo, and 4.8% with oral finasteride.
- Pregnant women, women of childbearing potential, and minors must not come into contact with the solution; concomitant use with minoxidil on the same area has not been studied, and alopecia progresses again after discontinuation.
- The Finjuve SmPC requires no dose adjustment in renal impairment, but no studies have been conducted in such patients. In CKD, the author recommends documenting baseline sexual symptoms and mood and assessing the effect no earlier than after three to six months.

14.12 Axial spondyloarthritis: early diagnosis, imaging, and individualized treatment

Published August 15, 2026 · [online version \(Slovak\)](#)

Abstract. Axial spondyloarthritis has no single diagnostic test and is often recognized only after years of symptoms. The review summarizes the revised ASAS-SPARTAN 2025 classification criteria derived from the CLASSIC cohort of 1,015 patients (sensitivity 79.5%, specificity 90.4%), under which bone marrow edema on magnetic resonance imaging alone is no longer sufficient, and the ACR, SAA, and SPARTAN 2026 recommendations, which place TNF and IL-17 inhibitors on an equal footing as the first biologic choice. For nephrology practice, the kidney risk of NSAIDs and earlier consideration of targeted therapy in CKD are important.

Summary

- Suspicion arises with back pain beginning before age 45 and lasting at least three months. Neither HLA-B27 nor C-reactive protein confirms or rules out the diagnosis, and classification criteria remain a research tool, not a diagnostic algorithm.
- The initial imaging in adults is an anteroposterior pelvic radiograph and, if the finding is equivocal, magnetic resonance imaging of the sacroiliac joints without gadolinium; minor bone marrow edema also occurs in athletes, after childbirth, and with overuse. In children and adolescents, magnetic resonance imaging is the first examination.
- The cornerstones are supervised physiotherapy and smoking cessation, whereas manipulative therapy, opioids, systemic glucocorticoids, and conventional synthetic DMARDs are not recommended for axial disease. NSAIDs are the first pharmacologic choice, but the GESPIC cohort data on slowing progression are observational, with a small effect.
- TNF and IL-17 inhibitors are equivalent first biologic choices, JAK inhibitors are generally used later because of their safety profile, and IL-23 inhibitors are not recommended. In active inflammatory bowel disease, IL-17 inhibitors should generally be avoided.
- NSAIDs reduce renal prostaglandin synthesis and can lead to a decline in glomerular filtration rate, hyperkalemia, and acute kidney injury, particularly in combination with an ACE inhibitor or ARB and a diuretic. In at-risk patients, blood pressure, creatinine, eGFR, and potassium should be checked after treatment initiation, and NSAIDs should be temporarily withheld during dehydration.
- Kidney manifestations may include AA amyloidosis, IgA nephropathy, and drug-induced injury, but hematuria requires the standard nephrologic differential diagnosis. In CKD, when long-term NSAID use is not safe, the case for earlier targeted therapy is stronger.

14.13 Borderline personality disorder and physical health: higher morbidity cannot be explained by the psyche alone

Published August 15, 2026 · [online version \(Slovak\)](#)

Abstract. Borderline personality disorder (BPD) is associated with premature mortality and higher physical morbidity, but most of the evidence is observational. The largest population-based cohort, from Hong Kong

(2026), compared 3,092 people with BPD with 902,927 persons and found an HR of 5.65 for all-cause death and 2.02 for death from natural causes, with an estimated loss of 13.01 years of life. Neither causal biologic mechanisms nor an organ benefit of psychotherapy have been proven. In practice, the key is to prevent diagnostic overshadowing and to monitor metabolic, drug-related, and kidney risks.

Summary

- In the Hong Kong cohort, the HR for death from external causes was 30.35; external causes accounted for 40.4% and natural causes for 37.3% of deaths, and among somatic diagnoses, epilepsy carried the highest risk (HR 7.58). The associations with physical diseases remained significant mainly in women and in people diagnosed at a younger age; this is a population average, not a biologic constant.
- Data on cardiovascular risk are weaker: the sixfold higher likelihood of cardiovascular mortality from the Baltimore ECA concerns the entire cluster B, 244 participants, and DSM-III criteria. In a community study of 1,295 adults, the association of borderline traits with cardiometabolic risk weakened to borderline significance after socioeconomic factors were taken into account.
- Diagnostic overshadowing, that is, attributing physical symptoms to a psychiatric diagnosis without examination, can lead to missed coronary syndrome, embolism, or acute kidney injury. Structured psychotherapy is the core of treatment; there is no licensed drug for BPD, and olanzapine, clozapine, and quetiapine require monitoring of weight, blood pressure, glucose, and lipids.
- BPD is not a proven independent kidney risk factor; the risk is mediated by obesity, diabetes, hypertension, smoking, intoxications with rhabdomyolysis, and the combination of an NSAID, a diuretic, and a renin-angiotensin system blocker. When kidney risks are present, creatinine, eGFR, electrolytes, and UACR are assessed.
- In CKD, the doses of gabapentin, pregabalin, and some opioids must be adjusted, and when the risk of overdose is high, the quantity per prescription should be limited; lithium requires monitoring of levels, kidney function, and sodium. A diagnosis of BPD alone does not exclude a patient from a transplant program; individual adherence, suicide risk, and the possibility of stabilization are assessed.

14.14 Diagnosing ADHD in adults: what it takes to avoid mistaking distractibility for a diagnosis

Published August 20, 2026 · [online version \(Slovak\)](#)

Abstract. A diagnosis of ADHD in adults requires a persistent pattern of symptoms with onset in childhood, functional impairment in multiple settings, and exclusion of other causes; inattention alone is not sufficient. The review evaluates diagnostic tools: in a validation study of 154 respondents, the six-item ASRS screening questionnaire had a specificity of 99.5% but a sensitivity of only 68.7%, so a negative result does not rule out ADHD. In CKD, treatable causes of attention impairment should be considered first, and after the diagnosis is confirmed, blood pressure and heart rate should be monitored during pharmacotherapy.

Summary

- The diagnosis rests on a clinical interview according to DSM-5 criteria, supplemented by a collateral childhood history; rating scales and neuropsychological tests are only supportive. The full 18-item ASRS had a sensitivity of 56.3% and a specificity of 98.3%, agreement on individual items varied (kappa 0.16 to 0.81), and the scale is tied to DSM-IV.
- The DIVA validation study concerns version 2.0 based on DSM-IV and 40 outpatients; the current DIVA-5 is based on DSM-5. Retrospective reconstruction of childhood is unreliable, so when the supporting information is weak, the conclusion should be more cautious.
- The differential diagnosis must exclude depression and anxiety, sleep disorders, substances and medications, anemia, thyroid disorders, B12 deficiency, and cognitive impairment; new-onset attention impairment in a previously unaffected adult warrants a medical workup. Digital distraction can mimic ADHD, but causality has not been demonstrated.

- In CKD, uremia, anemia, sleep apnea, restless legs syndrome, depression, polypharmacy, and fluctuations in the internal milieu during dialysis contribute to attention impairment; their assessment should proceed in parallel with the psychiatric evaluation.
- A meta-analysis of 22 studies with 46,107 participants showed that methylphenidate increases heart rate and systolic blood pressure compared with placebo ($P < 0.001$), with no difference in cardiac events during short follow-up, but the CKD population was not included. The author recommends measuring blood pressure and pulse before treatment and after every dose change and basing dosing in reduced kidney function on the summary of product characteristics of the specific drug.

14.15 The JAK1 inhibitor upadacitinib in severe alopecia areata: what the new phase 3 trials show and how to read media claims correctly

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. Severe alopecia areata (SALT ≥ 50) has few systemic treatment options. Two parallel double-blind phase 3 RCTs, UP-AA1 and UP-AA2, with 1,399 adults and adolescents evaluated the selective JAK1 inhibitor upadacitinib at 15 mg and 30 mg daily. The primary endpoint of SALT ≤ 20 at week 24 was achieved by 44.6% to 55.0% of patients versus 1.5% to 3.4% with placebo. However, the primary endpoint was not complete hair regrowth, and the media figure of 23% for SALT 0 was not confirmed in the primary summary. In CKD, dosing and interactions should follow the SmPC.

Summary

- The population was aged 12 to less than 64 years, with a mean baseline SALT of 83.9. In UP-AA1, SALT ≤ 20 was achieved by 45.2% with 15 mg, 55.0% with 30 mg, and 1.5% with placebo, and in UP-AA2 by 44.6%, 54.3%, and 3.4%.
- SALT ≤ 20 means at least 80% scalp hair coverage, not complete regrowth. SALT 0 was a multiplicity-controlled secondary endpoint and, according to open sources, was met versus placebo, but the JAMA summary does not report the percentages; the figure of around 23% on the EMA website refers to non-segmental vitiligo.
- Serious adverse events occurred in 1.6% of patients with 15 mg, 2.3% with 30 mg, and 0.4% with placebo; common events were respiratory tract infections, acne, and creatine kinase elevation. The class risks of JAK inhibitors (serious infections, thromboembolism, cardiovascular events, malignancies) apply even though no new signals were identified.
- The European Commission authorization of July 29, 2026, is valid throughout the EU from age 12, but reimbursement in Slovakia is a separate decision. Limitations include a horizon of only 24 weeks and the absence of older patients and kidney outcomes.
- No dose adjustment is needed in mild and moderate renal impairment; at an eGFR of 15 to less than 30 mL/min/1.73 m², the dose for alopecia areata is 15 mg daily, and at an eGFR below 15 mL/min/1.73 m², the drug is not recommended. Upadacitinib is metabolized by CYP3A4, so interactions, infection risk, and hydration must be monitored given polypharmacy in CKD.

14.16 Rome criteria for disorders of gut-brain interaction (DGBI) and when to “test further” in IBS: lessons from a study of medical students

Published September 3, 2026 · [online version \(Slovak\)](#)

Abstract. The Rome criteria allow irritable bowel syndrome to be established as a positive diagnosis in the absence of alarm features, not as a diagnosis of exclusion. A multicenter cross-sectional vignette study of 238 medical students from 14 Latin American countries showed that 76% rated a typical presentation as functional, yet 70% still ordered initial tests and 53% continued after normal results. Knowledge of the cri-

teria therefore does not guarantee diagnostic restraint. In CKD, investigations should not be escalated without alarm features, but anemia or bleeding should not be attributed to the kidneys.

Summary

- Rome IV for IBS requires recurrent abdominal pain at least one day per week over the last three months with two of the following features: relation to defecation, change in stool frequency, and change in stool form; Rome V (2026) adds discomfort and lowers the threshold to three days per month. A 2013 RCT showed that a positive diagnostic strategy is noninferior to a strategy of exclusion.
- Only 46% of the students defined IBS as a positive diagnosis, 61% escalated testing for more severe pain without new alarm features, and 66% showed high diagnostic overuse. Among those with greater knowledge, 68% versus 83% ordered initial tests ($p = 0.010$); only 45% correctly identified nocturnal diarrhea as an alarm feature, and 30% considered pain relieved by defecation to be a warning sign.
- Limitations include the use of vignettes, a single IBS presentation, and voluntary participation, so transfer to Slovak practice is not automatic.
- According to ACG 2021, celiac serology and fecal calprotectin are appropriate in the diarrhea-predominant phenotype, whereas CT and colonoscopy without an indication are not; ACG does not recommend routine colonoscopy under age 45. In CKD, gastrointestinal symptoms are common (uremic nausea, constipation due to phosphate binders, gastroparesis), but iron deficiency anemia, bleeding, weight loss, or nocturnal diarrhea require further investigation.

14.17 Incretin-based agonists in MASLD and MASH: liver protection and relevance for the nephrologist

Published September 13, 2026 · [online version \(Slovak\)](#)

Abstract. Semaglutide, tirzepatide, and retatrutide differ in their level of evidence and regulatory status in MASLD and MASH and cannot be ranked by the number of receptors they target. The phase 3 ESSENCE RCT showed resolution of steatohepatitis without worsening of fibrosis after 72 weeks in 62.9% of patients with semaglutide 2.4 mg weekly versus 34.3% with placebo; tirzepatide has histologic evidence from phase 2, and retatrutide only an imaging-based reduction in liver fat. Kayshild has conditional marketing authorization in the EU for noncirrhotic MASH with F2 to F3 fibrosis. For the nephrologist, hydration, glycemia, nutrition, and the kidney-related restrictions of the specific product are decisive.

Summary

- In ESSENCE (interim analysis of 800 patients), fibrosis improved in 36.8% versus 22.4%, and body weight decreased by 10.5% versus 2.0% ($p < 0.001$); prevention of decompensation, transplantation, or death has not yet been demonstrated. The results cannot be extrapolated to cirrhosis or dialysis.
- In SYNERGY-NASH (190 patients, 52 weeks), MASH resolution was achieved by 44%, 56%, and 62% with tirzepatide 5, 10, and 15 mg versus 10% with placebo ($p < 0.001$); however, Mounjaro does not have a MASH indication in the EU. Retatrutide reduced liver fat on MRI by 42.9% to 82.4% without a histologic endpoint and remains investigational.
- All three drugs were administered subcutaneously, not orally, as the original article erroneously states. In addition to Kayshild, Rezdiffra (resmetirom) also has conditional marketing authorization for MASH in the EU.
- Kayshild requires no dose adjustment in mild and moderate renal impairment but is not recommended at an eGFR < 30 mL/min/1.73 m², including end-stage kidney failure. With Mounjaro, the dose is not adjusted even in end-stage kidney failure, but caution is recommended.
- Gastrointestinal symptoms can lead to hypovolemia and worsening kidney function, insulin or a sulfonyleurea increases the risk of hypoglycemia, and in advanced CKD muscle strength and protein intake should be monitored.
- In 3,533 patients with type 2 diabetes and CKD, FLOW demonstrated a 24% reduction in the composite kidney outcome with semaglutide 1 mg (HR 0.76; 95% CI, 0.66 to 0.88); this separate evidence cannot be extrapolated to tirzepatide or retatrutide.

14.18 Steatotic liver disease: why it belongs in the risk assessment of patients with chronic kidney disease

Published September 13, 2026 · [online version \(Slovak\)](#)

Abstract. Metabolic dysfunction-associated steatotic liver disease (MASLD) is associated with cardiovascular and kidney risk. A narrative review by Marshall and coauthors proposes the concept of a cardiovascular, renal, hepatic, and metabolic (CRHM) syndrome, but this is a conceptual proposal, not a new study or an accepted classification. A key NHANES cohort of 14,818 adults showed the worst prognosis with coexisting CKD and advanced fibrosis but does not prove causality. For the nephrologist, this implies the need for targeted fibrosis triage with FIB-4 and elastography, not blanket treatment of steatosis.

Summary

- In NHANES 2007 to 2018 (1,142 persons with both diseases), MASLD was determined by a fatty liver index, fibrosis by FIB-4, and CKD from single measurements; the groups differed markedly in age, diabetes, and obesity. This is an observational association without proof of causality.
- The US AHA/ACC/ADA/ASN 2026 guideline recommends FIB-4 every 1 to 2 years in CKM, diabetes, or at least two cardiometabolic risk factors; the review authors propose FIB-4 in every patient with CKD and at least one risk factor, which is not what the guidelines state.
- A FIB-4 below 1.3 indicates a lower likelihood of advanced fibrosis (threshold 2.0 above age 65), 1.3 to 2.67 requires elastography or the ELF test, and above 2.67 requires hepatology assessment. In CKD, the result is distorted by nonhepatic thrombocytopenia and by lower aminotransferases on hemodialysis.
- Elastography below 8 kPa supports a low risk of advanced fibrosis, but venous congestion also alters the result; in a pilot study of 41 patients on hemodialysis, approximately 10% exceeded the 8 kPa threshold after dialysis, so the timing of measurement and volume status should be standardized.
- SGLT2 inhibitors, RAS blockade, and finerenone are not treatments for MASH; the FIDELITY analysis of 13,026 patients showed a preserved cardiorenal benefit of finerenone with neutral liver parameters. Targeted MASH treatment consists of Rezdiffra and Kayshild, with conditional marketing authorization for noncirrhotic MASH with F2 to F3 fibrosis; the European SmPC for Rezdiffra recommends no dose adjustment at an eGFR of 15 to 89 mL/min, and the US threshold of 45 mL/min/1.73 m² is not a European contraindication.
- The author recommends verifying the etiology, including alcohol, and involving a hepatologist in advanced fibrosis.

14.19 Pioglitazone in MASLD and MASH: histologic benefit vs. cardiorenal risk

Published September 24, 2026 · [online version \(Slovak\)](#)

Abstract. Pioglitazone, a PPAR- γ receptor agonist, shows a randomized histologic signal in MASH, but in CKD the decision rests mainly on its fluid-related and skeletal risks. This review appraises the PIVENS RCT and the trial by Cusi and colleagues, the AASLD and EASL-EASD-EASO guidelines, and the IRIS trial. Among patients with prediabetes or type 2 diabetes, 58% reached the primary histologic endpoint and 51% had resolution of NASH, yet evidence of an antifibrotic effect or of an effect on clinical liver events is lacking. In practice, the drug is for a carefully selected patient with confirmed MASH and a low risk of congestion.

Summary

- In PIVENS (247 adults without diabetes, 96 weeks), the primary histologic improvement was achieved in 34% of patients receiving pioglitazone 30 mg versus 19% receiving placebo, but $p = 0.04$ did not meet the prespecified threshold of $p < 0.025$; steatosis and lobular inflammation improved, fibrosis did not.
- In the single-center trial by Cusi and colleagues (101 patients with prediabetes or type 2 diabetes, pioglitazone 45 mg, 18 months), the difference in fibrosis score was only -0.5 points (95% CI, -0.9 to 0.0), and body weight increased by 2.5 kg more than with placebo.

- AASLD (2023) allows pioglitazone for NASH in the setting of type 2 diabetes without evidence of anti-fibrotic benefit, whereas EASL-EASD-EASO (2024) does not regard it as standard targeted therapy for MASH because phase 3 trials are lacking.
- In IRIS (3,876 patients without diabetes after ischemic stroke or TIA), the incidence of stroke or myocardial infarction fell from 11.8% to 9.0% (HR 0.76), but edema (35.6% vs. 24.9%) and serious fractures (5.1% vs. 3.2%) increased.
- In CKD, the main problem is not low eGFR but sodium and water retention, heart failure, a fall in hemoglobin, and fractures; pioglitazone is not a kidney-protective pillar and does not replace an SGLT2 inhibitor, RAS blockade, finerenone, or a GLP-1 receptor agonist.
- The author recommends considering the drug only in diabetes or prediabetes with confirmed MASH, without congestion, high fracture risk, or active bladder cancer, and monitoring body weight, edema, dyspnea, HbA1c, and macroscopic hematuria.

CHAPTER 15

Artificial intelligence, digital medicine, and experimental research

Artificial intelligence, wearable sensors, regulation of digital tools, and experimental research from organoids to regenerative medicine. Number of articles: 19.

Key messages of the chapter

This chapter documents a marked asymmetry between the technical performance of artificial intelligence and the evidence of its clinical benefit. A review of machine learning in nephrology cites models predicting AKI up to 48 hours in advance and prognostic models of CKD progression with C statistics of approximately 0.79 to 0.84, yet a randomized trial with 6,030 patients did not demonstrate a benefit of electronic health record alerts on length of stay or mortality (15.7). Closest to clinical practice is automated measurement of total kidney volume in ADPKD, which agrees closely with expert segmentation, but predictive and genetic algorithms do not replace the Mayo Imaging Classification or the PROPKD score, and every result must be visually verified (15.13). The iBox system has been externally validated for predicting graft failure (15.7), whereas AI-assisted POCUS remains a triage tool, its source has a conflict of interest, and its transfer to nephrology is the author's reflection (15.9). According to a review, regulatory clearance is not evidence of improved clinical outcomes, because most devices passed through the 510(k) pathway based on equivalence (15.16).

A recurring theme is the limitations of the tools themselves. At a hands-on workshop, identical prompts in different language models led to different answers and lines of reasoning; this is a description of an educational experience, not a validation study (15.8). Reviews consistently cite bias in the data, distribution shift, lack of transparency, and unclear accountability (15.7, 15.16). A WHO report from 2022, a scoping review with predominantly low-quality included studies, showed higher use of digital health among younger, more educated, and economically stronger groups and warned that a CKD progression model may misread low use of care as low need (15.15). Wearable devices have the strongest benefit in diabetes, heart failure, and arrhythmias, but they measure neither eGFR nor albuminuria, and without a response protocol they remain expensive data collectors (15.6).

Opinion pieces add an ethical and relational dimension. O'Hare's essay questions whether time saved by AI will automatically bring more humane care and emphasizes shared decision-making about dialysis or conservative management (15.4). A commentary on the emotional persuasiveness of chatbots proposes targeted history questions and motivational interviewing instead of moralizing (15.5). When a patient arrives with a diagnosis from AI, the patient should be reminded that CKD according to KDIGO requires an abnormality lasting at least three months, and the consultation should follow six steps from excluding an emergency to a concrete plan (15.14). With AI scribes, the physician remains fully responsible for the documentation of doses, eGFR, and electrolytes, and the patient's consent must be recorded (15.10). A reflective piece on combining clinical judgment with programming, together with economic scenarios to 2030, which the authors explicitly do not regard as forecasts, frame these changes without study data (15.1, 15.18).

Experimental research forms a separate block that is further from practice. Selective autophagy in tubules and podocytes may promote both repair and fibrosis during the AKI-to-CKD transition, but biomarkers of autophagic flux are lacking (15.3). Synthetic WNT organizers improved the spatial organization of kidney organoids, which, however, still lack vascularization and function (15.11), and kidney-on-a-chip does not yet replace animal models (15.17). A healing model in the jellyfish *Clytia*

hemisphaerica and SagA inhibition enhancing the effect of vancomycin against resistant *Enterococcus faecium* are preclinical studies in which renal elimination and toxicity remain open questions (15.12, 15.19). Clinical research is shifting toward precisely defined diseases and biomarkers in phase II, which according to a sponsored review does not yet change routine practice (15.2).

This yields a framework for the outpatient clinic: use AI as a support tool with mandatory verification of the output by the physician, prefer tools with external validation in a similar population, and when implementing, start with a silent deployment and an audit (15.7, 15.13, 15.16). For digital services, maintain telephone and in-person channels, measure differences between subgroups, and do not regard nonuse as lack of interest (15.15). Conduct the conversation about a patient's chatbot-derived information empathetically, separate facts from hypotheses, and record the new plan; identifying data do not belong in unverified tools (15.14, 15.10). Follow preclinical findings as a direction for future drug and biomarker development, not as a basis for treatment decisions (15.3, 15.11, 15.19).

15.1 When nephrology meets code

Published April 25, 2026 · [online version \(Slovak\)](#)

Abstract. Nephrology and dialysis generate large volumes of data from laboratory testing and from monitoring of blood pressure, pulse, and ultrafiltration. This short reflective piece highlights the arrival of predictive models that estimate the trajectory of eGFR decline or the risk of intradialytic hypotension. The author stresses how rare it is to combine a physician's clinical judgment with a programmer's algorithmic thinking, a combination that makes it possible to build custom tools for analyzing patient trends and automating administration in the dialysis center. It is an opinion piece without trial data.

Summary

- Predictive models can use seemingly unrelated variables to predict eGFR decline or the risk of intradialytic hypotension.
- A physician proficient in PHP, JS, Python, HTML, and CSS can build tools tailored to a dialysis center.
- The author raises the question of how these innovations can be put into real use in Slovakia.

15.2 Kidney research is changing: from large trials to more precise and targeted nephrology

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. Clinical drug development in nephrology has long been slow, expensive, and of limited regulatory success. This commentary on an analysis by the Clinical Trials Arena portal describes a shift from large trials targeting CKD or the dialysis population in general toward precisely defined diseases such as IgA nephropathy, FSGS, ADPKD, and Alport syndrome. Biomarkers in phase II, patient-centered design, and a geographic shift to the Asia-Pacific region are gaining a key role. For everyday practice, this is not yet an immediate change but a more rational approach to drug development.

Summary

- A more precise diagnosis allows better setting of entry criteria, control arms, endpoints, and the expected treatment effect; eGFR, creatinine, and proteinuria remain important but change slowly and are nonspecific.
- Biomarkers of podocyte, tubular, vascular, complement, fibrotic, and immune pathway injury matter most in phase II, where the dose is selected; selection based on surrogate parameters risks underdosing with respect to kidney protection.
- According to the analysis, nephrology still lags well behind oncology in the use of biomarkers.
- Protocols are designed with the involvement of patient organizations to reduce the burden on patients and caregivers, because demanding logistics reduce retention and trial quality.

- Sponsors are moving to regions with higher prevalence and better recruitment, such as South Korea, China, Taiwan, and Singapore, but protocols must respect local standards of care and the availability of dialysis.
- Successful programs prepare global implementation nine to twelve months in advance; in rare indications, early dialogue with regulators about surrogate endpoints is easier.
- This is an overview of trends from a sponsored source, not new clinical data, and the author notes that it brings no immediate change to routine care for now.

15.3 Selective autophagy in the kidney: when cellular recycling determines both injury and repair

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. The proximal tubule and podocytes are exceptionally sensitive to disturbances of mitochondria, lipids, the endoplasmic reticulum, and lysosomes. A review in *Nature Reviews Nephrology* describes selective autophagy as a network of specialized pathways, chiefly mitophagy, lipophagy, and ER-phagy, that determine both repair and maladaptation after kidney injury. Their effect depends on magnitude, timing, and cell type, and their failure can promote ferroptosis, senescence, and fibrosis during the transition from AKI to CKD. This is basic research without a direct treatment recommendation.

Summary

- Proximal tubular cells depend on oxidative phosphorylation, so mitophagy and lipophagy dominate when they are injured; impaired lipophagy may contribute to lipotoxicity and CKD progression, including diabetic kidney disease.
- Podocytes chiefly require control of proteostasis, and ER-phagy may dampen the stress response that would otherwise lead to apoptosis, proteinuria, and glomerular injury.
- Autophagy is not automatically protective: when flux is insufficient, excessive, or mistimed, it can promote ferroptosis, cellular senescence, immune remodeling, and fibrosis during the transition from AKI to CKD.
- In diabetic kidney disease, the autophagic response may become exhausted under long-term metabolic overload, with accumulation of damaged mitochondria and disordered lipid metabolism.
- Less studied pathways, namely lysophagy, Golgiphagy, and nucleophagy, may complete the picture of the kidney stress response, but their significance in kidney disease remains unclear.
- The authors reject a crude approach of “increase or block autophagy” and call for cell-specific, temporally targeted modulation with stratification by biomarkers of autophagic flux, which are not yet available.
- For practice, this is not a treatment guide but a biological framework for future development of drugs and biomarkers in AKI, diabetic kidney disease, and CKD.

15.4 Will artificial intelligence make medicine more human again?

Published May 19, 2026 · [online version \(Slovak\)](#)

Abstract. Artificial intelligence tools promise less administration and more time for the patient. An interview with Ann O'Hare, MD, of the University of Washington for Healio, discussed here from a nephrology perspective, challenges the assumption that time saved will automatically bring more humane care. It emphasizes the values of geriatrics and palliative medicine, shared decision making about dialysis or conservative care, and the risk that an AI scribe will strip personal details from the record. This is an opinion piece; according to the author, the outcome depends on the people who use the technology, not on the technology itself.

Summary

- O'Hare advocates bringing the values of geriatrics and palliative medicine into nephrology: knowing the patient's priorities, helping with decisions about dialysis, conservative care, or transplantation, and supporting the family at the end of life.

- She warns that an AI scribe that deletes seemingly unimportant parts of a conversation, such as the name of the patient's dog, may strip the record of the human dimension that helps restore the relationship and trust.
- Real help is often hidden in ordinary problems such as unavailable transportation, a medication error, or conflicting advice from several physicians, and the nephrologist should not dismiss them as outside the specialty.
- In her view, robotic medicine did not begin with AI; technology can free up space for conversation or, conversely, further accelerate and standardize care and distance patients from physicians.
- New kidney-protective drugs are a major advance over the era of ACE inhibitors and ARBs, but uncertainty remains at the individual level, so partnership-based communication about benefits, risks, and uncertainty is essential.
- This is the opinion of a single author without empirical data; her conclusion is that the future of nephrology depends above all on what kind of physicians we choose to be.

15.5 “The most dangerous thing” about AI: not hallucinations, but how AI can become emotionally persuasive

Published May 27, 2026 · [online version \(Slovak\)](#)

Abstract. Assessments of chatbot risks focus on hallucinations and less on their emotional persuasiveness. This commentary on a Medscape article describes how an AI that appears empathetic and is always available can create intimacy, reinforce distorted beliefs, and become a confidant for a vulnerable person. It is an opinion piece without trial data. The author proposes stricter rules for relational design, targeted questions in the medical history, warning signs, and working with the patient through motivational interviewing instead of moralizing.

Summary

- The risk lies not only in incorrect information: AI can praise, mirror emotions, and reassure, first creating intimacy and then influencing beliefs, and in severe cases reinforcing rather than challenging them.
- The author calls for systems that do not imply their own sentience or emotional investment in the user, do not encourage romantic attachment, have stricter safeguards around self-harm and suicidality, and distinguish validated tools from entertainment products.
- The clinician should ask sensitively and without judgment whether the patient uses a chatbot for emotional support, how often, whether they feel attachment or dependence, and whether they spend more time with the bot than with people.
- Warning signs include calling the bot a friend or partner, preferring AI to relationships, attributing emotions to the bot, social withdrawal, and distress at the idea of ending use; adolescents, people with psychotic spectrum disorders, and socially isolated people are at greatest risk.
- Motivational interviewing with validation and reflective listening is recommended, along with distinguishing productive from relational use and gradually shifting sources of support to the human environment.

15.6 Wearables in chronic disease: more data is not enough; protocols and clinical purpose are what matter

Published June 10, 2026 · [online version \(Slovak\)](#)

Abstract. Smartwatches, wristbands, and continuous glucose sensors provide data from the everyday lives of patients with chronic diseases. This review, drawing among other sources on a systematic review in JMIR mHealth and uHealth, assesses where wearable devices make clinical sense. The strongest benefit is in diabetes, heart failure, and arrhythmias, but the devices can generate more than 5 GB of data per patient per

week. Without validated measurements, clinical protocols, and clear responsibility for responding, they remain merely expensive data collectors.

Summary

- In diabetes, continuous glucose monitoring provides trends, time in range, and nocturnal hypoglycemia; combined with digital education, it may help adults with type 2 diabetes manage their disease better.
- In heart failure, smartwatches can support remote monitoring for exacerbations, and data from consumer devices showed good agreement with cardiopulmonary exercise testing.
- Validity, accuracy, and interpretation vary by diagnosis, parameter, and device, so their potential does not amount to a universal recommendation; poor-quality inputs can lead AI to false alarms or to a missed signal.
- Useful deployment requires a trained patient, definition of clinically relevant values, a protocol stating who responds to deviations, and integration of the data into the medical record; otherwise, patient anxiety and physician overload follow.
- Unresolved interoperability, privacy protection, and the question of who acts when decompensation or an arrhythmia is detected can make even accurate measurement organizationally dangerous.
- In CKD, watches cannot measure eGFR or albuminuria, but they can track blood pressure, activity, pulse, sleep, and body weight; in patients on dialysis, interdialytic trends are for now only a supplement to, not a substitute for, examination and assessment of volume status.

15.7 Artificial intelligence in nephrology: what we already know, where the limits lie, and where it is heading

Published June 17, 2026 · [online version \(Slovak\)](#)

Abstract. Nephrology works with a large volume of structured data but lags in the adoption of artificial intelligence. This review summarizes the use of machine learning in AKI, CKD, dialysis, transplantation, and imaging and pathology. Models predict AKI up to 48 hours before dialysis, but a randomized trial of 6,030 patients showed no benefit of EHR alerts on length of hospital stay or mortality. The main obstacles are bias, data quality, opacity, distribution shift, and accountability. AI should extend the nephrologist's capabilities, not replace the nephrologist.

Summary

- In AKI, extreme gradient boosting outperformed traditional models in a retrospective analysis of 2,010 patients after vascular and thoracic procedures; a model developed in 703,782 patients had a higher rate of false-positive predictions and questionable transferability because of a predominance of men.
- In CKD, logistic regression predicted progression of diabetic nephropathy with an accuracy of approximately 71%, a CKD progression model achieved C-statistics of approximately 0.84, 0.81, and 0.79 at horizons of 1, 2, and 5 years, and KidneyIntelX approximately 0.77.
- In dialysis, AI is being tested for prediction of intradialytic hypotension, hospitalization, and mortality and for NLP of EHR data, but apart from anemia management it is not yet widely used in routine care because of regulation, privacy, and difficult integration.
- The iBox system predicted the long-term risk of graft failure better than nephrologists and was externally validated in the United States and Europe; other tools assist decisions on marginal offers using the Kidney Donor Profile Index.
- In pathology and imaging, CNNs are applied to lupus nephritis lesions, differentiation of lesions on MRI, and volume segmentation in ADPKD.
- Limitations include bias in the data, such as race adjustments in GFR estimation, missing and nonstandardized data, the black box, distribution shift, and unclear liability for harm.
- The author stresses the need for further validation and for educating clinicians about the data life cycle, with AI meant to raise the quality of the nephrologist's work, not to replace judgment.

15.8 AI in nephrology in practice: what the “Hands-On Primer” offers for clinical reasoning and safe integration

Published June 21, 2026 · [online version \(Slovak\)](#)

Abstract. Generative AI is changing how clinicians search for information and synthesize evidence. The article AI in Nephrology Hands-On Primer describes experience from a practical workshop in New York in March 2026, where nephrologists tested LLMs and image-based AI on real tasks. The main finding was inconsistency: identical prompts led to markedly different answers and reasoning paths in ChatGPT, Gemini, and Claude. This is a description of an educational experience, not a validation study; outputs must always be verified, and responsibility must remain with the human.

Summary

- LLMs can summarize publications within seconds, help with documentation, and support decision making, but they rely on probabilistic pattern recognition without true understanding, with hallucinations, contextual errors, and unstable outputs.
- In literature synthesis, AI handled summarization and extraction of key results well and saved time, but the outputs had to be checked for both accuracy and completeness.
- In assessing diet from photographs and the catheter exit site in peritoneal dialysis, AI sometimes achieved high agreement with an expert, but it was sensitive to image quality and, without clinical context and validation in the given population, cannot replace expert judgment.
- Tools are grouped by function into literature synthesis (NotebookLM), documentation (DoxGPT, Microsoft Copilot), visualization (Canva, Napkin AI), and evidence-based support (OpenEvidence); the human should remain the final decision-making authority.
- When clinical scenarios were tested, different LLMs produced different recommendations and different logical structures of argument, so outputs should be cross-checked and compared with relevant sources and one's own clinical experience.
- NephroResource is a web platform with curated nephrology information, and ASN is developing workshops and AI communities for the responsible use of AI.

15.9 AI-assisted POCUS speeds up cardiac decisions by NPs and PAs, and what it could mean for nephrology

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. In dyspnea and edema, the bottleneck is the availability of echocardiography, which can take hours to days. This commentary on a Medscape article describes AI-assisted point-of-care ultrasound for nurse practitioners and physician assistants: the cloud-based AISAP platform provides automated measurements and diagnostic suggestions for structural heart disease, valvular abnormalities, and heart failure. It is a triage tool, not a substitute for formal echocardiography. Its transfer to nephrology is the author's reflection, not evidence, and the source has a declared conflict of interest.

Summary

- The main delay arises not in image acquisition but in interpretation, when subtle valvular abnormalities and quantitative measurements require an expert; AI is meant to shorten the path to a decision on escalation or an outpatient plan.
- At Sheba Medical Center, some focused examinations are performed by trained staff after several days of theoretical instruction and supervision, and an examination can take approximately 2 minutes.
- AISAP is one of several tools focused on hardware, AI analysis of echocardiography, guidance of nonexperts during image acquisition, or AI-enabled digital stethoscopes.
- In patients on dialysis and in cardiorenal syndrome, the tool could theoretically speed up identification of a cardiac cause of dyspnea; the author recommends introducing it only with local protocols, quality audits, and rules for referral to formal echocardiography.

- The platform is meant to answer whether urgent cardiology evaluation is needed, whether signs of heart failure or significant valvular disease are present, and whether the patient can remain in outpatient care; for APPs, rapid and reliable interpretation remains the main challenge.
- Robert Klempfner, cited in the source, is a cofounder and the medical director of AISAP and therefore has a financial relationship with the technology being evaluated.

15.10 Legal pitfalls of using AI scribes in the outpatient clinic and what they mean for nephrology practice

Published July 2, 2026 · [online version \(Slovak\)](#)

Abstract. AI scribes reduce the administrative burden, which in nephrology is exceptionally high because of chronic disease, drug dosing, and laboratory trends. This commentary on a Medscape article summarizes the legal risks of introducing them in the outpatient clinic under US law, including HIPAA. The risks center on three points: the vendor's regulatory compliance in storing and accessing records, documented patient consent, and the physician's full responsibility for the accuracy of the record, since AI can hallucinate. The author proposes a safe minimum for nephrology practice.

Summary

- Before deployment, it is necessary to know where recordings are stored, who at the vendor has access, and how long they are retained; if privacy protection cannot be explained, the legal risk grows from the outset.
- The patient must be informed in advance and have the option to consent or refuse, and consent must be reliably documented, because there have been cases of patients claiming they were recorded without consent.
- The physician remains responsible for the content, which in nephrology is sensitive for drug doses, creatinine and eGFR, proteinuria, electrolytes, and dialysis treatment records; the entire medical content should be checked after every visit.
- The source recommends addressing contracts and internal policies with a lawyer and following the rules on recording and retaining records, in the United States under state and federal laws and HIPAA.
- In nephrology, documentation is exceptionally demanding because of the chronic course, dosing, and laboratory trends, so an inaccurate record can directly affect long-term decision making and treatment safety.
- The author recommends an approved vendor selection, a standard consent script, mandatory physician review, and a rule that an unverified record does not enter the medical record without correction.

15.11 Synthetic Wnt-secreting organizers improve the spatial organization of human kidney organoids

Published July 5, 2026 · [online version \(Slovak\)](#)

Abstract. Kidney organoids derived from human pluripotent stem cells contain the correct cell types, but their spatial arrangement is often chaotic. Experimental work by Faust and colleagues in Science used spatial transcriptomics to identify nephron polarity at the WNT11 and WNT9B boundary and mimicked it with synthetic Wnt-secreting organizers. The result was promotion of distal nephron differentiation and morphogenesis oriented toward the signal source. This is basic research that improves experimental models, not a functional or transplantable kidney.

Summary

- Cells respond not only to the presence of a signal but also to its local concentration, direction, duration, and position, so a signal delivered diffusely in the medium acts differently from a local signaling field.
- Spatial transcriptomics of human kidney development showed that nascent nephrons are oriented from the region near the collecting duct toward more distant segments, with the axis lying at the WNT11 and WNT9B signaling boundary.

- After introduction of synthetic organizers, distal nephron programs, which are weaker in standard protocols, were strengthened, and nephrons were oriented toward the source of the WNT signal.
- Self-organization need not be replaced by engineering; it is enough to supply the missing local developmental information, which may improve reproducibility across organoids and laboratories.
- A model with podocytes and tubular cells may still not faithfully mimic organ biology if anatomical continuity of segments is lacking, so organoid quality should not be judged solely by the presence of the correct cell markers.
- The potential lies in research on congenital anomalies of the kidney and urinary tract, tubulopathies, and ciliopathies, in nephrotoxicity testing, and in models derived from an individual patient's cells.
- Organoids still lack full vascularization, a drainage system, adult organ architecture, and physiologic function, so their impact on the treatment of CKD, glomerulonephritis, or polycystic kidney disease is for now only indirect.

15.12 A jellyfish that heals wounds without scars: what a simple model can teach regenerative medicine

Published July 7, 2026 · [online version \(Slovak\)](#)

Abstract. In mammalian wound healing, hemostasis, inflammation, angiogenesis, and fibrotic remodeling overlap, which makes it difficult to study basic epithelial repair. An experimental study in Molecular Biology of the Cell used the transparent jellyfish *Clytia hemisphaerica*, which closes small wounds within minutes and larger ones in approximately an hour without a scar. The authors described three mechanisms, lamellipodial crawling, actomyosin contraction, and collective migration, the choice among which is governed by the availability of the basement membrane. This is a basic model, not a therapeutic guide for human wounds.

Summary

- *Clytia hemisphaerica* is a model without a vascular system and without a mammalian inflammatory phase, with a single layer of large epithelial cells, which makes it possible to follow the mechanics of wound closure microscopically in real time.
- In smaller wounds, cells extend lamellipodia over the exposed basement membrane; when they meet or the substrate is missing, an actomyosin cable with purse-string contraction is activated.
- In large defects, the entire epithelial sheet is recruited, and after the fronts meet, closure proceeds through the same final sequence: lamellipodia, their retraction, and contraction.
- The basement membrane functions as an information interface: its availability decides among migration, arrest, contraction, and collective movement, and when the substrate is damaged, contraction assists closure.
- Microwounds passing through the interior of a single cell closed in approximately 3 to 5 minutes, with cells distinguishing their own protrusions from those of neighbors and restoring cell boundaries.
- The absence of a scar results from the lack of a vascular network, a mammalian inflammatory phase, and collagen remodeling; human skin is multilayered, vascularized, and immunologically active, so direct transfer is not possible.
- For nephrology, the relevance is indirect, for example in impaired healing of surgical wounds, vascular accesses, and defects in patients with CKD, diabetes, and on dialysis, and in the development of biomaterials that mimic the basement membrane.

15.13 Artificial intelligence in ADPKD: current possibilities and limits of clinical use

Published July 13, 2026 · [online version \(Slovak\)](#)

Abstract. The course of ADPKD is highly variable, and eGFR may remain preserved for a long time despite cyst growth, so estimation of individual risk is key. A review in Nephrology Dialysis Transplantation evaluates the use of AI in imaging, prognostics, genetics, and clinical research. Closest to clinical use is auto-

mated measurement of total kidney volume (TKV), with high agreement with expert segmentation. Predictive models and genetic algorithms are still under development and validation and do not replace the Mayo Imaging Classification, the rate of eGFR decline, or the PROPKD score.

Summary

- KDIGO 2025 recommends the Mayo Imaging Classification based on htTKV and age; subclasses 1C to 1E are among the criteria for rapid progression when tolvaptan is considered, but MIC applies only to typical morphology and should not be used with a pathogenic variant outside PKD1 and PKD2.
- Deep learning automatically segments the kidneys on MRI or CT, shortens analysis, and improves reproducibility, but performance is degraded by differing protocols, very large kidneys, exophytic cysts, or a previous procedure, so visual review is necessary.
- Models combining imaging, clinical data, and genotype are promising but may not transfer to other populations, and a validated model predicting response to tolvaptan is not yet part of standard care.
- Algorithms can prioritize variants, including in less common genes such as ALG5, ALG9, DNAJB11, GANAB, IFT140, and NEK8, but they should not determine pathogenicity on their own.
- Repeated TKV measurement at short intervals is not an established way of assessing treatment response, because the change may be small and affected by measurement error.
- Before wider implementation, external validation, transparency, and protection of genetic data are required; the greatest risks are a false sense of precision and automation bias.
- The authors recommend technically verifying every TKV, assessing the suitability of MIC, and, for tolvaptan, weighing the overall risk, contraindications, and patient preferences.

15.14 When the patient arrives with a diagnosis from AI: what it changes in the nephrology outpatient clinic

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. Patients increasingly arrive with a diagnosis or treatment proposal from generative AI, which may be inaccurate, outdated, or unduly confident. The text is based on a Medscape News Europe news article and a discussion by the French Medical Council, supplemented by WHO and KDIGO 2024; it is not a clinical study. In nephrology the risk is high, because a single eGFR or albuminuria value without its time course does not confirm CKD. The author proposes a six-step consultation approach: urgent triage, independent verification, and a clear plan without ridiculing the patient.

Summary

- KDIGO defines CKD as an abnormality of structure or function lasting at least three months, so a chatbot's conclusion of kidney failure based on a single eGFR of 52 mL/min/1.73 m² is unfounded; eGFR is an estimate influenced by muscle mass, diet, or medications and can be misleading in AKI.
- Albuminuria fluctuates with infection, fever, exercise, or hyperglycemia, yet a normal eGFR does not exclude disease in the presence of persistent albuminuria, hematuria, or abnormal urine sediment.
- Before discussing the source, an urgent condition must be excluded, for example anuria, pulmonary edema, symptoms of severe hyperkalemia, hypertension with neurologic symptoms, or fever with pain over the graft.
- The approach has six steps: exclude an urgent condition, obtain the exact prompt and answer, identify the patient's concern, separate facts from hypotheses, assess independently, and conclude with a specific plan, including the medications that remain unchanged.
- An AI-generated hypothesis should be judged by the evidence, not by its origin, because both patient and physician are subject to anchoring; a qualitative study of 34 patients confirmed the importance of transparency and an empathetic relationship but is not directly transferable.
- Name, national identification number, and complete reports do not belong in unverified chatbots, and removing the name does not necessarily mean anonymization; a substantial influence of AI on the patient's decision should be briefly documented.

- The source does not allow the prevalence of AI use or the accuracy of the models to be determined, and the claim that physicians without AI will be replaced is only one discussant's opinion.

15.15 Digital health can also deepen inequities: what the evidence base says and what it means for nephrology practice

Published August 19, 2026 · [online version \(Slovak\)](#)

Abstract. Digital tools are presented as a path to more equitable care, but they can also deepen inequities. A 2022 review report by the WHO Regional Office for Europe summarized 22 reviews and meta-analyses and showed consistently higher use of digital health in cities and among people who are better educated, economically better off, younger, and members of the majority language group. It is a scoping review without quality appraisal and with limited transferability to lower-income countries. For nephrology, it follows that equity should be measured just as clinical benefit is.

Summary

- The report distinguishes three dimensions of digital health, namely access, use, and engagement, and ten equity domains according to PROGRESS Plus; according to WHO, digital literacy is a key factor in use and engagement.
- Digitally mediated care produced favorable outcomes for rural communities, but barriers of infrastructure, connection cost, and skills accumulate there; only two of the 22 reviews took an intersectional perspective.
- Algorithms trained on historical data may misinterpret low use of care as low need, so a CKD progression model can entrench inequity and lend it an appearance of objectivity.
- As good practice, WHO cites the NICE framework with 21 standards for digital technologies, one of which is equity and which also covers mitigation of algorithmic bias.
- Limitations: only 12 of the 22 reviews critically appraised quality, which was low, primary studies often enrolled only people with access to technology, and the search ended in May 2022, before generative AI became widespread.
- In CKD, transplantation, and especially home dialysis, the author recommends measuring differences between subgroups, testing with patients who face barriers, keeping telephone and in-person channels, not equating nonuse with lack of interest, and involving nurses, social work, and the family.

15.16 Regulation of medical artificial intelligence: what physicians should know and what it means for nephrology

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. According to AMA data, more than 80% of physicians use AI professionally, compared with 38% in 2023, but regulation lags behind. This review examines an August 2026 FDA CDRH discussion paper on generative AI devices, which is neither a draft nor binding guidance. It proposes a two-axis risk framework based on the system's autonomy and the severity of harm, and a competency assessment with benchmarking, clinical validation, and postmarket monitoring. For nephrology, it stresses that regulatory clearance is not evidence of improved clinical outcomes.

Summary

- The FDA does not regulate AI as such but medical devices and their individual functions, so the same model can have a different status depending on its intended use, and the term “approved AI” is imprecise.
- The framework combines an axis of action, ranging from nondirective information to fully autonomous action, with an axis of severity of harm; directiveness is a continuum, for example from general information to an instruction to increase lisinopril from 10 mg to 20 mg daily.

- CDRH considers measurement and signal functions, such as laboratory-derived eGFR, to be higher risk because the user cannot assess their basis; a nephrology tool in the hands of a nonspecialist may also carry increased risk.
- Clinical validation includes retrospective evaluation, silent deployment, simulated patients, adjudication, and a prospective study; after deployment, drift and changes to a third-party foundation model must be monitored.
- Most AI devices went through the 510(k) pathway based on equivalence to an existing device, so clearance need not rest on large prospective studies, and good discrimination does not mean lower mortality.
- Liability is contested: a proposal by the Washington Medical Commission placed it entirely on the physician, whereas the AMA chief executive officer proposed assigning it according to who best understands the tool's risk.
- In dialysis and transplantation, performance may be worse in underrepresented subgroups; the author recommends a checklist, data from the vendor, and starting with silent deployment before clinical use.

15.17 Kidney-on-chip and kidney organoids: mechanical forces open new possibilities for research into kidney disease

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. Animal models and static cultures cannot separate mechanical from biochemical effects, even though the kidney is a mechanically active organ. A review in the Journal of the American Society of Nephrology evaluates kidney-on-chip and iPSC-derived organoids for modeling nephrotoxicity, development, and pathophysiology. Microfluidic systems allow shear stress, pressure, and matrix stiffness to be controlled, but organoids remain immature and insufficiently vascularized. This is a review, with no evidence that these models can already replace animal studies; for now they change neither diagnosis nor treatment in clinical practice.

Summary

- Shear stress influences the polarity, cytoskeleton, primary cilia, and transporter expression of the tubular epithelium, but higher flow is not automatically more physiologic, and permeability in a glomerular chip is not equivalent to glomerular filtration.
- Organoids derived from a patient's iPSCs with an isogenic control can clarify the pathogenicity of rare variants, but they resemble fetal tissue more closely, and their response does not yet predict the clinical response of an individual patient.
- In the assessment of nephrotoxicity, barrier integrity, transport, mitochondrial function, and injury biomarkers can be monitored, but the concentration in the medium may not correspond to exposure in the kidney, and lipophilic substances may be absorbed into the chip polymers.
- The FDA Modernization Act 2.0 and the FDA plan of April 2025 support organoids and organs-on-chip as an alternative to animal testing, but this is neither an abolition of animal testing nor automatic acceptance of every model.
- The principal obstacles are cellular immaturity, the absence of vascularization and immunity, batch-to-batch variability, material artifacts, the lack of standards, and incomplete clinical validation.
- Machine learning will not remove the biological invalidity of the input model; a realistic approach combines pharmacokinetic modeling, clinical databases, and, where needed, animal studies.

15.18 Artificial intelligence through 2030: what different economic scenarios may mean for wages, work, and healthcare

Published September 9, 2026 · [online version \(Slovak\)](#)

Abstract. The working paper Economic Scenarios for Transformative AI, published by The Anthropic Institute, is not a peer-reviewed publication but an analytic framework: a simple task-based model translates a small number of assumptions about the development of artificial intelligence into implications for pro-

ductivity, wages, and employment through 2030. In the extreme scenario, GDP is 32.4% above the trajectory without artificial intelligence, while the labor share of income falls from 60% to 45%. The authors emphasize that the scenarios are not forecasts and that no probabilities are assigned to them.

Summary

- The model divides workers into cognitive occupations that artificial intelligence can directly affect and all others; by 2030 it adds 1.6%, 8.3%, or 32.4% to GDP, and the labor share of GDP falls to 59.4%, 56.1%, or 45.2%.
- In the extreme scenario, cognitive wages are 11.5% lower than they would be without artificial intelligence, which therefore does not automatically mean an absolute decline relative to today; wages in occupations not directly affected are 34% higher, and almost all of the divergence between scenarios occurs only after 2027.
- The innovation channel is small in the model: labor productivity gains from accelerated research amount to considerably less than one percent in all three scenarios, which the authors consider a lower bound.
- A survey of 10,980 adults in the United States yielded, after the median responses were entered, a result close to the substantial scenario: GDP 8% higher and cognitive employment 4% lower by 2030.
- The authors acknowledge that the model omits individual workers, economic policy, cycles, demand from infrastructure construction, and research feedback loops, and that it addresses the US economy; the muted data to date are compatible both with the mild scenario and with the onset of a more extreme trajectory.
- For healthcare, the author expects a mixed model and points out that an occupation's exposure to artificial intelligence does not mean the disappearance of jobs; in nephrology, every tool must be validated in the specific population and clinical setting.

15.19 SagA inhibition may enhance the effect of vancomycin against resistant *Enterococcus faecium*

Published September 12, 2026 · [online version \(Slovak\)](#)

Abstract. A preclinical study published in June 2026 in Nature Communications examined whether inhibition of the peptidoglycan hydrolase SagA enhances the effect of vancomycin against vancomycin-resistant *Enterococcus faecium*. The experimental compound pghi-4 reduced the minimum inhibitory concentration of vancomycin in the ERV165 strain by up to eightfold and, in a short murine experiment, reduced the bacterial burden in the liver and spleen. These are preclinical data that do not demonstrate efficacy or safety in patients; for nephrology, exposure, renal elimination, and toxicity remain open questions.

Summary

- In VanA- or VanB-type resistance, the bacterium produces precursors terminating in D-Ala-D-lactate and the affinity of vancomycin falls by approximately three orders of magnitude; genetic deletion of sagA disrupted cell growth and division and increased binding of labeled vancomycin while the VanA machinery was preserved.
- From a library of 616 compounds, five covalent inhibitors were identified; pghi-4 belongs to the beta-chloroalkenylsulfonyle fluorides and also inhibited the related hydrolase PGH2, so the effect cannot be attributed exclusively to selective SagA inhibition.
- In a combination microdilution assay with the ERV165 strain, the vancomycin MIC fell by up to eightfold depending on the inhibitor concentration; EUCAST version 16.1, valid from June 24, 2026, gives breakpoints for enterococci of $S \leq 4$ mg/L and $R > 4$ mg/L, so a fold reduction alone does not render resistant isolates susceptible.
- In a murine experiment with six animals per group and unblinded analysis, the combination of vancomycin 100 mg/kg and pghi-4 25 mg/kg with a second dose after 24 hours attenuated weight loss and reduced bacterial counts in the liver and spleen; the single-dose regimen had no significant effect.

- The *sagA* gene was present in all 559 *E. faecium* genomes analyzed (395 resistant and 164 susceptible), representing 99 sequence types, which supports the prevalence of the target rather than an identical treatment response in every isolate.
- Vancomycin is substantially eliminated by the kidneys and excessive exposure increases the risk of acute kidney injury; for *pghi-4*, renal elimination, dialyzability, and interactions with immunosuppressants need to be clarified, while treatment of severe VRE_{fm} infection continues to be guided by susceptibility, source control, and collaboration with an infectious disease specialist.

CHAPTER 16

Congresses, the profession, and medicine in society

Congress reviews, research methodology, physicians' working conditions, and the ethical and social context of medicine. Number of articles: 9.

Key messages of the chapter

This chapter differs in character from the clinical parts of the compendium: commentaries, reports, and reflections predominate rather than original studies, and its common denominator is the question of how medicine talks with patients, how it organizes itself, and how it withstands external pressures. Several texts themselves point out that they are personal positions or educational materials that do not replace individual clinical decision-making (16.4, 16.5, 16.8).

The most coherent theme is communication and the lived experience of illness. A commentary on condescending tone distinguishes health literacy from intelligence and warns that infantilization, including elderspeak, undermines trust and the willingness to ask questions; this is a professional reflection and patient experience, not a quantitative study (16.2). A reflection on the difference between disease as a measurable disorder and illness as a lived experience reminds us that despite relatively stable laboratory values, a patient with CKD or on dialysis may experience marked functional limitation (16.4). A methodological review of qualitative research shows that interviews reveal the context of decisions about modality or conservative care; however, the example of 20 US nephrologists, who often viewed dialysis and hospice as mutually exclusive, cannot be converted into population prevalence (16.7). According to reviews, the evidence on narrative medicine is cautiously encouraging, but without clear evidence of lasting behavior change, and high heterogeneity precluded meta-analysis (16.8). All these texts converge on the recommendation to ask open questions about concerns and treatment burden (16.2, 16.4, 16.7).

The second theme is knowledge transfer and the quality of evidence. The post-congress analysis of ERA 2026 deliberately does not cite specific studies and instead proposes its own protocol with clear decision points for cardiovascular risk in CKD, transplantation, dialysis, and CKD follow-up (16.6). A literature review emphasizes distinguishing biologic plausibility, observational association, and evidence of clinical benefit, because an improvement in a biomarker need not mean lower mortality (16.8). A report from the ECO 2026 congress recalls the role of obesity in albuminuria, hyperfiltration, and obesity-related glomerulopathy, as well as open questions about the safety of incretin-based therapy at a reduced glomerular filtration rate (16.1).

The third theme concerns the profession. The Medscape survey, in which 59% of physicians declared a willingness to consider lower pay for better balance, reflects a stated preference with self-selected respondents, and the 46% figure conflates burnout with depression, which are distinct constructs (16.9). Meta-analytic data are partly contradictory here: in practicing physicians, organization-directed interventions were more effective than individual ones, whereas in physicians in training, individual interventions had a small effect and organizational ones did not show a significant effect (16.9). For nephrology, limited cover in small centers is decisive (16.9). A commentary on events at the ADA is a personal position whose claims should be verified in primary sources, and it links politicization and research cuts with a possible slowing of progress in diabetic kidney disease (16.5). A BBC report on threats to remove the kidneys of kidnapped migrants in Libya relies on testimonies and photographs but could not fully verify organ removal (16.3).

This yields a framework for the outpatient clinic: speak clearly without infantilization and check understanding continuously, take the patient's description of their experience as a clinically relevant signal, ask what is hardest for the patient today, and reflect the answer in the plan (16.2, 16.4, 16.7). Translate congress news into predefined decision points and verify it in primary records (16.6, 16.5). In victims of violence with oliguria, hematuria, or changes in creatinine, consider urgent nephrologic evaluation with forensic documentation (16.3). At the unit level, prioritize teams with mutual cover, predictable scheduling, and reduced administrative burden over individual advice on resilience (16.9).

16.1 ECO 2026: the European Congress on Obesity will be held in Istanbul

Published May 12, 2026 · [online version \(Slovak\)](#)

Abstract. A brief report on the 33rd European Congress on Obesity, ECO 2026, organized by the European Association for the Study of Obesity and to be held from May 12 to 15, 2026, in Istanbul. The text outlines the range of expected topics, from epidemiology and the pharmacologic treatment of obesity to bariatric surgery and the relationship between obesity and chronic kidney disease, and explains why the meeting is relevant to nephrology practice and the management of risk factors.

Summary

- The meeting will be held from May 12 to 15, 2026, in Istanbul as the 33rd European Congress on Obesity, organized by EASO, and is intended for physicians, researchers, public health specialists, dietitians, psychologists, and pharmacists.
- The expected topics include the epidemiology and prevention of obesity, pharmacologic treatment, metabolic and cardiovascular complications, the relationship with type 2 diabetes, obesity and chronic kidney disease, bariatric and metabolic surgery, nutrition and physical activity, new drugs, and public health strategies.
- The author notes that obesity contributes to the development of hypertension, type 2 diabetes, albuminuria, glomerular hyperfiltration, and obesity-associated glomerulopathy, and that in advanced kidney disease it affects transplant preparation, dialysis treatment, and nutritional status.
- For practice, the author points out that body mass index in patients on dialysis may not reliably reflect nutritional status, sarcopenia, or inflammatory burden, and that with incretin-based treatment, questions remain about safety at a reduced glomerular filtration rate and about the effect on albuminuria.

16.2 A condescending tone in medicine: when plain language stops being respectful

Published June 5, 2026 · [online version \(Slovak\)](#)

Abstract. A commentary by Aryu Anthony Kamyab on the Medscape portal examines when the justified simplification of medical communication turns into infantilization of the patient. The author distinguishes health literacy from intelligence, gives examples of a condescending tone including elderspeak, and argues that communication style affects not only comprehension but also trust, willingness to ask questions, and cooperation with treatment. This is a professional reflection combined with patient experience, not a quantitative study. The practical message is to tailor the level of detail to the individual patient and to verify comprehension as the conversation proceeds.

Summary

- Avoiding jargon has a clear rationale: a patient who understands neither the diagnosis nor the discharge instructions is at risk of a treatment error, and technical words can mislead when, for example, the patient understands the word chronic to mean severe or serious.
- The author distinguishes health literacy from intelligence: an educated person need not know the terminology, yet instead of the term paresthesia it is enough to explain tingling or pins and needles related to the nerves, which is comprehensible without being childish.

- As an example of condescension, the author cites explaining chemotherapy as treatment of naughty cells; in the author's view this may be well intentioned, yet to an adult patient it comes across as demeaning and affects both trust and willingness to ask questions.
- A distinct form is elderspeak, that is, diminutives, forms of address such as sweetie, needlessly raised volume, or the plural as in “are we ready to go for our examination,” which the author reports may, according to studies, increase resistance to care and worsen cooperation.
- The author recommends tailoring the explanation to the individual and beginning with the question of what the patient would like to know and whether they want a brief explanation or a more detailed overview of treatment options.
- The practical conclusion for the outpatient clinic: speak clearly and appropriately, sometimes use a technical term and explain it immediately, verify comprehension as you go, and do not confuse clarity with infantilization.

16.3 BBC: kidnappings of migrants in Libya and threats of kidney removal

Published June 16, 2026 · [online version \(Slovak\)](#)

A condensed summary of this article is not included in this edition of the compendium. The full text is available as article 16.3 in the English edition of SK Nefro Báza 1 and online (in Slovak).

16.4 When we feel ill: what medicine misses and why it matters in nephrology too

Published June 26, 2026 · [online version \(Slovak\)](#)

Abstract. This review reflects on the difference between disease as a measurable disorder and illness as the lived experience of not being oneself. The text argues that the two dimensions often coincide, yet neither fully depends on the other, and that a diagnosis illuminates part of the picture while obscuring another part. For nephrology, the author draws concrete implications for conducting conversations and evaluating the patient's account in chronic kidney disease (CKD) and dialysis.

Summary

- The text distinguishes disease, a nameable disorder of organ function, from illness, a disruption of ordinary functioning, routine, sense of security, and relationship to one's own body; when patients say they are ill, they usually describe the latter.
- Recognition of illness often precedes the diagnostic process, so the patient has already lived for some time in a state that becomes actionable information for medicine only later.
- The author points to the paradox that marked distress may have no clear-cut finding; absence of evidence does not diminish the reality of suffering, and a patient with relatively stable laboratory values may experience substantial functional limitation.
- In nephrology, this means treating the patient's account as a clinically relevant signal rather than an addendum to the history, and deliberately assessing factors that cannot be reduced to a single parameter, such as sleep quality, treatment tolerance, and functional capacity.
- No single diagnostic label captures the combination of limitations, fatigue, fluctuating symptoms, and psychological burden in CKD and dialysis, so the author recommends asking what is hardest for the patient today, what has changed, and what barriers have emerged.
- The text concludes by calling for a balance between objective measurement and attention to lived experience; it is educational and awareness-raising material and does not replace individual clinical decision-making by the treating team.

16.5 The new ADA as a prompt for reflection: research grants, political interference, and the impact on nephrology

Published June 27, 2026 · [online version \(Slovak\)](#)

Abstract. A commentary on an article by Irl B. Hirsch on the Medscape portal that critically describes the direction of the American Diabetes Association and events at the ADA Scientific Sessions, including the confiscation of several attendees' badges in connection with the distribution of an editorial from Diabetes Care. It is the author's personal opinion, not a clinical study. The text links these events to the broader question of how research cuts and politicization affect innovation in diabetology and nephrology.

Summary

- Irl B. Hirsch describes actions by ADA leadership that led to attendees' badges being revoked and confiscated, and links them to the distribution of a Diabetes Care editorial on the effects of NIH research cuts on the diabetes pipeline.
- As background, he notes that Jay Bhattacharya was to have been the keynote speaker and was subsequently replaced by Richard Woychik, and the text also has a broader political dimension concerning research priorities.
- The author's thesis is that interference with the grant environment may slow discovery over the long term, reduce the number of projects, and shift priorities away from basic and translational research toward short-term or politically favored directions.
- He draws indirect implications for nephrology: progress in diabetic kidney disease depends on research into biomarkers and mechanisms of progression, and fewer grants may slow risk stratification and delay updates to clinical guidelines.
- The author also stresses that this is his own perspective and account of events, so conflicting claims about who acted, when, and why should be verified against the organization's primary materials and independent reports.
- The practical message for the professional community is to maintain stable long-term research funding, academic freedom, transparent peer review, and the mission-driven nature of professional societies and journals.

16.6 Post-congress analysis of ERA 2026 for nephrology: how to translate the highlights into practice in CKD, dialysis, transplantation, and cardiovascular risk

Published July 1, 2026 · [online version \(Slovak\)](#)

Abstract. A deliberately high-level post-congress analysis of ERA 2026 that cites no specific trials or doses and focuses on how to translate the main findings into everyday decision-making in four areas: cardiovascular risk in chronic kidney disease (CKD), transplantation, dialysis, and CKD monitoring itself. The author recommends developing one's own post-congress protocol and verifying specific recommendations directly in the congress recordings. The goal is to translate the main findings into clear decision points and a shared logic for the nephrology and dialysis team.

Summary

- In CKD, cardiovascular risk is often dominant and decisions do not rest on eGFR alone; the author recommends defining which patients are cardiovascular-dominant based on history of events, blood pressure, albuminuria, disease course, and metabolic factors.
- He proposes creating a CKD-CV check for every visit that also covers comorbidity and risk profile, and eliminating inconsistent parallel plans across nephrology, diabetology, and cardiology.
- In transplantation, he recommends focusing on monitoring and prevention: specifying what is assessed when graft function worsens, not settling for the creatinine trend alone, and setting clear escalation decision points according to the local algorithm.
- In dialysis, the author considers three areas decisive: patient stability (interdialytic symptoms, hypotensive episodes, and tolerance), safety (including infections, vascular access, and electrolyte fluctuations),

and long-term goals such as hospitalizations, nutritional status, anemia, and mineral and bone metabolism.

- A chairside dialysis protocol should clearly define which parameters require intervention and what the first-line adjustment is, so that the dialysis and nephrology teams work within the same logic.
- In CKD monitoring, the author warns against measuring everything without knowing what it changes, and recommends defining in advance which results signal a change in therapy, when and for what purpose a test should be repeated, and what to do when a target is not achieved.

16.7 Qualitative research in nephrology: what numbers do not reveal about decision-making in CKD

Published July 15, 2026 · [online version \(Slovak\)](#)

Abstract. This methodological review explains which questions qualitative research in nephrology answers and how to read it critically. The main source is a professional news commentary, not an original study or a systematic review, so the author verified the methodological claims against reviews and the COREQ and SRQR guidelines. As an example, the author cites a secondary analysis of interviews with 20 US nephrologists about hospice care, whose findings cannot be converted into population prevalence.

Summary

- Qualitative research answers a different type of question than an experiment or a registry: it explores meanings, experiences, and the context of decision-making, such as treatment burden, choice of dialysis modality, acceptance of vascular access, or decisions about conservative care.
- Interviews and focus groups are data collection methods, not a methodology; the framework may draw on phenomenology, grounded theory, ethnography, or a qualitative descriptive approach, and adding an open-ended question at the end of a questionnaire does not create a sound mixed-methods design.
- Sample size is judged not by a power calculation but by the information power of the data; thematic saturation is neither a universal nor the sole criterion of quality, and reflexivity means documenting the researcher's influence on the data.
- COREQ and SRQR help authors report the essential elements of their work transparently, but they are not scoring systems that confirm methodological quality; warning signs include claims of efficacy or prevalence derived from themes and selective quotations.
- In the cited example, a secondary analysis of semistructured interviews with 20 US nephrologists showed that they often viewed dialysis and hospice as mutually exclusive models and were unsure who should lead the conversation and coordinate care; participants came from two tertiary centers and four community practices in a single city, 85% were categorized as White, and 75% were men.
- The author notes that palliative care is not synonymous with hospice and can be provided concurrently with dialysis, and that US reimbursement rules for hospice and dialysis do not apply to Slovakia.
- For the outpatient clinic, he recommends open-ended questions, such as what the patient fears most about treatment, and translating the answer into a concrete plan.

16.8 Twelve books that can help physicians better understand illness, the patient, and themselves

Published August 2, 2026 · [online version \(Slovak\)](#)

Abstract. An overview of twelve titles compiled for Medscape by journalist Ted Spiker based on recommendations from people who bridge medicine and literature. It is neither a scientific ranking nor a mandatory canon. The text supplements the selection with a critical appraisal of the evidence on narrative medicine and, for individual books, corrects popular historical oversimplifications, for example about Lister and Sem-

melweis. For nephrology, it emphasizes the role of narrative competence in long-term therapeutic relationships and suggests asking three questions about one's own clinical assumptions for each book.

Summary

- The author describes the evidence on narrative medicine as cautiously encouraging rather than definitive: a 2019 review of 36 publications found measurable changes in engagement, attitudes, knowledge, and skills, but no clear evidence of lasting change in behavior or in interactions with patients.
- A 2026 critical review included 17 studies of arts-based education with 835 participants, of which only four evaluated narrative medicine, and high heterogeneity precluded meta-analysis; the authors therefore recommend locally facilitated programs with realistic expectations.
- The text corrects historical oversimplifications: Lister began using antisepsis systematically in 1865, but he was not the sole originator of modern infection prevention, and the public demonstration of ether anesthesia had already taken place in 1846; for Semmelweis, the historical record does not allow a reliable retrospective diagnosis of his decline.
- For the book *Rigor Mortis*, the author stresses distinguishing among biological plausibility, experimental results, observational associations, and evidence of clinical benefit, because an improvement in a biomarker does not necessarily mean lower mortality, fewer hospitalizations, or better quality of life.
- A natural experiment is not automatically equivalent to a randomized trial, and its validity depends on data quality, the identification strategy, and control of bias; population-level findings should not be used to evaluate an individual physician or patient.
- For nephrology, the author sees the value of narrative competence in conversations about choosing among home therapy, in-center dialysis, transplantation, and conservative management, about the real burden of the regimen, adherence, fatigue, pain, pruritus, and loss of roles, and about dialysis withdrawal and end-of-life care.
- The text concludes that literature is neither a clinical guideline nor epidemiologic evidence, and by itself makes no one a better physician.

16.9 Physicians are not just asking for higher pay. Increasingly, they want control over their own time back

Published August 5, 2026 · [online version \(Slovak\)](#)

Abstract. A Medscape survey found that 59% of physicians surveyed would consider a lower salary in exchange for better work-life balance, compared with 63% in the previous survey, and 46% reported burnout, depression, or both. The text critically frames this finding as a stated preference rather than a decision and supplements it with meta-analytic evidence that a substantial part of the solution lies at the level of the workplace, not the individual.

Summary

- About three quarters of respondents attached great importance to time with family and vacation, and about eight in ten considered leisure activities important for mental health; the problem is not ignorance of a healthy lifestyle but the inability to practice it at work.
- The author points out that the 46% figure combines distinct constructs: burnout is a syndrome of chronic occupational stress with emotional exhaustion and cynicism, not a standalone diagnosis, so the figure does not indicate the proportion of physicians with diagnosed depression.
- An automatic pay cut is not an intervention against burnout but merely a pay cut; a meaningful trade-off requires shorter or more flexible working hours, predictable scheduling, and limits on after-hours administrative work.
- A meta-analysis of controlled interventions published by Maria Panagioti and colleagues included 20 comparisons from 19 studies with 1,550 physicians and showed a small but significant reduction in burnout (standardized mean difference, -0.29), with organization-directed interventions achieving -0.45 (95% CI, -0.62 to -0.28) and individual-focused interventions only -0.18 .

- A more recent meta-analysis in physicians in training (33 studies, 2,536 participants) showed a small effect of individual interventions on emotional exhaustion (Cohen's d , -0.25) and depersonalization (-0.17), whereas organizational interventions showed no significant effect and accounted for less than a quarter of the studies.
- Limitations of the survey include incomplete reporting of respondent selection and response rate, self-selection, self-report, a hypothetical question that did not specify the size of the pay cut, and limited transferability of US data.
- In nephrology, limited cover capacity is decisive: if a center operates with two nephrologists, protected time off for one shifts the workload to the other, so the author calls for teams with adequate cover and staffing standards.

APPENDIX

Cheat sheets, checklists, and algorithms for practice

Concise overviews, checklists, and algorithms for quick orientation at the bedside and in the outpatient clinic. Number of articles: 11.

A.1 Anemia of CKD 2026 in practice: an algorithm from diagnosis to ESA and HIF-PHI according to the KDOQI US Commentary

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. A practical version of the KDOQI US Commentary on the KDIGO 2026 guideline for anemia in chronic kidney disease, restructured as a step-by-step algorithm for both the outpatient clinic and the dialysis unit. The approach begins with confirming anemia and searching for its cause, proceeds to targeted iron repletion with clear safety thresholds, and only then addresses ESA or HIF-PHI, so as to minimize risk and the number of unnecessary transfusions. The author emphasizes that hemoglobin is not the sole decision criterion.

Summary

- Evaluation is warranted for worsening exercise tolerance, fatigue, dyspnea, a declining hemoglobin on follow-up, or a change in clinical status; the basic panel comprises a complete blood count with MCV and RDW, reticulocytes, ferritin, and TSAT, supplemented as needed by assessment of inflammation, hemolysis, nutritional factors, and screening for bleeding.
- In patients on hemodialysis, the commentary supports a more proactive intravenous approach and gives as a practical trigger the combination of ferritin ≤ 500 ng/mL and TSAT $\leq 30\%$, because oral iron is less effective and harder to manage.
- The safety stop signal is to withhold routine iron administration when ferritin is > 700 ng/mL or TSAT is $\geq 40\%$; outside hemodialysis, the choice between oral and intravenous iron is individualized according to tolerance, availability, the required speed of correction, and whether the deficiency is absolute or masked by inflammation.
- The highest risk of hypophosphatemia is associated with ferric carboxymaltose; therefore, unexplained weakness or bone pain after intravenous iron should prompt consideration of phosphate, which should be included in monitoring of at-risk patients receiving repeated doses, and confirmed hypophosphatemia should lead to reconsideration of the iron formulation.
- ESA and HIF-PHI are appropriate only after correctable causes have been addressed and within shared decision-making that includes transfusion risk; for CKD G5D, the commentary gives initiation at a hemoglobin of approximately ≤ 9.0 to 10.0 g/dL and, during maintenance, keeping hemoglobin below an upper limit of approximately 11.5 g/dL.
- Once correctable causes have been addressed, ESA is generally the first-line option, and HIF-PHI is chosen more cautiously according to risks and the clinical situation; if hemoglobin is too high, the dose should be adjusted and it should be rechecked whether another cause of anemia has been overlooked.

A.2 Checklist for practice: anemia of CKD according to KDIGO 2026 and the KDOQI US Commentary

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. This checklist summarizes the practical management of anemia in patients with chronic kidney disease according to KDIGO 2026 and the KDOQI US Commentary in eight steps: confirmation of anemia,

the iron laboratory profile, the search for a correctable cause, the decision on iron with safety thresholds, special attention to ferric carboxymaltose and phosphate, the decision on ESA or HIF-PHI, monitoring intervals, and documentation that allows a retrospective audit. It emphasizes that anemia of CKD is often multifactorial.

Summary

- The first step is to confirm anemia on the basis of hemoglobin and its trend, to add a complete blood count with MCV and RDW and reticulocytes, and to clinically assess fatigue, dyspnea, infection, inflammation, bleeding, and nutritional status.
- Ferritin and TSAT are interpreted together with the clinical picture and the inflammatory status, not in isolation; if the picture does not fit anemia of CKD, nutritional or hematologic causes should be considered, and suspected bleeding warrants a planned search for its source.
- In hemodialysis, it is reasonable to start intravenous iron at the combination of ferritin ≤ 500 ng/mL and TSAT $\leq 30\%$, whereas routine iron should be withheld at ferritin > 700 ng/mL or TSAT $\geq 40\%$; outside dialysis, the route of administration is individualized.
- For ferric carboxymaltose, the checklist recommends building phosphate monitoring into the plan in advance, particularly with repeated doses, and replacing phosphate and adjusting the approach if symptoms of hypophosphatemia occur.
- ESA or HIF-PHI comes into consideration only after correctable causes have been addressed and within shared decision-making; in CKD G5D, ESA is typically considered at a hemoglobin of approximately ≤ 9.0 to 10.0 g/dL, and during maintenance hemoglobin is kept below an upper limit of approximately 11.5 g/dL.
- Hemoglobin, ferritin, and TSAT are monitored every 3 months in non-dialysis CKD and in CKD G5PD and every 1 to 3 months in CKD G5HD, with reassessment in the event of infection, bleeding, hospitalization, or deteriorating nutrition.
- Documentation for audit should include current values, their interpretation and the rationale for the chosen approach, the thresholds for starting or withholding iron, the target hemoglobin with a monitoring plan, and, for ferric carboxymaltose, a phosphate monitoring plan.

A.3 Checklist for practice (one A4 page): anemia of CKD (HD and non-HD)

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. A one-page form for quick use in the outpatient clinic and on dialysis guides the user from confirmation of anemia and the iron laboratory profile to the choice of treatment by patient group, special attention to ferric carboxymaltose and phosphate, the decision on ESA or HIF-PHI, and monitoring intervals. Its purpose is also to have a clear written record of why a particular approach was chosen. It includes fields for recording ferritin and TSAT values, the formulation used, and the date of the next follow-up.

Summary

- The initial steps comprise hemoglobin and its trend with dates, a complete blood count with MCV, RDW, and reticulocytes, symptoms and exercise tolerance, and a rapid assessment of inflammation, infection, bleeding, and nutrition, with a plan for further workup if the picture does not correspond to anemia of CKD.
- In CKD G5HD, the form specifies starting intravenous iron at ferritin ≤ 500 ng/mL and TSAT $\leq 30\%$ and withholding routine iron at ferritin > 700 ng/mL or TSAT $\geq 40\%$, with the formulation and dosing schedule recorded; in CKD without hemodialysis, oral or intravenous iron is chosen according to ferritin, TSAT, tolerance, and the need for rapid correction.
- For ferric carboxymaltose, the form records the phosphate monitoring plan, the occurrence of symptoms such as weakness or bone pain, and the date of the phosphate check; ESA or HIF-PHI is considered after correctable causes have been addressed, in CKD G5D typically at a hemoglobin of approximately ≤ 9.0 to 10.0 g/dL, with maintenance below an upper limit of approximately 11.5 g/dL.

- Hemoglobin, ferritin, and TSAT checks are scheduled every 3 months in non-HD CKD and CKD G5PD and every 1 to 3 months in CKD G5HD, with reassessment in the event of infection, hospitalization, or a rapid decline in hemoglobin.

A.4 Checklist for practice: anemia of CKD for dialysis and the outpatient clinic

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. A pair of parallel one-page A4 forms for anemia of chronic kidney disease, one for dialysis (hemodialysis and peritoneal dialysis) and the other for the non-dialysis outpatient clinic. Both follow the same logic: confirm anemia, assess iron status, identify correctable causes, and then make a safe treatment decision, with a header identifying the patient, the physician, and the purpose of the record, which is initiation, adjustment, or monitoring. The documentation section supports a retrospective audit of decisions.

Summary

- The dialysis version starts with hemoglobin and its trend, a complete blood count with MCV, RDW, and reticulocytes, symptoms, and questions about inflammation, infection, bleeding, or nutritional decline, with a plan for additional workup if the results do not fit anemia of CKD.
- The decision on intravenous iron follows the threshold of ferritin ≤ 500 and TSAT ≤ 30 for initiation and the stop signal of ferritin > 700 or TSAT ≥ 40 for withholding or not escalating routine iron, with the formulation, dosing schedule, and plan for follow-up laboratory tests recorded.
- For ferric carboxymaltose, the form confirms a phosphate monitoring plan according to the local protocol and the clinical picture, and symptoms such as weakness or bone pain require an immediate phosphate check and adjustment of the plan.
- ESA or HIF-PHI is chosen after correctable causes have been addressed and within shared decision-making; in CKD G5D, ESA is typically considered at a hemoglobin of approximately ≤ 9.0 to 10.0 g/dL, and during maintenance hemoglobin is kept below an upper limit of approximately 11.5 g/dL, with the reason for the choice stated.
- The outpatient version additionally requires justification of the preferred iron route according to tolerance, speed of correction, availability or cost, and the laboratory picture; initiation of ESA according to symptoms, hemoglobin trend, and risks rather than a number alone; and reconsideration of intravenous administration if oral iron is ineffective or not tolerated within 1 to 3 months.
- Monitoring is set at 1 to 3 months in CKD G5HD, or according to the local protocol in G5PD, and at 3 months in the outpatient clinic, with reassessment in the event of infection, hospitalization, bleeding, or a rapid decline in hemoglobin.

A.5 Practical appendix: an algorithm for the management of IgA nephropathy according to KDIGO 2025, taking into account the KDOQI US Commentary

Published June 19, 2026 · [online version \(Slovak\)](#)

Abstract. An algorithm for the management of IgA nephropathy according to KDIGO 2025, taking into account the KDOQI US Commentary, condensed into a concise one-page A4 practical appendix. It leads from recognition of suspected disease in persistent hematuria or proteinuria, through confirmation of the diagnosis by biopsy and estimation of the risk of progression, to treatment targeting disease mechanisms and slowing the loss of kidney function, including a documentation sentence for the medical record. It is a framework approach that does not replace individual clinical decision-making.

Summary

- The diagnosis of IgA nephropathy is confirmed by kidney biopsy, which should be considered in an adult with proteinuria ≥ 0.5 g/day or equivalent and should not be regarded as optional if the result would substantially change treatment and there are no contraindications.

- After confirmation, risk is determined mainly by eGFR and proteinuria, using the Oxford classification; the goal is to reduce proteinuria as early and as much as possible, at least to below 0.5 g/day and ideally to below 0.3 g/day, if achievable.
- The foundation for all patients is sodium intake below 2 g/day, cessation of smoking and vaping, weight control, aerobic activity as tolerated, a target blood pressure of approximately $\leq 120/70$ mm Hg, RAS blockade with an ACE inhibitor or ARB if indicated and tolerated, and consideration of SGLT2 inhibition.
- Specific therapy should be considered in patients at higher risk of progression, particularly with persistent proteinuria despite basic kidney-protective treatment, taking into account eGFR, histology, comorbidities, and treatment availability; proteinuria is assessed by its trend, not by a single value.
- If Nefecon is available, the algorithm recommends considering a 9-month course in at-risk patients according to KDIGO and explaining to the patient that the main assessment is the longer-term trend in eGFR and proteinuria, and that after discontinuation only a partial persistent effect (legacy effect) on proteinuria may remain.
- If Nefecon is not available, a reduced-dose systemic glucocorticoid regimen with risk-based prophylaxis is considered, with attention to infection risk, gastroprotection, and bone protection.
- Monitoring tracks proteinuria as the main indicator of response, together with eGFR and blood pressure, and safety parameters during glucocorticoid therapy; an unexpected course should prompt reassessment of adherence and the differential diagnosis.

A.6 Acid-base: cheat sheet

Published June 23, 2026 · [online version \(Slovak\)](#)

Abstract. A compact aid for the systematic interpretation of arterial acid-base status in steps: pH, primary disorder, compensation, anion gap, delta ratio, and clinical context. It includes reference ranges, compensation formulas including Winter's formula, albumin correction of the anion gap, the GOLD MARK mnemonic for high anion gap acidosis, differentiation of normal anion gap acidosis by the urine anion gap, and classification of metabolic alkalosis by urine chloride. The aid also refers to an acid-base calculator.

Summary

- Reference values: pH 7.35 to 7.45, pCO₂ 35 to 45 mm Hg, HCO₃⁻ 22 to 26 mmol/L, base excess -2 to +2 mmol/L, and anion gap 8 to 12 mmol/L; a normal pH with an abnormal pCO₂ indicates a mixed disorder.
- Deviation from the expected compensation indicates an additional disorder: in metabolic acidosis, Winter's formula applies, $pCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$; in metabolic alkalosis, pCO₂ rises by about 0.7 mm Hg per 1 mmol/L of HCO₃⁻.
- In respiratory acidosis, HCO₃⁻ rises by 1 acutely and by 3.5 to 4 chronically for every 10 mm Hg increase in pCO₂; in respiratory alkalosis, it falls by 2 and 4 to 5, respectively.
- The anion gap is calculated as $Na^+ - (Cl^- + HCO_3^-)$ and should always be corrected for albumin: it rises by about 2.5 mmol/L for every 10 g/L below 40 g/L; otherwise, an elevated gap may remain masked.
- For high anion gap acidosis, the GOLD MARK mnemonic applies (glycols, oxoproline, L-lactate, D-lactate, methanol, aspirin, renal failure, ketoacidosis); in normal anion gap acidosis, the urine anion gap distinguishes extrarenal HCO₃⁻ loss from renal tubular acidosis.
- The delta ratio $(AG - 12) / (24 - HCO_3^-)$ reveals a hidden mixed disorder: below 0.4 indicates hyperchloremic acidosis, 0.4 to 1.0 a mixed form, 1.0 to 2.0 pure HAGMA, and above 2.0 alkalosis or chronic respiratory acidosis.
- Metabolic alkalosis is classified by urine chloride: below 20 mmol/L it is chloride-responsive (vomiting, nasogastric suction), and above 20 mmol/L chloride-resistant (hyperaldosteronism, hypokalemia); clinical context takes precedence over the numbers.

A.7 Diuretics: cheat sheet

Published June 23, 2026 · [online version \(Slovak\)](#)

Abstract. This quick reference on diuretics covers drug classes and their sites of action, from loop diuretics to vaptans, dose equivalence of loop diuretics, thiazides, and mineralocorticoid receptor antagonists, principles for managing diuretic resistance, and what to monitor with each class. It identifies finerenone as a nonsteroidal mineralocorticoid receptor antagonist used primarily as a kidney-protective agent in chronic kidney disease (CKD), not as a diuretic. It also covers the bioavailability of oral loop diuretics and the oral-to-intravenous dose ratio of furosemide.

Summary

- Loop diuretics act on NKCC2 in the thick ascending limb, thiazide and thiazide-like diuretics on NCC in the distal convoluted tubule, mineralocorticoid receptor antagonists and ENaC blockers in the collecting duct, acetazolamide in the proximal tubule, and tolvaptan at the V2 receptor.
- Equivalent doses are furosemide 40 mg, torsemide 10 to 20 mg, and bumetanide 1 mg; the bioavailability of furosemide is approximately 50% and variable, versus 80 to 100% for the other two, and the oral-to-intravenous ratio for furosemide is 2:1.
- In diuretic resistance, the threshold dose of the loop diuretic must be exceeded, a thiazide added for sequential nephron blockade, and a continuous infusion considered instead of boluses; metolazone and chlorthalidone retain part of their effect even at eGFR below 30 mL/min/1.73 m².
- Monitor for hypokalemia and hypomagnesemia with loop diuretics, the typical hyponatremia with thiazides, hyperkalemia with mineralocorticoid receptor antagonists and ENaC blockers, particularly in combination with RAAS blockade, metabolic acidosis with acetazolamide, and the rate of serum sodium correction with tolvaptan.

A.8 Electrolytes: cheat sheet

Published June 23, 2026 · [online version \(Slovak\)](#)

Abstract. This cheat sheet on disorders of sodium, potassium, calcium, magnesium, and phosphate lists serum reference ranges, the main causes of each disorder, typical ECG findings, safe correction limits, and four golden rules. It emphasizes that in sodium disorders the rate of correction, not the target value, is decisive, and that magnesium must be corrected first, because without it refractory hypokalemia and hypocalcemia often fail to resolve. A formula for albumin-corrected calcium completes the overview.

Summary

- Serum reference values: sodium 135 to 145 mmol/L, potassium 3.5 to 5.0 mmol/L, total calcium 2.15 to 2.55 mmol/L, ionized calcium 1.15 to 1.30 mmol/L, magnesium 0.70 to 1.00 mmol/L, and phosphate 0.80 to 1.45 mmol/L.
- In hyponatremia, tonicity and volume status should be determined first, and the correction rate kept at ≤ 8 to 10 mmol/L per 24 h, or ≤ 6 in high-risk patients, to prevent osmotic demyelination syndrome; in hypernatremia, the limit is ≤ 10 to 12 mmol/L per 24 h.
- Hypokalemia manifests on the ECG as flattened T waves, U waves, and a prolonged QT interval; intravenous potassium is given peripherally at up to 10 mmol/h, always with magnesium replacement. In hyperkalemia with ECG changes, the first step is intravenous calcium, which by itself does not lower serum potassium.
- Calcium is assessed either corrected for albumin using the formula $\text{Ca corr} = \text{Ca total} + 0.02 \times (40 - \text{albumin g/L})$ or as ionized calcium; hypocalcemia prolongs the QT interval, hypercalcemia shortens it, and in magnesium deficiency magnesium is corrected first.
- Hypomagnesemia causes refractory hypokalemia and hypocalcemia, severe hypophosphatemia below 0.3 mmol/L carries a risk of muscle weakness, rhabdomyolysis, and respiratory failure, and hyperphosphatemia in CKD is managed with diet and phosphate binders.

A.9 Intravenous fluids: cheat sheet

Published June 23, 2026 · [online version \(Slovak\)](#)

Abstract. This overview of the composition and selection of intravenous fluids lists electrolyte concentrations and osmolarity per liter for common products, the practical consequences of tonicity, a comparison of balanced crystalloids with 0.9% NaCl, and fluid choice by clinical situation, from resuscitation to maintenance therapy. It is complemented by a list of pitfalls, including colloids not recommended in critical illness and the risk of hospital-acquired hyponatremia with hypotonic fluids. For the comparison with 0.9% NaCl, it cites the SMART, BaSICS, and PLUS trials.

Summary

- Composition per liter: 0.9% NaCl contains Na⁺ and Cl⁻ 154 each at approximately 308 mOsm/L; Ringer's lactate contains Na⁺ 130, Cl⁻ 109, K⁺ 4, Ca 1.5, and lactate 28 at approximately 273 mOsm/L; Plasma-Lyte 148 contains Na⁺ 140 and Cl⁻ 98 at approximately 295 mOsm/L; and 3% NaCl contains 513 at approximately 1,027 mOsm/L.
- 5% dextrose is nearly isotonic in vitro, but after glucose is metabolized it provides free water and acts as a hypotonic fluid; hypotonic fluids are a common cause of hospital-acquired hyponatremia, so isotonic fluids are preferred for maintenance therapy.
- The high chloride content of 0.9% NaCl causes hyperchloremic metabolic acidosis when large volumes are given and may impair renal perfusion, so balanced crystalloids are preferred for resuscitation; the SMART, BaSICS, and PLUS trials favor them.
- Ringer's lactate contains calcium, so it is not given in the same line as citrated blood products or with ceftriaxone in neonates; in severe hyperkalemia, balanced fluids are tolerated, whereas 0.9% NaCl may worsen serum potassium through acidosis.
- Choice by situation: a balanced crystalloid for hypovolemia, 5% dextrose or 0.45% NaCl for hyponatremia, 3% NaCl in boluses of 100 to 150 mL for symptomatic severe hyponatremia with a limit of 8 mmol/L per 24 h, and an isotonic fluid for hypercalcemia; hydroxyethyl starch colloids are not recommended in critical illness and sepsis.

A.10 Complement and kidney disease: cheat sheet

Published June 24, 2026 · [online version \(Slovak\)](#)

Abstract. This cheat sheet on the complement system in nephrology summarizes the three activation pathways and their triggers, interpretation of serum C3 and C4 in relation to individual glomerulopathies, an overview of complement-mediated kidney diseases with their key findings, the regulators and genetics of atypical hemolytic uremic syndrome and C3 glomerulopathy, and complement-targeted drugs, including a reminder about vaccination before C5 inhibition. The interpretation is supplemented by an overview of the typical entities associated with each C3 and C4 pattern.

Summary

- The classical pathway is triggered by immune complexes via C1q, the lectin pathway by pathogen carbohydrates via MBL and ficolins, and the alternative pathway by spontaneous tick-over involving factor B, factor D, and properdin; all converge on C3, C5, and the membrane attack complex C5b-9.
- Low C3 with normal C4 points to the alternative pathway, that is, C3 glomerulopathy or poststreptococcal glomerulonephritis; low C3 and C4 point to the classical pathway in lupus nephritis and cryoglobulinemic glomerulonephritis, whereas normal C3 and C4 are typical of IgA nephropathy and ANCA-associated nephritis.
- In C3 glomerulopathy, the key findings are C3-dominant immunofluorescence and C3 nephritic factor; in atypical hemolytic uremic syndrome, they are thrombotic microangiopathy with thrombocytopenia, hemolysis, and acute kidney injury (AKI), and mutations in CFH, CFI, MCP, C3, or CFB.

- Among targeted drugs, the anti-C5 agents eculizumab and ravulizumab are first-line therapy for aHUS, avacopan blocks the C5a receptor in ANCA-associated vasculitis, iptacopan targets factor B, and pegcetacoplan targets C3; meningococcal vaccination is mandatory before C5 inhibition.

A.11 Membranous nephropathy: cheat sheet

Published June 24, 2026 · [online version \(Slovak\)](#)

Abstract. This cheat sheet on membranous nephropathy, the most common cause of nephrotic syndrome in nondiabetic adults, summarizes the target podocyte antigens and their proportions, the distinction between primary and secondary forms, the diagnostic workup from anti-PLA2R antibodies to biopsy, and KDIGO risk stratification with the corresponding treatment. The disease is driven by subepithelial immune complexes and complement activation. Supportive care includes kidney protection, thromboprophylaxis when serum albumin is below 25 g/L, and monitoring of the anti-PLA2R titer as a marker of response.

Summary

- PLA2R accounts for approximately 70% of primary cases, and the serum antibody correlates with activity and prognosis; THSD7A accounts for approximately 1 to 3%, NELL1 for approximately 5 to 10%, often secondary, and EXT1 and EXT2 are associated with lupus and autoimmunity.
- The primary autoimmune form accounts for approximately 75 to 80% of cases, with IgG4-dominant immunofluorescence; secondary causes include solid tumors, warranting age-appropriate screening especially with NELL1 and THSD7A, class V lupus, hepatitis B and C, and drugs such as NSAIDs or penicillamine.
- Anti-PLA2R positivity supports the primary form and often allows diagnosis without biopsy; a falling titer precedes remission, and a persistent titer indicates a higher risk of relapse. Biopsy shows subepithelial deposits, spikes on silver staining, and granular IgG.
- According to KDIGO, low-risk disease is managed with supportive kidney protection and monitoring, rituximab or a calcineurin inhibitor is considered at moderate risk, rituximab or cyclophosphamide with glucocorticoids is used at high risk with eGFR below 60 or proteinuria above 8 g/day, and cyclophosphamide with glucocorticoids is started promptly at very high risk.

List of abbreviations

A selection of the most common abbreviations. Abbreviations specific to individual trials are explained in the full text of the article.

ABI	ankle-brachial index
ACC	American College of Cardiology
ACE	angiotensin-converting enzyme
ACEi	angiotensin-converting enzyme inhibitor
ACR	American College of Rheumatology
ADA	American Diabetes Association
ADPKD	autosomal dominant polycystic kidney disease
AHA	American Heart Association
AI	artificial intelligence
AKI	acute kidney injury
ANCA	antineutrophil cytoplasmic antibodies
APOL1	apolipoprotein L1
ARB	angiotensin receptor blocker
ASCVD	atherosclerotic cardiovascular disease
ASN	American Society of Nephrology
AVF	arteriovenous fistula
AVG	arteriovenous graft
BMI	body mass index
BNP	B-type natriuretic peptide
CGM	continuous glucose monitoring
CI	confidence interval
CKD	chronic kidney disease
CKD-MBD	chronic kidney disease-mineral and bone disorder
CKM	cardiovascular-kidney-metabolic (syndrome)
CRP	C-reactive protein
CT	computed tomography
CVC	central venous catheter
CVOT	cardiovascular outcome trial
DKA	diabetic ketoacidosis
DKD	diabetic kidney disease
DOI	digital object identifier
EASD	European Association for the Study of Diabetes

ECG	electrocardiogram
eGFR	estimated glomerular filtration rate
EMA	European Medicines Agency
ERA	European Renal Association
ERBP	European Renal Best Practice
ESA	erythropoiesis-stimulating agent
ESC	European Society of Cardiology
ESKD	end-stage kidney disease
FDA	US Food and Drug Administration
FSGS	focal segmental glomerulosclerosis
GDMT	guideline-directed medical therapy
GFR	glomerular filtration rate
GIP	glucose-dependent insulintropic polypeptide
GLP-1	glucagon-like peptide 1
GLP-1 RA	GLP-1 receptor agonist
HbA1c	glycated hemoglobin
HD	hemodialysis
HDF	hemodiafiltration
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
HIF-PHI	hypoxia-inducible factor prolyl hydroxylase inhibitor
HLA	human leukocyte antigen
HR	hazard ratio
IgAN	IgA nephropathy
KDIGO	Kidney Disease: Improving Global Outcomes
KDOQI	Kidney Disease Outcomes Quality Initiative
KRT	kidney replacement therapy
LDL	low-density lipoprotein
LN	lupus nephritis
MACE	major adverse cardiovascular events
MASH	metabolic dysfunction-associated steatohepatitis
MASLD	metabolic dysfunction-associated steatotic liver disease
MCO	medium cut-off (dialysis membrane)
MGRS	monoclonal gammopathy of renal significance
MRA	mineralocorticoid receptor antagonist
NNT	number needed to treat
NSAID	nonsteroidal anti-inflammatory drug
nsMRA	nonsteroidal mineralocorticoid receptor antagonist
OL-HDF	online hemodiafiltration

OR	odds ratio
PD	peritoneal dialysis
PMID	PubMed identifier
POCUS	point-of-care ultrasound
PPI	proton pump inhibitor
PTH	parathyroid hormone
RAAS	renin-angiotensin-aldosterone system
RAASi	renin-angiotensin-aldosterone system inhibitor(s)
RCT	randomized controlled trial
RR	relative risk
SGLT2	sodium-glucose cotransporter 2
SmPC	summary of product characteristics
TMA	thrombotic microangiopathy
TMAO	trimethylamine N-oxide
TSAT	transferrin saturation
TTP	thrombotic thrombocytopenic purpura
UACR	urine albumin-to-creatinine ratio
UPCR	urine protein-to-creatinine ratio
UTI	urinary tract infection
WHO	World Health Organization
ŠÚKL	State Institute for Drug Control (Slovakia)

About the author

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Alongside clinical work, he writes specialist and literary prose, translates medical texts between Slovak and English, lectures, consults, and develops digital medical tools. He is the author of the nefro.polascin.net portal, which besides specialist articles offers clinical calculators, diagnostic tools, and cheat sheets for nephrology practice.

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387 specialist articles from the nefro.polasci.net portal condensed into abstracts and key points.

Each of the 16 chapters opens with a synthesis of its main messages. Chronic kidney disease and kidney protection, cardiovascular-kidney-metabolic syndrome, diabetes and incretin-based therapy, glomerular diseases, dialysis, transplantation, clinical pharmacology, nutrition, and artificial intelligence in medicine.

Quick orientation for the outpatient clinic, the ward, and the dialysis center, with numbering identical to the full edition of SK Nefro Báza 1.